CCB & VCS PROJECT DESCRIPTION:

CCB Version 3, VCS Version 3

ATELES REDD+

Document prepared by Biofílica Ambipar Environmental Investments S.A.

Project Title	Ateles REDD+
Version	v.1
Issue Date	November 11, 2023
Project Location	Pará, Brazil
Project Proponents	Biofílica Ambipar Environmental Investments S.A – Plinio Ribeiro (CEO), phone: +55 11 3073-0430; e-mail: plinio@biofilica.com.br AgroSB Agropecuária S.A. – Cristiano Soares (CEO), tel: +55 11 989907040; e-mail: cristiano.soares@agrosb.com.br
Prepared by	Biofílica Ambipar Environmental Investments S.A
Validation Body	EcoLance Private Limitated
Project Lifetime	July 1, 2022 to June 30, 2062 – 40 years
GHG accounting	July 1, 2022 to June 30, 2062 – 40 years
CCB Status History	First validation attempt
Gold Level Criteria	GL3 Gold Seal – Exceptional Biodiversity Benefits
Expected Verification Schedule	

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1. SUMMARY OF PROJECT BENEFITS

1.1. Unique Project Benefits

The summary results or impacts of the expected benefits in the Ateles REDD+ Project are in Table 1. below.

Table 1: Summary of expected benefits in the Ateles REDD+ Project.

Outcome or Impact Estimated by the End of Project Lifetime	Section Reference
1) Expected Climate Benefits: with the Ateles REDD+ Project, it is expected that after its life cycle, based on the first baseline defined for the Project, it will help mitigate climate change with a total avoided emissions of 32,121,834 tCO2eq. The deforestation avoided in the scenario with the Project is 12,631 hectares during the first 10 years of the Project and an average of 803,046 tCO2eq of reduced emissions.	3
2) The benefits for the communities located in the Project Area and other interested actors will be directed to the promotion of regional socioeconomic development, through the encouragement of alternative sustainable practices to deforestation, the expansion of value chains and the strengthening of the local market. Therefore, it is intended to influence social issues and the living conditions of community members, reducing social vulnerability and rural exodus, generating value in adapting to climate change, increasing the level of socioeconomic conditions and the quality of life of families. In addition, the Project aims to obtain and strengthen partnerships that aim to collaborate in the aggregation of generation of goods and services that promote economic and social well-being.	4
3) Expected benefits for Biodiversity: The Ateles REDD+ Project provides for the maintenance of fauna and flora in the Project Area, ensuring the protection and conservation of habitats and local biodiversity, including endemic species and with some degree of threat according to the IUCN RedList. In addition, the Project Area is located in the Tapajós Endemism Center, a region with a high concentration of rare and endemic species.	5

1.2. Standardized Benefit Metrics

Various metrics for estimating the net benefit that the Ateles REDD+ Project aims to achieve over the life of the Project are shown below (Table 2).

Table 2: Estimated net benefit for different metrics over the life of the Ateles REDD+ Project.

Category	Metric	Estimated by the End of Project Lifetime	Section Reference
GHG emission reductions or removals	Estimated net removals of emissions in the Project area, measured against the scenario without the Project.	Not applicable	-
	Estimated net emission reductions in the Project area, measured against the without-project scenario	32,121,834 tCO2eq	3
Forest Cover	For REDD projects: Estimated number of hectares of reduced forest loss in the Project area, measured against the without-project scenario	88,489 hectares	3
	For ARR projects: Estimated number of hectares of forest cover added in the Project area, measured against the without-project scenario	Not applicable	-
Improving Land Management	Number of hectares of forest land with production, on which IFM (Improved Forest Management) forest management practices are expected to occur as a result of Project activities, measured against the without-project scenario	Not applicable	-
	Number of hectares of non-forest land where improvements in land management practices are expected to occur as a result of Project activities, measured against the without-project scenario	Not applicable	-
Training	Total number of community members who must have enhanced skills and/or knowledge resulting from the trainings provided as part of the Project activities	To be mapped	
	Number of female community members who must have enhanced skills and/or knowledge resulting from the trainings provided as part of the Project activities	To be mapped	
Employment	Total number of people expected to be employed in Project activities, expressed as number of full-time employees	To be mapped	
	Number of women expected to be employed as a result of Project activities, expressed as number of full-time female employees	To be mapped	
Livelihoods	Total number of people who must have improved livelihoods or income generated as a result of Project activities	To be mapped	
	Number of women who must have improved livelihoods or income generated as a result of Project activities	To be mapped	

Category	Metric	Estimated by the End of Project Lifetime	Section Reference
Health	Total number of people for whom health services are expected to improve as a result of Project activities, measured against the without-project scenario	To be mapped	
	Number of women for whom health services are expected to improve as a result of Project activities, measured against the without-project scenario	To be mapped	
Education	Total number of people for whom access or quality of education is expected to improve as a result of Project activities, measured against the without-project scenario	To be mapped	
	Number of women and girls for whom access to or quality of education is expected to improve as a result of Project activities, measured against the without-project scenario	To be mapped	
Water	Total number of people who are expected to experience increased water quality and/or improved access to drinking water as a result of Project activities, measured against the without-project scenario	Not applicable	-
	Number of women expected to experience increased water quality and/or improved access to drinking water as a result of Project activities, measured against the without-project scenario	Not applicable	-
Wellness	Total number of community members whose well-being is expected to improve as a result of Project activities	To be mapped	
	Number of women whose well-being is expected to improve as a result of Project activities	To be mapped	
Biodiversity Conservation	Expected change in the number of hectares managed significantly better by the Biodiversity Conservation Project, measured against the without-project scenario	Not applicable	-
	Expected number of critically endangered or endangered species (according to the IUCN list of threatened species) which benefit from threat reduction as a result of Project activities, measured against the without-project scenario	1 specie	5

2. GENERAL

2.1. Project Goals, Design and Long-Term Viability

2.1.1. Summary Description of the Project (G1.2)

The Ateles REDD+ Project is a partnership between Biofílica Ambipar Environment and AgroSB Agropecuária S.A., whose purpose is to promote forest conservation and mitigate climate change by reducing Greenhouse Gas (GHG) emissions from land use changes. In addition to climate benefits, the Project also intends to generate environmental benefits through actions aimed at conserving local biodiversity, and social benefits, based on sustainable economic development practices and improving the well-being of surrounding communities.

The Project is being developed in the areas of Fazenda Lagoa do Triunfo, has a total area of 106,921 hectares, composed of a relevant portion of Amazon forest, located in the Municipality of São Félix do Xingú, southeast of the State of Pará. The Project is inserted, in its entirety, in the APA Triunfo do Xingu, a conservation unit with the greatest pressure for deforestation in Brazil, which over the last decades has been suffering great threats to the maintenance of its forest remnants. In addition, the region has great relevance in relation to its biodiversity, as it is located in the Tapajós Endemism Center, a place with a large concentration of rare and endemic species.

The natural plant formations present in the Fazenda Lagoa do Triunfo set are the Submontane Dense Ombrophilous Forest with emergent canopy, the Submontane Open Ombrophilous Forest with vines, and the Alluvial Dense Ombrophilous Forest with uniform canopy. Among the identified species, twelve were classified as endemic and thirteen species were categorized under some level of threat. Endangered species are, for the most part, timber species that experience population decline due to habitat reduction and forest exploitation. As for fauna, several species with some degree of threat were identified according to the IUCN Red List, with emphasis on the white-faced spider monkey (Ateles marginatus), which inspired the name of the project and is a species classified as "Endangered (EN)".

Deforestation was identified as the main illegal activity that may negatively impact the future development of the Project. Small farmers were identified as the main drivers of this illegal deforestation, who use the felling and burning system for annual plantations, and subsequent implementation of pasture for small livestock, in addition to the extensive cattle breeding practiced by medium and large producers. Other deforestation agents are identified in the figure of land grabbers, who appropriate the territory to expand pasture areas for extensive cattle breeding. In addition, around the project area we can identify sawmills, logging and mining activities as vectors of deforestation.

From this context, the activities of the Project were designed to ensure the conservation and protection of biodiversity and natural resources, through mitigating and preventive measures such as the strengthening of heritage surveillance, social inclusion and regional socioeconomic development through alternative practices to deforestation and the development and strengthening of value chains, the establishment of a biodiversity conservation program, promoting the in situ monitoring of fauna and flora in the Project Area, the development of strategies that seek to bring land security and territorial belonging, in addition to the strengthening of local governance, stimulated mainly by the engagement and involvement of stakeholders.

Thus, the Ateles REDD+ Project hopes to improve the well-being of communities and generate exceptional benefits for local biodiversity, in addition to the climate impact by reducing GHG emissions by 3,809,086 tCO_2 in 10 years, equivalent to an average annual reduction of 380,908 tCO_2.

2.1.2. Project Scale

Project Scale	
Project	
Large project	x

2.1.3. Project Proponent (G1.1)

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2.1.5. Physical Parameters (G1.3)

The Ateles REDD+ Project is located in the municipality of São Félix do Xingu, in the state of Pará. Access is through two main highways, BR155 and PA279. The property borders Igarapé Triunfo and is located in the middle Xingu region, at the intersection with the Iriri River, which flows downstream into the Xingu River, being its main tributary. In addition, the farm is located in the vicinity of Terra do Meio Ecological Station, Integral Protection Conservation Unit, which has an excellent state of preservation. (Figure 1).

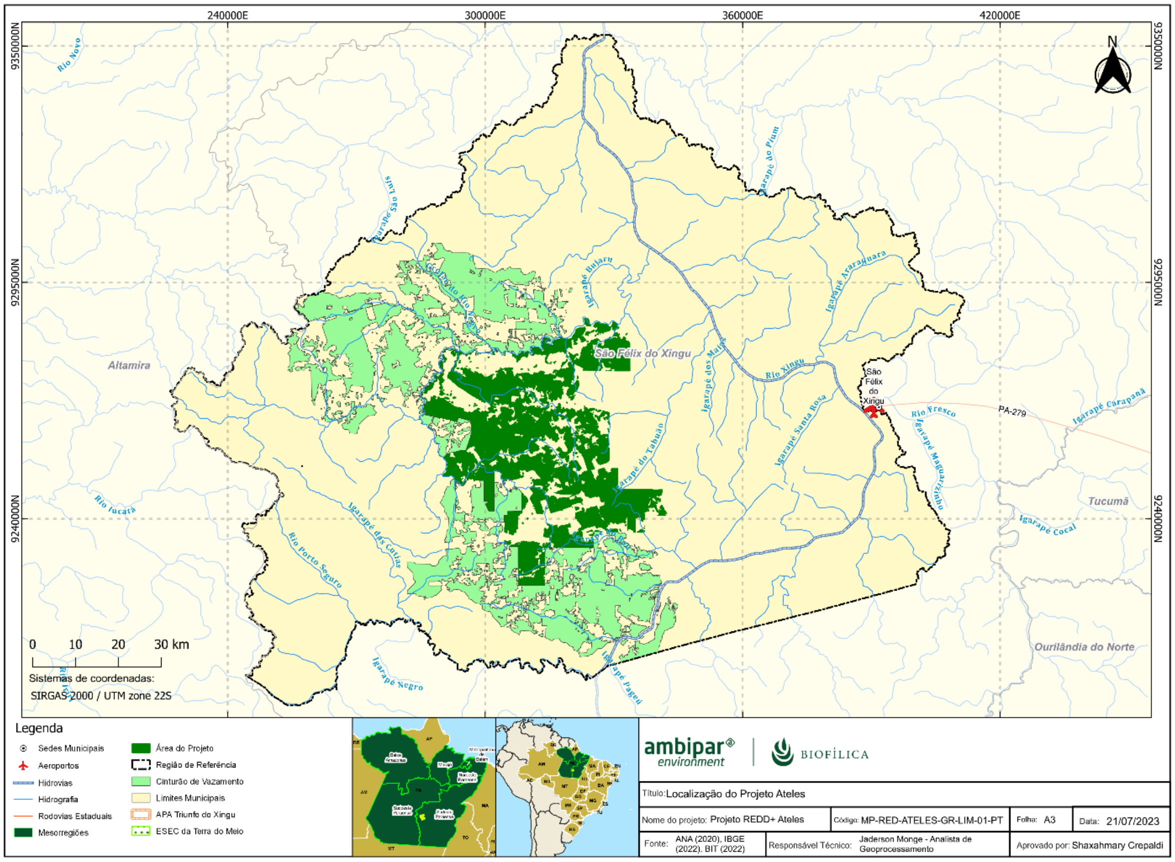

Figure 1: Location of the Ateles REDD+ Project Area.

The geological, geomorphological, pedological, climatic and hydrological parameters were evaluated for the Reference Region (described in 3.1.3) and are presented below:

<u>Geological Aspects</u>

According to the database of the Geological Map of Pará (from CPRM, VASQUEZ et al., 2008[1]; Figure 2.7), the area of interest is located in the Central Amazon Province, Iriri-Xingu Domain, in the Iriri Group, Salustiano Formation and its extreme southern portion belongs to the Sobreiro Formation, superimposed on the Iriri Group referring to the Iriri-Xingu Domain.

According to Forman et al. (1972[2]), the Salustian Formation covers the effusions of explosive acid volcanism, related to Uatumã magmatism, consisting of rhyolites and dacytes, having as a type-section the course of the Salustian igarapé, a tributary of the Tocantins River.

The volcanic rocks of the Sobreiro Formation, according to Forman et al. (1972), present varied colors, according to the facies, but generally present gray, greenish gray and dark gray.

[1] VASQUEZ, Marcelo Lacerda; ROSA-COSTA, Lúcia Travassos da (orgs.). Geology and mineral resources of the state of Pará. Belém: CPRM, 2008. Scale 1:1,000,000. Program Geology of Brazil - PGB. Available at: https://rigeo.cprm.gov.br/handle/doc/10443

[2] FORMAN, J.H.A.; NARDI, J.I.S.; MARQUES, J. P.M.; LIMA, M.I.C. pesquisa mineral no iriri/curuá. Belém: sudam/geomineração, 1972. 62 p.

Geomorphological Aspects

Based on the Geomorphological Map of the State of Pará (IBGE, 2008[3]), the geomorphology of Pará consists of plains, plateaus and depressions, and specifically in the municipality of São Félix do Xingu are present the Jamanxim Depression, the Middle Xingu Depression, the Amazonian Plain, the Residual Plateaus of Southern Pará, the Serra dos Carajás, the Serras de São Félix and the Serras do Pardo.

The Jamanxim Depression has a Colinous Differential Dissection, marked by evident structural control, with variable top shapes and the same drainage pattern carved in different types of rocks. They are defined by shallow valleys, with slopes of medium to smooth slope, notched by grooves and first-order drainage headwaters and a density of coarse drainage. In the southern part of the Jamanxin region Colinous Differential Dissection at latitudes below 8ºS the dissection characteristics and shape resemble those above, but with very weak incision deepening and an average drainage density (IBGE, 2008).

The Middle Xingu Depression has a dissection pattern marked by evident structural control and density controlled by tectonics and lithology, with a set of tabular relief shapes, forming features of gently inclined ramps and spines, carved in sedimentary and crystalline rocks, are the shallow valleys, with low to medium slope slopes, varying their forms of deepening of the incisions and drainage density (IBGE, 2008).

The Amazon Plain is a unit of Brazilian relief that is located in the northern region of Brazil. The Brazilian Amazon region has great scientific and didactic interest for the comprehensiveness of its biodiversity and ecosystems. The plain of 14 to 35 kilometers wide borders the Amazon River, in the middle of low areas, and the relief is divided into floodplains, rough or river terraces and dry land. The long Amazonian plain, from the bar of the Negro River to the vicinity of the Marajoara Gulf, has its own strip and its own ecosystem. The Amazon River follows the middle of the great plain, which was formed during the Holocene. The floodplains of the Amazon Plain follow the lower areas, being usually flooded by the floods of the Amazon River. They are rich soils in their flood area and have a small topographic gradient. The Amazonian Plain also presents the river terraces, which are flat or slightly inclined surfaces on the banks of the Amazon River, which were formed according to the water level over several years (DANTAS & TEIXEIRA, 2013[4]).

The Residual Plateaus of the South of Pará (encompassing the denominations Residual Plateaus of the South of the Amazon and Mountains of the South of Pará, according to IBGE, 1995) exhibit a diverse set of relief patterns composed of imposing mountainous alignments, with a mountainous aspect, groupings of hills and low mountains and a set of isolated plateaus arranged in different lithostructural arrangements immersed in the vast Flattened Surfaces of the South of the Amazon. These sets of rugged reliefs have sloping slopes, sometimes strongly dissected by a high-density drainage network, sometimes preserving flattened tops on sparse planaltic surfaces, remnants of old erosion surfaces. This whole set of relief forms demonstrates deep dissection and razing of an intensely eroded plateau, presenting a residual character, amid the vast flattened surfaces of the South Amazon Shield (DANTAS & TEIXEIRA, 2013).

Serra dos Carajás located in the state of Pará is the largest mineralogical province on the planet. It houses the largest mined iron ore deposit in the world. In addition to iron, it concentrates a large amount of manganese, copper, gold and nickel. Reaching an area of approximately 1 million square kilometers, the rich mineral region covers the southwest of the state of Pará, the west of Maranhão and the north of Tocantins. Like most lands rich in metallic minerals, the Serra dos Carajás region is located in lands of

[3] IBGE. Geomorphological Map of the State of Pará. IBGE, 2008. Available at: https://geoftp.ibge.gov.br/informacoes_ambientais/geomorfologia/mapas/unidades_da_federacao/pa_geomorfologia.pdf

[4] DANTAS, M.E.; TEIXEIRA, S.G. Origin of Landscapes. In: JOÃO, X. S. J; TEIXEIRA, S. G.; FONSECA, D.F.F Geodiversity of the state of Pará. Belém: CPRM, 2013. p.25-52. Available at: https://rigeo.cprm.gov.br/handle/doc/16736

crystalline origin (shields and cratons). The mineral deposits there originated in plutonic rocks, formed from the solidification of magma inside the crust (AGUIAR et al., 2019[5]).

Serra do Pardo has a Planing Model with a degraded pedestrian denuded and a partially conserved planing surface, losing continuity due to the morphogenetic system, dissected and separated by scarps and bumps, in addition to other models of a subsequent system, denuded by exhumation of a sedimentary layer or removal of pre-existing cover (IBGE, 2008).

The Bacajá Depression is considered the Middle Xingu Depression, which has Colinous Differential Dissection, northern part of the municipality, with evident structural control, with top shapes and deepening of the incisions, with drainage pattern, and tectonic control, with convex tops, carved in different types of rocks. They are shallow valleys, slopes of medium to smooth slope, in grooves and first-order drainage headwaters. It also exhibits a marked dissection pattern by evident structural control, and same top shapes and drainage pattern. In addition to the Modeled dissection whose interpretation metasedimentary and crystalline rocks modeled by relief shapes of narrow and elongated tops, with structural control, and embedded valleys. The sharp-looking tops result from the interception of slopes of high slope, carved by grooves and ravines (DANTAS & TEIXEIRA, 2013).

Pedological Aspects

According to data from IBGE and EMBRAPA, the study area is mainly covered by dystrophic red-yellow argisol, eutrophic red nitosol, dystrophic litholic neosol and dystrophic clayluvic plinthosol. Subsequently, these soils are described according to EMBRAPA.

The dystrophic red-yellow argisol has a yellowish red color due to the presence of the mixture of iron oxides, hematite and goethite, it is a soil that varies between deep and very deep and is well drained, in general it presents low to very low fertility, as it is a dystrophic soil, requiring correction to obtain agricultural productivity. It is formed through sediments of the Barreiras group and crystalline rocks and has medium/medium and medium/very clayey EMBRAPA texture (2021b[6]).

The eutrophic red nitosol is formed predominantly by basic and ultrabasic rocks, such as gabbro and basalt, this soil usually is in rugged reliefs, so they have a high rate of erosion, as they derive from these rocks has a high natural chemical fertility, (eutrophic) being considered as one of the most productive, previously called "Purple Earth" having horizons with little differentiation and clayey texture EMBRAPA (2021c[7]).

Regarding the dystrophic litholic neosol, it is possible to mention that it is a soil of recent formation that is usually present in more sloping reliefs and is characterized by being shallow, where the sum of the horizons to the rock does not exceed 50 cm, has little fertility because it is a dystrophic soil and low levels of EMBRAPA phosphorus (2021d[8]).

[5] AGUIAR, P.F.; EL-ROBRINI, M.; GUERREIRO, J.S.; FREIRE, G.S.S. Mappings for the analysis of geomorphological aspects using geoprocessing in the municipality of Altamira, Pará, Brazil. NAEA Paper 2019, Volume 28, No. 2 (414), 2019. DOI: 10.18542/papersnaea.v28i2.8108

[6] EMBRAPA. EMBRAPA Portal, 2021b. Red-Yellow Argisol. Available at: <https://www.embrapa.br/en/agencia-de-informacao-tecnologica/territorios/territorio-mata-sul-pernambucana/caracteristicas-do-territorio/recursos-naturais/solos/argissolos-vermelho-amarelos>. Access on: Feb. 11, 2023.

[7] EMBRAPA. EMBRAPA Portal, 2021c. Nitosols. Available at <https://www.embrapa.br/en/agencia-de-informacao-tecnologica/tematicas/solos-tropicais/sibcs/chave-do-sibcs/nitossolos>. Access on: Feb. 11, 2023.

[8] EMBRAPA. EMBRAPA Portal, 2021d. Litholic Neosols. Available at: <https://www.embrapa.br/en/agencia-de-informacao-tecnologica/tematicas/solos-tropicais/sibcs/chave-do-sibcs/neossolos/neossolos-litolicos>. Access on: Feb. 11, 2023.

The dystrophic clayluvic plinthosol has variable drainage and excess water may occur during the year, and they have little natural fertility because they are dystrophic, in addition, they have a horizon or layer with accumulation of clay below EMBRAPA's surface horizon A (2021e[9]).

Climatic Aspects

The State of Pará has a tropical rainy climate, ranging from always humid to with occurrences of summer rainfall (savanna climate, with dry season), also passing through a tropical monsoon climate (ANDRADE et al., 2017[10]). When it comes to climatology, rainfall is also a fundamental element, since the intensity, frequency and duration of rainfall are determinants in the balance of the systems.

The Xingu River basin covers part of three mesoregions of the State of Pará: Southwest, Southeast and Lower Amazonas. As latitude is one of the factors that influences climate dynamics, it is important to highlight that the basin is approximately between latitudes 1°30'S and 15°00'S, which shows that due to its extent, some differences are evident when analyzing rainfall in its sub-regions. Thus, the highest values of monthly rainfall of the entire basin during the rainy season, which peaks in March, are observed in the Pará portion (Middle and Lower Xingu). The period with the lowest monthly average, especially in the Mato Grosso portion, with its minimum in July. Figure 2 shows the distribution of average annual rainfall in the Xingu River basin.

[9] EMBRAPA. EMBRAPA Portal, 2021e. Plintossolos Argilúvicos. Available at: <https://www.embrapa.br/en/agencia-de-informacao-tecnologica/tematicas/solos-tropicais/sibcs/chave-do-sibcs/plintossolos/plintossolos-argiluvicos>. Access on: Feb. 11, 2023.

[10] ANDRADE, V.M.S. et al. Considerations on climate and edaphoclimatic aspects of the northeastern mesoregion of Pará. In: CORDEIRO, Iracema Maria Castro Coimbra et al (ed.). Northeast of Pará: Overview and sustainable use of secondary forests. Mainland: Edufra, 2017. p. 59-96. Available at: https://ainfo.cnptia.embrapa.br/digital/bitstream/item/162429/1/Livro-Nordeste-2.pdf. Access on: April 19th 2022.

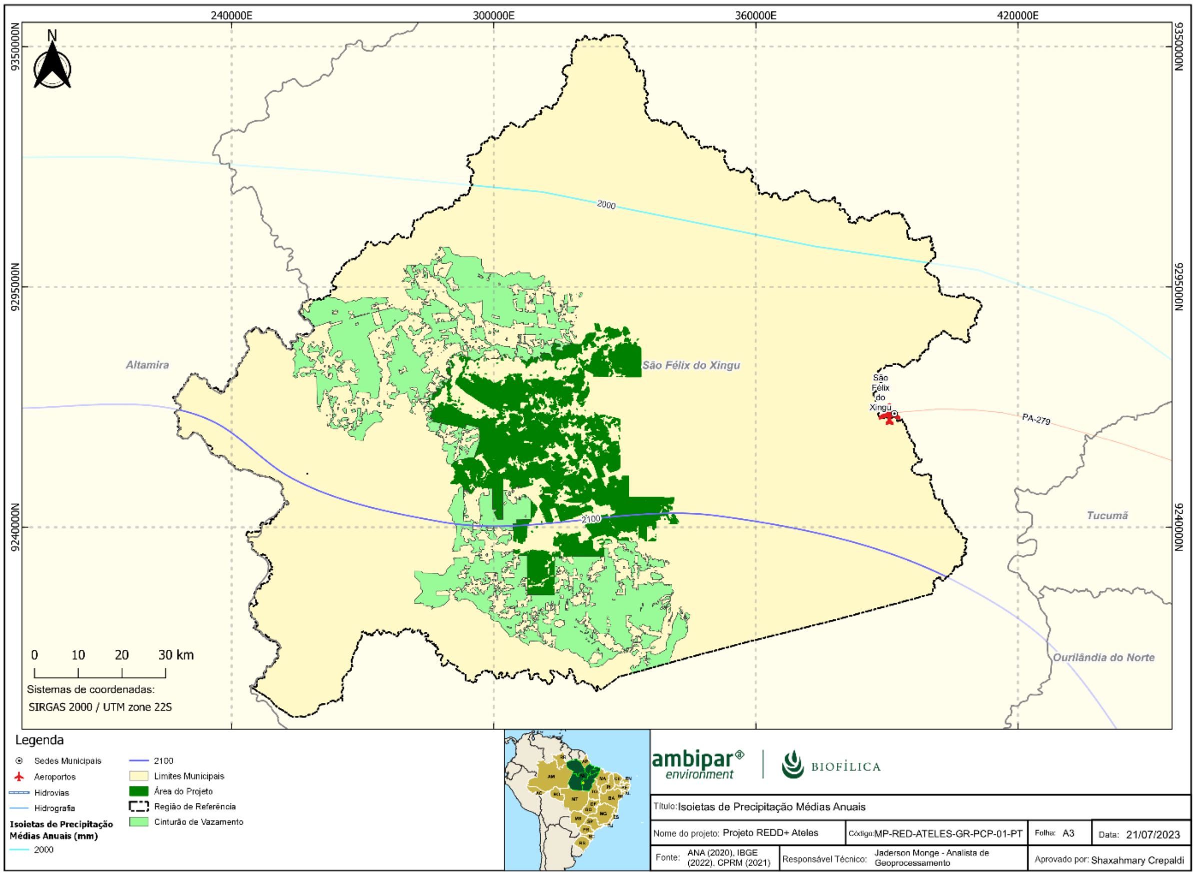

Figure 2: Distribution of average annual rainfall in São Félix do Xingu.

Rainfall has a well-defined rainfall during the year and it is observed that the variation over the months is characterized by the occurrence of highs during the months of December to March (Summer - Autumn) and lows in the months of June, July and August (Winter). The beginning of the rainy season in the basin occurs in October, in the portion between Upper and Middle Xingu, and moves in the South-North direction to the Lower Xingu region, parking in May. This “march” of rainfall in the Xingu River basin occurs due to the direct influence of the two major meteorological systems, namely, the South Atlantic Convergence Zone (SACZ) and the Intertropical Convergence Zone (ITCZ), also affecting the river's surface runoff (CPRM, 2021).

The seasonal variation of the Amazon is also influenced by the ENOS phenomenon – El Niño-Southern Oscillation. El Niño/La Niña are the positive/negative phases of the ENOS phenomenon and represent the anomalous warming/cooling of the Sea Surface Temperature (SST) in the tropical Pacific region (DE SOUZA et al, 2000[11]).

In addition, studies (FU et al 2001[12]) presented observational results on the influence of the Pacific on rainfall in the eastern Amazon and found that, during La Niña events, rainfall patterns are configured with regional patterns above the climatological average during the rainy season, while, during El Niño events, rainfall patterns are essentially inverse.

[11] DE SOUZA, E.; KAYANO, M.; TOTA, J.; PEZZI, L.; FISCH, G.; NOBRE, C. On the influences of the El Niño, La Niña and Atlantic dipole pattern on the Amazonian rainfall during 1960-1998. Acta Amazônica, v. 30, n. 2, p. 305-318, 2000.
[12] FU, R.; DICKINSON, R.E.; CHEN, M., WANG, H. How do tropical sea surface temperatures influence the seasonal distribution of precipitation in the equatorial Amazon? Journal of Climate, v. 14, p. 4003–4026, 2001.

Another important atmospheric system is the phenomenon of the South Atlantic Convergence Zone (SACZ), which is defined as a persistent band of cloudiness and precipitation with a Northwest-Southeast orientation, extending from the south and east of the Amazon to the southwest of the South Atlantic Ocean (KODAMA, 1992[13], 1993[14]; SATYAMURTI et al., 1998[15]; CARVALHO et al., 2002[16], 2004[17]). The cloud band can remain semi-stationary for days in a row, favoring the occurrence of flooding in the affected areas. Thus, the South of the state of Pará (Middle and Lower Xingu) and North of Mato-Grosso are influenced by the South Atlantic Convergence Zone (SACZ), which causes high rainfall levels, especially in the Southwest region of Pará (Altamira, for example). According to MASTER (2017[18]), the ZCAS tends to position itself further north at the beginning of summer, moving later to the south and may vary up to 10°S and 15°S latitude (Alto Xingu). Figure 3 shows a graph referring to cloudiness in the municipality of São Félix do Xingu, in the interior of the state of Pará (Brazilian Climatological Normal 1991-2020; INMET, 2021[19]).

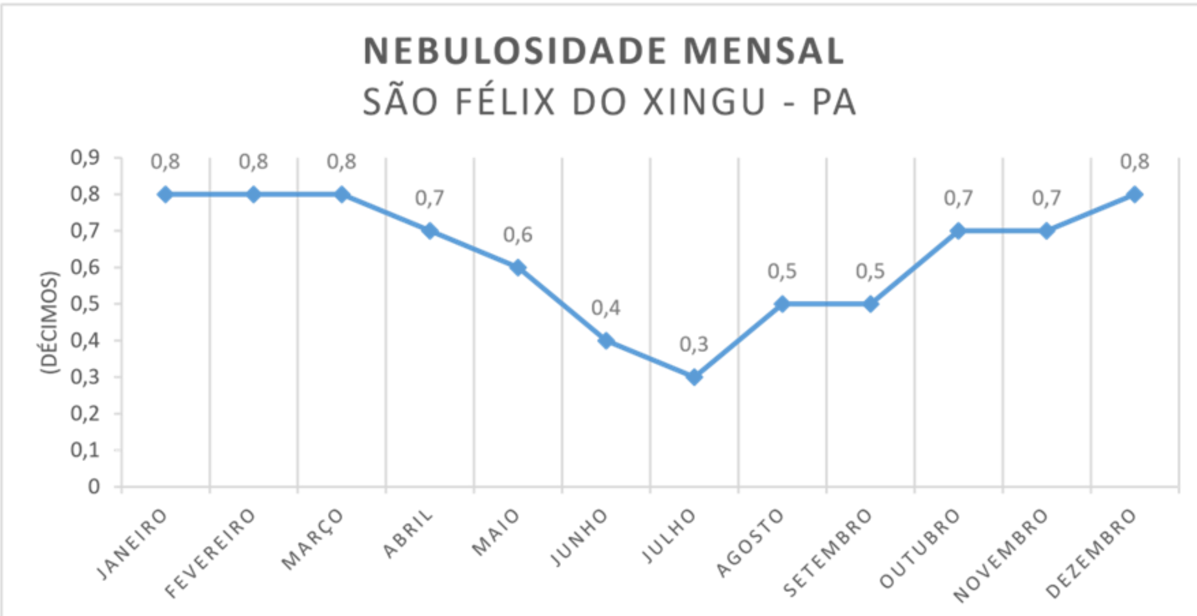

Figure 3: Monthly cloudiness of São Félix do Xingu, Pará. Source: INMET (2021).

<u>Hydrological Aspects</u>

[13] KODAMA, Y. M. Large-Scale Common Features of Subtropical Precipitation Zones (the Baiu Frontal Zone, the SPCZ, and the ZCAS). Part I: Characteristics of Subtropical Frontal Zones. Journal of the Meteorological Society of Japan, 70, 1992, p.813-835.

[14] KODAMA, Y. M. Large-scale common features of sub-tropical precipitation zones (the Baiu Frontal Zone, the SPCZ, and the SACZ). Part II: Conditions of the circulations for generating the STCZs. J. Meteor. Soc. Japan, 71, 1993, p.581-610.

[15] SATYAMURTI, P.; NOBRE, C.; SILVA DIAS, PL South America. Meteorology of the Southern Hemisphere, D. J. Karoly and D. G. Vincent, Eds., Amer. Meteor. Soc., p. 119–139, 1998.

[16] CARVALHO, L. M. V.; JONES, C.; LIEBMANN, B. Extreme Precipitation Events in Southeastern South America and Large-Scale Convective Patterns in the South Atlantic Convergence Zone. Journal of Climate, v. 15, p. 2377-2394, 2002.

[17] CARVALHO, L. M. V.; JONES, C.; LIEBMANN, B. The South Atlantic Convergence Zone: persistence, intensity, form, extreme precipitation and relationships with intraseasonal activity. J. Climate, 17, p. 88-108, 2004.

[18] MASTER. Meteorology Applied to Regional Weather Systems. Institute of Astronomy, Geophysics and Atmospheric Sciences. Universidade de São Paulo. South Atlantic Convergence Zone. 2017. Available at: <http://master.iag.usp.br/pr/ensino/sinotica/aula14/>. 2017.

[19] INMET – NATIONAL INSTITUTE OF MEOROLOGY OF BRAZIL. Brazilian Climatological Normal 1991-2020. Brasília – DF. INMET, 2021. Available at: https://portal.inmet.gov.br/normais.

The Xingu hydrographic region occupies an area of 26.9% of the state of Pará, consisting of the Xingu river basin, encompassing as main drainages the Xingu, Iriri, Caeté, Chiche, Xinxim, Carajás, Ribeirão da Paz, Fresco and Petita rivers. The region consists of the following municipalities: São Félix do Xingu (municipality targeted by this study), Cumarú do Norte, Bannach, Ourilândia do Norte, Água Azul do Norte, Tucumã, Senador José Porfírio, Anapu, Vitória do Xingu, Altamira, Brasil Novo, Medicilândia, Uruará, Placas, Rurópolis, Trairão, Itaituba, Novo Progresso and Porto de Moz. The Xingu hydrographic region is formed by the following sub-regions: Hydrographic Sub-Region of the Fresco River; Hydrographic Sub-Region of the Iriri River; Hydrographic Sub-Region of the Lower Xingu; and Hydrographic Sub-Region of the Upper Xingu (SEMA, 2012[20]).

The Xingu River is born at the meeting of Serras Formosa and Roncador and is basically divided into three compartments: Upper Xingu, Middle Xingu and Lower Xingu. In the Upper Xingu: Rio Ferro, Culuene and Sete de Setembro. It enters the Xingu Indigenous Park, where it receives other important tributaries such as the Suiá-Miçu, Manissaua-Miçu and Arraias (CPRM, 2021). The Middle Xingu receives, among others, contributions from the Fresco River, in the municipality of São Félix do Xingu, and further downstream, from the Iriri River, the most important tributary of the Xingu. In the Lower Xingu, the river receives the contribution of another large tributary, the Bacajá River, and after a stretch with many rapids, it opens into a lake, to its mouth in the Amazon (CPRM, 2021).

In the study area, in São Félix do Xingu, it is possible to notice a very large contribution from the Xingu River, with drains corresponding to the sub-basins of Igarapé do Triunfo, Igarapé do Columbizinho, Grotão do Rio Negro, Igarapé do Tabuão. In Figure 4 it is possible to notice the drainage networks in place.

[20] SEMA, 2012. Water Resources Policy of the State of Pará /State Secretariat for the Environment. – Belém: SEMA, 2012. Available at: https://www2.mppa.mp.br/sistemas/gcsubsites/upload/41/POLITICA_DE_RECURSOS_HIDRICOS_DO_ESTADO_DO_PARA.pdf

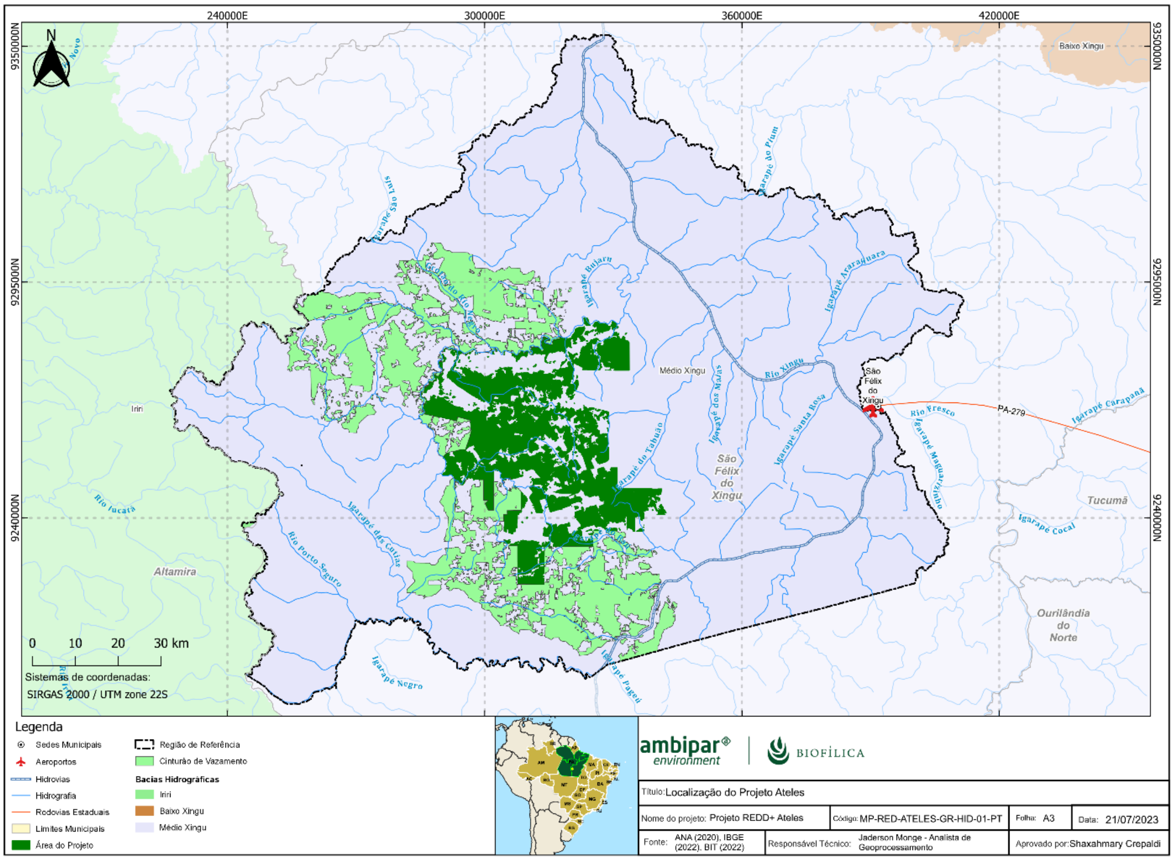

Figure 4: Hydrography of the study area. Source: ANA (2020), IBGE (2021, BIT (2022).

<u>Vegetation</u>

The natural plant formations present in the Reference Region are predominantly Dense Ombrophilous Forest and Open Ombrophilous Forest (according to Table 3), while in the Fazenda Lagoa do Triunfo (FLDT) complex, the project area, these phytophysiognomies represent 99, 9% of the plant formations, more specifically we have the Submontane Ombrophilous Dense Forest with emergent canopy, the Submontane Ombrophilous Open Forest with lianas and the Alluvial Ombrophilous Dense Forest with uniform canopy (figure 5).

Table 3: Classification and percentage of phytophysiognomy in the Ateles REDD+ Project Reference Region.

Class	Vegetation Area in the Reference Region (ha)	% of Vegetation in Relation to the Reference Region
Agriculture with Cyclical Crops	7,239,441	0,43%
Continental body of water	36,344,971	2,15%
Submontane open ombrophilous forest with lianas	487,045,021	28,87%
Submontane open ombrophilous forest with palm trees	98,012,137	5,81%
Alluvial Dense Ombrophilous Forest	6,998,886	0,41%
Alluvial Dense Ombrophilous Forest with uniform canopy	29,402,476	1,74%
Submontane Dense Ombrophilous Forest	3,977,074	0,24%

Class	Vegetation Area in the Reference Region (ha)	% of Vegetation in Relation to the Reference Region
Submontane Dense Ombrophilous Forest with emergent canopy	849,931,087	50,38%
Submontane dense ombrophilous forest with uniform canopy	21,510,601	1,28%
Livestock (pastures)	96,468,851	5,72%
Wooded Savannah without gallery forest	30,899,477	1,83%
Park Savannah without gallery forest	18,678,476	1,11%
Secondary vegetation without palm trees	443,101	0,03%
Total	**1,686,951,599**	**100,00%**

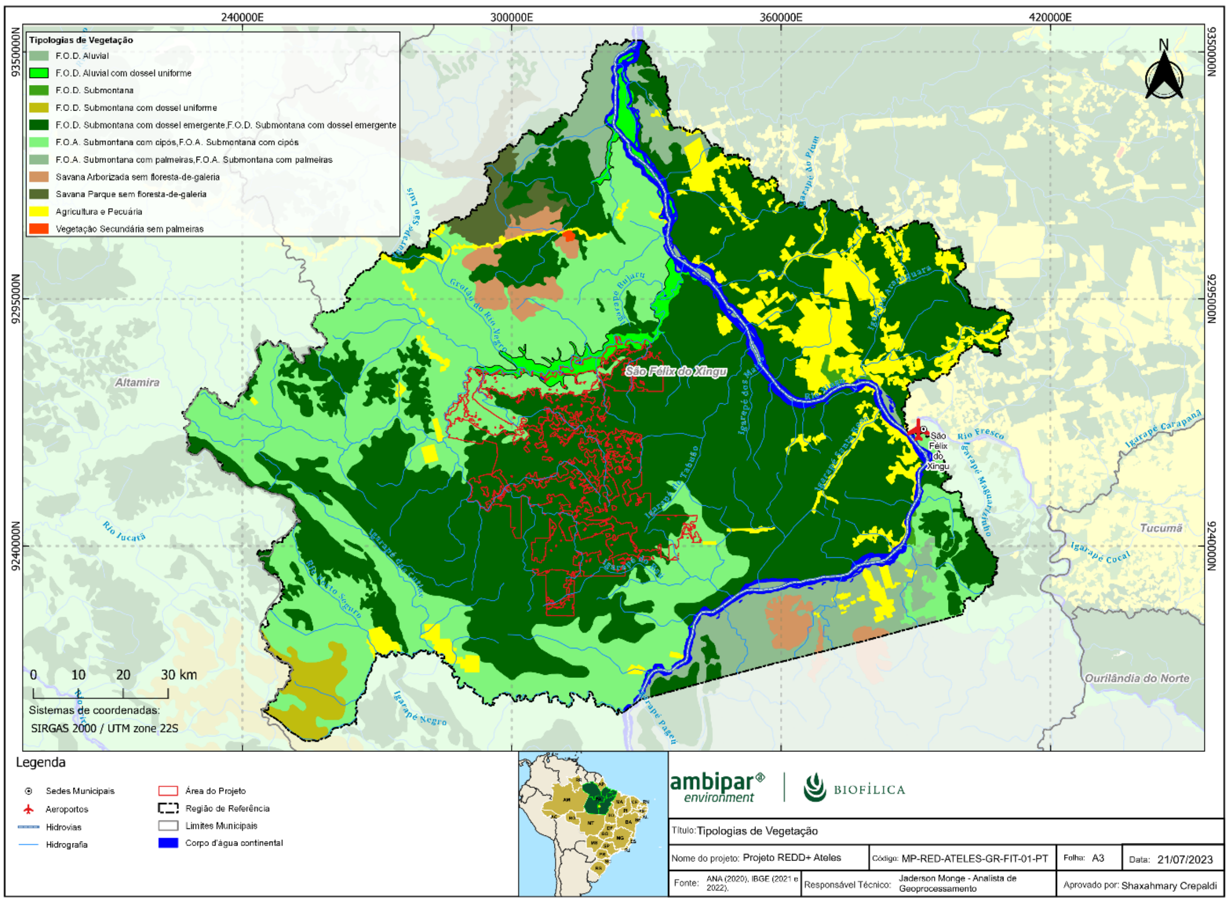

Figure 5: Plant typologies present in the project area.

The Alluvial Dense Ombrophilous Forest (FODau) occurs to the North of the Farm near the tributary of the Xingu River in a strip, which extends almost the entire length of the northern limit of the properties Fazenda Lagoa do Triunfo I and II. In this forest typology, trees with DBH greater than 10 cm have an average height of 12 m. The canopy has vertical stratification, with trees with DBH up to 20 cm having an average height of 10 m, those with DBH between 20 cm and 50 cm having an average height of 14 m and 16 m and larger trees, with DBH ≥ 50 cm, having an average height of 20 m. In conglomerates installed in areas with FODau, the above-ground biomass of living trees was estimated at 167 ton/ha. In FODau there are species such as cedar (*Erisma uncinatum*), laurel (*Licaria brasiliensis*), matamatá (*Eschweilera coriacea*), maçaranduba

(*Manilkara bidentata*), tachi (*Tachigali glauca*), urucurana (*Sloanea guianensis*) and rubber tree (*Hevea brasiliensis*).

While the Submontane Open Forest with vines (FOAsc) occupies the Northeast region of the property, including a large fraction of the FLDT I property and part of the FLDT III property. This forest typology also occurs at the eastern end of Fazenda Nova Caracol. In this typology, trees with DBH >10 cm have an average height of 11 m, with the stratum formed by a tree with DBH between 10 cm and 20 cm with an average height of 8 m, trees with DBH between 20 cm and 30 cm and from 30 cm to 50 cm have an average height of 12 m and 16 m, respectively. Trees with DBH ≥ 50 cm have an average height of 21 m. In the conglomerates that focused on the FOAsc area, the biomass above ground of the living trees was estimated at 126 t/ha. The main species of the FOAsc forest typology are pitch (***Protium*** sp.), embaúba (*Cecropia sciadophylla*), sumaúma (*Ceiba pentandra*) and tamarindo (*Dialium guianense*), among others.

And finally, the Submontane Dense Ombrophilous Forest (FODse) occupies the other areas of the real estate complex of Fazenda Lagoa do Triunfo: FLDT IV, FLDT V, FLDT VI and Fazenda Nova Caracol. In this forest typology, the average height of trees with DBH >10 cm is 13m, and trees with DBH between 10 cm and 20 cm have an average height of 9 m, trees with DBH between 20 cm and 30 cm and between 30 cm and 50 cm have an average height of 14 m and 18 m, respectively. The average height of trees with DBH ≥ 50 cm is 22m. In the conglomerates drawn in the FODse typology, the above-ground biomass of the living trees was estimated at 205 ton/ha. The main species of FODse are abiu (*Pouteria oblanceolata*), amapá (*Brosimum parinarioides*), angelim-pedra (*Hymenolobium petraeum*), pitcher (*Protium* sp.), cacao (*Theobroma cacao*), **chestnut** (*Bertholletia excelsa*), embaúba (*Cecropia sciadophylla*), enviras (Annonaceae sp.), fava (*Parkia nitida*), guajará-faia (*Pouteria* sp.), ingá (*Inga* sp.), ipê (*Handroanthus* sp.), laurel (*Licaria brasiliensis*), buckwort (*Ocotea longifolia*), laurel pepper (*Licaria armeniaca*), macucu (Licania sp.), marupá (*Simarouba amara*), pau-mulato (*Calycophyllum spruceanum*), pente-de-macaco (*Apeiba albiflora*), periquiteira (*Guazuma ulmifolia*), sucupira (*Diplotropis* pururea*)* and tanimbuca (*Terminalia tetraphylla*).

2.1.6. Social Parameters (G1.3)

The municipality of São Félix do Xingu is located in the southeast of Pará. Work and usufruct related to natural resources are an immanent part of the local collective memory and are part of the history of all those who currently occupy the region. Considered an agricultural frontier in the Amazon, São Félix do Xingu showed significant demographic growth, accompanied by urban growth and environmental degradation (KAHWAGE[21]. 2002). The territory where São Félix do Xingu is located went through accelerated occupation processes, the opening of highways in the 1970s and 1980s by the military government, resulted in an intense spontaneous migration perpetrated by several peasant families from other municipalities and states, to the southeastern region of Pará (SCHMINK et al, 2017[22]; CASTRO et al, 2002[23]).

The migratory process and the advance of the border had in its core the movement of small rural producers and landless workers who wanted to find work or lots to settle; as well as fleeing from drought and latifundia in the northeast, southeast and south of the country. The large volume of these migrants arriving in the region came mainly from Maranhão, Minas Gerais, Pernambuco, Ceará, Piauí and Rio Grande do Sul. The availability of land, the quality of the soil, the low prices of the lots found in São Félix do Xingu and the

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[21] KAHWAGE, Claudia. Political organization of peasants in São Félix do Xingu – PA: Strategies, Identities and Associations. Master's Dissertation in Family Farming and Sustainable Development. Federal University of Pará, Belém, 2002, p. 23.

[22] SCHMINK, Marianne et al. From contested to 'green' frontiers in the Amazon? A long-term analysis of São Félix do Xingu, Brazil. The Journal of Peasant Studies, v. 46, n. 2, p. 377-399, 2019.

[23] CASTRO, Edna Ramos; MONTEIRO, Raimunda; CASTRO, Carlos Potiara. Actors and social relationships in new frontiers in the Amazon. Novo Progresso, Castelo de Sonhos and São Félix do Xingu. 2002. Doctoral Thesis. World Bank Groupe-Banque Mondiale.

cheap labor were certainly central stimuli to attract capitalized people amid the migratory processes of peasants (KAHWAGE, 2002).

The opening of the PA-279 road brought to the popular composition of São Félix do Xingu another type of peasant category: the "settlers". Together with the waves of squatters, the settlers arrived more intensely forming the so-called "colonies" (KAHWAGE, 2002). Faced with land conflicts and the conquest of the right to housing on the land, social identities were structured and some squatters began to self-identify as "settlers". In the occupation model of the 1980s, the predominant pattern stood out with the large farmers who appropriated the land in the municipality, around 1985, consolidating the system of immense properties focused on agriculture and, especially, on livestock (CASTRO et al. 2004)[24].

The advance of livestock in São Félix do Xingu had its rhythm influenced, directly and indirectly, by government incentive policies that transformed the area into one of the largest agricultural centers in the country (KAHWAGE, 2002). Despite this, due to the late opening of the road, the development of livestock in the municipality is seen as recent and a little different from the "nobler" livestock areas of the southern center of the country. Livestock in São Félix do Xingu was able to trigger other segments of the economy of the urban sector by moving the trade in seeds, fertilizers, fences, and other agricultural inputs. However, agricultural penetration in the Xingu territory left in its wake high rates of deforestation of its forest resources (KAHWAGE, 2002).

Currently, the municipality's economy is based on beef cattle. This significant number of head of cattle explains the significant expansion of pasture areas and accompanied by high rates of deforestation (Trindade et al., 2019). [25]For the authors, profitability and land availability are the foundations that underpin the cattle breeding system in the region. Although São Félix do Xingu has reached the municipal position holding the largest cattle herd in Pará and Brazil, with 2.46 million head of cattle, according to the IBGE Municipal Livestock Survey for 2021, other vectors that drive its economic growth are in agricultural production. In this productive environment, there are cocoa and banana crops, in addition to temporary crops of corn, beans and cassava. As for the mining industry, it focuses on the extraction of cassiterite, nickel and cobalt (STATE OF PARÁ, 2021).[26]

Regarding the demographic characterization of the municipality where Fazenda Lagoa do Triunfo is located, according to the census data provided by IBGE, São Félix do Xingu has an estimated population of 135,732 inhabitants (IBGE, 2021[27]). Before the 1980s, the population was estimated at 4,982 due to the low migratory flow, reduced number of rural settlements and logging accompanied by agricultural enterprises that boosted its economy (Costa, 2013). [28]Currently, 18 settlements are present in the municipality, totaling an area of about 360,200 hectares. The support capacity of these settlements is 4,455 families, however, the total number of settled families corresponds to 86.0% of this capacity. According to

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[24] CASTRO, Edna Ramos de; MONTEIRO, Raimunda; CASTRO, Carlos Potiara. Social actors in the most advanced border of Pará: São Félix do Xingu and Terra do Meio. Paper of the Center and High Amazonian Studies (NAEA), 2004.

[25] TRINDADE, Ariadne Reinaldo; SOUTO, Jefferson Inayan de Oliveira; BELTRÃO, Norma Ely Santos. Landscape changes in the municipality of São Félix do Xingu: a study addressing the impacts of anthropogenic factors between 1985 and 2015. Encyclopedia Biosfera, Centro Científico Conhecer – Goiânia, v. 16 n. 29, 2019, p. 6-7.

[26] STATE OF PARÁ. PLAMUS: Sustainable Urban Mobility Plan. Municipality of São Félix do Xingu, 2021, p. 14-15. Available at: www.sfxingu.pa.gov.br. Accessed on Wednesday, December 14, 2022.

[27] IBGE, 2021. Available at: https://cidades.ibge.gov.br/brasil/pa/sao-felix-do-xingu/panorama. Accessed on Sunday, December 15, 2002.

[28] COSTA, André Luis Souza da**.** EFFECTIVENESS OF MANAGEMENT OF the TRIUNFO DO XINGU ENVIRONMENTAL PROTECTION AREA: challenges of consolidating a Conservation Unit in the region of Terra do Meio, State of Pará. Doctoral Thesis

the MAPA data, of the total settlements, only two of them are settlements in installation, while the others are in the created settlement condition.

According to research carried out at the Land Institute of Pará, there is no land titled as quilombolas in the municipality of São Félix do Xingu. However, there are six indigenous lands with part of their territories included in the municipality. The only indigenous territory contained entirely in the municipality of São Félix is that of Pyterewa, with a declared and unregulated situation, and with an extension of 773,400 ha. The other territories are all regularized. The territories with the largest extension are Kayapó (3.2 million ha), Menkragnoti (4.9 million ha) and Trincheira Bacajá (1.6 million ha) (ITERPA, 2021[29]).

Regarding the health system, in 2021, the municipality of São Félix do Xingú had 64 establishments registered with SUS, 52.4% more than what was registered in 2015. The distribution by type of establishment shows that there was an increase in the number of polyclinics, health care centers, Basic Health Unit, and in the number of general hospitals in the municipality. The municipality has 20 other establishments, including laboratories, clinics and offices, and an indigenous health care unit (Ministry of Health, 2021[30]).

With regard to education, the municipality had, until 2021, 122 registered schools, with 271 teachers. Regarding the number of students enrolled in the final years of elementary school and high school, there was an increase in 2021 compared to 2016. However, there was a reduction in the number of enrollments in the initial years of elementary school and in Youth and Adult Education, of 11.63% and 88.52%, respectively. The performance of high school students is very worrying, as there is a negative combination of results, in which the approval rate decreased in the period under analysis, from 66.00% in 2015 to 49.00% in 2021, 17 percentage points of reduction of the indicator. Failure remained constant in the years under analysis. Meanwhile, abandonment grew 16.6 percentage points, from 20.2% in 2015 to 36.8% in 2021. In the case of grade-age distortion in the 3rd year of elementary school, the rate remained constant (QEdu, 2021[31]). In terms of infrastructure present in schools located in the municipality, it is observed that there are few physical and pedagogical resources to leverage the teaching-learning process. In summary, the management of the municipality's educational sector needs to carry out infrastructure, teacher qualification and management actions of the municipality's educational system (QEdu, 2021).

The formal employment stock in the municipality has an increasing trajectory in recent years, growing 20.7% and 26.7%, when compared to 2021, with 2010 and 2015, respectively. The activities that showed strong growth in the number of jobs in 2021 compared to 2010 were: mineral extraction, manufacturing industry, trade, service, public administration, agriculture, plant extraction, hunting and fishing. The average compensation perceived by workers registered with Rais, in nominal values, in 2021 was BRL 2,422.00. The highest salaries were paid in the activities of public administration, industrial services of public utility, mineral extraction, manufacturing industry, services, commerce, civil construction and agriculture, plant extraction, hunting and fishing (FAPESPA, 2022[32]).

In São Félix do Xingú, 37.85% of the territorial extension in the municipality is used for forests or natural forests intended for permanent preservation or legal reserve, 1.23% for forests and/or natural forests and 0.02% for planted forests. Aggregation of these forest areas represents 39.1%, this represents 11.43% of the total area of the municipality. Land use with agroforestry systems represents 0.87% of the area of establishments in the municipality. This system takes permanent crops as a structuring basis, integrating

[29] INSTITUTO DE TERRAS DO PARÁ (ITERPA), 2021. Available at: http://portal.iterpa.pa.gov.br/banco-de-dados/. Accessed on Sunday, January 15, 2023.
[30] MINISTRY OF HEALTH, 2021. DATE/SUS. Available at: http://www.saude.pa.gov.br/coronavirus/. Accessed on Wednesday, January 25, 2023.
[31] QEDU, 2021. Available at: https://qedu.org.br/sobre/conteudos. Accessed on Sunday, January 15, 2023.
[32]AMAZON FOUNDATION FOR STUDIES AND RESEARCH SUPPORT (FAPESPA) Pará Municipal Statistics: São Félix do Xingu. / Directorate of Statistics and Technology and Information Management. – Belém, 2022.

them with the breeding of animals and other crops. According to Santana *et al*. (2020[33]), agroforestry systems have a high degree of security to generate income for small producers, in addition to using a large amount of labor, having good profitability, little dependence on chemical inputs and being a sustainable production model compared to monoculture systems. The main permanent crops in the municipality of São Félix do Xingu in the last 10 years are coconut, cocoa and bananas, while cassava and corn (in grains) are the main temporary crops in the years analyzed (FAPESPA, 2022).

According to the results of the 2017 agricultural census carried out by IBGE, there were 6,375 rural establishments in the municipality of São Félix do Xingu, in which 1,499 (23.5%) of them were classified as non-family farming and 4,876 (76.5%) linked to family farming and agricultural production. The average area of family farming productive establishments is 91.3 ha, while the average area of the group of non-family farming establishments is 1,448.8 ha, almost fourteen times more than the average of family producers (FAPESPA, 2022).

Regarding the distribution and use of land, the largest portion of this area is used for pastures (58.3%), whether natural (3.1%), pastures in good condition (51.8%) and pastures planted in poor condition (3.5%). The high extension of pastures in the municipality represented 17.1%, in 2017, supports the 2.5 million head of cattle, which, in a simple relationship with the pasture area, reaches almost 1.7 head/ha, an indicator higher than that observed in the state of Pará. The areas used for the cultivation of permanent and temporary crops are less than 1.0%, with 0.45% and 0.21% for each of them, respectively (FAPESPA, 2022). This portion of the area for crops shows that the municipality is focused on cattle production, with a focus on meeting the international demand for beef. Future scenarios point to further increases in demand for meat until 2050, mainly sustained by increased supply and demand from developing countries (BUCHHOLZ[34], 2021).

The high number of cattle in the municipality can be explained, in part, by the fact that establishments with more than 500ha are equipped with infrastructure with electricity, animal handling practices, use of information and communication technologies (ICT), in addition to managers being literate and herd productivity being high (Olímpio *et al.,* 2022[35]). However, the micro-region of São Félix do Xingu is one of those with the lowest proportion of establishments with conservation practices and low sustainability of pastures (Santana *et al*., 2019[36]). According to (Olímpio *et al*., 2022), livestock productivity in Pará is low (less than one head per hectare), generating the need for large areas of pasture to meet the expansion of cattle supply in recent years.

[33] SANTANA, Antônio Cordeiro et al. Bioeconomy applied to agribusiness: market, externalities and natural assets. 2020.

[34] BUCHHOLZ, K. (2019). The world's largest beef exporters. Available at: https://www.statista.com/chart/19122/biggest-exporters-of-beef/.

[35] OLIMPIO, Silvia Cristina Maia; GOMES, Sergio Castro; SANTANA, Antônio Cordeiro de. Patterns of production and sustainability of cattle ranching in the state of Pará-Brazilian Amazon. Semina ciênc. agrar, p. 541-560, 2022.

[36] SANTANA, A.C. et al. Territorial occupation and government management in the municipalities surrounding the protected areas of Carajás In: CAMPOS, J.C.F. (Org.) Carajás: natural and socioeconomic aspects around protected areas. Belo Horizonte: Rupestre, v. 1, p. 77-108, 2019.

2.1.7. Project Zone Map (G1.4-7, G1.13, CM1.2, B1.2)

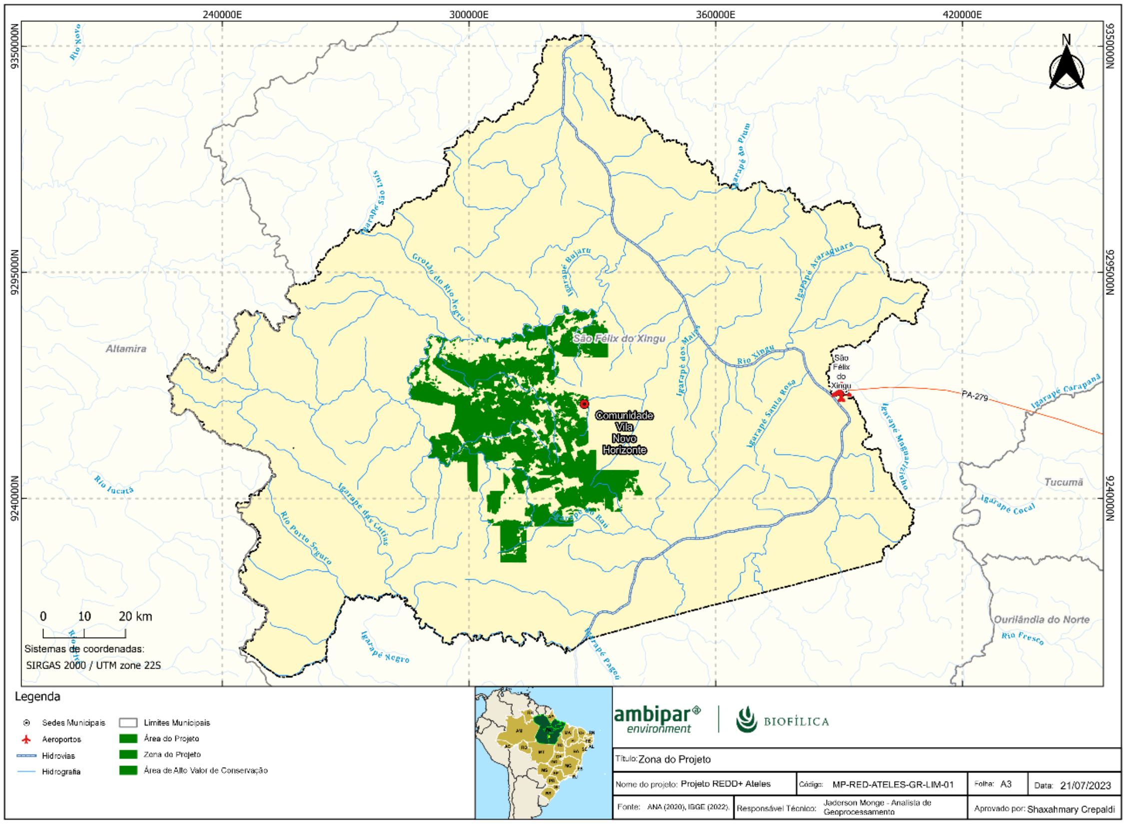

Figure 6: Project Zone Map.

2.1.8. Stakeholder Identification (G1.5)

The identification of stakeholders was based on the elaboration of the Socioeconomic Diagnosis of the Region of the Ateles REDD+ Project, developed by the Peabiru Institute.

The methodological component of this diagnosis involved different qualitative and quantitative approaches divided into four main aspects: (i) collection and treatment of field data; (ii) collection and analysis of secondary socioeconomic data; (iii) collection and treatment of geospatial environmental information; (iv) conducting perception techniques about the AgroSB Company with a focus group of its employees. These procedures are anchored in the VCS VM0015 and CCB-Standards - Climate, Community and Biodiversity methodology. At all stages of the fieldwork, the local managers of the AgroSB companies were supported and monitored, in the case of employees, and representatives of the Association of Farmers of Calumbi and Tucunaré – AGRICATU in the case of settlers.

In the diagnosis, a field survey was carried out with the settlers and AgroSB employees living in the areas of Fazenda Lagoa do Triunfo. The approach was developed through participant observation techniques; application of a questionnaire with a script of quantitative and qualitative questions; interviews with a script of semi-structured questions; and the construction of a timeline to contextualize these interactions within the APA Triunfo do Xingu.

Field collection was carried out from November 24 to 29, 2022. The survey took place in five clusters, in which 21 settlers were interviewed. The largest concentration of interviews, 75.5%, took place in Vila Novo Horizonte, due to the fact that many settlers have a residence there. The other interviews (25.5%) were carried out as described in Table 4.

Table 4: Location of interviews with settlers

Interview location	Cases	%
Vila Novo Horizonte	16	75.5
Fazenda California	1	4.9
Fazenda Santa Rita I	1	4.9
Sítio Novo Horizonte	1	4.9
Not informed	1	4.9
Sítio Estrela D'Alva	1	4.9
Total	21	100

It should be noted that the data collection methodology adopted with farmers considered the free and spontaneous willingness to participate in the interviews. In some properties of medium and large size, it was not possible to carry out the survey due to the refusal of the farmers approached. In addition, due to the climate of tension in the territory and aiming at preserving the physical integrity of the researchers, no research was carried out in places, pointed out by the leaders as susceptible to violence.

The average time of application of the questionnaire ranged from 30 to 40 minutes, and in the case of leaders this time reached 1:00h. The application of the interviews took place mostly with one of the spouses, with the majority being answered by a male person.

In addition to the interviews conducted with the consent of the interviewees, a questionnaire divided into the following blocks of questions was applied: productive aspects, school education, existing infrastructure in the communities, access to public policies, among others. These questionnaires contributed with objective and quantitative information, which assisted in the evaluation of the current situation of the communities, in addition to providing important references and directions for future planning in the region covered by the research.

Research on the relationship with workers involved qualitative and quantitative approaches. The qualitative research was carried out during the fieldwork from the mapping and diagnosis of the networks of employees of the Company AgroSB – Lagoa do Triunfo, through the application of questionnaires, work meetings and participant observation techniques. The quantitative research was developed from the application of a questionnaire with closed questions according to the scale developed by Walton. 49 AgroSB employees were interviewed, and the stratification of the categories of position and/or function of the organization was defined according to the dynamics of the research in the field. It is worth mentioning that participation in the research was optional and voluntary.

2.1.9. Stakeholder Descriptions (G1.6, G1.13)

At Fazenda Lagoa do Triunfo, groups of settlers were identified, who declared themselves as family farmers, small producers, squatters, farmers or medium producers. During the activities of the socioeconomic and environmental diagnosis, developed by the Peabiru Institute, 21 people were interviewed, with 91% represented by men (19 people). The average age of the interviewees is 54 years, and the most frequent level of education is incomplete elementary school. Considering the age of the interviewees, it is observed that little education generated low knowledge from the studies, with practical knowledge, in the handling of their daily activities, agriculture and livestock, being their main source of knowledge. Income generation is based on the versatility of activities, mainly in the breeding and marketing of dairy cattle and small-scale cheese manufacturing.

On the properties of the 21 farmers interviewed, an average of 1.57 families were found, and 4.4 people per family. By extrapolating data, based on the number of properties registered in the SICAR database and the population declared in the interview, a total of 104 families and 458 people residing in the project area are estimated on average.

These community members settled in the area between 2001 and 2003, where the average residence time at the interview site is 19 years. Most of them originate from the states of Tocantins, Goiás and Maranhão. This training results, to a large extent, from the internal and external migration process to the state, caused by the financing granted by the agencies that promote the public policy of land occupation in the Amazon and transport and port infrastructure.

Settlers have a very heterogeneous production scale, with properties ranging from 50 to more than 1000 hectares, classified as minifundia, small, medium and large properties, according to INCRA criteria. As their main economic activity, they carry out livestock farming, and to a lesser extent they carry out agroextractivist activities, collect Brazil nuts and açaí, cultivate cocoa and have small fields of beans, corn, cassava and other annual crops. For the most part, they can be classified as small rural producers, with an average income of 1 to 2 minimum wages. However, it is relevant to mention that it is not only low-income people who enjoy the land and need it for subsistence, we have some representatives who have income between 10,500 and 19,800 reais.

Most of the settlers are organized around the Association of Farmers of Calumbi and Tucunaré (AGRICATU). They are represented and guided through this association. According to them, participation in AGRICATU enabled access to Pronaf, enabled social security insurance, access to rural credit, facilitation of the purchase and sale of production, feel represented in relation to defense and claim in favor of properties, receive guidance in productive activities, in addition to strengthening the bonds of members.

In addition, referring to the workers of Fazenda Lagoa do Triunfo, of the company AgroSB, a total of 144 active employees were identified in 2022. Table 4 presents a summary of the employees and work positions on the farm. The age profile of the employees is 33 years on average, and in terms of gender, 85.7% are men and 14.3% are women. About the educational level of the employees, 44.9% said they had completed and/or incomplete primary education, 24.5% had completed and/or incomplete secondary education, 28.6% had completed and/or incomplete higher education, and 2% had a technical degree. Among those interviewed, 42.9% said they were in a stable union and/or married, 53.1% were single and 4.1% were divorced. About the number of family members living with the interviewee, the average is 3.

The majority (69.4%) said they lived close to work and walked. When asked about their satisfaction with the company, all of them said they liked the company they worked for. Regarding the interaction between employees and hierarchical superiors on the São Félix do Xingu farms, in general, it was reported that there is a good relationship between them and that not-so-friendly relationships are an exception within the farm and happen on a small scale. It was also reported that the relationship between collaborators from the same hierarchical groups is respectful and collaborative because due to the isolation and exile caused by the location of the farm and their jobs, they maintain that there is a building of cohesion and solidarity between them (collaborators) to alleviate the longing for their families, their hometowns and for the protection of their colleagues in their jobs, especially in cattle handling activities.

Table 5: Number of employees of Fazenda Lagoa do Triunfo, distributed by positions.

Position	Number of employees
Administrative Assistant	1
Administrative coordinator	1
Agricultural Machine Operator I	7
Animal Slaughterer	1
Cowboy	35
Production Fencer	14
Carpenter's Helper	2
Lead Fencer	3
Warehouse Assistant II	2
Pasture Supervisor	1

Infrastructure Analyst	1
Foreman	12
Support Foreman	1
General Foreman II	1
General Foreman I	1
Cook (O)	12
Master builder	5
Bricklayer's Helper	2
Rural Worker II	2
Aprendiz74	7
Livestock Coordinator	1
Pec Control Assistant	2
Insemination Leader	3
Pec Control Supervisor	1
Esp In Pec Reproduction	1
Pec Nutrition Supervisor	1
Electrician	1
Mechanic	1
Heavy Machine Operator II	1
Motor Mechanization Supervisor	1
Heavy Machine Operator	3
Train Driver	1
Service Inspector	1
Lead Cook	3
Calf Manager	2
Carpenter	2
Biker	1
Topographer	1
Nurse Technician	1
Teacher	1
Cleaning Assistant	1
Personal Dept. Assistant	1
Tec Occupational Safety	1
Joint Manager	1
TOTAL EMPLOYEES	144

In addition to the community members (settlers) and employees of Fazenda Lagoa do Triunfo, other interested parties were identified, as follows:

- AgroSB;
- Biofílica Ambipar Environment;
- Public Agencies;
- Social Actors - Public, private, non-governmental organizations, associations, among others.

All stakeholders should be invited to be part of the discussions of the Ateles REDD+ Project, in order to have a space for articulation and communication between AgroSB, the settlers and other stakeholders involved in the Project. The evaluation of rights, interests and relevance of each group of actors in relation to the Project was carried out, together with AgroSB employees and is specified in Table 5 below.

Table 6: Description of the actors involved in the Ateles REDD+ Project.

Group of actors involved in the Project	Rights regarding the Project	Interests in their participation in the Project	Relevance of participation in the Project
AgroSB	Holder of the right of credits, responsible for the investments, development and implementation of the Project. Execution and local management of social activities. It is also the organization responsible for managing financial resources.	Ensure the inclusion of communities (settlers) in the activities of the Project and that the activities of Technical Assistance and Rural Extension also incorporate a look at issues such as education, health, guarantee of human rights, environment, culture and generation of employment and income.	High – Due to its great influence in the region, it becomes a primordial component in the containment of deforestation, in addition to generating opportunities for settler families through the activities already carried out and those foreseen in the Project.
Families residing in the Project area	Beneficiaries of social activities	Access alternatives of rural and socioeconomic technical assistance services to improve their living conditions.	High – They are essential components of social activities, deforestation control and the development of a local economy model based on sustainable and harmonious practices with the forest.
AgroSB workers	Beneficiaries of social activities	With constant presence in the project area, workers help contain deforestation, in addition to accessing alternative rural and socio-economic technical assistance services to improve their living conditions.	High – many workers are also community workers, in this sense, they are essential components of social activities and prevention of deforestation, due to their constant presence in the project area.
Public Agencies	Coordinate with other actors to improve the implementation and permeability of public policies, support complementary actions to implement the Project	Bring the government closer to the community demands and strengthen governmental relations, which currently present themselves as fragile. Participate in monitoring the development of private and voluntary REDD+ initiatives, cooperate with the development of public policies.	Medium – The players are officially responsible for developing and implementing socio-environmental and economic public policies

Public, private, non-governmental organizations, associations and others	Not applicable.	Develop partnerships, provide technical assistance, promote job creation, expand operations, improve local governance, carry out research, produce and disseminate knowledge, promote forest management for sustainable production, ensure the permanence of traditional communities, develop and publish scientific works, guarantee of an area with a rich socio-economic and environmental context, etc.	High – In order for the social activities of the Project to have maximum effectiveness in practice, the Project will seek to establish partnerships in the region, whether public, private, third sector and/or organizations. Through partnerships, it is expected that there will be greater strengthening and involvement of all stakeholders, as well as mutual collaboration and exchange of expertise in the demands mapped by the Project.

2.1.10. Sectoral Scope and Project Type

- **Sectoral scope:** 14 – Agriculture, Forestry and Other Land Uses (AFOLU);
- **Project Category:** Reducing Emissions from Deforestation and Forest Degradation (REDD);
- **Activity Type:** Avoided Unplanned Deforestation (AUD);
- **Grouped Project:** No.

2.1.11. Project Activities and Theory of Change (G1.8)

The theory of change is a hypothesis about how project activities will enable the achievement of certain objectives, including social and biodiversity benefits. Result chains can help address these challenges by helping project teams articulate strong and clear theories of change, focus and organize efforts, prepare for effective performance monitoring, learn what works and what doesn't, understand important conditions for success, and practice adaptive management.

For the elaboration and design of this Project, the result chain guided the use of the theory of change and was considered an important step in the process of developing the activities to help proponents clarify how they support the proposed strategic approach and will affect the change in the identified problem(s).

In this sense, the Ateles REDD+ Project aims to promote joint actions with the objective of generating net benefits for the climate, communities and biodiversity. Thus, the activities were designed and the actions proposed seeking to ensure the conservation and protection of biodiversity and natural resources, the reduction of unplanned deforestation and greenhouse gas emissions, local socioeconomic development, social inclusion, engagement and involvement of stakeholders and the promotion of adaptive and assertive governance.

This set of interconnected actions will allow the generation of financial resources, mainly from the commercialization of REDD+ credits registered in the VCS (*Verified Carbon Standard*) associated with social development and the conservation of natural resources, seeking to ensure adequate financing for the fulfillment of the objectives mentioned above, in addition to allowing its maintenance throughout the life cycle of the Ateles REDD+ Project.

To this end, the following series of activities were proposed for the development of the Project, after carefully evaluating and weighing the ideas and the wealth of knowledge accumulated for the Project Zone. In addition, the activities defined are divided by scopes for better understanding, namely:

a) General Scope

Initial Studies and Joints

The beginning of the Project included the articulations that resulted in the first on-site visit, for recognition of the area, in the signing of the contract, where a long-term partnership was defined aiming at environmental conservation and socioeconomic development of the region, in addition to meetings with technical partners to start the specific diagnoses of the Project.

Diagnoses are initial studies that provide technical support for the preparation of the project management plan. Among the studies carried out are: the survey of the forest carbon stock estimate and the elaboration of the deforestation baseline, which results in the understanding of the direct climate impacts; the Socioeconomic Diagnosis and Consultation with the communities, which deepened the studies already carried out in the area and resulted in the understanding of the direct social impacts; and the Environmental Diagnoses that, as well as the Socioeconomic Diagnosis, supported the construction of actions to stimulate economic development and sustainable use of natural resources, as well as ensuring the conservation of the standing forest, supporting the proposed activities and resulting in direct impacts on communities and biodiversity.

Implementation, monitoring and evaluation of activities developed by the Project

As a fundamental part of obtaining net benefits from a REDD+ Project, it must be monitored within the activities, indicators and monitoring plan proposed in this document. For this to occur, adequate Project management is necessary, ensuring that activities are carried out within a well-constructed and realistic schedule, that possible risks are mitigated and, in addition, that there is continuous and adaptive improvement for the effective implementation of the activities proposed by the Project.

Briefly, this activity will focus on monitoring and improving the management of the Project, adapting to the real scenario, verifying failures and risks, and finally, developing a space dedicated to managing, documenting, organizing and making official the historical milestones of the Project, the changes that have occurred and the adaptations made throughout the credit generation period. Always associated with making efficient and better guided decisions during the implementation and monitoring stages. This will bring more effective adaptive management, promoting more assertive decisions, as well as aligned with the assumptions drawn in the PDD.

Thus, the activity "implementation, monitoring and evaluation of the activities developed by the Project" has its foundations based on conducting efficient management throughout the Project's life cycle, thus achieving the expected positive impacts. In order for the objectives to be achieved, actions will be developed such as the production of reports on the Project's activities, the evaluation of indicators and results, the definition of the planning schedule and task priorities for all years, tactical plans to be followed and the implementation of tools that can express the historical milestones and the adaptive changes found for later evaluation of the activities with the interested parties.

If well conducted, the actions will lead to a greater understanding of the impacts generated and opportunities to redirect demands at priority levels, establishing procedures and protocols to be followed by stakeholders. That said, constant alignment and common strategies among stakeholders are indispensable so that this activity actually occurs and demonstrates, in detail, the efforts of the actors in the implementation and monitoring of the Project. All this assists the Project proponents in the continuous improvement of the implementation of the proposed activities, due to the identification of bottlenecks, ineffectiveness and failures.

Theme	Description of Activity	Expected for climate, community, and biodiversity		
		Outputs (short term)	Outcomes (medium term)	Impacts (long term)
Initial Studies	Estimate of carbon stock	- Estimate the carbon stock for the Forest class and its reservoirs, through the Forest inventory in the Project area; - Generate the technical report.	- Generate knowledge about biomass and carbon stocks in the project area and surroundings; -Enable the monitoring of reservoirs and adjustments in stock projections throughout the useful life of the project; - Contribution to the accounting of reduced emissions.	- Generation of inputs for long-term forest monitoring; - Identification of priority areas for stock conservation.
	Baseline determination	- Conducting the study to determine the spatial limits of the project and determine the deforestation baseline; - Generation of technical report; - Modeling of future deforestation	- Generation of knowledge about the dynamics of deforestation in the region; - Contribution to the accounting of reduced emissions; - Determination of the area of greatest risk to conduct field actions.	- Generation of inputs for long-term forest monitoring; - Generation of relevant data to be used by the government in the projection of future jurisdictional systems.
	Socio-economic diagnosis	- Elaboration of the contextualization of the municipalities, communities, and workers of Fazenda Lagoa do Triunfo; - Conducting the Socioeconomic Study; - Generation of technical reports	- Generation of the socioeconomic context of the updated region; - Supply of inputs for the design of proposed activities in line with the local context; - Providing input to the work of other stakeholders.	- Improvement of socio-economic conditions; - Long-term prevention of deforestation in the Project zone; - Ensure proper management of forest areas and the use of other natural resources.

Theme	Description of Activity	Expected for climate, community, and biodiversity		
		Outputs (short term)	Outcomes (medium term)	Impacts (long term)
	Biodiversity Diagnosis	- Elaboration of the contextualization of aspects of local biodiversity; - Conducting the Biodiversity Study; - Generation of technical report.	- Update of biodiversity studies; - Supply of inputs for the design of the proposed *in situ* biodiversity monitoring activities; - Providing input to the work of other stakeholders.	- Improving knowledge about local biodiversity; - Long-term prevention of deforestation in the Project zone; - Ensure proper management of forest areas and the use of other natural resources.
	Situational Alignment and Feedback with Communities	- Inform the interested parties of the REDD+ Project; - Identification, understanding and prioritization of the problems encountered in the region communities; - Conduct interviews and workshops with communities directly and indirectly involved, so as to design and present the Project activities; - Generation of technical reports.	- Allow for adaptable management of the project to incorporate the needs and reality of families; - Definition of Parameters to measure the benefits and impacts of the Project on the communities; - Share information about REDD+ and promote community involvement.	- Strengthening communication among stakeholders; - Improvement of quality of life and socio-economic aspects of communities; - Empowerment of communities regarding their rights, duties and importance of involvement in the Project.
Implementation, monitoring and evaluation of activities developed by the Project	The Project aims to monitor the implementation of proposed activities, to mitigate risks and ensure continuous improvement of activities, increasing the	- Production of reports on the Project's activities; - Evaluation of the indicators and results of the actions implemented; - Definition of activity planning for the following year; - Implementation of milestones for	- Greater understanding of the impacts generated with stakeholders; - Opportunity to redirect the activities of the Project to those of higher priority; - Refinement of data collection methods.	- Efficient management and implementation of the Project; - Scope of the impacts expected by the Project.

Theme	Description of Activity	Expected for climate, community, and biodiversity		
		Outputs (short term)	Outcomes (medium term)	Impacts (long term)
	quality of Project Management.	evaluation of Project activities with stakeholders.		

b) Climate Scope

<u>Structuring an Asset Surveillance Program</u>

The asset surveillance activity is related to climate benefits and its main objective is to mitigate and prevent the occurrence of unplanned deforestation in the Project Area as well as the consequent reduction of greenhouse gas emissions. Currently, Fazenda Lagoa do Triunfo does not have an effective protection program for its forest reserves.

Thus, the proposed activity intends to structure the processes related to surveillance actions, including training for workers in which they are trained to take the necessary measures in cases of illegal activities in the area, monitor border areas, register police reports, as well as preventive measures to prevent the unwanted entry of third parties. In the same sense, the Project will seek to strengthen local partnerships, especially with supervisory bodies, to assist in combating illegal activities within the area and facilitate communication for possible reports of complaints.

Within this scope, the Project proposes to assist field activities by interconnecting the monitoring activity via satellite image (described below) with the asset surveillance activity. Thus, the products of the satellite image monitoring activity will be used by the heritage surveillance team to evaluate the areas detected in the monitoring in the field, understanding the context of greater deforestation pressure in the area and being more assertive in the actions to prevent and combat illegal activities. In addition to this field detection, it will assist in the Project's strategy to verify sensitive areas and propose measures that mitigate deforestation in the Project area.

<u>Monitoring of Deforestation and Heat Spots</u>

Currently, no remote monitoring of deforestation or hot spots is carried out continuously at Fazenda Lagoa do Triunfo, this information is only obtained from public data.

That said, this activity provides for the elaboration of a remote monitoring of deforestation and hot spots, developing reports that monitor and record the deforested areas in the Project Zone, as well as the hot spots, in addition to strengthening and formalizing communication with surveillance actions.

In addition to this monitoring, the Project will seek to monitor deforestation through other available data as well as other high-resolution satellite images. This monitoring may result in bulletins with deforestation points and hot spots that will be forwarded to stakeholders and, in this way, this additional remote monitoring will provide better support for the strategic field patrol plan.

These actions are directly related to the containment of deforestation and invasions, maintenance of forest cover and, consequently, improvement of adaptations to climate change and, consequently, maintenance of the benefits for the climate, community and biodiversity foreseen by the scenario with the Project.

Theme	Description of Activity	Expected for climate, community, and biodiversity		
		Outputs (short term)	Outcomes (medium term)	Impacts (long term)
Structuring an Asset Surveillance Program	The Project aims to structure the inspection activities inside the farm, by training surveillance teams to contain illegal deforestation and invasions, as well as the maintenance of forest cover and, consequently, the maintenance of benefits for the climate, community and biodiversity by the scenario with the Project.	- Map invasion vulnerable points to install surveillance posts - Develop field intervention procedures - Training and qualification of security guards - Develop communication strategies between the surveillance team and remote monitoring team	- Greater understanding of the dynamics and agents of deforestation; - Conducting effective asset surveillance; - Reestablish the relationship with the environmental inspection agency and report illegal activities;	- Mitigation and prevention of deforestation; - Reduction of emissions from deforestation and forest degradation.
Monitoring of Deforestation and Heat Spots	The Project aims to remotely monitor the entire area of the farm and its surroundings, through the processing and analysis of satellite images, to assess human pressure, and prepare bulletins informing the points of deforestation and hot spots to strengthen surveillance actions.	- Implementation of deforestation monitoring systems and hot spots; - Monitoring and evaluation of deforestation areas remotely and hot spots; - Generation of deforestation reports and hot spots;	- Greater understanding of the dynamics of deforestation to conduct more effective asset surveillance; - Supply of inputs for design and improvement of interventions in the field;	- Mitigation and prevention of deforestation; - Reduction of emissions from deforestation and forest degradation.

c) Scope Biodiversity

Biodiversity Monitoring

Fazenda Lagoa do Triunfo is located in one of the eight Amazonian Endemism Regions, the Tapajós Endemism Center, a region delimited by the right bank of the Tapajós River and the left bank of the Xingu River, where approximately 20% of its original primary vegetation had already been degraded by 2017 (Braz et al., 2017). In this context, the natural reserve areas of the farm play a key role in the preservation of various populations of species endemic to this region.

Since the central theme of the Ateles REDD+ Project is the containment of deforestation in the area in which it was inserted, the maintenance and preservation of local biodiversity will also be promoted through the project's activities. Thus, one of the activities identified in the development of the theory of change that is proposed for biodiversity in the Ateles REDD+ Project is the “Biodiversity Monitoring”. Activity that aims to: assess the responses of populations and ecosystems to conservation practices through a REDD+ Project and the impacts of external factors, such as habitat loss, landscape changes, over-exploitation of species and climate change.

In addition, we recognize the importance of engaging the communities that live there. The active participation of these communities is fundamental to the success of the project and to promote biodiversity conservation in a sustainable way. We will make efforts to engage local communities by raising awareness of the importance of biodiversity and empowering them to participate in monitoring.

Thus, the long-term objective of this activity is to generate a series of positive impacts on biodiversity, such as conservation of already diagnosed species, including those that present some degree of threat, preservation of local habitats, conservation of HCVAs, generation and dissemination of knowledge and environmental awareness regarding biodiversity, dissemination of results to stakeholders, permanence of ecosystem services, mapping of new areas of great relevance for conservation and maintenance of connectivity in the landscape

Sanitary precautions with domestic animals

According to diagnostics carried out in the project area, the existence of many domestic animals was verified, mainly in order to increase the safety of the residents of the region. However, the abundance of these animals endangers local biodiversity, especially wild animals, due to the fact that dogs and cats are potential carriers of several etiological agents of risk to native fauna, being agents of transmission of more than 30 zoonoses (Heliodoro G et al. 2020[37]).

From this information, it is evident the importance of actions aimed at the welfare of domestic animals, so we developed the activity "Sanitary precautions with domestic animals", which seeks to monitor the abundance of domestic animals in the project area and ensure that their interactions and dynamics do not have negative impacts on local biodiversity. In addition, we seek to raise awareness among residents of the region about the importance of adopting adequate sanitary measures for the health and well-being of domestic animals.

Cattle predation assessment

Fazenda Lagoa do Triunfo has livestock operations, in addition to the forest areas that are the focus of our REDD+ Project, and another challenge encountered during studies in the project area was the losses caused by cattle predation by jaguars. Although no traces of hunting activities have been found, we

[37] Heliodoro G et al. Domestic Animals and the Risk of Transmission of Pathogens to Wildlife in the Surrounding Area of Tijuca National Park. Brazilian Biodiversity, 10(2): 133-147, 2020

understand that the relationship of stakeholders with the jaguar may become conflictual due to the financial damage caused by predation.

With this challenge in mind, the activity "Evaluation of livestock predation" will be implemented in the life cycle of the project, which aims to map the predation that occurs on the farm and the main causes, for this we need to understand the dynamics and movement patterns of predators in the region, the attack records, in addition to being able to separate the normal mortality of livestock from that resulting from predation. And, after understanding the scenario, to be able to present what would be the possible improvements in cattle management aimed at reducing predation.

Theme	Description of Activity	Expected for climate, community, and biodiversity		
		Outputs (short term)	Outcomes (medium term)	Impacts (long term)
Biodiversity monitoring	This activity aims to monitor the fauna and flora in the project area over time, contributing to the conservation and protection of local biodiversity during the Project. To this end, the activity seeks to engage stakeholders in this monitoring.	- Organization and implementation of long-term biodiversity monitoring in the Project Area; - Articulation for stakeholder involvement in biodiversity monitoring throughout the project; - Evaluation of the possibility of stakeholder involvement in monitoring.	- Increased knowledge on regional biodiversity; - Obtaining data that tracks population dynamics and species behavior, including endangered, endemic and invasive species; - Local environmental engagement and awareness about biodiversity and its importance; - Further clarification of the difficulties encountered in this activity, providing adjustments and control of changes in the implementation of the Project.	- Stakeholder engagement and empowerment on local biodiversity; - Robust database on local biodiversity; - Balanced ecosystem and for resident biodiversity; - Maintenance or addition of endangered and endemic species on site. - Consolidated knowledge about invasive species for correct decision-making
Sanitary precautions with domestic animals	The purpose of this activity is to monitor the abundance of domestic animals in the project area, in addition to ensuring that their dynamics do	- Collection of data on abundance, health and those responsible for domestic animals in the project area - Articulate partnerships for health awareness campaigns for domestic animals	- Development of “policies” for the insertion of new domestic animals in the project areas with stakeholders	- Raise awareness among residents of the region about the influence of domestic animals on local biodiversity

Theme	Description of Activity	Expected for climate, community, and biodiversity		
		Outputs (short term)	Outcomes (medium term)	Impacts (long term)
	not negatively influence local biodiversity.			
Cattle predation assessment	This activity is aimed at reducing the losses resulting from livestock predation.	- Collection of records and monitoring of cattle predation - Understanding of cattle management actions already carried out by the farm, aiming to reduce predation	- Consider improvements in livestock management - Propose refinements in the record of cattle mortalities - Monitoring of the movement patterns of predators near the areas of operation	- Decrease conflicts between farm staff and predators - Reduce the damage caused by livestock predation

d) Social scope

An estimated 104 families of settlers residing in the area of the Ateles REDD+ Project were estimated. Most of the community members are affiliated with the Association of Farmers of Calumbi and Tucunaré (AGRICATU). Through it, these settlers are represented and guided in several areas of knowledge, in addition, it was from it that the ties between the project and the community were established. In this way, initiatives and negotiations can be presented and leveraged through AGRICATU. The proposed activities were defined incorporating and addressing the main focal problems associated with the way of life of these people, in line with the premises necessary for the development of a REDD+ Project.

In addition, AgroSB employees and their families will also be able to access training courses, alternative rural and socio-economic technical assistance services, and whatever else is of interest to the group, since these community members are important players in containing deforestation and supporting the development of the activities proposed within the scope of climate and biodiversity.

Strengthening local governance

The construction of cooperation and articulation mechanisms that stimulate a network of cohesive social relations between the different actors that make up a territory is one of the fundamental conditions to boost local development. Still, in the case of a long-term project, construction and maintenance over time of a good relationship between stakeholders and proponents is essential to enable the implementation and monitoring of the activities proposed by the Project, in addition to allowing effective follow-up of response to demands and results obtained, contributing to the peaceful and fair resolution of possible conflicts and mitigating possible risks.

Therefore, this activity aims to strengthen local governance, not only in actions to promote a good relationship between stakeholders and proponents of the Project, but also by encouraging partnerships and good relations between communities in the Project Zone and other governmental and non-governmental social actors. These actions go through the strengthening of the social organization of the settlers with the empowerment of local leaders, the creation of safe environments for discussion and exchange of knowledge, as well as a mechanism for transparency and communication that will be implemented throughout the Project.

The strengthening of relations between bidders and stakeholders will be carried out by structuring procedures that guarantee transparency in the management of the activities developed by the Ateles REDD+ Project, such as communication and consultation channels, participatory councils, assemblies, among other formats described in detail in Section 2.3.8. These mechanisms will be fundamental to the success of the Project and will be encouraged and reinforced by this activity. In addition, the activity also aims to promote and strengthen local social organization, structured in the figure of the Association of Farmers of Calumbi and Tucunaré (AGRICATU), especially as a strategy to address issues of collective interest, such as environmental, economic and land activities, in the search for basic community rights.

Nevertheless, Lagoa do Triunfo Farm is located in the Triunfo do Xingú Environmental Protection Area, which is a Conservation Unit for sustainable use. The APA Triunfo do Xingú, which is in the development phase of its Management Plan, aims to reconcile the multiple uses and subjects that are inserted in its area, as well as to ensure the enjoyment of its natural resources for present and future generations. In this sense, the effective management of this area is vital for the development of the Project. To this end, the Ateles REDD+ Project aims to develop action strategies that strengthen the improvement of APA management, including the strengthening of partnerships between the project, the APA management council and the State Secretariat for the Environment and Sustainability of Pará (SEMAS), supporting the implementation of the Management Plan for APA Triunfo do Xingu.

In view of the actions described, the long-term objective of this activity is to strengthen cohesion in social relations between stakeholders, stimulating mutual trust, empowerment of vulnerable groups, representativeness and equality in decision-making.

<u>Development and strengthening of value chains</u>

Most of the communities that are part of the Project Area are made up of family farmers, who also carry out livestock farming. Therefore, there are problems intrinsic to this social group, mainly related to planning, both territorial and production, in addition to problems related to marketing, such as product pricing and difficulties in accessing the market.

This activity will seek to promote actions focused on improving the management of family agricultural production units, especially in the process of planning, producing, valuing and marketing livestock products. For this, the structuring of technical assistance and rural extension programs will be fundamental to promote efficiency and diversification of production, through the adoption of integrated and sustainable production systems.

The Ateles REDD+ Project can contribute to increasing the quality of beef production and its derivatives, through the transfer of scientific knowledge and technologies through the vast experience of the proposing company, AgroSB. Thus, the proposed initiatives are expected to facilitate access to different marketing channels, promote the valorization of local production and assist in the construction or strengthening of value chains and local productive arrangements.

Thus, the long-term goal of this activity is to improve socioeconomic conditions and the well-being of community members. The proposed actions seek to encourage food security, autonomy and financial resilience provided by the development and strengthening of local production chains.

<u>Promoting sustainable production practices</u>

Based on the socioeconomic diagnosis, it was found that the settlers living on the farm, when developing their agricultural and livestock practices, act as drivers of deforestation in the Project area, being one of the main agents identified. The deforestation carried out is related to culturally established agricultural practices, which impair soil fertility, result in low productivity and are associated with poorly diversified production models. In this context, the settlers lead the opening of new production areas in search of more fertile soils and practice burning for the cleaning of new pasture areas and new production fields.

From this context, the activity proposes to act mainly around small and medium-sized rural producers, family farmers and associates of AGRICATU, promoting actions such as encouraging the implementation of systems and models of agricultural and livestock production, which are adaptive, sustainable and assimilate crops already established and widespread in the region, such as the production of cocoa, corn, cassava, bananas, the production of milk and its derivatives. Stimulating production in agroforestry systems, structuring land use in order to guarantee small rural producers a minimum income resulting from the diversity of activities on the ground, in addition to providing the recovery of degraded areas, the sustainable use of primary or secondary forest areas and mitigating the advance over forest areas.

Sustainable agricultural and livestock production practices and models can provide diversification of production, increase productivity, discourage rural exodus, reduce pressure for deforestation and the effects of land speculation. The necessary actions for this initially involve building trust and cooperation relationships among the target audience, establishing strategic partnerships for in locu actions, promoting training and technical development and structuring networks of technical assistance and rural extension.

Thus, based on the actions described, the long-term objective of this activity is to enable food security combined with the diversification of income sources, promoting forest conservation through the strengthening of sustainable agricultural and livestock production practices. Thus, the proposed actions encourage the permanence of small producers and family farmers in the countryside and increase their resilience in relation to their migration to activities that are vectors of deforestation in the region, such as mining, illegal timber trade or extensive livestock farming carried out through predatory and disorderly occupation of forest areas. Furthermore, encouraging the permanence of vulnerable families in the countryside and reinforcing their sense of belonging is also a strategy to reduce the effects of land speculation and break the cycle of deforestation due to predatory occupation.

<u>Land security and social welfare</u>

It is known that the lack of definition of land rights in the Amazon is admittedly a serious obstacle to the advancement of sustainable development policies in the region. Land security guarantees access to credit, government programs and technological innovations, in addition to promoting business competitiveness and sustainability, without neglecting environmental conservation. In the context of the Ateles REDD+ Project, land security is one of the main demands pointed out by the socioeconomic diagnosis. At this juncture, the project proposes to collaborate so that the communities obtain their rights, helping in the delineation of the useful and productive areas of these families, thus allowing the assured use of these lands, so that there is a structuring of the social fabric of this community and the feeling of territorial belonging in the area is strengthened.

As a measure to support these communities that depend on the land for their subsistence, it is intended to develop cooperation agreements through the Association of Farmers of Calumbi and Tucunaré (AGRICATU), in which these settlers acquire land security and, in return, collaborate for the preservation of forest remnants. In this sense, the project aims to propose a payment program for environmental services (PES), through sharing the benefits of carbon credits, following the principles of the protector-receiver. Stakeholders and beneficiaries of this PES program will also be encouraged to report new invaders and prevent future leaks, as it will be from actions to contain deforestation that the generation of carbon credits will take place, and will result in direct financial returns for these community members.

Also with a view to improving the quality of life of these people living in the project area, bringing more social well-being, infrastructure and qualities of essential services, it is intended to strengthen partnerships with the municipality of São Félix do Xingu and the state of Pará to maximize efforts aimed at raising the level of education, technology, health and transport infrastructure in the area and surrounding the project.

Thus, the long-term objective of this activity is to improve the socioeconomic conditions and well-being of the community, promoting facilitation actions to obtain land security. The proposed actions seek improvements in public infrastructure and services, access to rural credit benefits and other government programs, autonomy and financial resilience, provided by the development and strengthening of the community social fabric.

<table>
<tr><th rowspan="2">Theme</th><th rowspan="2">Objectives</th><th colspan="3">Expected for climate, community, and biodiversity</th></tr>
<tr><th>Outputs (short term)</th><th>Outcomes (medium term)</th><th>Impacts (long term)</th></tr>
<tr><td>Strengthening Local Governance</td><td>The objective of this activity is to strengthen cohesion in social relations between stakeholders, stimulating mutual trust, empowerment of vulnerable groups, representativeness and equality in decision-making.</td><td>- Strategic partnerships signed with socio-environmental projects already developed in the municipality;
- Meetings, meetings and dialogues with community leaders and other stakeholders to carry out the defined interventions;
- Constant approximation of the Target Audience and the establishment of relationships of trust;
- Implementation and consolidation of the communication channel with stakeholders.
- Space resumed within the APA's management council, collaborating with conservation and restoration actions.</td><td>- Local participatory governance structures built and strategic partnerships in place;
- Increased collective engagement in claiming basic rights and services;
- Mapping and resolution of suggestions and complaints from interested parties;
- Creation of opportunities for exchange of experiences;
- Stakeholders with access to Project information and actively collaborating in the proposed activities;
- Communication channels consolidated and widely used by stakeholders, bringing positive returns for closer relationships.</td><td>- Efficient management and implementation of the Project;
- Empowerment of community members and strengthening of social capital;
- Strengthening communication between stakeholders, helping the performance of the Project;
- Improvement in the social organization of the territory and in the relationships between stakeholders.</td></tr>
</table>

Development and strengthening of value chains	The objective of this activity is to improve socioeconomic conditions and the well-being of community members, through actions aimed at encouraging food security, autonomy and financial resilience provided by the development and strengthening of local production chains.	- Improvement in the management of useful livestock areas, stratified by families or community groups; - Mapping of market opportunities carried out with a focus on directing the production of community members to meet the demands of the local market. - Training, technical and technological assistance aimed at trade and production diversification, in progress; - Production developed with greater added value; - Agreements signed with strategic partners;	- Expansion of the trade network and strengthening of the production chain; - Production partnership between AgroSB and the community; - Empowered community leaders and technically educated community; - Diversified production, reaching new markets and increasing producers' income.	- Improving the well-being of communities; - Food and financial security in communities; - Technical autonomy of rural producers;
Promoting sustainable production practices	The objective of this activity is to enable allied food security, diversification of income sources, strengthening of the value chain and forest conservation through the strengthening of sustainable agricultural practices.	- Mapping of rural properties carried out and the zoning of the defined useful production areas; - Training and courses aimed at sustainable agricultural and livestock production practices, in progress; - Training and courses aimed at the management	- Relationships of trust and cooperation between the established target audience; - Implemented sustainable production systems and models; - Decrease in fires; - Decrease in deforestation; - Increased agricultural and	- Strengthening sustainable and adaptive livestock and agricultural practices; - Food security; - Diversification and increase in income; - Conservation of forest resources and ecosystem services; - Reduction of emissions from deforestation and forest degradation.

		and control of fire, in progress. - Access to technical assistance and rural extension; - Community members technically trained through courses and able to conduct sustainable land use practices - Mapped agroextractivist activities; - Mapped degraded areas; - Agreements signed with strategic partners;	livestock quality and productivity, and diversification of production; - Strengthening human capital and increasing local knowledge in sustainable practices.	
Land security and social welfare	The objective of this activity is to improve the socioeconomic conditions and well-being of the community, promoting facilitation actions to obtain land security and the improvement of basic services to serve the population.	- Design of useful areas of production and occupation by families; - Agreements signed guaranteeing community members the permanence and use of land; - Cooperation agreements with financial returns established with the community, through the principle of protector-recipient; - Collective inputs available to community members; - Participatory workshops in progress;	- Sense of belonging and reduction of agrarian conflicts; - Active participation of community members in the containment of deforestation; - Improvement in basic infrastructures and services to serve the population; - Increased income and satisfaction with the project's actions.	- Reduction of deforestation rates; - Improvement in the quality of life of community members; - Diversification and increase in income; - Land security;

		- Improvements in community infrastructure; - Agreements signed with the government, for the development of improvement actions in the basic services of service to the population.		

2.1.12. Sustainable Development

The Ateles REDD+ Project is focused on promoting sustainable development in the region, building joint actions with all stakeholders, under the facilitation and encouragement of AgroSB. Such actions are drivers of net benefits for climate, biodiversity and local communities. Based on this support and in line with the expected impacts, the project will contribute to the following UN Sustainable Development Goals (SDGs):

SDG 2 – END HUNGER, ACHIEVE FOOD SECURITY, IMPROVE NUTRITION and PROMOTE SUSTAINABLE AGRICULTURE 2.4 - By 2030, ensure sustainable food production systems and implement resilient agricultural practices that increase productivity and production, help maintain ecosystems, strengthen the capacity to adapt to climate change, extreme weather, droughts, floods and other disasters, and progressively improve land and soil quality

National Goal (Brazil): Goal maintained unchanged, in relation to the official text of goal 2.4, due to the fact that it is very comprehensive and, therefore, contemplates the specificities of the Brazilian reality.

Justification for application in the project: The project combines the socio-economic demands of communities with local opportunities aimed at more resilient economic activities, through the "Promotion of sustainable agricultural practices". To this end, the project enables training and technical development actions aimed at communities, in association with different partners and extension agents, in order to promote knowledge about the importance of reconciling good production practices with the preservation and maintenance of natural resources and encourage the adoption of sustainable and adaptive production systems.

SDG 3 – ENSURE A HEALTHY LIFE AND PROMOTE WELL-BEING FOR ALL, AT ALL AGES

United Nations (UN) Goals: 3.9 - By 2030, substantially reduce the number of deaths and diseases from hazardous chemicals, air and soil water contamination and pollution.

National Goal (Brazil): 3.9 - Goal maintained without change. The original wording was maintained, as there is still no reference that allows the country to stipulate what percentage of mortality reduction due to the causes mentioned in the target would be recommended for the 2015-2030 period.

Justification for application in the project: The project enables and encourages the development of sustainable and organic agricultural practices, without the use of chemical pesticides, exploring diversified and biodynamic crops, such as the adoption of agroforestry systems. Thus contributing to the maintenance of the biodiversity of local fauna and flora, and preserving the sources of contamination by chemicals harmful to health. The use and dissemination of technologies for the application of biobased resources for the generation of bioproducts, bioinputs and also the use of renewable energy sources, such as biogas, to the detriment of the use of oil, will be encouraged. To this end, the project has the support and collaboration of specialized partners in order to ensure the effectiveness and engagement of stakeholders. These practices promoted by the Project contribute positively to improvements in the health and quality of life of community residents in the project area and its surroundings, reducing the risks of death and disease caused by the indiscriminate use of hazardous chemicals, contamination and pollution of air and soil water.

SDG 4 – ENSURE INCLUSIVE and QUALITY EDUCATION and PROMOTE LIFELONG LEARNING OPPORTUNITIES FOR ALL United Nations (UN) goals: 4.4 - By 2030, substantially increase the number of young people and adults who have relevant skills, including technical and professional skills, for employment, decent jobs and entrepreneurship.

4.7- By 2030, ensure that all apprentices acquire the knowledge and skills necessary to promote sustainable development, including, but not limited to, through education for sustainable development and sustainable lifestyles, human rights, gender equality, promotion of a culture of peace and non-violence, global citizenship, and appreciation of cultural diversity and of culture's contribution to sustainable development.

National Goal (Brazil): 4.4- By 2030, substantially increase the number of youth and adults who have the necessary skills, especially technical and vocational skills, for employment, proper work and entrepreneurship.

4.7 - Goal maintained unchanged, in relation to the official text of goal 4.7, due to the fact that it is very comprehensive and, therefore, contemplates the specificities of the Brazilian reality.

Justification for application in the project: The project enables and encourages access to education through technical courses and training focused on the environmental and socio-economic areas, especially on resilient agricultural production practices, development and strengthening of value chains, incentives for social organization and environmental education for preservation of endangered species. In addition, it provides specific training for the property surveillance team in the Project Area. For this, it has the support and collaboration of specialized partners, in order to guarantee the effectiveness and engagement of stakeholders. These training activities promoted by the project make it possible to strengthen human capital, consolidate the feeling of belonging, access to information, better employment conditions and income diversification; mainly for small rural producers, resulting, consequently, in the maintenance of the forest and its resources.

SDG 6 - ENSURE the AVAILABILITY and SUSTAINABLE MANAGEMENT OF WATER AND SANITATION FOR ALL and ALL
United Nations (UN) goals: 6.6 - By 2030, protect and restore water-related ecosystems, including mountains, forests, wetlands, rivers, aquifers and lakes.

National Goal (Brazil): 6.6 - By 2030, protect and restore water-related ecosystems, including mountains, forests, wetlands, rivers, aquifers and lakes, reducing the impacts of human action.

Justification for application in the project: The main objective of a REDD+ Project is the conservation of forest cover and its respective carbon stocks by curbing deforestation and forest degradation. An important project activity is the strengthening of asset surveillance, ensuring protection of the Project Area's forest cover, in addition to other activities that encourage the adoption of sustainable production practices as alternatives to deforestation. Thus, the maintenance of forests is essential for the provision of water ecosystem services and, consequently, the availability of water for all. They contribute to the process of regulating the hydrological cycle, influencing some factors such as rainfall, water availability and purification, soil protection, lakes and water courses.

SDG 12 – ENSURE SUSTAINABLE CONSUMPTION and PRODUCTION PATTERNS United Nations (UN) goals: 12.2 - By 2030, achieve sustainable management and efficient use of natural resources.12.8 - By 2030, ensure that people everywhere have the information and awareness relevant to sustainable development and lifestyles in harmony with nature.

National Goal (Brazil): 12.2 - Goal kept unchanged in relation to the official text of global goal 12.2

12.8 - By 2030 ensure that people everywhere have the relevant information and awareness for sustainable development and lifestyles in harmony with nature, in accordance with the National Environmental Education Program (PRONEA).

Justification for application in the project: The Project encourages the "Promotion of sustainable agricultural practices" with actions aimed at identifying potential activities related to resilient subsistence agriculture, extractivism and management of forest products; according to the demand and profile of local communities. In this sense, the project works to disseminate knowledge, instructions and experiences focused on the efficient use of natural resources and environmental preservation; in addition to the "Development and Strengthening of Value Chains", encouraging sustainable business chains through greater integration between stakeholders and regional markets, thus generating income, well-being and cultural identity for the communities fostered. With this, learning, engagement and predisposition of these families in activities to improve productive and extractive practices, increase the project's governance and help maintain the forest cover and preserve its ecological aspects.

SDG 13 – TAKE URGENT MEASURES TO COMBAT CLIMATE CHANGE AND ITS IMPACTS

United Nations (UN) Goals: 13.3 - Improve education, raise awareness and human and institutional capacity on climate global mitigation, adaptation, impact reduction and early warning.

National Goal (Brazil): 13.3 - Improve education, raise awareness and human and institutional capacity on climate change, its risks, mitigation, adaptation, impacts, and early warning.

Justification for application in the project: The activities developed by the project are focused on sustainable practices, which contribute to the reduction of unplanned deforestation and forest degradation, and consequently, the reduction of greenhouse gas emissions. The Project has the potential to reduce 6,717,438 tCO_2 of GHG emissions in 10 years, with prevention of deforestation in native forest. The asset surveillance strengthening activity will be carried out through the presence of agents in the Project Area, in an integrated manner with remote and continuous monitoring activities of deforestation and forest fires, allowing refinement of prevention actions, combating illegal activities and maintenance of the forest.

SDG 15 – PROTECT, RESTORE AND PROMOTE THE SUSTAINABLE USE OF TERRESTRIAL ECOSYSTEMS, SUSTAINABLY MANAGE FORESTS, COMBAT DESERTIFICATION, HALT AND REVERSE LAND DEGRADATION AND HALT BIODIVERSITY LOSS

United Nations (UN) Goals: 15.1 - By 2020, ensure the conservation, recovery and sustainable use of inland terrestrial and freshwater ecosystems and their services, in particular forests, wetlands, mountains and arid lands, in accordance with the obligations arising from international agreements.

15.2 - By 2020, promote the implementation of sustainable management of all types of forests, halt deforestation, restore degraded forests and substantially increase afforestation and reforestation.

National Goal (Brazil): 15.1 - By 2020, areas such as Permanent Preservation Areas (PPA's), Legal Reserves (LR's) and indigenous lands with native vegetation, at least 30% of the Amazon, 17% of each of other terrestrial biomes and 10% of marine and coastal areas, mainly areas of special importance for biodiversity and ecosystem services will be conserved, through systems of conservation units provided for in the Law of the National System of Conservation Units (SNUC), as well as other categories of areas officially protected, ensuring and respecting the demarcation, regularization and effective and equitable management, with a view to ensuring interconnection, integration and ecological representation in broader terrestrial and marine landscapes.

15.2 - By 2030, zero illegal deforestation in all Brazilian biomes, expand the area of forests under sustainable environmental management and recover 12 million hectares of forests and other forms of degraded native vegetation, in all biomes and preferably in Permanent Preservation Areas (APPs) and Legal Reserves (RLs) and, in areas of alternative land use, expand the area of planted forests by 1.4 million hectares.

Justification for application in the project: The main objective of a REDD+ Project is the conservation of forest cover and its respective carbon stocks by curbing deforestation and forest degradation. The Project is part of and protects a large part of a region classified, by the Ministry of Environment (Ordinance No. 463, of 12/18/2018), as a Priority Area for the Conservation, Sustainable Use and Sharing of Benefits of Brazilian Biodiversity and serves as a ecological corridor for preserved areas in the region, protecting endangered fauna and flora species. In addition, the Ateles REDD+ Project is part of the Tapajós Endemism Center, a region with a large concentration of endangered and endemic species. From this context, the Project aims to minimize habitat loss, landscape changes, over-exploitation of species and climate change. To this end, it seeks to engage, involve and raise awareness of all stakeholders regarding the importance of fauna and flora biodiversity in providing ecosystem services, maintaining landscape connectivity, controlling environmental degradation and limiting the excessive use of natural resources. Also, the systematic in situ monitoring of fauna and flora, the studies on the natural and socio-economic resources already carried out, necessary to comply with the CCB certification, will also contribute to integrate the values of ecosystems and biodiversity into national and local planning, in the development processes and in poverty reduction and biodiversity conservation strategies.

2.1.13. Implementation Schedule (G1.9)

The articulation between the stakeholders of the REDD + Ateles Project began in 2021, with the beginning of the first negotiations of the proposal. In addition, in this first stage, the area and the social context were recognized, in order to verify the technical feasibility of implementing a REED+ Project in the areas of Fazenda Lagoa do Triunfo. This activity was conducted in a partnership between the teams of Biofílica, AgroSB and the technicians of the Peabiru Institute.

On July 1, 2022, the contract was signed by the bidders, and from this milestone the Ateles REDD+ Project began to be planned and developed effectively.

The schedule with the main dates and milestones for the development of the Ateles REDD+ Project will be presented below in table 6. The detailing and management of project activities was conducted from an implementation, validation and verification schedule, as described.
After the registration stage of the Project and credits, more detailed schedules will be developed for the monitoring of management and monitoring activities, including the monitoring of socio-environmental management activities, deforestation, emissions, biodiversity (fauna and flora) and High Conservation Value Areas (HCVA), in addition to new credit checks, conducting the credit marketing process, and whatever else is relevant for the efficient conduct of REDD+ Ateles, over the 40 years of the Project's useful life.

Table 7: Detailed implementation schedule of the main activities related to the Ateles REDD+ Project.

Stage	Date	Activity
Management Plan	2nd half of 2022	Formalization of the Partnership Agreement
		Presentation of the Project Implementation Plan
		Start of Planning definition of Operational Management Guidelines
		Prospecting partners and suppliers
		Formalization of strategic partnerships and contracting of suppliers
		Conducting the 1st Workshop with partners and suppliers involved
	1st half of 2023	Socioeconomic Diagnosis
		Environmental Diagnosis
		Carbon Stock Inventory
		Holding of the 2nd Workshop with partners and suppliers involved
	2nd half of 2023	Baseline Study
		Situational alignment with the community
		Feedback with community and other stakeholders
Certification		Preparation of the Project Design Document (DCP)
		Preparation of the Monitoring Report
		Public Consultation with communities and other stakeholders
		Audit (Validation/Verification)
	From 2024 to 2052	Project and Credit Registration
Management and Monitoring Plan		Elaboration of an action plan for the implementation of project activities
		Development and Monitoring of socio-environmental management activities
		Monitoring of deforestation and emissions

		Monitoring Biodiversity (fauna and flora) and High Conservation Value Areas (HCVA)
		Verification of credits (Selection and contracting of the verification body; Production of Follow-up Bulletins for verification; Field audit monitoring; Registration of credits)
		Conducting the credit marketing process
		Other relevant activities for efficient conduction of REDD+ Ateles- Detailing to be defined throughout the Project verifications.
		and June 2052- End of project

<table>
<tr><th>Stage</th><th>Management Plan</th></tr>
<tr><td>Date</td><td>2nd half of 2022</td></tr>
<tr><td colspan="2">Activity:
• Formalization of the Partnership Agreement
• Presentation of the Project Implementation Plan
• Start of Planning definition of Operational Management Guidelines
• Prospecting partners and suppliers
• Formalization of strategic partnerships and contracting of suppliers
• Conducting the 1st Workshop with partners and suppliers involved</td></tr>
<tr><td>Date</td><td>1st half of 2023</td></tr>
<tr><td colspan="2">Activity:
• Socioeconomic Diagnosis
• Environmental Diagnosis
• Carbon Stock Inventory
• Holding of the 2nd Workshop with partners and suppliers involved</td></tr>
<tr><td>Date</td><td>2nd half of 2023</td></tr>
<tr><td colspan="2">Activity:
• Baseline Study
• Situational alignment with the community
• Feedback with community and other stakeholders</td></tr>
<tr><th>Stage</th><th>Certification</th></tr>
<tr><td>Date</td><td>2nd half of 2023</td></tr>
<tr><td colspan="2">Activity:
• Preparation of the Project Design Document (DCP)
• Preparation of the Monitoring Report
• Public Consultation with communities and other stakeholders
• Audit (Validation/Verification)
• Project and Credit Registration</td></tr>
<tr><td>Date</td><td>1st half of 2024</td></tr>
</table>

Activity: • Project and Credit Registration	
Stage	Management and Monitoring Plan
Date	2024 to 2062
Activity: • Elaboration of an action plan for the implementation of project activities • Development and Monitoring of socio-environmental management activities • Monitoring of deforestation and emissions • Monitoring Biodiversity (fauna and flora) and High Conservation Value Areas (HCVA) • Verification of credits (Selection and contracting of the verification body; Production of Follow-up Bulletins for verification; Field audit monitoring; Registration of credits) • Conducting the credit marketing process • Other relevant activities for efficient conduction of REDD+ Ateles- Detailing to be defined throughout the Project verifications.	

2.1.14. Project Start Date

The start date of the Ateles REDD+ Project was set at July 1, 2022, as it represents the moment of formalization of the partnership agreement between AgroSB Agropecuária S.A. and Biofílica Ambipar Environment for the development of the Project.

2.1.15. Benefits Assessment and Crediting Period (G1.9)

The accreditation period of the Ateles REDD+ Project will refer to the complete period of 40 years, starting on July 1, 2022 and ending on June 30, 2062.

The monitoring of the benefits to climate, communities and biodiversity in the Ateles REDD+ Project will be continuous, being subject to verification with the CCB, ideally every three years, throughout the duration of the Project.

2.1.16. Differences in Assessment/Project Crediting Periods (G1.9)

The accreditation period of the Project is marked by the formalization of the partnership agreement between AgroSB Agropecuária S.A. and Biofílica Ambipar Environment for the development of the Project, as mentioned in Section 2.1.14. After the formalization of this partnership, the Project begins and, consequently, the first major investments for the development of technical studies such as the determination of the baseline and socioeconomic and environmental diagnoses.

Then, activities related to climate, community and biodiversity are carried out, together with the monitoring of the corresponding attributes. At that moment, the collection of the first carbon credits commercialized takes place, which come from the first verification of the Project by VCS certification in order to financially encourage the development of these activities and monitoring.

In this way, the assessment of changes related to climate, community and biodiversity benefits begins in a period shortly after the start of the crediting period of the Project.

2.1.17. Estimated GHG Emission Reductions or Removals

Table 8: Estimated GHG emission reductions or removals in the Ateles REDD+ Project.

Year	Estimated reduction of GHG emissions (tCO2e)
2023	356.187
2024	237.546
2025	314.819
2026	339.590
2027	378.483
2028	259.517
2029	446.283
2030	492.183
2031	433.235
2032	551.244
Total estimated ERs	**3.809.086**
Total number of crediting years	**40**
Average annual ERs	**803.046**

2.1.18. Risks to the Project (G1.10)

Under development.

2.1.19. Benefit Permanence (G1.11).

That said, in order to maintain and improve the benefits for the climate, the community and biodiversity during the 40 years of the project, as well as beyond that time, several activities of the Project are focused on improving local capacities for governance, management, education, fostering, decision-making and greater awareness and capacity for sustainable resource management for all stakeholders. These activities will have short-, medium- and long-term impacts and will therefore help empower and guide all stakeholders to self-determine sustainable pathways to achieve benefits for climate, community and biodiversity well beyond the life of the Project.

The strategies associated with each activity for the benefits to occur during the life cycle of the Project and after this period are:

- **Asset surveillance program:** through the provision of tools such as remote monitoring by high-resolution satellite images, the acquisition of support equipment, and the provision of training for the asset surveillance team, the Project aims to structure a program of asset surveillance operations. In this way, surveillance operations will be able to carry out territorial monitoring and management, which should directly reflect on the maintenance of climate benefits beyond the life of the Project;

- **Monitoring of deforestation**: through monitoring and recording of deforested areas in the Project Zone, in addition to strengthening and formalizing communication with surveillance actions, this axis is a crucial measure to maintain and improve the climate benefits of the carbon project beyond its useful life. By providing up-to-date and accurate information, we can take proactive steps to protect forests, preserve biodiversity, and combat climate change effectively.

- **Biodiversity Monitoring**: the Project has as its axis of action the implementation of biodiversity monitoring in the Project Area throughout its life cycle, but, going forward, it will seek to provide feedback and improve the engagement of stakeholders in this activity, seeking environmental involvement and awareness. In this way, the benefits associated with these actions are expected to help, in some way, to change the view of stakeholders on local biodiversity, providing greater knowledge for people and thus reducing the impacts caused between human conflict and nature beyond the lifetime of the Project.

- **Sanitary precautions with domestic animals:** in addition to monitoring, the project also seeks to actively intervene in the problems identified during the diagnosis of biodiversity, such as the abundance of domestic animals present in the project area. To this end, this activity was developed with the aim of bringing benefits to local biodiversity and empowering residents of the region on the importance of the health and well-being of domestic animals.

- **Alternatives for cattle predation:** the Project has as its peculiarity the interaction of jaguars with the livestock operation (carried out by the proponent of the AgroSB project), due to which this activity was developed, which seeks, through various actions, to reduce the losses caused by jaguars to the operation of the Farm, in addition to reducing the conflict between employees and jaguars. Although no hunting activity was identified in the project, it is important to make the community aware of the importance of the jaguar for local biodiversity, in addition to proposing improvements in cattle management, thus avoiding predation.

- **Strengthening local governance:** the strengthening of local governance allows the construction of a network of solid relationships between the various factors that make up the territory, not only restricting the construction of a good relationship between the stakeholders and the proponents of the Project, but expanding partnerships and good relations between the communities located in the municipality and other governmental and non-governmental social actors. Based on the engagement of stakeholders and the social articulations promoted by the project's activities, it is expected that the benefits associated with these actions will empower the existing local association (AGRICATU) and boost regional development, through the construction of robust strategies to face issues of collective interest, such as environmental, social, economic and land activities, seeking to expand and consolidate the rights acquired by community members.

- **Promotion of sustainable production practices:** the activities proposed in this axis of the Project foster and encourage, through training and other means, the implementation of systems and models of agricultural and livestock production that are adaptive, sustainable and assimilate crops already established and widespread in the region. In this sense, it is expected that there will be an adhesion to these new production models, which provide product diversification, increase productivity, discourage rural exodus, reduce pressure for deforestation and reduce the effects of land speculation.

- **Development and strengthening of value chains:** the actions proposed in this activity seek to develop and strengthen productive value chains of the main crops in the region, mainly in the livestock activity, encouraging social and productive organization. The proposed initiatives are expected to facilitate access to different marketing channels, promote the enhancement of local production and assist in the construction and strengthening of value chains and local productive arrangements.

- **Land security and social welfare:** By strengthening the community social fabric, through sharing the benefits of carbon credits, and stimulating the achievement of land security, the actions proposed in this axis seek to reduce deforestation rates, encourage improvements in public infrastructures and services, expand access to rural credit benefits and other government programs and bring financial autonomy in a robust and permanent way.

2.1.20. Financial Sustainability (G1.12)

The proponents of the Project have a solid partnership signed in 2022 with the objective of enabling Investments in the conservation of the forest reserves of the Lagoa do Triunfo farm through the commercialization of environmental assets. The Ateles REDD+ Project will be an initiative that should enable, in the medium and long term, the continuous investment of resources focused solely on conservation and sustainable development in the region.

Considering the current assumptions of the carbon market and the potential to generate GHG Emission Reductions, the financial flow of the Ateles REDD+ Project presents very attractive results. In this model, proponents expect to recover the investment made in the Project when the commercialization of GHG Emissions Reductions begins.

Other information related to the financial analyses of the Ateles REDD+ Project and financial health statements of the partner institutions (project proponents) are considered commercially sensitive information and were shared with the audit team on a confidential basis.

2.1.21. Grouped Projects

Not applicable

2.2. Without-project Land Use Scenario and Additionality

2.2.1. Land Use Scenarios without the Project (G2.1)

To determine the land use scenario in the absence of the Project (baseline scenario) the approved methodology VCS VM0015 version 1.1 was used together with the approved VCS tool “VT0001 - Tool for the Demonstration and Assesment of Additionality in VCS Agriculture, Forestry and Other Land Use (AFOLU) Project Activities”, version 3.0. The analysis of deforestation, vector agents and hidden causes, as well as the probable land use scenarios in the absence of the Project were performed based on the baseline scenario, so further details can be found in Section 3.1.4.

2.2.2. Most-Likely Scenario Justification (G2.1)

The Reference Region has an area of 1,686,952 hectares and has a historical deforestation rate, verified through PRODES deforestation data, with August 2012 (inclusive) and July 2022 of 167,497 hectares of deforested land, which corresponds to a reduction of 16.1% in the existing forest in August 2012 and during the years 2012 and 2022, an average deforestation rate of 16,750 hectares per year (2.1% per year) was observed.

Among the alternative realistic land use scenarios that would occur within the Project boundaries in the absence of the AFOLU Project activity registered in the VCS, the following were considered:

I) Continuation of land use prior to the Project (baseline scenario):

The Project Area is inserted as a private area in the middle of the Triunfo do Xingu APA, a conservation unit with the greatest pressure for deforestation in Brazil, which has suffered over the last decades great threats to biodiversity maintenance with a history of degradation related to the intense occupation of the region since 1990, which resulted in the opening of a large number of secondary roads, in addition to the indirect impacts resulting from the productive activities of São Félix do Xingu and land grabbing that increases violence in the rural area of the municipality, so that, despite the protection determined under the law, the security and surveillance of these areas are extremely fragile, enabling such illegal activities.

Deforestation is caused, in general, by the advance of the agricultural frontier and by the historical practice of production based on felling-burning, or slash-burning system, for annual plantations, and subsequent implementation of pasture for small livestock. In addition to the search of medium and large rural producers for new areas to expand pasture areas for extensive cattle breeding.

Between 2013 and 2022, 176,010 hectares were deforested in the Reference Region for these activities. In the deforestation projection for the next decade (2023-2032) in the Reference Region, a deforestation of 185,908 ha is estimated while the total deforestation projected for the Project Area in the credit period was 88.489 ha, with an annual average of 2,212. In this scenario, this process tends to remain until much of the forest cover is changed, not contributing to the mitigation of climate change and generating an immeasurable impact on local biodiversity, in addition to deepening even more with social and economic problems.

2.2.3. Community and Biodiversity Additionality (G2.2)

The current scenario, which considers the absence of the Project, would be limited in generating climate, community and biodiversity benefits. This panorama without the activities of REDD+ Project tends to induce and enhance illegal practices, such as exploitation of wood of commercial value by lumber companies, sawmills and charcoal plants; associated with converting land to subsistence agriculture under conventional “slash and burn” practices, agricultural production of grains, raising beef cattle on extensive pastures and the possibility of implementing and maintaining infrastructure and logistics projects. As a result, environmental degradation would be leveraged by the increase in deforestation pressure in the reference region, gradually advancing to the limits of Fazenda Lagoa do Triunfo. This context is further presented, described and explored in Sections 2.3.4 (Community costs, risks and benefits), step 3 (Analysis of agents, determinants and underlying causes of deforestation and its probable future development), 4.1.4 (Without-project scenario: Community) and 5.1.3 (Without-project scenario: Biodiversity).

The scenario with the development of the Ateles REDD+ Project, through activities focused on climate, biodiversity and local socio-economy; would be positive from an environmental, social and economic point of view. Fostering resilient agricultural production practices associated with sustainable extractivism is an important and appropriate path for the conservation of forests on private properties and for the economy of communities and families in the region, and the project seeks to monitor and improve agricultural production techniques, animal husbandry, extraction of non-timber products and moderation of production chains. For this, in the field of agriculture and livestock; agroecological production techniques with reduced damage and impacts on natural resources could stimulate increased productivity and increased income. Likewise, the sustainable and official extraction of non-timber forest products would provide the strengthening of production networks, providing socioeconomic improvements for the population of the region in line with the environment.

In this sense, the support and incentive for education and training in the scenario with the REDD+ Project are crucial and extremely relevant, since access to courses, workshops and informative lectures should provide better employment and income conditions, as well as raising awareness about the importance of maintaining forest cover and its ecological aspects, engagement, empowerment and territorial bond. In addition, stimuli for resilient practices of agricultural production and sustainable forest management of multiple use help reduce pressures on the forest.

At the same time, articulation initiatives, development and consolidation of partnerships between stakeholders and local agents ensure effectiveness and continuous improvement in the implementation of activities proposed by the project, taking into account the reality and local demands. This integration and engagement allows the constant and thorough evaluation of the project's actions, strategic decision-making, the involvement of all actors and the strengthening of teamwork.

Therefore, the REDD+ Project, through a set of technical, governance and management mechanisms; aims to guarantee the preservation of the standing forest, consequently providing benefits to the conservation of biodiversity, maintenance of ecosystem services, climate regulation and local socio-economic development. Therefore, remaining conserved forests move towards anthropized and degraded areas, through deforestation, in the without-project scenario. This is different from the context with the implementation and execution of REDD+ Project activities, as demonstrated above.

In view of the scenarios presented "with" and "without" the REDD+ Project, and through the primary and secondary information collected and ascertained in the environmental and socioeconomic diagnoses, the importance of the implementation and development of the Project at Fazenda Lagoa do Triunfo is clear. Therefore, considering that the impacts of the Ateles REDD+ Project are essentially due to avoided deforestation, improvements in production practices, extractivism and forest management; monitoring of deforestation and biodiversity, environmental education, heritage surveillance, strengthening of governance and other activities carried out during its term, the main net benefits of the community and biodiversity project that would not occur in its absence are:

For Communities:

- Access to training and capacity building services on sustainable agricultural and extractive practices through partnerships;
- Improvement in land quality and agricultural productivity through the introduction of new production techniques;
- Strengthening human skills, knowledge and capacities on economic productivity and sustainable use of resources;
- Promotion and generation of new sustainable businesses, increase and diversification of income in the surrounding communities and integration with new markets;
- Environmental awareness and permanence of families on their lands;
- Establishment of partnerships, strengthened social organization, efficient communication and improvement in joint work.

For Biodiversity:

- Direct action against habitat loss and forest fragmentation;
- Promoting the conservation of the biodiversity of fauna and flora;
- Conservation of diagnosed species and high conservation value attributes (HCVAs), including those presenting some degree of threat;
- Reduction of extinction risks, ensuring genetic diversity;
- Stimulating, deepening and improving knowledge about local biodiversity through long-term monitoring;
- Monitoring of exotic species in order to ensure local biodiversity;
- Mapping of new areas of relevance for conservation

As described in Section 2.5.7, there are various laws, regulations, and ordinances (federal and state) that address issues related to the conservation of environmental and ecological heritage and respect for the rights of traditional peoples and communities. Among these regulations, we highlight, for example, Federal Laws 13.123 (2015) and 12.651 (2012). However, as described in the common practice scenario, these legislations, in short, are not applied in practice. That is, there is legal weakness and inefficiency regarding access to and use of land in Brazil and the protection of natural ecosystems and their resources, as well as the protection of endangered species of fauna and flora, reinforcing that the existence of such regulations does not guarantee their effective length and execution.

In addition, the analysis of the without-Project scenario demonstrates the existence of several barriers to the implementation of activities that may have positive impacts, such as those mentioned above. Further details on the additionality of the Project to community and biodiversity can be obtained and consulted in Sections 4.1.4 and 5.1.3.

2.2.4. Benefits to be used as Offsets (G2.2)

There was a demonstration of additionality for biodiversity and the community, as presented in Section 2.2.3. It should be noted that the benefits generated will not be used in other compensation programs, as the Ateles REDD+ Project aims to produce only offsets related to Reduced Emissions by Avoiding Deforestation, as described in Section 3 – Climate.

2.3. Stakeholder Engagement

2.3.1. Stakeholder Access to Project Documents (G3.1)

The communication of the Ateles REDD+ Project with stakeholders will be done by three main means: Virtual, written and face-to-face/oral. The main objective is to ensure access to relevant documents and information regarding the Project by all stakeholders.

Virtual communication: the project design document and/or referral links to access it, important information to stakeholders, such as monitoring reports and other relevant documents, will be available virtually on the Biofílica and Verra websites. News and news about the Project will also be released on social media (LinkedIn and Instagram).

Written communication: a printed version of the summary of the Project design document and versions of the respective monitoring reports will be made available at the headquarters of Fazenda Lagoa do Triunfo for consultation with all interested parties.

Face-to-face/oral communication: information, news and updates of the Project will be passed on through meetings, events and other face-to-face meetings.

In addition, the public consultation event at Verra will be widely disseminated to stakeholders, who will have access to the draft Project Design Document.

2.3.2. Dissemination of Summary Project Documents (G3.1)

The documents referring to the Ateles REDD+ Project will be made available and disseminated to all interested parties through virtual means on the official website of Biofílica Ambipar Environment and on the website of the Verra registration platform. In addition, a printed summary of the Project Design Document will be made available in the community and at the headquarters of Fazenda Lagoa do Triunfo, as well as the monitoring reports, for free access and consultation of communities and other stakeholders.

As mentioned in the previous section, news and news will be made publicly available through Biofílica Ambipar Environment's social networks, on its LinkedIn and Instagram account. In addition, all information and news will be reported orally through meetings, lectures and other face-to-face meetings for communities, partners, proponents and other stakeholders.

2.3.3. Informational Meetings with Stakeholders (G3.1)

In order to develop social activities appropriate to the communities, a Socioeconomic and Environmental Diagnosis (DSEA) was carried out in the area of the Ateles REDD+ Project and its surroundings, which was conducted by the Peabiru Institute and the proponents of the project, from November 24 to 29, 2022. The approach used was through the collection of secondary data, dialogs, interviews and application of questionnaires. The methodology used was described in detail in Section 2.1.9. In this sense, this first activity (DSEA) aimed to identify the key actors, their way of life, the opportunities and risks for the communities that will be the main beneficiaries of these actions, and to define, based on the results obtained, the possible paths to be followed in pursuit of sustainable development in the region.

In order to manage expectations and engage stakeholders, strengthening relations between proponents and community residents in the Project area, a second field activity was developed on June 27, 2023, called Situational Alignment. In the situational alignment, the concepts and certification processes of a REDD+ Project (Reduction of Emissions from Deforestation and Forest Degradation) were presented through lectures, conducted by Biofílica. As an additional form of dissemination, communication materials were distributed that illustrate the objective of the Ateles REDD+ Project and emphasize the CCB principle of Free, Prior and Informed Consent. In addition, in this activity, the results obtained from the socioeconomic diagnosis were shared by the Peabiru Institute with stakeholders. In addition, on the part of the AgroSB proponent, some positive actions were taken to increase the engagement of the community members and strengthen the ties between them, these actions are aligned with the interests of the community members, according to the information presented in the DSEA. Such actions included lectures and practical activities addressing the topics of safety, health and well-being of community members. In a complementary way, a lunch and an afternoon coffee were organized, providing the opportunity for interaction between the parties and informal chats. Also at this same event, feedback, suggestions and perspectives were collected on activities carried out and the next ones that will be developed throughout the project, in order to align expectations and improve actions, seeking the long-term success of the project and always including the vision of the impacted community.

The third field activity took place through the feedback process and stakeholder consultation, between the 27th and 28th of September 2023. The main objective of the feedback is to answer questions about the project, inform about its implementation status, present the communication channels developed (Section 2.3.8), inform participants about the audit processes, in addition to presenting the activities carried out for the Project (Section 2.1.11) to communities and other interested parties. The activity also contains a participatory workshop, developed by the Peabiru Institute, and a new positive action by the AgroSB team, with the aim of transferring the company's technical knowledge to community members through lectures and training.

As mentioned previously, this field activity was divided into two days. On the first day, the Peabiru Institute, in partnership with the project developers, contributed a participatory workshop with the communities, with the main focus being a better understanding of land issues that encompass the reality of the community residing in the Project area. A conversation circle was held, where participants explained their desires and highlighted their needs on this topic. From this survey it will be possible to develop more assertive strategies for the project's activities regarding the land security of settlers. On the second day, in the morning, a meeting was held focused on the presentation and proposal of the REDD+ Ateles project with employees from the company AgroSB, this meeting with the participation of the main leaders of the Lagoa do Triunfo farm. Publicity materials and declarations of knowledge about the project were distributed, in addition to the commitment to make these leaders disseminators of knowledge, so that the understanding of the project reaches the different hierarchical levels.

In the afternoon, on September 28th, the biofílica team, from Peabiru and AgroSB, returned to Vila Novo Horizonte, to continue the proposed activities. Firstly, the Biofílica team held an oral presentation, with spaces for exchange and the opportunity to clarify doubts about what the REDD+ Ateles project is about. Subsequently, the main activities that will be included in the scope of biodiversity and climate were presented, given the due emphasis on the social scope. The project's official communication channels were also announced during the presentation, including a WhatsApp number, email address and a suggestion box, made available to assist those who do not want or cannot access the other virtual channels.

Also in this presentation, the audit process, its dynamics, data and presentation of the responsible company were discussed. Finally, a macro schedule was presented to inform the community of the project's implementation status. All the material used in the presentation was made available to the communities present at the meeting, as well as the distribution of complementary material called the Ateles REDD+ Dictionary, which was developed with the aim of bringing a little more clarity to the technical terms used in carbon projects. At the end of the oral presentation, a questionnaire was carried out with some communities that signed up to participate, to infer their level of understanding and satisfaction.

To conclude the activities proposed in this activity, the resource manager of Fazenda Lagoa do Triunfo gave a lecture on pasture management and the use of fire. It was a moment of great exchange of knowledge between AgroSB and the settlers. During the interaction, educational folders were distributed to complement the training and further disseminate knowledge. The exchanges and doubts continued during the afternoon coffee, and the meeting ended in a very friendly atmosphere between the field technical team and the communities of Vila Novo Horizonte.

It is worth mentioning that these field activities presented above were considered the most relevant, but that other meetings, gatherings and workshops between the proponents of the Ateles REDD+ Project and stakeholders took place and will continue to take place during all phases of design, development, execution and monitoring of activities, in order to propose dialogues and open space for contributions on the planning, structuring and reporting of carbon project initiatives with the community and other stakeholders. These events will happen more intensely in the first years of the project, and then will follow the cadence that is deemed necessary for the full and effective development of the activities to occur. All meeting planning will be previously disclosed through the delivery of invitations and letters in hand to interested parties, as well as via email, Whatsapp and calls. In addition, the means of Communication developed by the project and informed in Section 2.3.8 may also be used.

2.3.4. Community Costs, Risks, and Benefits (G3.2)

As explained above, a Socioeconomic and Environmental Diagnosis (DSEA) was carried out in the area of the Ateles REDD+ Project involving the communities residing on site (details in Section 2.1.9 – Description of Actors). The approach used to understand the context of this population was through dialogues, interviews and questionnaires.

Among the strengths of the communities, there is a history of involvement in projects for the recovery of degraded areas and practices aimed at the use of production systems that consider intercropping, such as the agroforestry system, thus enabling new income alternatives with low impact on forest remnants. To a lesser extent, the community uses non-timber forest products, through the agroextractivist practice of collecting açaí and Brazil nuts. In addition, a representative social organization of the communities (AGRICATU) was identified, which plays an important role and can be an opportunity to form potential partnerships for the project's actions, in addition to access to new markets.

As for the negative points, difficulties in accessing some basic public health and education services were identified, which, when they exist, are precarious in serving the population. This also compromises the quality of life of Fazenda Lagoa do Triunfo's employees.

In addition, it is important to highlight the low application of environmental legislation, conflicts related to property rights, the low technical level of agricultural production, the difficulty of marketing production, the absence of technical assistance and rural extension services, in addition to the conflicting historical relationship between AgroSB and the settlers. The main threats relate to the deforestation of forest areas for the opening of new fields and pasture areas, in addition to the unmanaged use of fire and other illegal activities.

Throughout the project more appropriate and relevant information on potential costs, risks and benefits to communities should be discussed and presented through new field activities, meetings, events and workshops. This theme will be revisited in all phases of the project, from its conception, development, execution and monitoring. In addition, on all possible and pertinent occasions, we seek to make it clear to the community that participation in the Project is voluntary and the decision to participate or not is not definitive or results in some kind of restriction.

2.3.5. Information to Stakeholders on Validation and Verification Process (G3.3)

Potentially participating Project communities and other stakeholders will be informed about CCB validation and verification through the available means of communication described in Section 2.3.8, in advance of the event.

Virtual channels, such as through social networks, the Biofílica and AgroSB websites will be used to inform other stakeholders and the general public. In addition, the events will also be disseminated through messages in email groups and the WhatsApp messaging app.

In addition, as explained in the previous sections, information about the Project has already been disclosed during the Socioeconomic Diagnosis (SDR), situational alignment and feedback of the results raised and presentation of the proposed activities to interested parties.

2.3.6. Site Visit Information and Opportunities to Communicate with Auditor (G3.3)

The auditor's visit to the Project area will be previously informed to the communities, proponents, partners, farm employees and other stakeholders participating in the Ateles REDD+ Project. This communication will take place via emails, official letters, WhatsApp messages, calls and other appropriate and available means of communication (Section 2.3.9), with the necessary advance notice to the event.

In addition, during the situational alignment, the feedback of the results and validation of the activities proposed by the Project, carried out with the interested parties, the steps and next steps of the Project were mentioned, as well as the audit event.

Communication between the auditor, the communities and the other actors of the Project, as well as the dissemination of information will be facilitated through the distribution of folders, printed dissemination materials, virtual materials for dissemination on whatsapp, social media, Biofílica and AgroSB website, in addition to the entire communication structure developed by the Project, directed to stakeholders. During the audit event all stakeholders will have broad access and opportunity to communicate with the auditor.

2.3.7. Stakeholder Consultations (G3.4)

Fazenda Lagoa do Triunfo has some conflicting backgrounds between the residents of Vila Novo Horizonte and the proponent AgroSB, due to territorial disputes since the company's installation in the region. In addition to this troubled relationship, there are many activities related to unbridled and unplanned deforestation carried out by these communities that live there. In this sense, the companies AgroSB Agropecuária S.A. and Biofilica started discussions about the development of a REDD+ Project, in which their main mission is to contain deforestation and forest degradation in the Lagoa do Triunfo property. The articulation between the stakeholders of the REDD + Ateles Project began in 2021, with the beginning of the first negotiations, through submission and revisions of the proposals. In addition, in this first stage, the Biofílica team and the partners of the Peabiru Institute recognized and verified the technical feasibility of implementing a REED+ Project, which was marked by closer ties and positive communication between settlers and technicians.

In this sense, the doors were open and the feeling that the project would be feasible to be developed was present, resulting in the signing of the contract by the bidders, Biofílica and AgroSB, on July 1, 2022.

After the formalization of the agreement, the next step was the identification of actors and partnerships to hire the consultancies that develop the technical studies and that provide subsidies for the writing and development of the Project. The process of defining partnerships for technical studies began in the second half of 2022. With the technical partners defined, diagnostics were started in December 2022 and completed in June 2023. The kickoff workshop was held on July 20, 2022, a workshop to present the Project to the Technical Partners on November 11, 2022 and the workshop to present the preliminary results on March 8, 2023, all initiatives were developed with the objective of sharing knowledge between the parties, aligning the main problems identified and describing the scope of activities and their casual relationships.

During the workshops, participants were divided into working groups in which technical and descriptive points were discussed on the certification standards and their requirements regarding fauna, flora, socioeconomic, physical environment, carbon inventory and baseline determination. These working groups defined, among the strategic actions of the Project, the Field Work Plan and a prior evaluation of the communities that would be selected to be directly involved in the first phase of the Project.

The first field campaign with consultation with the communities initially selected for the Project was held from November 24 to 29, 2022. This activity led to the report developed by the Peabiru Institute, which gathered the results of the socioeconomic diagnosis (SDR) carried out in the field (Section 2.1.6 for more details). The DSE conducted on-site interviews to collect information, with the results of participatory workshops where the strengths and weaknesses of stakeholders (communities and AgroSB workers) were listed, aligned and related to their potential and opportunities and which resulted in the activities and actions proposed by the Project. The main results found in this research that were considered in the construction of the Project are described in Section 2.3.4.

The combination of this information gave rise to the main points included in the Project's activities regarding the direct and indirect impact on communities through the strengthening of governance, implementation, monitoring and evaluation of the activities developed, described in Section 2.1.11, with the main objective of stimulating sustainable economic development through the promotion of sustainable practices, promoting the strengthening of local governance, diversifying productivity, developing initiatives that bring greater territorial belonging to families, inserting new production techniques and technologies and also improving the communication process and the quality of life of residents.

With the conclusion of the technical studies, a new field activity called situational alignment was carried out on June 27, 2023, where the focus of the action was directed to the engagement and management of community expectations, bringing information about what a REDD+ Project is, the development phases, and the next steps, in addition to the presentation of the results of the DSE by the Peabiru Institute and some positive actions related to health and safety at work, which were developed at the initiative of AgroSB.

Subsequently, feedback was carried out with the communities and interested parties from September 27 to 28, 2023. During this field meeting, the Project development phases, communication and consultation channels, and field audit dates were presented again. , in addition to the proposed activities, which will be conducted by AgroSB and Biofílica throughout the project. Additionally, two more activities were conducted, a participatory workshop focusing on land issues, led by the Peabiru Institute, and an educational lecture on the use of fire and pasture management, led by the AgroSB Livestock Manager. Field activities were proposed to strengthen ties between stakeholders, generating a safe environment for participants to ask questions and suggestions about the design of the Project. During the events, all participants had the opportunity to express their ideas and opinions on the content presented, in order to improve the proposed activities. The contact with the communities took place from the sending of electronic invitations through Whatsapp through community leaders and telephone calls. We emphasize to everyone that participation was voluntary and the decision to participate, or not, is not definitive or results in some kind of restriction.

Photographic records, attendance list and application of a voluntary questionnaire to the participants were carried out, as a way to ensure the understanding of what was being passed, as well as to have a space for criticism and suggestions about the Project and the dynamics followed in the situational and feedback alignment. The documents have all been scanned and will be made available to the audit team.

Additionally, all participants received Project folders and banners were made available for display in the community and at the headquarters of Fazenda Lagoa do Triunfo. Another initiative carried out with the aim of increasing publicity of the Project to the public in the region was the dissemination of the poster virtually, through WhatsApp. of complementary material called the Ateles REDD+ Dictionary, which was developed with the aim of bringing greater clarity to the technical terms used in carbon projects. In addition to the field actions, the expansion of stakeholder consultation was reinforced by the sending of letters to relevant local institutions in the states of Pará, among others that have direct and indirect involvement in the forest conservation and environment sector, such as unions, non-governmental organizations (NGOs), the private sector, the State Public Prosecutor's Office and other governmental and federal agencies, where updated information on the status of the project, the communication channels used and an invitation to participate in the Public Consultation as described in Section 2.5.8 were presented.

2.3.8. Continued Consultation and Adaptive Management (G3.4)

The communication plan was developed to continuously meet the communication and consultation between project proponents, community members and other stakeholders, following the following structure:

Communication channels: communication will be carried out through the email created for the Project (sustentabilidade@agrosb.com.br) and by a telephone number with access to the WhatsApp (63 99966-

0569) according to the folder for the communication channels illustrated below (Figure 7), suggestion boxes will also be allocated in strategic places at the farm's headquarters and in Vila Novo Horizonte, where questions, suggestions and complaints may be registered by interested parties. In addition, meetings, lectures and other face-to-face events will also be held in order to listen to the community.

Figure 7: Folder promoting the communication channel with stakeholders.

Social mobilization strategy: social mobilizations will be carried out to carry out face-to-face actions, meetings, participatory activities, lectures, training, among other meetings. The invitations will be made and disclosed in advance, by email, telephone, WhatsApp and other means of communication that may be necessary. In each event, it will be defined which parties will be mobilized depending on the theme and negotiations.

Communication procedures: through a defined structure, the demands of the interested parties will be received by the means of communication described, recorded in a standard form, analyzed and forwarded to the areas responsible for resolution. The responsibility and time of the response will depend on the complexity of the demand, and can be met by the person responsible for communicating the Project, by the specific areas or by the direction of AgroSB.

Conflict management: conflict management will be based on the peaceful resolution of opposition of interests directly linked to the Ateles REDD+ Project, seeking all possible means of negotiation, and decision-making will harmoniously serve all parties involved, taking into account the well-being of all. They will be carried out through dialogues using the aforementioned means of communication, prioritizing those in which the parties involved feel most comfortable and with the shortest resolution time.

It is worth mentioning that the communication plan of the Ateles REDD+ Project can be adapted over the life of the project, as necessary, in order to improve communication between stakeholders and responses to demands.

2.3.9. Stakeholder Consultation Channels (G3.5).

The Project activities are designed and implemented considering the desires, characteristics and limitations of the community involved, as defined and verified during the data collected in the Socioeconomic and Environmental Diagnosis (DSEA), in the situational alignment and in the feedback meeting. Situational alignment and feedback were also targeted activities for sharing information about the Project and consulting with stakeholders, as described in Section 2.3.6.

This communication and accessibility for discussion on the progress of the Project activities between stakeholders and proponents will take place continuously throughout the duration of the Project through the available communication channels, as described in more detail in Section 2.3.8, participatory meetings, among other remote and in-person activities, according to demand.

2.3.10. Stakeholder Participation in Decision-Making and Implementation (G3.6)

The processes related to decision-making and implementation, as well as the various activities related to the Project, are open to community participation. In addition, the project will seek the equal participation of young people and women in decision-making and implementation of the activities of the Ateles REDD+ Project. Involvement in the design, implementation, monitoring and evaluation of the Project will take place through the available communication channels (Section 2.3.8), virtual and face-to-face activities, as well as informative meetings, where all stakeholders are previously communicated, and have the opportunity to participate.

2.3.11. Anti-Discrimination Assurance (G3.7)

Agro SB has a solid policy related to human rights and social responsibility, with a Code of Ethics and an integrity policy. The Code of Ethics serves as a guideline and its main objective is to guide the conduct of the company's members in relation to contact with the internal and external public, providing the dissemination and sharing of values and stimulating the improvement of behaviors and attitudes free of prejudice.

In the document referring to Agro SB's Code of Ethics, the subject of anti-discrimination, fair treatment, always respecting the individuality of each subject, is guided. The company guides its employees to outline their decisions and attitudes with respect, without prejudice, discrimination or individual preferences. Thus promoting a work environment that reflects this conduct. In addition, it is advised that any prejudiced or discriminatory attitude experienced by employees, whether directed to themselves or by a co-worker, be denounced. Not practicing or colluding with any type of moral or sexual harassment, aggressive behavior or violent act. Respecting political-partisan opinions, the religious belief of each one, as well as the right not to participate in religious manifestations, cultural or ideological differences, opinions, disabilities, gender, color, ethnicity, sexual orientation, among other individual characteristics.

Each Employee is responsible for informing the company of any circumstance that they witness or suspect to be a violation of the AgroSB Code of Ethics. In order to ensure an open space for people inside and outside the company to report such acts, AgroSB provides a confidential channel for making complaints, as well as complaints and suggestions. The company considers that the omission of this liability is as serious as the breach itself. Information regarding any violation should be forwarded to the Confidential Channel (Telephone: 08007505528; email: agrosbconfidencial@deloitte.com; website: ethicsdeloitte.com.br/agrosbconfidencial.) or, if you prefer, directly to the Ethics Committee (etica@agrosb.com.br). In addition, AgroSB values an environment open to suggestions and opportunities for improvement and correction of possible problems, not admitting, in any way, retaliation or intimidation to Employees or Business Partners who present doubts, complaints or complaints in good faith regarding any concept or situation of violation of the Code of Ethics and other regulations and guidelines established by AgroSB. The investigation of complaints is coordinated by Internal Audit & Integrity and will always be done carefully, respecting the rights of the whistleblower and the accused.

The communication channels of the Ateles REDD+ Project (Section 2.3.8) will also be available to report any attitude that goes against the values described by the Code of Ethics. AgroSB has the responsibility to behave ethically in the Communities where it operates, aware of the economic and social role they play. It seeks to promote the socioeconomic development of the regions where its farms are located, through educational and social inclusion initiatives. There is a process of encouraging employees to take an active personal role in organizations dedicated to public service, as well as social and humanitarian movements and projects, which can contribute to reducing social inequality and promote effective changes in people's lives.

Biofílica Ambipar Environment, belonging to the Ambipar Group, has a code of ethics and conduct that represents all companies in the Group. Thus, the document serves as a primordial instrument to guide the conduct of all parties involved, in the adoption of good practices in relationships and in doing business, guiding attitudes of respect for diversity, to combat any form of prejudice.

2.3.12. Feedback and Grievance Redress Procedure (G3.8)

AgroSB has a confidential and independent reporting channel, so that any employee or third party can report situations or suspected violations of the Code of Ethics, laws and other internal rules.

The whistleblower can make a report through the following channels:

- Telephone: 08007505528;
- E-mail: agrosbconfidencial@deloitte.com;
- website: ethicsdeloitte.com.br/agrosbconfidencial.

As provided for in the Code of Ethics, and guided by AgroSB's integrity policy, anonymity and non-retaliation to the whistleblower in good faith is guaranteed. To ensure this anonymity, this confidential channel is independent, being operated by Deloitte, one of the largest auditing and consulting companies in the world.

Deloitte performs the prior treatment of all reports received and, among other actions, removes from the report any information that can identify the whistleblower, unless it is the whistleblower's will to identify himself. After this treatment, Deloitte makes the report available to AgroSB so that the calculation can be carried out.

The reports follow the following treatment macroflow shown below (Figure 8):

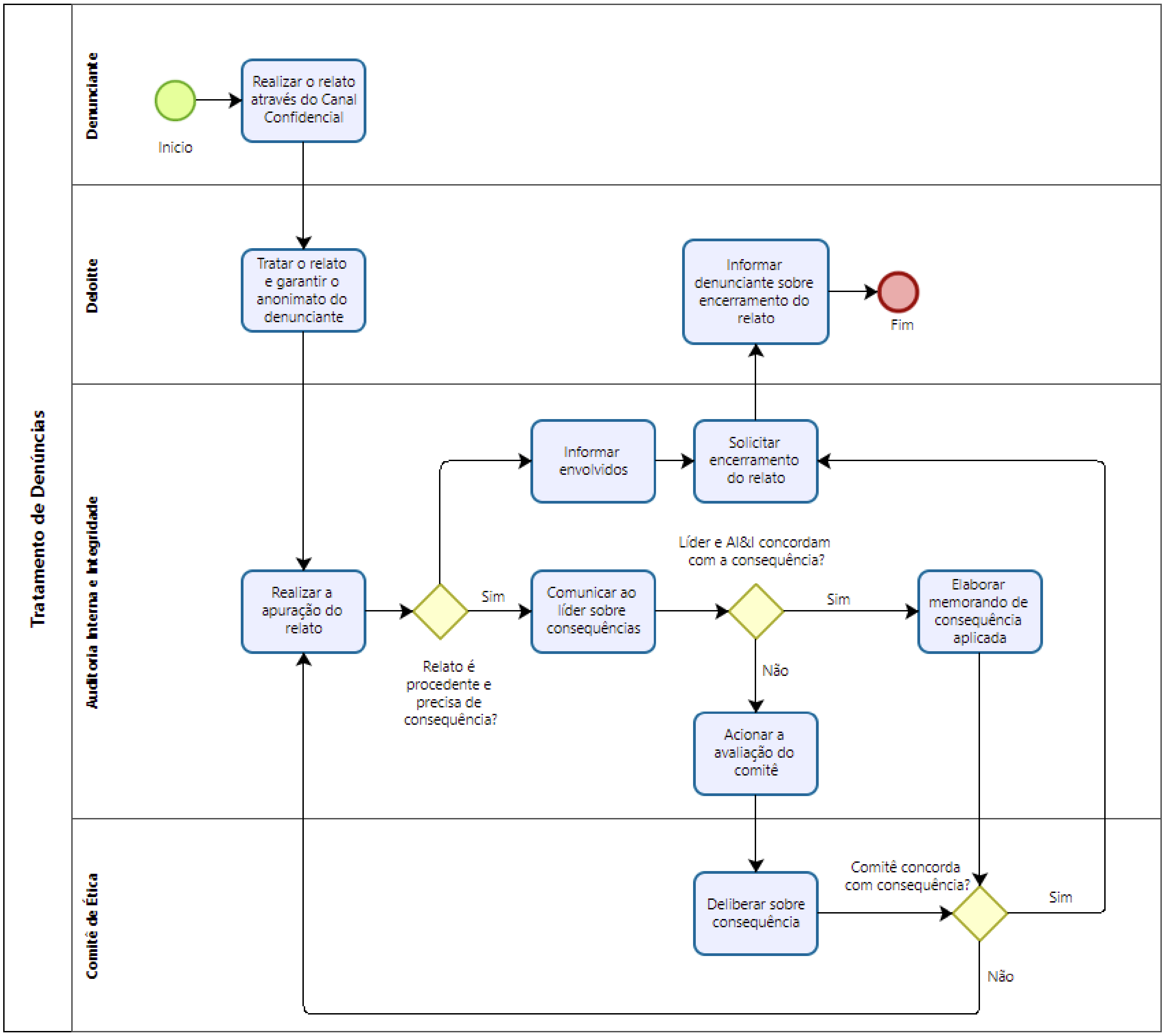

Figure 8: Complaint handling flowchart.

The AgroSB Ethics Committee is an advisory body formed by a minimum of 3 and a maximum of 5 employees. The composition must have the CEO, an employee with knowledge in Compliance/Integrity and an employee with knowledge in Human Resources.

All reports, whether valid or not, are communicated to the Ethics Committee through a memorandum of inquiry. If the Committee does not agree with the closure or consequence to be applied, it may request the reopening of the case or resolve on a consequence different from that decided by the leadership.

2.3.13. Accessibility of the Feedback and Grievance Redress Procedure (G3.8)

As mentioned in the previous topic, the Ateles REDD+ Project will adopt the receipt of feedbacks, complaints and complaints in addition to the conflict management and Feedback procedure (Communication Plan Section 2.3.8 and Section 2.3.12), in telephone number (63) 999660569 and e-mail (sustentabilidade@agrosb.com.br). Its developers will ensure that the entire history of complaints, doubts, suggestions, compliments, claims and requests, and their respective developments in procedures and decision-making, are disclosed and accessible in a clear and transparent way to all individuals, community memebers and institutions, both in physical files (for example notebook/folder of records) and virtual files.

It is worth reinforcing that, in all the channels for receiving feedbacks, complaints and denunciations, the records may be anonymous, except in cases in which the individuals require identification. In general, the time limit for return of up to 3 weeks will be considered.

2.3.14. Worker Training (G3.9)

The qualification and empowerment of local actors are essential to assure quality in the implementation of the actions proposed by the Project, as well as to ensure the permanence of results and positive impacts in the long term. It is understood that, to ensure effectiveness in the implementation of the Ateles REDD+ Project, it is essential to work on the generation of local human capital, focused mainly on the responsible management of natural resources. Thus, among the various actions proposed by the Project (as detailed in Section 2.1.11), one part involves training and engagement of local actors. The main proposals that aim to promote the training of local actors, income generation and direct and indirect jobs are described below.

- Implementation of asset surveillance: the activity must involve the training and qualification of employees who will act in the asset surveillance of Fazenda Lagoa do Triunfo. The training and capacities aim to develop asset surveillance techniques that will assist in the containment of invasions and illegal activities;

- Strengthening local governance: involves the strengthening and expansion of social actions already developed by AgroSB in the region and the formation of partnerships, mainly for on-site activities. The Project, through this activity and its objective of strengthening social capital and community engagement in relation to the search for basic rights, can stimulate the creation of indirect jobs and attract investments to the region;

- Strengthening of sustainable agricultural and livestock farming practices: the activity involves the promotion of training within the lines mapped by the Project, in addition to mapping stakeholders with potential to adhere to actions to promote sustainable practices and technical development. In addition, the activity promotes the improvement of the practices developed by the community, promoting the strengthening of human capital as one of the means for sustainable development;

- Development and strengthening of value chains of agricultural, livestock and forestry products: the activity involves the promotion of training for the valorization and commercialization of agricultural, livestock and forestry products, promoting sustainable development. In addition, the Project proposes to map the main opportunities and implement partnerships for the development of activities, increasing investments and improving the income of families that depend on these activities for their food and financial security;

- Biodiversity conservation program: the activity involves the in situ monitoring of biodiversity carried out by local partners, in addition to the participation of stakeholders, mainly community and workers from Fazenda Lagoa do Triunfo, in environmental education actions, promoting technical development and environmental awareness, in addition to strengthening the management of the territory and promoting biodiversity conservation.

In order to guarantee the efficiency and permanence of these actions listed above, proponents must seek the best techniques and procedures to conduct training for those people involved in the activities, always seeking to ensure successful qualification of the team to work with the communities and meet project goals. These processes will follow all relevant laws and regulations related to worker rights, as described in Section 2.3.16, bringing team involvement to meet the project schedule and goals, with the objective of optimizing investments and avoiding loss of human capital due to staff turnover. Other measures adopted to avoid the loss of acquired capacity will be the constant recording and reporting of procedures and monitoring of results obtained, since, in the event of staff turnover, the procedures can be easily reproduced, mitigating impacts on the implementation of the project plan.

Far from the above, the Project intends to seek partnerships with institutions trained to develop the proposed activities.

All activities are open to the participation of residents of the communities located in the Project area and surroundings. The participation of women, young people and marginalised people is stimulated by the proponents. In addition, the process of hiring workers from Fazenda Lagoa do Triunfo involves all people residing in the communities, without distinction (Section 2.3.15).

2.3.15. Community Employment Opportunities (G3.10)

The employment opportunities offered by AgroSB are equal to the surrounding communities, encompassing management positions, if the requirements for the vacancy are met. All work positions generated locally by the Project follow the Recruitment and Selection process, belonging to AgroSB, which performs the entire process according to the need and availability of vacancies. The procedure for recruiting and selecting new employees will always be in line with the company's values and in synergy with the organization's policies and practices.

Recruitment and selection, hiring, access to training and promotion are based on skills, capacities, qualities and medical fitness required for the available jobs and are in compliance with current legislation.

AgroSB does not support or use any type of discrimination in its hiring process, whether of race, national or social origin, social class, birth, religion, disability, sex, sexual orientation, family responsibilities, marital status, association with unions, political opinion, age, and anyone can be part of the company's staff, and must only fulfill the requirements necessary to perform the intended function. The hiring of children under 18 years of age is prohibited, except young apprentices (between 16 and 24 years of age).

2.3.16. Relevant Laws and Regulations Related to Worker's Rights (G3.11)

Table 9: Laws and regulations protecting worker rights in Brazil.

Laws, Rules and Decrees	Provisions
BRAZILIAN CONSTITUTION, Chapter II - Social Rights, Articles 7 to 11	a) Minimum wage, b) Normal working hours, c) Guidance on vacations and weekly days off, d) Guidance on maternity and paternity leave, e) Recognition of collective bargaining and f) Prohibition of discrimination.
FEDERAL LABOR LAW, Consolidation of Labor Laws – CLT: DECREE-LAW No. 5.452, as of May 1, 1943	a) Hourly, daily, weekly and monthly working hours, b) Employment of minors and women, c) Establishes a minimum wage, d) Worker safety and safe working environments, e) Defines penalties for non-compliance by the employer, f) Establishes a work-related judicial process to address all worker-related issues.
FEDERAL LAW No. 5.889, as of June 8, 1973	It establishes regimental standards for the Rural Worker, created to specify equality between urban and rural workers, along with overtime compensation.
FEDERAL LAW No. 8.080, as of September 19, 1990	Conditions for the promotion, protection and recovery of health, the organization and operation of the corresponding services and other measures. Includes workers and their health in the Public Health System.
ORDINANCE OF the MINISTRY OF HEALTH (MS) No. 3.908, as of October 30, 1998	Establishes procedures for guidance and implementation of workers' health actions and services in the Unified Health System (SUS).
REGULATORY STANDARD No. 31 as of March 3, 2005:	It establishes the rules to be observed in the organization and in the work environment to ensure the safety, health and work environment of activities developed in agriculture, livestock, forestry, forestry and aquaculture.

In addition, Brazil is a founding member state of the International Labour Organization (ILO[38]) and a key partner in promoting the Decent Work Agenda. Founded in 1919, the ILO's mission is to promote opportunities for men and women to have access to decent and productive work, under conditions of freedom, equity, security and dignity. For the ILO, decent work is a fundamental condition for overcoming poverty, reducing social inequalities, ensuring democratic governance and sustainable development. Brazil has ratified 98 Conventions, including the eight Fundamental ones that provide for: freedom of association and effective recognition of the right to collective bargaining, elimination of any and all forms of forced or compulsory labor, effective abolition of child labor and elimination of discrimination in matters of employment and profession.

From the above, it is possible to ensure that all employees belonging to AgroSB, Biofílica, and service providers are legally hired in compliance with Brazilian labor legislation. In addition, the international agreements ratified by Brazil and issues related to the well-being of workers are respected.

Annually, there is compliance with the labor standards and laws applied by Biofílica by an audit, this is due to the fact that it is a S.A. company. Its financial statements are published on the website of Jus Brasil, the largest open and legal community in Latin America.

Also, according to the compliance policy, the Ambipar Group, to which Biofílica is a member, undertakes to promote working conditions that provide a balance between the professional, personal and family lives of all its employees; in addition to ensuring safety and health at work, providing all the conditions and all the necessary equipment for this purpose.

In both companies, after hiring and before the beginning of the worker's activities, training and qualification on technical procedures take place and empowerment is promoted regarding their rights and applicable legislation. In addition, employees are advised to join the institution responsible for defending their rights, the respective unions to the work area.

AgroSB conducts its activities based on honesty and commitment to the defense of the fundamental principles of respect for human life, laws, statutes and regulations, adopting practices to protect the well-being of all and the environment.

2.3.17. Occupational Safety Assessment (G3.12)

An important component of the Project involves strict and effective care for the safety of workers, taking into account the internal regulations and official norms established by the federal and state governments. In this context, AgroSB has a strict Occupational Health and Safety Policy, whose objective is to establish guidelines and principles, in addition to involving and committing all AgroSB employees, third parties and business partners to health promotion, respect for current legislation and the prevention of occupational accidents, in line with the Code of Ethics.

This Policy has as guidelines the search for continuous improvement of AgroSB's performance in Health and Safety, focusing on compliance with regulatory standards, current legislation, and the development of people, seeking to reduce risks and strengthen process safety. In addition, the standard aims to promote a healthy work environment, considering issues related to the physical and psychosocial work environment, resources for personal health and the company's involvement in improving community health.

A series of fundamental principles for its effectiveness are listed in the Occupational Health and Safety Policy. They are:

[38] International Labour Organization (ILO). Brazil - Cooperation with ILO. March 7, 2023. Available at: https://www.ilo.org/brasilia/publicacoes/WCMS_881632/lang--pt/index.htm. Access on: 5/30/2023

a) To ensure compliance with legal requirements, Health and Safety Program and other requirements applicable to Health and Safety;

b) To motivate, raise awareness, develop, train and qualify employees, making them responsible for ensuring and promoting a healthy and safe work environment;

c) To build and disseminate the Safety and Health culture as a value through leadership as an example, individual responsibility and shared surveillance;

d) To strengthen the Health and Safety culture with a focus on education, training and awareness and prevention;

e) To act in the anticipation and prevention of accident occurrences, understanding that every accident can and should be avoided;

f) To seek continuous improvement in processes, machinery and equipment, promoting control actions to eliminate or minimize the occurrence of incidents;

g) Training and educational actions must be permanent and all services developed on the company's premises must be performed by duly qualified and qualified people. All workers have involvement and participation, with the support of area managers, in the prevention of accidents, occupational diseases and environmental damage;

h) All accidents (occupational, environmental and occupational diseases) at AgroSB will be subject to analysis and investigation by the Occupational Health and Safety area. Measures to prevent recurrences will be taken;

i) The Occupational Health and Safety area must work in an integrated manner, involving occupational health and safety professionals and the environment, the Internal Commissions for the Prevention of Rural Occupational Accidents - CIPATR, area managers and employees.

As for responsibilities, it is understood that working safely is a fundamental principle of employment and hiring condition of each employee in the development of their activities. Thus, everyone must assume it as an individual responsibility. The leadership is responsible for the performance in safety, environment and occupational health, and each leader is directly responsible for the performance of their area and the safety of those who work with them, and must report to the Occupational Health and Safety area any unsafe act or condition in the workplace, as well as work-related accidents and diseases. Safety, environmental and occupational health observations must be carried out by all levels of the company to identify strengths and opportunities for improvement regarding the conditions of the facilities, the level of awareness of people and the effectiveness of existing programs.

After hiring, all employees go through the integration process at the time of their admission, where the principles and guidelines of the company's code of ethics and conduct are transferred to them. In addition, the Weekly Safety Dialogues (DSS), frequent training and campaigns related to occupational health and safety, and the Internal Week for the Prevention of Accidents at Rural Work – SIPATR are carried out during the routines of field activities. Through the Cuid@r – self-care program, AgroSB provides employees with access to online care (telemedicine) and medical consultations.

Those workers who work directly in field activities, the use of personal protective equipment is indispensable. PPE is provided by the company as provided in Art. 158 of the consolidation of Labor Laws, in addition, employees are periodically instructed by the Safety Technician on their correct use.

To date, no additional risks to workers arising from the activities proposed in the Ateles REDD+ Project have been identified. However, this topic will be revisited and reassessed whenever a new activity is implemented, aiming to preserve the physical health and mental integrity of the workers involved in the Project.

2.4. Management Capacity

2.4.1. Project Governance Structures (G4.1)

The management of the Project will be the responsibility of Biofílica Ambipar Environment and AgroSB Agropecuária S.A. The obligations and commitment of the parties are described below:

Biofílica Responsibilities: general coordination of socioeconomic and environmental diagnostics (DSEA), baseline studies and carbon stock; construction of the PDD (Project Design Description); remote monitoring of forest cover and implementation/coordination of additional actions aimed at reducing/mitigating greenhouse gas (GHG) emissions; conducting validation/verification audits; dissemination of the Project; commercialization of credits and co-management of the Project throughout its duration

AgroSB Responsibilities: Investments required to implement and validate the Project (Capex); Project co-management, as well as development of every activity related in environmental and social scopes, and infrastructure and logistic support to Biofílica and other professionals engaged in the Project. Moreover, all support required will be provided to audit process, dissemination material development and other business processes.

During the development of the project, other organizations (mentioned in Section 2.1.4) were involved to carry out the diagnostic studies. Accordingly, the responsibilities are described below:

- Ambiens Environmental Solutions: Development of environmental studies, such as characterization of the physical environment and biodiversity of the region (flora and fauna).
- Biodendro: Development of the carbon stock estimation study of the Ateles REDD+ Project.
- Peabiru Institute: Development of the socioeconomic study of the Ateles REDD+ Project

As presented, the Ateles REDD+ Project is supported by human resources that assisted in its development and implementation.

2.4.2. Required Technical Skills (G4.2)

The main technical skills required for the implementation of the Ateles REDD+ Project are knowledge about the development and management of forest conservation projects in the Amazon biome, experience in the implementation, development and assistance of programs for livestock and agroextractivist communities and implementation of effective land security and patrimonial surveillance. All proponents involved in the project have the necessary technical skills for the successful completion of the Ateles REDD+ Project.

Biofílica Ambipar Environment is a Brazilian company that promotes the multiple use of forests in the Amazon biome. The company has a specialized team and is a reference in the development of forest conservation projects, guaranteeing the quality and effectiveness of the REDD+ activities carried out. The company aims to reduce deforestation and carbon emissions into the atmosphere, conserve biodiversity and water resources and promote social inclusion and the development of communities living in the Amazon biome through the sale of credits for environmental services, development and funding of scientific research activities and the development of sustainable business chains. Biofílica aims to make environmental conservation an economically interesting activity for forest owners, communities and investors.

Founded in 2005, e AgroSB Agropecuária S.A. emerged with the purpose of being the largest producer in the agricultural and livestock market in the country. Agro moves the economy of the south/south-east of the state of Pará and is dedicated to the development of the herd by raising, recreating and fattening cattle on its farms. In 2011, AgroSB promoted the integration of farming with livestock through corn and soybean crops in order to maximize the use of its assets. The expansion of corn and soybean cultivation also includes cotton production in experimental centers in southeastern Pará.

With the Ateles REDD+ Project, the company aims to reduce deforestation in its forested areas and carbon emissions into the atmosphere, conserve biodiversity and water resources. In addition to promoting social inclusion and the development of communities living in the Amazon biome through the sale of credits for environmental services, development and financing of scientific research activities and the development of sustainable business chains.

In this sense, the owners of AgroSB wish to develop conservation projects and environmental services to ensure the long-term conservation of carbon stocks and local biodiversity and add value to forest assets.

2.4.3. Management Team Experience (G4.2)

The multidisciplinary technical team has postgraduate professionals with expertise of more than 10 years in the development of environmental projects for the conservation and management of biodiversity and its ecosystem services. In addition, the team is formed by members experienced in planning, executing and controlling administrative and financial resources, and with a management staff that has significant experience, of more than 5 years, in the development and implementation of AFOLU projects for the voluntary carbon market.

Below is the experience of the members of the Ateles REDD+ Project management team.

<u>Proponent</u>: Biofílica Ambipar Environment

Plínio Ribeiro – Executive Director

Plínio Ribeiro holds a degree in Business Administration from the Instituto de Ensino e Pesquisa - INSPER and a master’s degree in public administration and Environment from Columbia University and the Earth Institute (USA). He has participated in several conservation projects on the lower Negro River, through the Institute of Ecological Research – IPÊ since 2005 and was one of the producers of the documentary "Retorno à Amazônia", by Jean Michel Cousteau. He has been working at Biofílica since 2008, where he has led Projects, Operations and Business Management. He is co-founder and CEO of Biofílica.

Team:

Cláudio Padua – Scientific Director

Cláudio Pádua holds a degree in Business Administration and Biology, a Master's in Latin American Studies and a PhD in Ecology from the University of Florida in Gainsville (USA). Retired professor at the University of Brasilia, Pádua is currently dean of the Superior School of Conservation and Sustainability and vice-president of the Institute of Ecological Research (IPÊ). He is also Senior Associate Researcher at the Center for Environment and Conservation Studies at Columbia University (USA) and Director of International Conservation at the Wildlife Trust Alliance, as well as a consultant for the Brazilian Biodiversity Fund (FUNBIO) and WWF Brazil. Pádua represents Brazil before the International Advisory Group (IAG) of the G7 Pilot Program. In 2003, together with his wife, Suzana Pádua, he was named "Hero of the Planet" by Time magazine for his activities in favor of biodiversity conservation. Between 1997 and 2007, he won six conservation awards, three national and three internationals. Pádua has published two books and more than 30 articles in national and international scientific journals. Since 2008 he has directed Biofílica's involvement and scientific production as Scientific Director and advisor.

Paula Conde – Administrative Manager

Paula Conde has a degree in Business Administration from PUC of São Luís and a postgraduate degree in Accounting and Financial Administration from FAAP. She has extensive experience, most of it in one of the largest media and education groups in Latin America - Editora Abril, where she worked with Financial Control and Reporting, Treasury, Accounting and Financial Reconciliation, Accounts Payable and Receivable and Royalties. At Biofílica, she is responsible for administrative and financial activities, logistical support for the team and projects. She has been at Biofílica since 2015, being responsible for administrative and financial activities, specializing in REDD+ Carbon Market, intermediation, billing of these segments, logistical support to the team and projects.

Soraya Pires – Chief Operating Officer

Agronomist, graduated from Esalq/USP with specialization in strategic management and finance from FGV, with 15 years of experience in management and development of agribusiness businesses and career developed in large multinational companies in the sugar-energy sector (Adecoagro, BP and BP Bunge). Since 2022, she has been an executive director in the carbon market, carrying out activities related to business development, operational management, financial modeling, strategic planning and AFOLU project management.

Laion Pazian – Commercial Manager

Laion Pazian is an Economist from the University of São Paulo (USP/ESALQ) and an MBA in Commercial Management from Fundação Getúlio Vargas. He has been at Biofílica since 2016, and manages the commercial team of carbon credits, key-accounts, Biofílica's commercial policy and strategy. In addition, he monitors and directs the analysis of carbon market intelligence, being responsible for the area's pricing and planning policy.

Caio Gallego – Chief Operating Officer

Caio Gallego is a Forestry Engineer graduated from the University of São Paulo (USP/ESALQ). Specialist in geoprocessing and remote sensing focused on the area of environmental conservation, mapping and analysis of changes in land use. He has knowledge focused on Sustainable Forest Management, environmental modeling and the use of alternative GIS for forestry and agribusiness. He has advanced knowledge in the use of GIS software and analysis of changes in land use and land cover such as ArcGIS, QuantumGIS and DinâmicaEGO. Caio Gallego has more than 8 years of experience working with the development and implementation of AFOLU projects for voluntary carbon markets. He worked with project development, with a main focus on implementing one of the first REDD+ projects within the VCS program in the Brazilian Amazon. His main expertise has always been focused on the technical development of projects, having been responsible for more than 10 validation/verification processes. In more than 10 projects in which he was directly responsible for development, he worked on the application of socio-environmental safeguards, ensuring the generation of net positive impacts on climate, community and biodiversity, in addition to ensuring the maintenance of exceptional criteria applicable to each of these scopes.

Ricardo Cordeiro – Communication Coordinator

Ricardo Cordeiro is an advertiser, On and Offline art director with more than 10 years of experience, he has worked in digital agencies, trade and live marketing. Experience in UX, digital planning and strategies. Specialization in Digital Marketing and Web Project Management. At Biofílica, he acts as communication coordinator, responsible for digital marketing, branding and institutional communication actions.

Márcio Sales – Statistical Modeling Specialist

Márcio Sales is a statistician, graduated from the Federal University of Pará, Master in Geography from the University of California, Santa Barbara and PhD student in Production Ecology and Resource Conservation from the University of Wageningen, in the Netherlands. He specializes in data analysis and conducts research in geostatistical modeling of processes distributed in space and time. Works in Biophilic in the production of projections of GHG emissions from future deforestation for project baselines and satellite monitoring of deforestation

Shaxahmary of M.C. dos Santos – Project Coordinator

Shaxahmary dos Santos is a Forest Engineer, graduated from the University of São Paulo (USP/ESALQ) and specialist in Project Management. In his last professional experiences, he had an interface with themes of natural resource conservation, climate change, payments for environmental services, restoration and forest hydrology. Shaxahmary has approximately three years of experience in developing and monitoring REDD+ projects and currently operates as a carbon project coordinator.

Ricardo Cordeiro – Communication Coordinator

Ricardo Cordeiro is an advertiser, On and Offline art director with more than 10 years of experience, he has worked in digital agencies, trade and live marketing. Experience in UX, planning and digital strategies. Specialization in Digital Marketing and Web Project Management. At Biofílica, where he has been working since 2020, he acts as communication coordinator, responsible for digital marketing, branding and institutional communication actions.

Márcio Sales – Statistical Modeling Specialist

Márcio Sales is a statistician, graduated from the Federal University of Pará, Master in Geography from the University of California, Santa Barbara and PhD student in Production Ecology and Resource Conservation from the University of Wageningen, in the Netherlands. He specializes in data analysis and conducts research in geostatistical modeling of processes distributed in space and time. He works at Biofílica in the production of projections of GHG emissions from future deforestation for the baselines of the projects and in the monitoring of deforestation by satellite.

Kauanna D C de Andrade – Project Analyst

Kauanna Andrade is a forest engineer, graduated from the Federal Rural University of Rio de Janeiro, and holds a master's degree in Tropical Forest Sciences, with an emphasis on Sustainable Forest Management, from the National Institute of Amazonian Research. Professional with expertise in forest inventory, environmental licensing, climate change and REDD+ projects. She has been carrying out activities for the preparation, development and monitoring of AFOLU Projects since 2023.

Ana Luiza Cabral – Project Analyst

Ana Luiza Cabral holds a degree in Biological Sciences from the Federal University of Lavras. In her professional career as an environmental consultant and analyst in projects such as: adaptation to climate change, environmental conservation, disaster risk management, environmental economics, territorial planning and sustainable development. She has been carrying out activities for the preparation, development and monitoring of AFOLU Projects since 2022.

Bianca de Oliveira - Project Analyst

Bianca de Oliveira is a Forest Engineer, graduated from the Federal Rural University of Rio de Janeiro, and a specialist in Integrated Environmental Management from the State University of Rio de Janeiro. Professional who has expertise in environmental licensing, forest restoration, climate change and conservation. She has been carrying out activities for the preparation, development and monitoring of AFOLU projects since 2023.

Amanda Rocha Fiallos - Project Analyst

Amanda Fiallos is a Forest Engineer, graduated from the University of São Paulo (USP/ESALQ). She has experience in forest restoration and conservation projects, environmental education, geoprocessing and remote sensing applied to the planning areas of planted forests. She has been working in Biophilic since May 2022, producing technical reports and reporting activities of the AFOLU Projects, and evaluating reports and products delivered by suppliers regarding the socio-environmental diagnoses made.

<u>Proponent:</u> AgroSB Agropecuária S.A.

Cristiano Soares – Lawyer at PUC/RJ and MBA in Finance at INSPER, with more than 10 years of experience in the agribusiness market. Currently CEO of AgroSB.

Team:

Juliana Correa - Lawyer at UERJ and postgraduate at FGV, with more than 10 years of experience in sustainability. Currently Legal and Sustainability Director at AgroSB.

Ericka Schelb - Lawyer at UFOP and postgraduate in Environmental Management at UFRJ, with more than 10 years of experience in Environmental Law. Currently Environmental and Regulatory Specialist at AgroSB

Ana Caroline Neres - Agronomist and Master from UPA, with a specialization in Plant Protection from UFV, with more than 10 years of experience in the agribusiness market. Currently ESG Specialist at AgroSB.

Anderson Gomes da Silva - Agricultural Technician, graduating in business administration, with more than 10 years in the administrative area in the agribusiness market. Currently Administrative Coordinator of the Lagoa do Triunfo Production Unit.

Rômulo Pires Sobral - Surveying technician, with more than 20 years of experience in the field of surveying. He currently works as a surveyor at the Lagoa do Triunfo Production Unit.

2.4.4. Project Management Partnerships/Team Development (G4.2)

The Ateles REDD+ Project has all the necessary and technically qualified partnerships for the construction, implementation and monitoring of the conservation activities of the environmental assets of the area. The partner institutions, already mentioned previously in topic 2.1.4, were responsible for preparing the Socioeconomic and Environmental Diagnostics, Carbon Stock and Baseline, which make up the Project Design Document, through service supply contracts. When other initiatives throughout the development of the Project require new technical knowledge and partners, the proponents of the Ateles REDD+ Project will articulate an association with governmental, non-governmental and private sector organizations, in order to enable the generation of net positive impacts on society and biodiversity.

2.4.5. Financial Health of Implementing Organization(s) (G4.3)

Biofílica Ambipar is a Brazilian company with 15 years of experience in the environmental services market in Brazil, through the generation and commercialization of carbon credits from nature-based solutions (NBS), having a diversified line of business and investors that support the company's business.

AgroSB Agropecuária S.A., founded in 2005, is one of the most active companies in the agricultural and livestock markets in the country with 764 employees working in 4 Production Units located in the Southeast of Pará in Cumaru do Norte, Redenção, Santa Maria das Barreiras, Santana do Araguaia, São Felix do Xingu and Xinguara. Currently, its economic activities involve raising, recreating and fattening cattle, undergoing intense genetic selection work, in addition to seeking in Crop/Livestock Integration to maximize the use of its assets. AgroSB also develops an important cocoa planting project in an area of 230 hectares and has been expanding its area of operation in soybean planting through increasing investment in the conversion of pasture areas for agriculture. In addition, Fazenda Lagoa do Triunfo I has the Onça Pintada Certificate, and a partnership with the Black Jaguar Institute, whose mission is to help implement the project known as the Araguaia Biodiversity Corridor through ecosystem restoration actions.

2.4.6. Avoidance of Corruption and Other Unethical Behavior (G4.3)

Biofílica Ambipar Environment supports annual financial audit processes ensuring that its resources are allocated responsibly and free of corruption. Financial statements and minutes of meetings relating to the company are published on the website of JusBrasil, the largest open legal community in Latin America.

AgroSB has strict anti-corruption and integrity policies. It prohibits and does not tolerate any practice of corruption, bribery, payment or receipt of bribes, whether with the national or foreign Public Administration, or with Private Companies, in compliance with national legislation, in particular the Anti-Corruption Law, and also the United Nations Global Compact to prevent and combat corruption in all its forms. The mere promise of undue advantages to Public Agents or persons related to them is already considered a harmful act, even if the unlawful act does not materialize.

AgroSB is aware that the practice of acts of corruption has severe consequences, since the penalties provided for in the Anti-Corruption Law may be administrative, such as a fine on gross revenue and extraordinary publication of the condemnatory decision in the media, and also judicial, such as the prohibition of receiving incentives or loans from public financial institutions or controlled by the government, decree of loss of assets and rights, repair of damage and even the suspension or dissolution of the company's activities.

In addition, the liability of the legal entity does not exclude the individual criminal liability of its managers, employees or any person who is the perpetrator or participant in the act of corruption. AgroSB's Senior Management is committed to Business Integrity and values ethics, transparency, equity, compliance and effectiveness in all activities carried out in the conduct of its business. To support this commitment, AgroSB has an Integrity System aimed at preventing, detecting and correcting conduct that is not consistent with the principles of ethics and business integrity.

2.4.7. Commercially Sensitive Information (*Rules* 3.5.13 - 3.5.14)

Some information required by the VCS and/or CCB standards is considered confidential or commercially sensitive and cannot be publicly disclosed by the Project proponents. This information was fully provided to the audit team during the validation process attached to this document, but was not included in the public version. Below is a list of information that has been made available:

- Land Documents and Legal Status;
- Financial Statements of AgroSB;
- Financial Statements of Biofílica Ambipar;
- Project Financial Performance Spreadsheet (budget) and other related documents;
- Agreements and contracts signed between the parties involved;
- Diagnostic Inventories.

2.5. Legal Status and Property Rights

2.5.1. Statutory and Customary Property Rights (G5.1)

AgroSB is the legitimate owner of Fazenda Lagoa do Triunfo, composed of Lagoa do Triunfo I to VI and Nova Caracol, where the Ateles REDD+ Project is located. AgroSB's properties cover the municipality of São Félix do Xingu, in the State of Pará, and were acquired through deeds and purchase and sale contracts, documents that guarantee ownership of the properties. In addition to the deeds, the registrations, which present the registration of ownership and the descriptive memorial of the properties, are attached to the DPD. Other official documents such as the Rural Property Registration Certificate (CCIR) and the Certificates of the Land Institute of Pará (ITERPA) will also be presented, which show that the definitive titles of the lots that make up the farm are being processed to be obtained. These documents are intended to attest to the process of buying and selling the properties, the ownership and location of the notaries where they were registered, proving the legitimacy of the property.

In Brazil, all land has been owned by the state since it was colonized. In other words, until the land is allocated by the government, it is public and anyone who occupies it has it. Over the years, the form of this allocation has undergone various changes, such as the granting of letters of sesmaria, parish (or vicar's) registry, legitimization of possession, provisional titles, and leasing titles, among others.

Therefore, disregarding legitimate possession would be to end the land regularization system itself, which in the absolute majority of cases is initially based on control that fulfills its social function. There are several examples of this, such as in the state of Pará, the main one being the environmental legislation itself, which provides for the possibility of obtaining an environmental license for production in areas of possession, dispensing with ownership as a requirement for certain activities, including agriculture (Decree No. 216/2011). In other words, the state legislation itself already recognizes squatter areas as legally productive, issuing environmental licenses for their development.

The demonstration of the right to use the Project area is respected according to the criteria of VCS Standard v4.4 (page 25):

"1) Project ownership arising or granted under statute, regulation or decree by a competent authority.

2) Project ownership arising under law.

3) Project ownership arising by virtue of a statutory, property or contractual right in the plant, equipment or process that generates GHG emission reductions and/or removals (where the project proponent has not been divested of such project ownership).

4) Project ownership arising by virtue of a statutory, property or contractual right in the land, vegetation or conservational or management process that generates GHG emission reductions and/or removals (where the project proponent has not been divested of such project ownership).

5) An enforceable and irrevocable agreement with the holder of the statutory, property or contractual right in the plant, equipment or process that generates GHG emission reductions and/or removals which vests project ownership in the project proponent.

6) An enforceable and irrevocable agreement with the holder of the statutory, property or contractual right in the land, vegetation or conservational or management process that generates GHG emission reductions or removals which vests project ownership in the project proponent.

7) Project ownership arising from the implementation[39] or enforcement of laws, statutes or regulatory frameworks that require activities to be undertaken or encourage activities that generate GHG emission reductions or removals."

In Portuguese:

1) Propriedade do projeto decorrente ou concedido ao abrigo de estatuto, regulamento ou decreto por uma autoridade competente.

2) Propriedade do projeto decorrente da lei.

3) Propriedade do projeto decorrente de um direito legal, de propriedade ou contratual na instalação, equipamento ou processo que gere reduções e/ou remoções de emissões de GEE (quando o proponente do projeto não tenha sido alienado da propriedade de tal projeto).

[39] Implemented in the context of this paragraph means enacted or introduced, consistent with the use of the term under the CDM rules on so-called Type E+ and Type E-policies.

4) Propriedade do projeto em virtude de um direito estatutário, de propriedade ou contratual na terra, vegetação ou processo conservador ou de gestão que gere reduções e/ou remoções de emissões de GEE (quando o proponente do projeto não tiver sido despojado de tal propriedade do projeto).

5) Um acordo executório e irrevogável com o titular do direito legal, de propriedade ou contratual na instalação, equipamento ou processo que gere reduções e/ou remoções de emissões de GEE que atribua ao proponente do projeto a propriedade do projeto.

6) Um acordo executório e irrevogável com o titular do direito estatutário, de propriedade ou contratual na terra, vegetação ou processo conservador ou de gestão que gere reduções ou remoções de emissões de GEE que confere a propriedade do projeto ao proponente do projeto.

7) Propriedade do projeto decorrente da implementação ou aplicação de leis, estatutos ou quadros regulamentares que exijam a realização de atividades ou incentivem atividades que gerem reduções ou remoções de emissões de GEE.

Fazenda Lagoa o Triunfo is composed of 14 properties certified and registered in a real estate registry office. All have descriptive memorial and georeferencing in accordance with Law 10.267/01[40]. See Table 10 below.

Table 10: List of registered properties that make up Fazenda Lagoa do Triunfo.

Denomination	Registration of the property	Area (ha)	SIGEF link
FAZENDA LAGOA DO TRIUNFO I - Part 1	4540	19990.30	https://sigef.incra.gov.br/geo/parcela/detalhe/e0845211-9f37-437d-91bd-8b815fa7de81/
FAZENDA LAGOA DO TRIUNFO II - Part 1	4568	20639.35	https://sigef.incra.gov.br/geo/parcela/detalhe/d945d2de-76e2-4818-a77d-8298f12f5e40/
LOT 09 SECTOR C - Part 1	4976	3081.02	https://sigef.incra.gov.br/geo/parcela/detalhe/78409852-3458-4c7c-92a8-fbc70bb50c12/
FAZENDA LAGOA DO TRIUNFO III - Part 1	4541	38558.79	https://sigef.incra.gov.br/geo/parcela/detalhe/cbe1af26-0d4c-4532-a5d2-8fe0e4e7aa82/
LOT 35 SECTOR B - 001	4,571 (current 6,933)	990.07	https://sigef.incra.gov.br/geo/parcela/detalhe/cf854d2b-32fc-4038-af17-182714cb82f2/
Legal Reserval Compensation Area_ Land B_Remaining - 001	6484	1498.28	https://sigef.incra.gov.br/geo/parcela/detalhe/99503f80-1fb6-48d6-89be-ceeb371f2dd6/
Legal Reserve Compensation Area_Land A - 001	4511	2446.24	https://sigef.incra.gov.br/geo/parcela/detalhe/23871f59-9060-494f-8c01-0136805e8609/

[40] http://www.planalto.gov.br/ccivil_03/leis/leis_2001/l10267.htm

Legal Reserve Compensation Area - Land B - 001	4516	2475.16	https://sigef.incra.gov.br/geo/parcela/detalhe/91e44cd4-ae5c-47b1-bb32-49b3568a8fa0/
Legal Reserve Compensation Area - Land B - 001	4517	2024.70	https://sigef.incra.gov.br/geo/parcela/detalhe/2f1cbe9e-2fe3-4541-a075-3caa36b22e41/
Legal Reserve Compensation Area – Land B - 001	4518	879.79	https://sigef.incra.gov.br/geo/parcela/detalhe/cc424683-c865-4542-80e5-86d5f94a9816/
Fazenda Lagoa do Triunfo IV-Remaining Part - Faz Lagoa do Triunfo IV	4586	12064.58	https://sigef.incra.gov.br/geo/parcela/detalhe/006b735a-399e-407b-ac01-37c09acd88c4/
FAZENDA LAGOA DO TRIUNFO V - FAZENDA LAGOA DO TRIUNFO V	4542	30444.62	https://sigef.incra.gov.br/geo/parcela/detalhe/b393c485-a602-4b14-9622-dcf652cf7396/
FAZENDA LAGOA DO TRIUNFO VI - Part 1	4543	6183.53	https://sigef.incra.gov.br/geo/parcela/detalhe/616b8fc9-3616-4dff-af9f-7389be4e2181/
Fazenda Ceita Corê - Part 1	4486	39551.48	https://sigef.incra.gov.br/geo/parcela/detalhe/bdd92e2c-5866-4a56-bf42-926fadbafe94/

As previously mentioned, AgroSB does not yet have the definitive titles of the properties, however, in accordance with State Law No. 4.584, as of October 8, 1975[41], AgroSB Agropecuária S.A. has the ITERPA Certificates of the lots that make up the properties, recognizing the validity of the land titles.

2.5.2. Recognition of Property Rights (G5.1)

The Ateles REDD+ Project recognizes and respects all property rights, complying with significant statutory and regular requirements, as well as having the necessary approvals from local authorities. The Project recognizes and supports any rights to lands, territories and resources, including the statutory and traditional rights of indigenous peoples, communities and other actors.

The project proponents act as mediators of potential conflicts, in addition to valuing good relationship with neighboring communities. In this way, the following aspects are described in detail:

- The company AgroSB is the owner of the rights of use and economic exploitation of the properties, as well as obtaining the right of access to the natural resources existing in them, under the terms of the Federal Constitution of Brazil and the Civil Code, by virtue of being the owner of the properties where the Ateles REDD+ Project will be carried out

2.5.3. Free, Prior and Informed Consent (G5.2)

[41] http://portal.iterpa.pa.gov.br/wp-content/uploads/2021/03/LEI-N%C2%B0-4.584-DE-08-DE-OUTUBRO-DE-1975.pdf

Free, Prior and Informed Consent will be carried out during all phases of the Project, always with a dialogue and consent approach between the parties involved. In addition, the Project does not intend to develop any activity on private properties owned by others, by traditional and indigenous communities or by the government. Regarding social and biodiversity activities, it is guaranteed that no activity will be carried out without the free, prior and informed consent of the parties involved.

Free, Prior and Informed Consent is a principle that ensures that affected communities are informed about the objectives, impacts and risks associated with these projects and have the opportunity to express their opinion and make conscious decisions about their participation or not. It is important to note that no activity related to the Project will cause the relocation of activities that are important for the culture or livelihoods of Property Rights Holders, nor will it aim at the relocation or involuntary removal of their lands or territories. Any proposed removal or relocation needs to take place only after obtaining the Free, Prior and Informed Consent from the appropriate Holders of Property Rights.

In addition, all actors who could be impacted in any way by the Ateles REDD+ Project were consulted. In the communities that are located in the Project area, face-to-face activities were carried out in order to pass on information regarding the Project and resolve possible doubts, as well as consultations regarding the opinions of the community members in relation to the Project as described in Section 2.3.6. These consultations will continue to be carried out throughout the Project's life cycle. In addition, all information about the Ateles REDD+ Project can be acquired through Biofílica's virtual channels, such as website and social media.

2.5.4. Property Rights Protection (G5.3)

The implementation and development of the Ateles REDD+ Project will not lead to the involuntary removal or relocation of any party, and activities important to the culture and livelihoods of communities residing within the boundaries of the Project area are respected and supported by the Project. Thus, the Project proposes social activities that seek to promote sustainable practices, strengthen and improve existing economic activities, in addition to discouraging the practice of illegal activities.

Also, the land regularization of the area is supported by AgroSB and supported by the responsible public institutions. As previously mentioned, possible judicial inconsistencies are resolved following all procedures established by the local jurisdiction.

2.5.5. Illegal Activity Identification (G5.4)

Considering that the Ateles REDD+ Project is, in its entirety, inserted in the Triunfo do Xingu Environmental Protection Area, which is currently considered the most pressured conservation unit in the Amazon, deforestation was identified as the main illegal activity that can negatively impact the future development of the Project. Small farmers were identified as the main drivers of this illegal deforestation in the project area, who use the felling and burning system for annual plantations, and subsequent implementation of pasture for small livestock, in addition to the extensive cattle breeding practiced by medium and large producers, who use extensive areas to produce, however, with low productivity. Other deforestation agents are identified in the figure of land grabbers, who seek to expand pasture areas for extensive cattle breeding. These individuals appropriate the territory, in newly deforested areas, using artifices such as land speculation, in the expectation that their claims to illegal land will be recognized in the future. In addition, around the project area, sawmills, logging and mining activities can be identified as vectors of deforestation. Thus, there are embargoed areas within the Treasury resulting from the inspection of this deforestation, but we understand that they are not an impediment to the realization of the Project. Since these areas suffered forest suppression and are not counted as project area, in addition to being already surrounded to comply with the embargo. The documents referring to these processes, as well as the report with the isolation of these areas will be made available to the auditing body.

The indiscriminate use of fire by settlers and land grabbers is also a relevant point of attention. These arson fires occur at different stages of deforestation, either to clear the land of vegetation after large trees have been felled or to weaken the forest before clear-cutting, but in both cases, the impact on the local climate and biodiversity is devastating.

Between 2012 and 2022, an average deforestation rate in the reference region of 16,750 hectares per year (2.1% per year) was observed. In the analyzed period, it is observed that the deforested area shows a considerable jump in the annual deforestation rate from 2019. From then on, annual deforestation equal to or greater than 25,000 hectares in the reference region predominated. Therefore, with the continuation of this dynamic, for the next 10 years (2023-2032), a loss of 185,908 hectares is projected in this scenario, of which 14,106 hectares are expected to be deforested in the Project area.

The Project seeks to control and combat these illegal activities commonly found in the Project region through mitigating and preventive measures, such as the implementation of a land and property inspection system. In addition, the project promotes and encourages the involvement of other actors and stakeholders, social inclusion and regional socioeconomic development through the generation of economic alternatives to deforestation and the use of fire.

Asset surveillance aims to curb illegal practices of deforestation, extraction of plant species, and criminal fires. With the implementation of these measures, it is expected to improve the well-being of communities without generating burdens for native forest and local biodiversity. The mechanisms and procedures for land inspection and mitigation and prevention of illegal activities are summarized in Table 11.

Table 11: Mechanisms and procedures for the prevention of illegal activities in the Ateles REDD+ Project.

Mechanisms and procedures for the prevention of illegal activities	
General conditions	- Carry out regular patrols in order to ensure the protection of the land heritage of AgroSB's Fazenda Lagoa do Triunfo; - Avoid deforestation, forest fires or other acts of aggression to the environment; - Prevent illegal timber extraction and trade; - Maintain good relationship with communities and other stakeholders; - Perform the entry and exit control of Fazenda Lagoa do Triunfo; - Promote environmental education and promote the practice of sustainable economic activities; - Request support from police and supervisory authorities, when necessary; - Displacement of a surveillance team to the place of occurrence to investigate the fact and application of appropriate measures; - Activation of the legal sector for measures; - Record of occurrences involving invasion of property, damage to property, fire outbreaks and illegal extraction of forest products; - Occurrences involving damage to the environment must be registered with the responsible bodies (IBAMA, Environmental Police, etc.); - In all situations involving land conflicts, it is necessary to avoid confrontation between the parties, respecting the laws in force in the country.
Forms of records	- Police report; - Photographic record of occurrences; - Continuous remote monitoring program; - Report on property security activities;

	- Report of general activities of the Project

2.5.6. Ongoing Disputes (G5.5)

There are some ongoing disputes in the lots that make up the Ateles REDD+ Project. These are, in the vast majority, public civil actions and claims for adverse possession. All of them are ethically monitored by AgroSB's legal department.

As for the claims, according to the judicial negotiations taken by AgroSB, the risk of loss of ownership is remote. In addition, AgroSB guarantees all documentary means of possession and use of these areas, which allows these specific conflicts to be resolved in a legal, transparent and safe manner. The legal opinion on the procedural situation was made available to the audit body.

As for the public civil actions in progress, these are actions for material and collective environmental damage, which in due legal measures are being taken and in half of the cases, there is a remote chance of loss of the action by AgroSB. Also, as mentioned in Section 2.5.1, the areas that are in legal proceedings that cannot yet be considered as resolved or resolved were disregarded in the composition of the Project Area.

In addition, the areas in which there are disputes were accounted for in the risk tool, presented in section 2.1.18.

2.5.7. National and Local Laws (G5.6)

Compliance with Laws, Statutes and other significant regulatory bodies for the Ateles REDD+ Project is related to livestock activity. In the State of Pará, the project's activities are being licensed by the Brazilian Institute for the Environment and Renewable Natural Resources (IBAMA), thus applying federal legislation. Subject to federal legislation, the laws at the state level applies.

Regarding REDD+ activities, a history of initiatives can be noted despite the construction and negotiation of this concept through agreements and meetings at the *United Nations Framework Convention on Climate Change* (UNFCCC). In December 2015, Brazil's National REDD+ Strategy (ENREDD+) was instituted by MMA Ordinance No.370, a document that formalizes to Brazilian society and to the signatory countries of the UNFCCC how the Brazilian government has structured its efforts and intends to improve them by 2020, contributing to the mitigation of climate change through the control of deforestation and forest degradation, promotion of forest recovery and the promotion of sustainable development. It is important to mention that the REDD+ needs to be updated and is currently being developed by the federal government. In this context in Brazil, Decree No. 11,548 of June 5, 2023, established the National REDD+ Commission (CONAREDD+) to coordinate, follow up, monitor, and review the National REDD+ Strategy and guide the development of requirements for access to payments for results of REDD+ policies and actions in the country.

At the same time, of a broadly relevant nature, so far, Bill No. 412/2022, which establishes the Brazilian Greenhouse Gas Emissions Trading System (SBCE) and makes other provisions, is currently under analysis. The bill creates the Brazilian Regulated Carbon Market and provides for interoperability between the voluntary market and the future regulated market. The creation of a Brazilian Emissions Reduction Market is provided for in Law No. 12.187, of December 29, 2009, which establishes the National Policy on Climate Change - PNMC, and is a recommendation of the Kyoto Protocol and the Paris Agreement, international treaties ratified by Brazil that provide for the reduction of the concentration of GHG on the planet.

In addition, the State of Pará, where the project is located, has two legislations that are significant for the development of a REDD+ project. Pará State Law No. 9.781/2022, which amends Pará State Law No. 9048/2020 (establishes the State Policy on Climate Change and creates the State REDD+ Steering

Committee) defining more clearly the guidelines for the implementation of REDD+ projects in the State of Pará.

In addition, the state of Pará, where the project is located, has two pieces of legislation that are significant for the development of a REDD+ project. State Law No. 9.048/2020, amended by State Law No. 9.781/2022, establishes the State Climate Change Policy and creates the State REDD+ Steering Committee, more clearly defining the guidelines for implementing REDD+ projects in the state of Pará.

After years of discussion, the Federal Executive Branch of the current administration is promoting and supporting Bill 412/2022. In addition to creating the Brazilian Emissions Trading System (Regulated Market), the bill will also influence several points regarding the Brazilian voluntary market.

Below in Table 12, the main relevant legislation and regulations at federal and state levels are listed and detailed. In addition, a brief analysis of the international climate agreements that has been guiding the creation and development of REDD+ initiatives around the world was carried out.

Table 12: Relevant legislation and regulations, at federal and state level, for the Ateles REDD+ Project.

Federal Legislation	Provision
Law No. 14.119, as of January 13, 2021	Establishes the National Policy on Payment for Environmental Services, amends Laws No. 8.212, as of July 24, 1991, 8.629, as of February 25, 1993, and 6.015, as of December 31, 1973, to adapt them to the new policy.
Law No. 12.727, as of December 17, 2012	Provides for the protection of native vegetation; amends Laws No. 6938, as of August 31, 1981; 9393, as of December 19, 1996; and, 11428, as of December 22, 2006; and revokes Laws No. 4771, as of September 15, 1965; and, 7754, as of April 14, 1989; Provisional Measure No. 2166-67, as of August 24, 2001, item 22 of section II of Art. 167 of Law No. 6015, as of December 31, 1973, and § 2 of Art. 4 of Law No. 12651, as of May 25, 2012.
Law No. 12.651, as of May 25, 2012	Provides for the protection of native vegetation; amends Laws No. 6938, of August 31, 1981; 9393, as of December 19, 1996; and, 11428, as of December 22, 2006; and revokes Laws No. 4771, as of September 15, 1965; and, 7754, as of April 14, 1989; Provisional Measure No. 2166-67, as of August 24, 2001, and other measures.
Law No. 12.187, as of December 29, 2009	Institutes the National Policy on Climate Change – PNMC and takes other measures.
Decree No. 11.548, as of 05/06/2023	Establishes the National Commission for Reducing Greenhouse Gas Emissions from Deforestation and Forest Degradation, Conserving Forest Carbon Stocks, Sustainable Forest Management, and Increasing Forest Carbon Stocks - REDD+.
Decree No. 5.975, as of 11/30/2006	Regulates Art. 12, final part, 15, 16, 19, 20 and 21 of Law No. 4771, as of September 15, 1965; Art. 4, section III, of Law No. 6938,as of August 31, 1981; Art. 2 of Law No. 10650, as of April 16, 2003; amends and adds provisions to Decrees Nos. 6514/08 and 3420/00, and other provisions.
Resolution CONAMA No. 16, as of 12/07/1989	Establishes the Legal Amazon Environmental Assessment and Control Integrated Program.

Resolution CONAMA No. 378, as of 10/19/2006	Defines undertakings potentially causing national or regional environmental impact for the purposes of the provisions of section III, § 1, Art. 19 of Law No. 4771, as of September 15, 1965, and other provisions.
Resolution CONAMA No. 379, as of 10/19/2006	Creates and regulates data and information system on forest management within the scope of the National Environmental System - SISNAMA.
MMA Normative Instruction No. 1, as of 09/05/1996:	Deals with the Mandatory Forest Replenishment and the Integrated Forestry Plan.
MMA Normative Instruction No. 02, as of 05/10/2001:	Provides for the economic exploitation of forests on rural properties located in the Legal Amazon, including Legal Reserve areas, except for permanent preservation areas established in current legislation, which will be carried out through sustainable forest management practices for multiple uses.
Regulatory Standard No. 31, as of 03/03/2005:	Approves the Regulatory Norm for Safety and Health at Work in Agriculture, Livestock, Forestry and Aquaculture
State Legislation	Provision
State Law No. 9.048, as of 05/04/2020:	Institutes the State Policy on Climate Change in Pará (PEMC/PA), and other measures.
State Law No. 7.389, as of 03/31/2010:	Defines activities with a local environmental impact in the State of Pará and other provisions.
State Law No. 7.381, as of 03/16/2010:	Provides for the restoration of vegetation cover, riparian forests of the State of Pará.
State Law No. 6.745, as of 05/06/2005:	Establishes the Ecological Economic Macrozoning of the State of Pará and other measures.
State Law No. 6.506 as of 12/02/2002:	Establishes the basic guidelines for carrying out the Ecological Economic Zoning (EEZ) in the State of Pará and other measures.
State Law No. 6.462, as of 07/04/2002:	Provides for the State Policy on Forests and other forms of vegetation.
State Law No. 5.977, as of 07/10/1996:	Provides for the protection of wild fauna in the State of Pará.
State Law No. 5.887, as of 05/09/1995:	Provides for the State Environmental Policy and other measures.
State Decree No. 941, as of 08/03/2020:	Institutes the *Amazônia Agora* State Plan (PEAA), creates the Plan's Scientific Committee and the Permanent Plan Monitoring Nucleus, and other measures.
State Decree No. 254, as of 08/08/2019:	Establishes the Paraense Forum on Climate Change and Adaptation (FPMAC).
State Decree No. 518, as of 09/05/2012:	Establishes the Paraense Forum on Climate Change and other measures.
State Decree No. 216, as of 09/22/2011:	Provides for the environmental licensing of agrosilvopastoral activities carried out in altered and/or underutilized areas outside the legal reserve area and permanent preservation area in rural properties in the State of Pará.

State Decree No. 2.436, as of 08/10/2010:	Regulates actions directly or indirectly linked to agrosilvopastoral activities, carried out within areas of alternative land use, considered to have low environmental impact.
State Decree No. 2,099, as of 01/25/2010:	Provides for the maintenance, recomposition, conduction of natural regeneration, compensation and composition of the Legal Reserve area of rural properties in the State of Pará and other measures.
State Decree No. 1,697, as of 06/05/2009:	Establishes the Plan for Prevention, Control and Alternatives to Deforestation in the State of Pará and other measures.
State Decree No. 1,148, as of 07/17/2008:	Provisions on the Rural Environmental Registry – CAR-PA, Legal Reserve area and other measures.
State Decree No. 2,592, as of 11/27/2006:	Establishes the Registry of Forest Product Explorers and Consumers in the State of Pará – CEPROF-PA and the System for Commercialization and Transport of Forest Products in the state of Pará SISFLORA-PA and its operational documents and other measures.
State Decree No. 2.141, as aof 06/31/2006:	Regulates provisions of Law No. 6462, as of July 4, 2002, which provides for the State Policy on Forests and other forms of vegetation.
State Decree No. 1,523, s of 07/25/1996:	Approves the Regulation of the State Fund for the Environment - FEMA, created by Law No. 5887, as of May 9, 1995.
Resolution No. 54, as of 10/24/2007 (ANNEX1):	It approves the list of threatened flora and fauna species in the State of Pará.
International Agreements	Provision
FCCC/CP/2005/Misc.1	Reducing emissions from deforestation in developing countries: an approach to spur action. Submission of the parties. COP 11, Montreal, 2005.
FCCC/CP/2007/6/add.1	Report of the Conference of the Parties on its thirteenth session, held in Bali from 3 to 5 December 2007. Addendum. Part Two: Action taken by the Conference of the Parties at its thirteenth session or “Action Bali Plan”. COP 13, Bali, 2007.
FCCC/CP/2009/Add.1	Report of the Conference of the Parties on its fifteenth session, held in Copenhagen from 7 to 19 December 2009. Addendum Part Two: Action taken by the Conference of the Parties at its fifteenth session or “Copenhagen Accord”. COP 15, Copenhagen, 2009.
FCCC/CP/2010/7/Add. 1	Report of the Conference of the Parties on its sixteenth session, held in Cancun from November 19 to December 10, 2010. Addendum. Part Two: Action taken by the Conference of the Parties at its sixteenth session or “Cancun Agreement”. COP 16, Cancun, 2010.
FCCC/CP/2011/9/Add. 1	Report of the Conference of the Parties on its seventeenth session, held in Durban from 28 November to 11 December 2011. Addendum. Part Two: Action taken by the Conference of the Parties at its seventeenth session. COP 17, Durban, 2011.

FCCC/CP/2012/8/Add.1	Report of the Conference of the Parties on its eighteenth session, held in Doha from 26 November to 8 December.Addendum. Parte Dois: Action taken by the Conference of the Parties at its eighteenth session.
FCCC/CP/2013/Add.1:	Warsaw Package for REDD+, which took place in Warsaw, Poland, from 11 to 22 November 2013, in particular the following decisions:
	Decision9/CP.19: Programa de trabalho em financiamento baseados em resultados para o progresso da implementação completa das atividades referidas na decisão 1/CP. 16, paragraph 70.
	Decision10/CP.19: Coordination of support for the implementation of activities related to mitigation actions in the forest sector by developing countries, including institutional arrangements.
	Decision12/CP.19 : The timing and frequency at which summary information is presented of how all safeguards referred to in decision 1/CP.16, appendix I, are being addressed and adhered to.
	Decision13/CP.19: Guide and procedures for technical evaluation of Parties' submissions on proposals for reference levels on forest emissions and/or forest reference levels.
	Decision14/CP.19: Modalities for measuring, reporting and verifying.
	Decision15/CP.19: Approach to deforestation and forest degradation vectors.
FCCC/CP/2015/Add.1	Relatório de Conferência das Partes sobre sua vigésima primeira sessão, ocorrida em Paris de 30 de novembro a 13 de dezembro. Addendum. Parte Dois: Action taken by the Conference of the Parties at its twenty-first session.
FCCC/CP/2015 Paris Agreement:	Global legally binding agreement establishing a global framework to prevent dangerous climate change by limiting global warming to well below 2°C and continuing efforts to limit it to 1.5°C. Entry into force on November 4, 2016.
FCCC/CP/2016 Decisions adopted by the Conference of the Parties (COP)	Especially decisions 1 (preparation for the entry into force of the Paris Agreement), 3 (Warsaw International Mechanism for Loss and Damage associated with the Impacts of Climate Change), 6 (National adaptation plans) and 7 (Long-term financing of the fight against climate change).
FCCC/CP/2017, FCCC/CP/2018, FCCC/CP/2019 Decisions adopted by the COP	Especially decision 1, which reports on the progress of the implementation of the Paris Agreement.
Nationally Determined Contribution – Brazilian NDC	Referred in September 2015 to the United Nations Framework Convention on Climate Change for Mitigation, Adaptation and Means of Implementation, in a manner consistent with the purpose of the contributions to achieving the ultimate objective of the Convention, pursuant to decision 1/CP.20, paragraph 9.

CITES, of 03/03/1973	"Convention on International Trade in Endangered Species of Wild Fauna and Flora", assinada em Washington D.C. em 03 de março de 1973, alterado em Bonn em 22 de junho de 1979.
Article 6 of the Paris Agreement (2021)	Decision 1/CP.21 mandated SBSTA to operationalize the provisions of this article by recommending a set of decisions to the COP, which functions as the meeting of the Parties to the Paris Agreement at its first session. At COP26, the Parties to the Paris Agreement, at their third session (CMA 3), adopted three main decisions related to Article 6: decision 2 (on Article 6.2), decision 3 (on Article 6.4) and decision 4 (on Article 6.8).
Glasgow Leaders' Declaration on Forests and Land Use (2021)	Signatories (including Brazil) pledge to reverse and end deforestation by 2030.
Brazilian Nationally Determined Contribution (NDC)	First Brazilian NDC submitted in September 2015 to the United Nations Framework Convention on Climate Change for mitigation, adaptation and means of implementation, consistently with the purpose of contributions to achieve the final objective of the Convention, pursuant to Decision 1/CP.20, paragraph 9. The updated Brazilian NDC was presented at COP26 on December 8, 2022.
CIM Resolution No. 5 (14/09/2023)	Determines that the Ministry of Foreign Affairs communicated to the UNFCCC the correction of Brazil's NDC, returning to the level of ambition presented in 2015, in the Paris Agreement, in terms of absolute values of greenhouse gas emissions.

2.5.8. Approvals (G5.7)

Project proponents have achieved recognition and approval on the implementation of the Ateles REDD+ Project with stakeholders through meetings, lectures and face-to-face meetings with the communities, partners, proponents and authorities mentioned in Section 2.1.9.

In addition to the mandatory feedback activity (described in section 2.3.6), the project also held an event called Situational Alignment, which took place on June 27, 2023 with the objective of presenting the Ateles REDD+ Project, its benefits and impacts for the communities living in the project area. This activity was also indispensable to collect relevant information for the design of the activities that will be developed throughout the project.

In addition to these face-to-face activities described above, the project will go through the public consultation event on the Verra registration platform for comments, suggestions and clarification of doubts about the Ateles REDD+ Project, which should take place at the beginning of November 2023. Such stakeholder engagement and collaboration relevance in this process was reinforced by submitting formal invitations to the ones directly and indirectly involved in forest conservation sector; including community association, non-governmental organizations (NGO´s), education institutions and private companies. Such invitation was made via mailing, with information about the project and invitation to participate in the public consultation.

In addition, official letters were sent to the relevant local institutions in the state of Pará; including the State Public Prosecutor's Office and other governmental and federal agencies, containing information about the public consultation, the context of the project and the communication channels used.

It is worth mentioning that despite the advances of the National Strategy for REDD+ of Brazil (ENREDD+), the processing of Bill No. 572/2020 and the resumption of the Paraense Forum on Climate Change and

Adaptation (FPMAC), demonstrated in section 2.5.6 - National and Local Laws, there are still no policies at the official national or jurisdictional level of REDD+. However, the Project proponents are always attentive to new information, always present in discussion forums of the federal and state governments in order to contribute to the formulation of these policies and regulations, being readily available to adjust the Project to the new officially established rules.

2.5.9. Project Ownership (G5.8)

AgroSB is the legitimate owner of the properties where the Ateles REDD+ Project is being implemented and developed, as detailed in section 2.5 – Statutory and Customary Property Rights. For the establishment of responsibility and rights over the Project, as well as the percentage of carbon credits allocated to each party, a contract was signed by he Project proponents.

2.5.10. Management of Double Counting Risk (G5.9)

The Ateles REDD+ Project generates benefits for climate, communities and biodiversity, but only net greenhouse gas reductions and removals will be commercialized after being properly recorded on a market platform.

2.5.11. Emissions Trading Programs and Other Binding Limits

Not applicable.

2.5.12. Other Forms of Environmental Credit

The Ateles REDD+ Project does not intend to generate any other form of environmental credits related to GHG emission reductions and removals claimed within the VCS (*Verified Carbon Standard) program.*

2.5.13. Participation under Other GHG Programs

The Ateles REDD+ Project did not receive or seek to be registered in any other GHG program, in addition to submitting the Project for validation and verification in the VCS (*Verified Carbon Standard*) and CCBS (*Climate, Community and Biodiversity Standard*) standards.

2.5.14. Projects Rejected by Other GHG Programs

The Ateles REDD+ Project has not undergone validation/verification of any other GHG program and is therefore not rejected by any other GHG program.

2.5.15. Double Counting (G5.9)

The Government of the State of Pará has brought the issue of REDD+ to debate since the beginning of discussions on the subject within the framework of international climate conferences. In 2009, the Paraense Forum for Climate Change and Adaptation (FPMCA) was created, and in 2019 it was reactivated through a Law Decree signed by the governor of the State of Pará[42]. The FPMCA, among its objectives, guides and subsidizes the preparation and implementation of the State Policy on Climate Change Law of Pará (PEMC/PA). One year after the reactivation of the FPMCA, the Law establishing the PEMC/PA[43] was published. Such law provides for the planning and execution of plans, actions and programs related to climate change, through policies, actions, research and technical studies focused on environmental services and Reduction of Emissions from Deforestation and Forest Degradation (REDD+)[44].

[42] FEDERATIVE REPUBLIC OF BRAZIL. Official Gazette No. 33948, of August 9, 2019. Belém-PA. Available at: http://www.ioepa.com.br/pages/2019/2019.08.09.DOE.pdf

[43] GOVERNMENT OF THE STATE OF PARÁ. Law No. 9048, of April 29, 2020. Available at: https://www.semas.pa.gov.br/legislacao/files/pdf/4093.pdf

[44] AGÊNCIA PARÁ. **Pará's Forum on Climate Change and Adaptation debates advances with society**. Available at: https://agenciapara.com.br/noticia/24012/.

Regarding REDD+, the FPMCA proposed the creation of a State REDD+ Strategy, aiming to organize and prioritize actions in the areas of deforestation and forest degradation, conservation and forest management. In this sense, in December 2021, the FPMCA approved the creation of the Technical Chamber of Bioeconomy, to compose the State Council on Climate Change; which, among several strategies, includes the regulation of the REDD+ jurisdictional system in Pará, training on REDD+ and the carbon market, and actions for eligibility for carbon certification[45].

Given this context, in February 2022 the 1st Seminar on Payments for Environmental Services (PSA) and Reduction of Emissions from Deforestation and Degradation (REDD+) was held in Pará, promoted by the Secretariat for the Environment and Sustainability (Semas) and the Institute of Environmental Research of Amazon (IPAM). The central objective of this event was to discuss the REDD+ jurisdictional system in Pará, taking into account the challenges and possible solutions in the areas of REDD+, such as payments for Environmental Services (PSA) and the carbon market in the State of Pará[46].

However, despite the initiatives, so far, the State of Pará does not have a defined REDD+ State Strategy.

Thus, it is the understanding of the proponents that there is no risk of double counting, since the Government of Pará does not have a structured legal program or any type of state regulation for Climate Change and REDD+ and does not carry out voluntary or unregulated market operations.

3. CLIMATE

3.1. Application of Methodology

3.1.1. Title and Reference of Methodology

Verified Carbon Standard (VCS) Approved Methodology VM0015 – Methodology for Prevented Planned Deforestation, version 1.1.

VT0001 Tool for the Demonstration and Evaluation of Additionality in the Activities of the VCS Agriculture, Forestry and Other Land Uses Project (AFOLU), version 3.0.

AFOLU Non-permanence Risk Tool, version 4.1.

3.1.2. Applicability of Methodology

For the Ateles REDD+ Project, the methodology approved by the VCS, code VM0015, was used, and is applicable according to the applicability criteria specified in Table 13, below:

Table 13: Criteria for applicability of the methodology for the Ateles REDD+ Project.

Applicability Criteria	Description of how the Project meets these criteria
(a) baseline activities may include planned or unplanned logging, firewood collection, charcoal production, agricultural and grazing activities as long as the category is unplanned deforestation according to the latest version of the VCS AFOLU Requirements.	Project baseline activities include unplanned deforestation as a result of agricultural and livestock activities, as per the recent version of the VCS AFOLU Requirements document.

[45] SMAS - SECRETARIAT FOR ENVIRONMENT AND SUSTAINABILITY OF THE STATE OF PARÁ. **State climate change forum creates Bioeconomy Technical Chamber**. Available at: https://www.semas.pa.gov.br/2021/12/17/forum-estadual-de-mudancas-climaticas-cria-camara-tecnica-de-bioeconomia/.

[46] AGÊNCIA PARÁ. **Semas and Ipam discuss REDD+ jurisdictional system and carbon market in seminar**. Available at: https://agenciapara.com.br/noticia/35041/..

(b) Project activities may be included in a category or a combination thereof defined in the methodology scope description.	Project activities include forest protection without logging, firewood use, or charcoal production, following the description of scope "A" of the methodology used (page 11, Table 1 and Figure 2-A in document VCS VM0015).
(c) The Project area may include different types of forest, including but not limited to primary forests, degraded forests, secondary forests, planted forests and agroforestry systems, obeying the definition of "forest".	The project area has different types of forests, mainly mature forests that fall under the national definition of "forest".
(d) At the start of the Project, the Project area shall only include areas qualified as "forest" for a minimum of 10 years prior to the Project start date.	The project area only includes areas classified as "forest" for a minimum period of 10 years prior to the project start date.
(e) The Project area may include floodplain areas (such as lowland forests, floodplain forests, mangroves) as long as they do not develop on peat. Peat should be defined as organic soils with at least 65% organic matter and minimum thickness of 50 cm. If the Project area includes lowland forests that develop into peat (e.g. peat forests), this methodology is not applicable.	Forest types found in the project area do not include forested wetlands or peatswamp forests.

3.1.3. Project Boundary

3.1.3.1. VM0015 Step 1 – Definition of Boundaries

3.1.3.1.1. Spatial Boundaries

Reference Region

According to the VCS VM0015 methodology, the reference region is the spatial boundary that contains the project area, the leak belt, the leak management areas and other relevant geographical areas to determine the project baseline (). The main criteria used to define the spatial limits of the reference region and thus demonstrate the compatibility conditions in the probability of future deforestation were:

a) Probable area of activity and influence of deforestation agents and drivers;

b) Landscape configurations and ecological conditions;

c) Socio-economic and cultural conditions

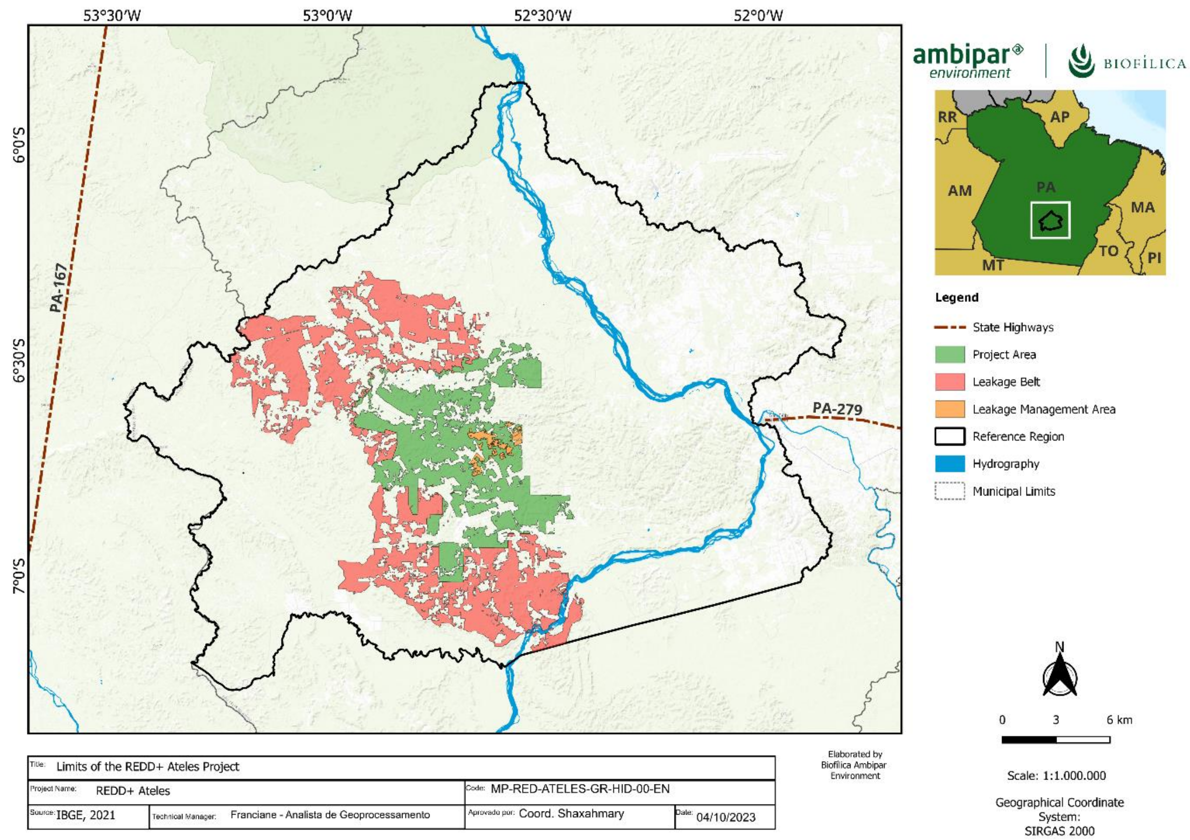

Figure 9: Boundaries of the Reference Region, Project Area, Leakage Belt and Leakage Management of the Ateles REDD+ Project.

Thus, the proposed reference region corresponds to an area of 1.686.952 hectares, equivalent to **15.8** times the Project Area.

To determine the reference region, we considered a set of sub-basins from the HydroBasins base, level 6 in its version 1 (LEHNER; GRILL, 2013). To the west of the Farm, the chosen sub-basins consist of those bordering the border of the municipalities of São Felix do Xingu and Altamira. The eastern limit is composed of sub-basins overlapping the bank of the Xingu River and the northern and southern limits by the Apyterewa, Kayapó and Menkragniti Indigenous lands. Thus, the hydrographic context together with the land situation of the areas surrounding Fazenda Lagoa do Triunfo were the main elements used to determine the geographical limit of the Reference Region.

It is also worth mentioning that in this region we observe the dynamics of deforestation typical of the deforestation arc in the Brazilian Amazon: areas with forest exploitation, forest degradation with the removal of wood of commercial value, followed by deforestation to attempt land regularization and creation of pastures with low productivity. The Reference region has only one stratum to analyze the performance of agents and vectors of deforestation and changes in land use and coverage.

In addition to the context described above, the following criteria established by VM0015 (pages 18 and 19 of the methodology), listed below, were analyzed:

a) **Infrastructure determinants:** There is no provision for the implementation of infrastructure such as highways and hydroelectric plants near the project area.
b) Landscape configuration and ecological conditions:

- **Condition 1**: At least 90% of the project area must have existing forest classes or vegetation types in at least 90% of the rest of the reference region.
- **Outcome 1**: The project area is 100% covered by Open Ombrophilous Forests or Dense Ombrophilous Forests, while in the Reference Region, this percentage is 88%, with the remainder being divided between Savannas and livestock areas. are 100% covered by Open Ombrophilous Forests or Dense Ombrophilous Forests (Table XXX of the physical environment).

- **Condition 2**: At least 90% of the project area must be within the elevation range of at least 90% of the remainder of the reference region.
- **Outcome 2**: 100% of the project area is in the elevation range of at least 90% of the remainder of the reference region, as indicated by the elevation and slope percentiles of theTable 14.

- **Condition 3**: The average slope of at least 90% of the project area must be within ±10% of the average slope of at least 90% of the rest of the reference region
- **Outcome 3**: 100% of the project area has an average slope within the range of 87% of the average slope of the reference region, according to the percentiles shown in Table 14.

Table 14: Elevation and slope percentiles for the project area and reference region.

Percentile	Elevation (m)		Slope (degrees)	
	Reference region	Project area	Reference region	Project area
0	139.0	179.0	0.0	0.0
5	193.4	198.5	0.9	1.3
10	201.6	212.5	1.3	1.9
15	205.5	226.5	1.9	2.1
20	209.5	238.5	2.1	2.9
25	217.5	252.5	2.6	3.4
30	221.5	264.5	2.9	3.9
35	225.5	274.5	2.9	4.2
40	229.5	284.5	3.7	4.7
45	237.5	296.5	3.9	5.4
50	241.5	304.5	4.7	6.2
55	249.5	314.5	4.9	7.1
60	257.5	322.5	5.6	8.4
65	269.5	330.5	6.5	9.9
70	281.5	340.5	7.4	11.6
75	293.5	348.5	8.8	13.4
80	313.5	358.5	11.1	15.6
85	333.5	370.5	13.9	18.1
90	361.5	388.5	17.4	20.9
95	409.5	414.5	22.1	24.9
100	736.8	554.0	61.0	52.9

c) Socioeconomic and cultural conditions:

- **Legal situation of the property**: the legal situation of the land of the Project Area in the base scenario can be observed in several locations in the reference region.

- **Land status**: The land situation of the Project area (private property) occurs in 91% of the reference region (land).
- **Land use**: The classes of current and projected land use and land cover types in the Project Area are the same throughout the reference region. They are**: *a) Forest* *and* *b) Equilibrium Anthropized Vegetation***.
- **Policies and regulations applied**: Project Area is governed by the same laws and regulations applied throughout the Reference Region.

Project Area

The Ateles REDD+ Project has an area of 106.921 hectares and its delimitation is shown in Figure 1. The limits of the project area were defined considering the existing forest area within Lagoa do Triunfo, owned by AgroSB Agropecuária. The description of the property, property rights and land documents were addressed in Section 2.5 The forest cover estimates in the project start year were based on data from PRODES/INPE. The areas planned for the implementation of the Project infrastructure must be excluded and the estimates presented in the certification process.

Leakage Belt

The Ateles REDD+ Project is not located within a jurisdictional project, so the VM0015 methodology recommends that an area called the Leakage Belt be defined. The Leakage Belt is the area around or adjacent to the Project Area in which baseline activities may be displaced due to project activities. The total area of the leakage belt corresponds to 179.526 hectares. Most of the deforestation in the reference region was carried out for the purpose of logging, attempted land regularization and subsequent extensive cattle raising activity and installation of fields (. Consequently, less than 80% of the deforested area in the region (or some of its layers) during the historical reference period occurred in areas where deforestation is profitable. Therefore, Mobility Analysis (Option II) was selected as the appropriate method for determining leak belt boundaries.

The multi-criteria approach was applied to delimit the Leakage Belt, based on the deforestation projection map and maps of private properties (Farms) located near the project area. The following criteria have been applied that facilitate and restrict the accessibility of deforestation agents:

a) Areas with a higher risk of deforestation have greater accessibility to deforestation agents (higher risk areas indicate greater accessibility of the deforestation agent).

b) Rural properties with controlled access and that have environmental characteristics similar to those observed in the project area (these properties have a certain level of access control to the interior of the forest, therefore, greater restriction of the deforestation agent).

The selection of farms in this initial analysis took into account the following cartographic bases:

- Limits of rural properties georeferenced by INCRA (Sigef Brasil)
- Area of the Ateles REDD+ Project

Leakage management

Under development.

Forest

The Forest area was identified based on the results of the Project for Monitoring Deforestation in the Legal Amazon by Satellite (PRODES) of the National Institute for Space Research (INPE). The forest area identified by PRODES in the Reference Region was 871.328 hectares in August 2022 (project start). The definition of forest adopted by PRODES is in accordance with the definition of forest contained in Annex I of VM0015 v1.1 (pg 124). The Figure 10 shows the areas covered by forests in the Reference Region in August 2022.

3.1.3.1.2. Time limits

a) **Start and end date of the Historical Reference Period**: 08/01/2012 and 07/01/2022. These dates were defined mainly considering the availability of PRODES data, used to generate land cover maps and meet the requirements of the VM0015 methodology.

b) **Start and end date of the first fixed baseline** period: The fixed baseline period is 10 years, according to VM0015 Methodology version 1.1.

c) **Monitoring Period**: The monitoring period of land use and use change will start from the start date of the Project, including the requirement of being at least 1 year.

The start date of the project activities and the accreditation period are described in sections 2.1.14 and 2.1.15.

VM0015 Step 1.3 – Carbon Reservoirs

The carbon reservoirs analyzed in the Ateles REDD+ project are shown in 14. Methodological details of the estimation of the carbon stocks considered can be found in the forest inventory report document carried out in the project area, made available to the validating/verifying body.

Define project boundaries and identify GHG sources, sinks and reservoirs relevant to the project and baseline scenarios (including leaks if applicable).

Table 15: Carbon reservoirs included and excluded in the analysis of the Ateles REDD+ Project and with respective justifications.

Scenario	Source	Gas	Included	Jutification/ Explanation
Baseline	Arboreal above ground	CO_2	Yes	Always significant
		CH_4	No	Not applicable
		N_2O	No	Not applicable
		Others	No	Not applicable
	Non-arboreal	CO_2	No	It is conservative to omit CO_2 emissions for the baseline scenario
		CH_4	No	Not applicable
		N_2O	No	Not applicable
		Others	No	Not applicable
	Below ground	CO_2	Yes	Significant according to the significance assessment tool
		CH_4	No	Not applicable
		N_2O	No	Not applicable
		Others	No	Not applicable
	Dead wood	CO_2	Yes	Significant according to the significance assessment tool
		CH_4	No	Not applicable

		N2O	No	Not applicable
		Others	No	Not applicable
	Timber products from logging	CO2	Yes	Not significant according to the significance tool
		CH4	No	Not applicable
		N2 O	No	Not applicable
		Others	No	Not applicable
	Litter	CO2	No	Not significant according to the significance tool
		CH4	No	Not applicable
		N2 O	No	Not applicable
		Others	No	Not applicable
	Soil organic carbon	CO2	No	Not significant according to the significance tool
		CH4	No	Not applicable
		N2 O	No	Not applicable
		Others	No	Not applicable
Project	Arboreal above ground	CO2	Yes	Always significant
		CH4	No	Not applicable
		N2 O	No	Not applicable
		Others	No	Not applicable
	Non-arboreal	CO2	No	It is conservative to omit CO2 emissions for the baseline scenario
		CH4	No	Not applicable
		N2 O	No	Not applicable
		Others	No	Not applicable
	Below ground	CO2	Yes	Significant according to the significance assessment tool
		CH4	No	Not applicable
		N2 O	No	Not applicable
		Others	No	Not applicable
	Dead wood	CO2	Yes	Significant according to the significance assessment tool
		CH4	No	Not applicable
		N2 O	No	Not applicable
		Others	No	Not applicable
	Timber products from logging	CO2	Yes	In line with the planning of logging in the project area
		CH4	No	Not applicable
		N2 O	No	Not applicable
		Others	No	Not applicable
	Litter	CO2	No	Not significant according to the significance tool
		CH4	No	Not applicable
		N2 O	No	Not applicable
		Others	No	Not applicable
		CO2	No	Not significant according to the significance tool

	Soil organic carbon	CH4	No	Not applicable
		N2 O	No	Not applicable
		Others	No	Not applicable

3.1.4. Baseline Scenario

3.1.4.1. VM0015 Step 2 – Analysis of historical land use and land cover changes

3.1.4.1.1. Survey of adequate data sources

For the mapping of land use and land cover classes, data from the PRODES program, provided by the National Institute for Space Research, were used. (PRODES 2005). The PRODES program uses images from the Landsat series of satellites and others to map annual deforestation and monitor forest remnants. The PRODES data in raster format provided by the Terra Brasilis system[47] were analyzed in the following thematic classes: forest, non-forest vegetation, hydrography and anthropized vegetation (deforestation). The images cover the period from 2012 to 2022 and correspond to the orbits/points.The maps produced by PRODES have methodology and estimate of accuracy of class mapping recognized by the national and international scientific[48] community[49].

Table 16: Data used for analysis of historical evolution of soil cover (Table 5 of VM0015).

Vector (Satellite or airplane)	Sensor	Resolution		Coverage	Acquisition date	Scene identifier	
		Spatial (m)	Spectral	(km^2)	(DD/MM/YY)	Path/ Latitude	Row/ Longitude
Landsat	TM	30	0.45 – 2.35 µm	34,225			
Landsat	TM	30	0.45 – 2.35 µm	34,225			

3.1.4.1.2. Definition of land use and land cover classes

The soil cover classes present in the Reference Region on the project start date are: Forest; Non-Forest Vegetation; Hydrography and anthropized vegetation in balance (Table 17). The description of each class and its existing area before the start of the project (2021) is presented below:

[47] Terra Brasilis – National Institute for Space Research (INPE). Available at: http://terrabrasilis.dpi.inpe.br/download/dataset/legal-amz-prodes/raster/PDigital2000_2021_AMZ_raster_v20211118.zip
[48] MAURANO, L.E.P.; ESCADA, M.I.S.; RENNO, C.D. Spatial patterns of deforestation and the estimation of the accuracy of PRODES maps for the Brazilian Legal Amazon. Ciência Florestal, Santa Maria, v. 29, n. 4, pp. 1763-1775
[49] KINTISCH, Eli. Better monitoring of tropical forests helps eliminate the fog of deforestation. Science (2007)

- **Forest** (ND[50] = 0): area of forest remnant belonging to different phytophysiognomies of the Ombrophilous Forest.
- **Equilibrium Anthropized Vegetation** (ND = 1): areas of deforested forests converted to other land uses (mosaic of different types of vegetation that includes pastures, gardens, plantations and secondary vegetation)
- **Hydrography** (ND = 2): water bodies (rivers, lakes, streams, among others).
- **Non-forest vegetation** (ND = 3): areas consisting of natural vegetation with a physiognomy different from the forest, regionally known as Campinarana, Savannah or Cerrado.

Table 17: Existing land use and land cover classes in the Reference Region (Table 6 of VM0015)

Class Identification		Carbon stock trend	Present in[1]	Baseline activity[2]			Description
IDcl	Name			LG	FW	CP	
1	Anthropized Vegetation, Equilibrium	Constant	RR, LM.	No	No	No	Forest areas deforested by clearcutting and with vegetation different from the Ombrophilous Forest.
2	Forest	Descending	RR, PA, LK.	Yes	Yes	Yes	Remaining forest.
3	Hydrography	-	RR	No	No	No	Bodies of water.
4	Non-forest vegetation	Constant	RR	No	No	No	Natural vegetation cover with non-forest phytophysiognomy.

1 - RR: Reference Region; PA: Project Area; LK: Leakage Belt; LM: Leakage Management Areas.
2 - LG: Logging. FW = Fuel-wood collection; CP = Charcoal Production (yes/no).

3.1.4.1.3. Definition of land use change and land cover classes

In the spatial modeling of future deforestation, the changes from areas with Forest (Class I1) to areas with Equilibrium Anthropic Vegetation (Class F1) within the Project Area and in the Leakage Belt (Table 18) were considered.

Table 18: Definition of land use and land cover change categories (Table 7.b of VM0015).

IDcl	Name	Carbon stock trend	Present in[1]	Activity in baseline scenario2			Name	Carbon stock trend	Present	Activity in the scenario with project		
				LG	FW	CP				LG	FW	CP
I1/F1	Forest	Descending	PA	Yes	Yes	Yes	Vegetation anthropized in balance.	Constant	RR. ML	No	No	No
I1/F1	Forest	Descending	LK	Yes	Yes	Yes	Vegetation anthropize	Constant	RR. ML	No	No	No

[50]ND = digital number of the ground use and cover raster.

							d in balance.					

1 - RR: Reference Region; PA: Project Area; LK: Leakage Belt; LM: Leakage Management Areas.
2 - LG: Logging. FW = Fuel-wood collection; CP = Charcoal Production (yes/no).

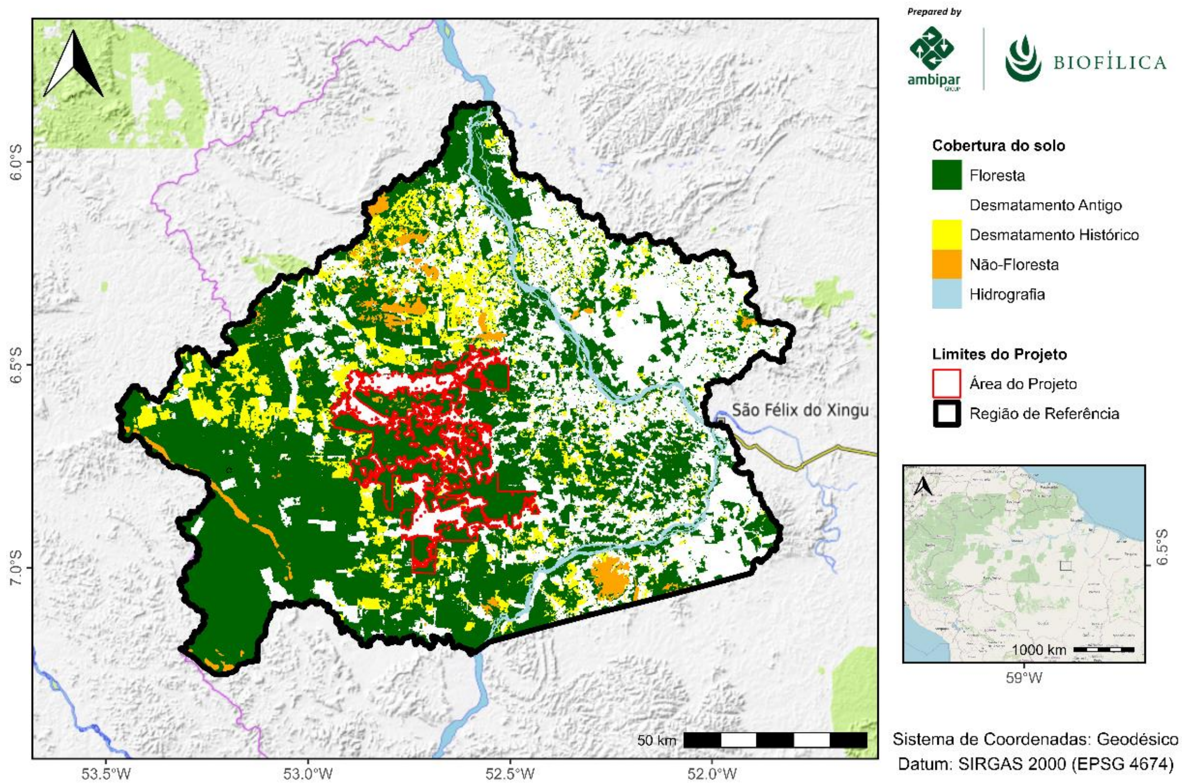

Figure 10: Spatial distribution of deforestation in the reference region. The project area is highlighted by the red polygon.

3.1.4.1.4. Analysis of the historical land use and land cover change

The data provided by PRODES was used to analyze the history of land use change. The main activities carried out by PRODES to map deforestation in the Brazilian Amazon are detailed below:

a) **Pre-processing**: The main **image** pre-processing **procedures** performed by PRODES consist of the steps of selecting images with less cloud cover, with acquisition date closer to the dry season in the Amazon and with adequate radiometric quality, previously georeferenced and orthoretified by NASA.

b) **Interpretation and Classification**: The satellite image classification method used by PRODES follows three steps[51]: i) these images are used to make a single composition per scene, of type 6R/5G/4B (Landsat-8 or 9), corresponding to mid-infrared (1.57 – 1.65 µm), near-infrared (0. 85 – 0.88 µm) and red (0.64 – 0.67 µm), respectively, which is used as the basis for mapping. This composition has contrast highlighted to facilitate the identification of new deforestation; ii) recent deforestation polygons are identified based on the interpretation of these highlighted images, based

[51] Almeida, Claudio Aparecido de, Luís Eduardo Pinheiro Maurano, Dalton de Morisson Valeriano, Gilberto Câmara, and Lúbia Vinhas. 2022. "Methodology used in PRODES and DETER systems - 2nd edition (updated)", 50.

on their tone, shape, texture and spatial context; iii) to certify that only new deforestation is accounted for, a mask is applied on all deforestation detected in previous years;

c) **Mapping Accuracy Assessment**: we evaluated the accuracy of land cover mapping by the most recent PRODES (2022) using reference data collected by interpreting high-resolution images over a set of points randomly distributed over the reference region. The reference data used in this step comes from the visual interpretation of a mosaic of high spatial resolution Planetscope images (July-October 2022). Using the land use and land cover reference points and the 2022 map, it was possible to evaluate the mapping performance through the analysis of the confusion matrix.

Table 19: PRODES 2020 data evaluation error matrix

Under development.

d) **Post-processing**: According to the **VM0015** Methodology, post-processing includes the use of non-spectral data to sub-stratify land use and land cover classes with heterogeneous carbon density into homogeneous carbon density subclasses.

The data required by step 2: a) Forest cover reference maps referring to the project start year and the beginning of the historical reference period b) Land use and land cover map referencing the project start year, c) Deforestation maps for the analyzed subperiods and d) Land use and land cover change matrix are all condensed in the single map of Figure 9. Product e) The land use and land cover change matrix is presented in Table 20 below.

3.1.4.1.5. Results of the analysis of historical land use and land cover change

The results of the analysis of the history of deforestation that occurred between August 2012 (inclusive) and July 2022 in the reference region are presented in Table 20. Calculating the area in the land use maps, an estimated 167,497 hectares of deforested land were estimated, which corresponds to a 16.1% reduction in the existing forest in August 2012.

Table 20: Matrix of land use change in the reference region between 2012 and 2022 (Table 7.a of the VM0015 methodology).

IDcl			Initial Class (2012)				Total (ha)
		Name	Forest	Deforestation	Non-forest vegetation	Hydrography	
			I1	I2	I3	I4	
Final Class (2022)	F1	Forest	871,328	0	0	0	871,328
	F2	Deforestation (Anthropized Vegetation in Balance)	167,497	569,641	0	0	737,138
	F3	Non-forest vegetation	0	0	43,943	0	43,943
	F4	Hydrography	0	0	0	34,541	34,541
Total (ha)			1,038,825	569,641	43,943	34,541	1,686,950

Between 2012 and 2022, an average deforestation rate of 17,601 hectares per year (1.7% per year) was observed. Figure 11 Presents a temporal variation of deforestation on an annual basis for the reference region, state of Pará and Legal Amazon between 2012 and 2022.

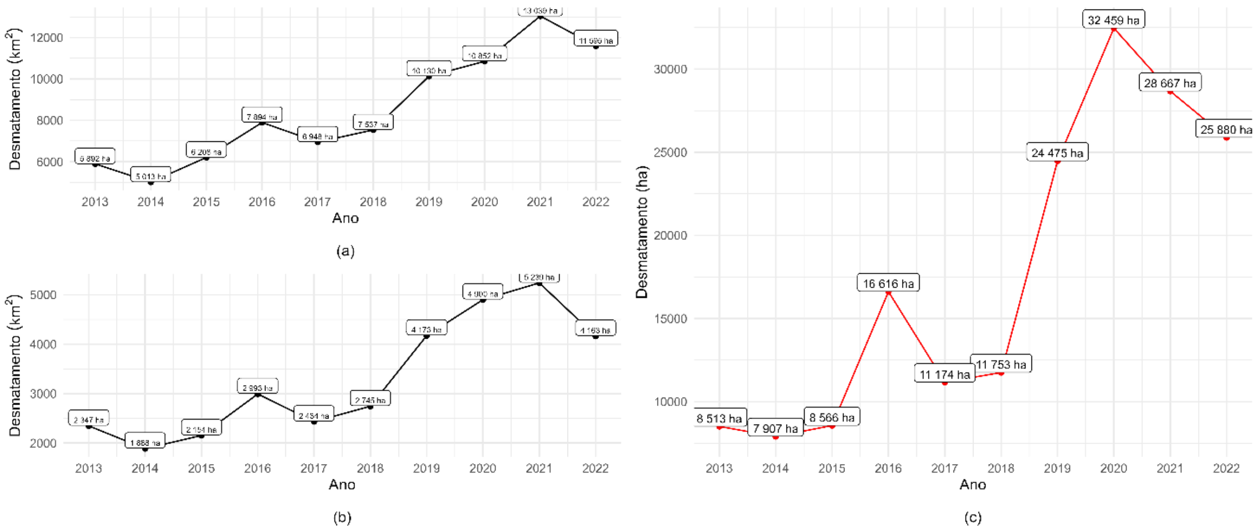

Figure 11: Annual deforestation in the historical period in the Legal Amazon (a), in the State of Pará (b) and in the Reference Region (c). Source: Terra Brazilis (http://terrabrasilis.dpi.inpe.br/app/map/deforestation. accessed on 08/07/2023).

In the analyzed period, it is observed that the deforested area shows a considerable jump in the annual deforestation rate from 2019. From then on, annual deforestation equal to or greater than 25,000 hectares predominated in the reference region (Figure 11c). When comparing the three graphs Figure 11, we found that the temporal behavior of the deforestation rate in the reference region is similar to that observed in the state of Pará (Figure 11b) and the Legal Amazon (Figure 11a).

The scenario reflects the current context of the government's lack of command and control in avoiding unplanned deforestation throughout the Brazilian Amazon, started from the political and economic instabilities of 2016[52] and intensified during the Bolsonaro administration[53]. Recent deforestation (since 2012) in the reference region is concentrated mainly in the north and northwest (Figure 11). The numerous occurrences of recent deforestation adjacent to the project area show the growing pressure on the forests in its interior. The region where the project area is located is surrounded by protection areas, APA Triunfo do Xingu encompasses the project area and ESEC Terra do Meio is adjacent, but this scenario does not shield this territory from anthropogenic pressures. The APA has already lost 35% of its forest cover (more than 500,000 ha) and currently occupies the 1st place of the most deforested conservation unit in the first half of 2023, with 60 km^2 of deforestation, while Esec wins the eighth place, with 2km^2 (Imazon, 2023).

APA Triunfo do Xingú was created in a region where there was already a large population concentration, great anthropic pressure, a degraded and altered territory, in addition to containing private areas within its limits. These aspects hinder the management of its area and the search for sustainable development (Costa, 2013)[54]. The Trans-iriri, a road that cuts through what is now the APA Triunfo do Xingu, was opened in the 1990s to supply mining and logging interests, which intensified the occupation of this region (ISA, 2012)[55]. The idea of creating the APA arose within this context with the intention of containing and ordering

[52] LEÃO PEREIRA, Eder Johnson de Área; SILVEIRA FERREIRA, Paulo Jorge; DE SANTANA RIBEIRO, Luiz Carlos; SABADINI CARVALHO, Terciane; DE BARROS PEREIRA, Hernane Borges. Policy in Brazil (2016-2019) threatens the conservation of the Amazon rainforest. Environmental Science & Policy, vol. 100, p. 8–12, October 2019. DOI 10.1016/j.envsci.2019.06.001. Available at: https://linkinghub.elsevier.com/retrieve/pii/S1462901119303818
[53] Fearnside, Philip Martin. 2019. "Setbacks under President Bolsonaro: A Challenge to Sustainability in the Amazon".
[54] Costa, A.L.S. Management Effectiveness of the Triunfo do Xingu Environmental Protection Area: challenges of consolidating a Conservation Unit in the Terra do Meio region, State of Pará. Belém, 2013.
[55] ISA- Instituto Socioambiental. Keeping an eye on the Xingu Basin. São Paulo, 2012.

this process of occupation growth and economic expansion in the region and, at the same time, minimizing environmental impacts (Costa, 2013).

The APA is located in a region with important remnants of Amazonian biodiversity. However, it also suffers great anthropic pressure due to the process of disorderly occupation of the territory that resulted in a high rate of degradation, a large number of roads, in addition to the indirect impacts resulting from the productive activities of São Félix do Xingu (Costa, 2013).

3.1.4.2. Step 3 of VM0015 – Analysis of agents, drivers and underlying causes of deforestation and their likely future development

3.1.4.2.1. Identification of agents of deforestation

a) **Deforestation agents in the reference region:** the main group of deforestation agents are family farmers, medium and large ranchers, land grabbers, sawmills, loggers and local mining companies.

b) **Relative importance of the amount of historical deforestation attributed to each agent or group**: family farmers, medium and large ranchers, land grabbers, sawmills, loggers and local mining companies are responsible for 100% of the unplanned deforestation observed in the reference region.

c) **Brief description**: Family farmers are agents of deforestation to the extent that they carry out their productive cultivation activities with the traditional cutting and burning technique, gradually deforesting areas over the years. The results of the study by Olímpio, et al. (2022) [56]show that rural establishments located in the micro-regions of São Félix do Xingu present management classified as traditional, in which accelerated land use change, conversion of natural ecosystems into cultivated areas and the practice of deforestation stand out.

   Other important vectors in the reference region are medium and large ranchers, who contribute to the deforestation of the area with extensive cattle production, deforesting areas as a way of expanding the size of the pasture, without applying sustainable animal management techniques and area turnover, which compromises productivity and degrades the soil.

   Deforestation in indigenous lands and protected areas shows the impetus of squatters, land grabbers and other agents interested in the illegal exploitation of natural resources in the municipality of São Felix do Xingu. According to the Rede Xingu+, which monitors deforestation in the Xingu Basin, about 30,000 hectares of forest were cleared in Apyterewa indigenous territory between January 2019 and September 2022. It is estimated that deforestation in the reserve hit a new record in 2022, with 8,189 hectares razed in the first nine months of the year alone. According to environmentalists, deforestation is driven by real estate speculation and carried out by loggers, ranchers and prospectors occupying the territory in the expectation that their claims to illegal land will be recognized.

   In addition, secondary roads, such as Trans-iriri, are opened by miners and illegal loggers, and can be considered important vectors of deforestation, as they enable the occupation of native forest areas at an accelerated rate.

d) **Brief assessment of the most likely population development of groups of deforestation agents in the Reference Region, Project Area and Leakage Belt**: The context evidenced in the reference region, which should follow the same trend in the Project area and in the leakage belt (in the baseline scenario), demonstrating that there are growth trends of the agents identified as family farmers,

[56] Olimpio, Silvia Cristina Maia; Gomes, Sergio Castro; Santana, Antônio Cordeiro de. Patterns of production and sustainability of cattle ranching in the state of Pará-Brazilian Amazon. Semina ciênc. agrar, p. 541-560, 2022. Available at: Patterns of production and sustainability of cattle ranching in the state of Pará - Brazilian Amazon | Semina ciênc. agrar;43(2): 541-560, Mar.-Apr. 2022. tab, maps | VETINDEX (bvsalud.org).

medium and large ranchers, land grabbers, sawmills, loggers and mining companies. The analysis of the most likely future trend of deforestation in the reference region and in the project area is still a continuation of the current pattern, where there has been an increase in deforestation in recent years, at different rates according to the related agents. Despite the trend of increasing deforestation for the coming years, Conservation Units and indigenous territories can act as physical barriers to this process. With the maintenance of the same cutting and burning pattern used by rural landowners, and the maintenance of population growth, there is a natural wear and tear of the soils and the need to advance in new areas of forest reserves, with the potential to further increase the rate of deforestation in the region.

e) **Historical deforestation statistics attributed to each agent in the reference region:** The deforestation process, at different intensities, occurs all around Fazenda Lagoa do Triunfo. In the south and west, it occurs more intensely while in the north and east, more moderately, at slower rates, over the analyzed period. Public areas, especially Conservation Units (UCs), tend to act over time as a barrier to the advance of deforestation, however, the Triunfo do Xingu Environmental Protection Area (APA), where the Ateles REDD+ Project is located, was the protected territory of the Amazon most pressured by deforestation in 2022 (ISA, 2022[57]). Also in that year, the APA Triunfo do Xingu was the Amazon conservation unit that burned the most. In August, it totaled 49.60% of the total outbreaks among all federal and state units in the biome, and the fire spread in September, keeping the APA at the top of the ranking of the number of fires in conservation units in the country (ISA, 2022).

The fire has a strong correlation with extensive cattle breeding in the municipality. São Félix do Xingu is the municipality with the largest cattle herd in Brazil, with almost 2.3 million head of cattle, according to the Brazilian Institute of Geography and Statistics (IBGE, 2022[58]). This pressure from livestock farming alarmingly affects the area of environmental protection. Fire, in this case, is used both to clean and renew pastures, and to expand areas and open new pastures for cattle.

The APA Triunfo do Xingu (APATX), created in 2006, is a conservation unit of the sustainable use group, which has been under great anthropic pressure in recent years, mainly due to logging, livestock and mining activities, and with many conflicts over land tenure (Costa and Reis, 2017[59]). According to a study by Costa (2013)[60], twelve existing community centers were identified in the APA region, consisting of more than 2,000 families. This population is unassisted by the government, without basic services such as education, health, sanitation and electricity.

Deforestation in the region presents itself with a specific dynamic. The stages of conversion of the forest associated with the expansion of livestock farming begin with the conversion of the forest cover to pasture implantation that, over time, suffers a process of degradation due to overcrowding of cattle, and the producer is forced to acquire and/or lease new land. The result is the expansion of deforestation to more remote areas, due to the great availability of land at low prices. (Doblas, 2015[61]).

The border with the APA ends up being a pressure factor on the Terra do Meio Ecological Station. With the reduction of this buffer zone created by the APA, illegal loggers invade Terra do Meio and

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[57] SOCIO-ENVIRONMENTAL INSTITUTE (ISA), 2022. Available at: https://uc.socioambiental.org/en-us/arp/4644. Access on: 8/24/2023

[58] IBGE, 2021. Available at: https://cidades.ibge.gov.br/brasil/pa/sao-felix-do-xingu/panorama. Accessed on Thursday, December 15, 2022.

[59] COSTA, A. L. S. A. and REIS, L. R. Contribution of APA Triunfo do Xingu to land planning in the region of Terra do Meio, state of Pará. Rev. Ciências Agrárias, v. 60, n. 1, p. 96-102, 2017.

[60] COSTA, ALS Management Effectiveness of the Triunfo do Xingu Environmental Protection Area: challenges of consolidating a Conservation Unit in the Terra do Meio region, State of Pará. Belém, 2013

[61] DOBLAS, J. R. Routes of looting: violations and threats to the territorial integrity of Terra do Meio (PA). São Paulo: Instituto Socioambiental, 2015. 31 p

other preserved areas in search of valuable timber. Meanwhile, speculators clear stretches of forest within the reserve in the hope that protections will be loosened and their ownership of these lands will be recognized later. In August 2022, ESEC was the second place in the number of hot spots, among full protection CUs, in all Brazilian biomes, with 904 hot spots (ISA, 2022[62]). According to the ISA, which monitors the Xingu Basin, fire in these protected areas is associated with the process of land grabbing and cattle ranching. The pressure involves some branches that invade the ecological station, towards the Iriri River and the Iriri State Forest. There are interests in establishing a land connection to BR-163, which further threatens the preserved areas (ISA, 2012[63]).

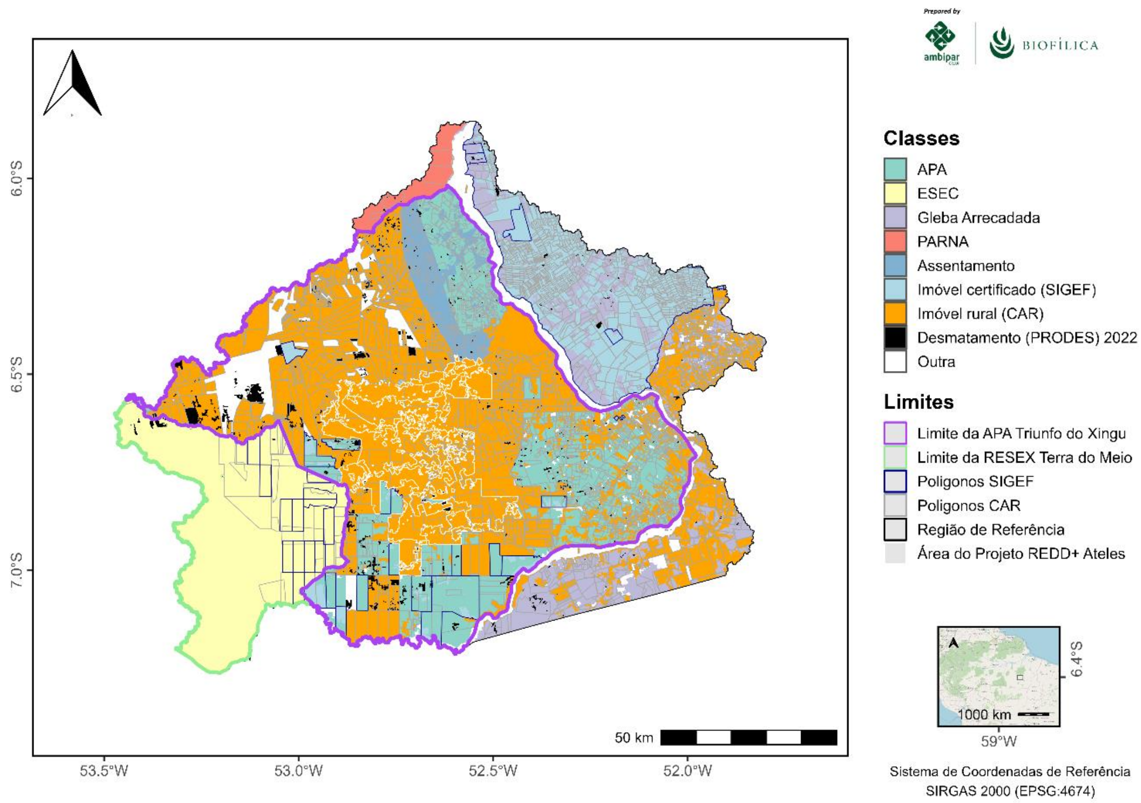

Figure 12: Distribution of the REDD+ Ateles Project Reference Region by Land Category.

Table 21:Deforestation in the Reference Region of Fazenda Lagoa do Triunfo, by Land Title Category

Category	Total Area (2022)	%	Deforested area (ha, 2012-2022)	%
APA	249275	14,8%	45881	26,1%
ESEC	247111	14,6%	4562	2,6%
Gleba Arrecadada	137315	8,1%	14652	8,3%
PARNA	21956	1,3%	264	0,1%
Certified Property (SIGEF)	164373	9,7%	8457	4,8%
Rural Settlement	46456	2,8%	12569	7,1%

[62] SOCIO-ENVIRONMENTAL INSTITUTE (ISA), 2022. Available at https://uc.socioambiental.org/pt-br/arp/4263. Access on: 8/24/2023
[63] SOCIO-ENVIRONMENTAL INSTITUTE (ISA). Keeping an eye on the Xingu Basin. São Paulo, 2012.

Rural property (CAR)	694991	41,2%	72609	41,3%
Other	125472	7,4%	17016	9,7%
Total	1.686.950	100,0%	176010	100,0%

Sources: National Registry of Public Forests (CNFP), Rural Environmental Registry (CAR), Land Management System (SIGEF)

3.1.4.2.2. Identification of deforestation drivers

a) Variables explaining the amount (hectares) of deforestation.

i) Population growth in the project area and reference region;

ii) Increased demand and price for land by small producers to raise production of cattle and other temporary and permanent crops;

iii) Increased demand and price for land by large producers to raise production of cattle and other temporary and permanent crops.

b) Cultural aspects and population growth:

i) Brief Description: It is expected that there will be maintenance of the growing population dynamics started in the 1980s when agricultural companies with national capital from the Center-South received tax incentives from the federal government through cheap credit, urban and road structure, for extensive occupation of southeastern Pará. With the advance of agricultural frontiers and the transformations caused by them, people in search of food security crowded in search of land on which they could produce, as occurred in São Félix do Xingu. It is noteworthy that the presence of these populations residing in agrarian regions results in the replacement of the existing forest with areas focused on agricultural systems.

ii) Impact on the Behavior of the agents**:** the population increase of the community and the population residing in São Félix do Xingu tends to increase deforestation to meet the demands of the advancement of extensive livestock and agriculture, based on cutting and burning practices, resulting in soil degradation and putting even more pressure on forest reserve areas. In addition, illegal logging can be intensified to meet the demands of sawmills and charcoal mills.

iii) Development forecast**:** if more sustainable alternative production techniques are not adopted by cattle ranchers and farmers in the region, the trend is to maintain the dynamics of deforestation. The traditionally used mode of production erodes the soil and forces small producers to expand their areas, associated with other determining factors such as population growth, scarcity of resources, geographical isolation, rural exodus, land speculation, migrant settlement, advancement of soybeans and livestock, and constant demand for commercially valuable timber, accelerate and stimulate deforestation in forest reserves conserved in the region where Fazenda Lagoa do Triunfo is located. Nevertheless, the growth of the world population happens concurrently with the increase in demand for beef, with a strong correlation in the expansion of cattle ranching, traditionally extensive in the region of São Félix do Xingu.

iv) Measures to be implemented**:** the actions planned to be implemented during the project management plan will have as their main objective the promotion of socioeconomic development in the field, offering alternatives for families to diversify and increase their production in a more sustainable way. By providing technical assistance and organizational support, the project aims to reduce the need for families to open new forest areas, develop responsible livestock and agricultural practices and reduce the predatory exploitation of natural resources, in addition to providing improvements in infrastructure, such as road maintenance, along with initiatives to motivate the establishment of man in the countryside, preventing land near forest reserves from being occupied by agents that cause environmental degradation.

- **Aspects related to the increase in demand and price for land by small producers to raise the production of cattle and other temporary and permanent crops;**

i) Brief Description: It results, to a large extent, from the deforestation carried out by family farmers who have a shortage of resources, in which burning is a necessary evil, and because it is more practical to tear down and burn, for the installation of their gardens and the opening of pastures for the creation of small herds. It should be noted that the family farmer does not clear a significant area of land, if the farmer places more than 2 or 3 ha, he cannot produce, as he does not have the manpower to carry out the required activities.

ii) Impact on the Behavior of agents**:** Brazil's current environmental legislation and the relaxation of deforestation inspection, associated with soil quality in the region, the increase in demand for beef and local agricultural and livestock production practices, can boost these rural producers to expand their pasture areas and grow in terms of production in corn, cocoa, banana, beans and cassava, suppressing forest reserve areas.

iii) Development forecast**:** the increase in world demand for beef and consequent increase in price, can drive the deforestation of new areas in the region. Since São Félix do Xingu is an important producer of this commodity. The arrival of new people in the municipality, attracted by new employment opportunities and income generation, can boost the agent in the opening of new areas for agriculture and livestock, succumbing to areas of forest reserves, where the search for noble species, of high value in the market, puts pressure on protected, private and community areas with remaining forests still existing. Added to this context, the local conditions of physical infrastructure and the low capacity of public agents to carry out inspection and control, favor the illegal action of these agents of deforestation.

iv) Measures to be implemented**:** the actions planned to be implemented during the project management plan will have as their main objective the promotion of socioeconomic development in the field so that rural producers can technically and quantitatively improve their production, without the need to expand the agricultural and livestock frontier. By offering technical assistance and rural extension, the project aims to reduce the advance of herds and agriculture of cocoa, banana, beans and cassava maize over the forest and develop responsible and efficient production practices, in addition to promoting articulation with the relevant public agencies to maintain an effective and responsive environmental policy.

- **Aspects related to the increased demand and price for land by large producers to raise the production of cattle and other temporary and permanent crops.**

i) Brief Description: The environmental policy of the Brazilian federal government in the last four years has leveraged deforestation in the Amazon region, mainly for the expansion of soybean and cattle cultivation in areas of protected areas and traditional communities. There was a relaxation in the enforcement of environmental laws that restricted uses and guaranteed greater safety and protection, both to the environment and to the traditional populations of the region. It is estimated that the sense of impunity for environmental crimes has grown in recent years. The search for an increase in areas for the production of these crops will continue to happen as demand for these increases in the international and national market, encouraging producers to put pressure on small producers to sell areas with or without forest areas.

ii) Impact on the Behavior of agents**:** the profitability associated with the exploitation of noble wood in lumber and sawmills and the growing demand for new areas for expansion of pastures and cattle raising; associated with the conditions of setback in environmental policies with the access of the federal government to the maintenance of illegality and inefficiency on the part of public agencies in combating illegal deforestation; provide the advance of environmental degradation in the region.

iii) Development forecast: Deforestation is positively correlated with the number of cattle in São Félix do Xingu. The increase in deforestation is also promoted by the reduction of control over deforestation agents in recent years, especially the previous administration of the federal government; the search for gains with the increase in commodity prices, deforestation as a way of expanding agricultural production, in which deforestation is carried out to produce pastures and then soybeans, the increase in the price of land, infrastructure projects, population growth and family dynamics. In this context, the trend is to maintain a scenario of degradation of forest remnants.

iv) Measures to be implemented: the actions planned to be implemented during the project management plan will have as their main objective the promotion of socioeconomic development in the field, offering alternatives for the mentioned agents to diversify and develop sustainable activities. Through the provision of technical assistance and organizational support, the project aims to encourage sustainable agricultural and livestock production practices, reducing the predatory exploitation of natural resources, in addition to encouraging and strengthening associations of rural producers. Concomitantly, the project aims to monitor changes in forest cover and articulate effective and restrictive environmental policies with the competent public bodies.

3.1.4.2.3. Control variables that explain the location of deforestation

Nine spatial variables were analyzed to identify the drivers that may represent the greatest influence on the location of deforestation in the reference region.

The relative importance of deforestation vectors was estimated using the *Evidence Weights* method, implemented in the *Dynamics EGO* software. In the reference region of the Ateles REDD+ Project, the results of the analysis of evidence weights (WoE) indicated values ranging from -1.66 to +1.66, where negative and positive values represent, respectively, less or greater influence of the vector on the occurrence of deforestation at a given pixel coordinate.

The spatial variables that represent the deforestation pattern in the model calibration period were:

i) distance from old deforestation;
ii) distance from settlements;
iii) distance to private property smaller than 1000 hectares;
iv) distance from rivers;
v) distance to roads;
vi) elevation,
vii) glebes
viii) slope and
ix) conservation units.

The results of the WoE analysis are presented in Figure 11, where the highest and lowest values indicate areas with higher or lower probability of deforestation, respectively.

The description of the variables analyzed to explain the occurrence of deforestation in the historical reference period is presented below:

a) **Distance from old deforestation**: areas of forest edges that represent the initial access of deforestation agents and vectors observed in the reference region during the 2013-2022 period.

b) **Distance to settlements**: in the reference region there are several locations with concentrated human occupation (settlements, villages, towns, communities, cities, etc.), and the proximity to the forest contributes to the greatest risk of new deforestation.

c) **Distance to private property smaller than 1000 hectares**: Areas smaller than 1000 registered in the CAR are often more likely to be deforested, especially those with several overlapping records.

d) **Distance from rivers**: rivers are commonly used for the flow of illegally harvested wood, extracted from areas that will later be deforested. Therefore, areas closer to access by rivers are also more susceptible to deforestation. The Project is inserted in the Middle Xingú Hydrographic Basin, the Xingu River cuts through the Project area in the southern region, however, historically, the main vector of deforestation is the Iriri River, widely used by illegal loggers and prospectors to connect São Félix do Xingu to Novo Progresso.

e) **Distance to roads**: the proximity of roads is one of the main vectors of deforestation in the Amazon. Transiriri, a road that cuts through what is now the Triunfo do Xingu APA, was opened in the 1990s to supply mining and logging interests, which intensified the occupation of this region. With 200 kilometers, the Trans-Iriri is an illegal road, which goes west, through Terra do Meio and from São Felix do Xingu.

f) **Elevation and slope**: Land in low altitude and/or high slope conditions sometimes acts as a barrier to deforestation.

g) **Glebes**: Land is public land with or without destination. Due to their weak land status, they are typically more susceptible to illegal deforestation.

h) **Conservation units**: Protected areas typically have a deforestation-restricting effect. The Ateles REDD+ Project is part of the Triunfo do Xingu Environmental Protection Area and borders the Terra do Meio Ecological Station Integral Protection Conservation Unit. Both conservation units are part of the great conservation mosaic of the middle Earth composed of PAs and indigenous territories.

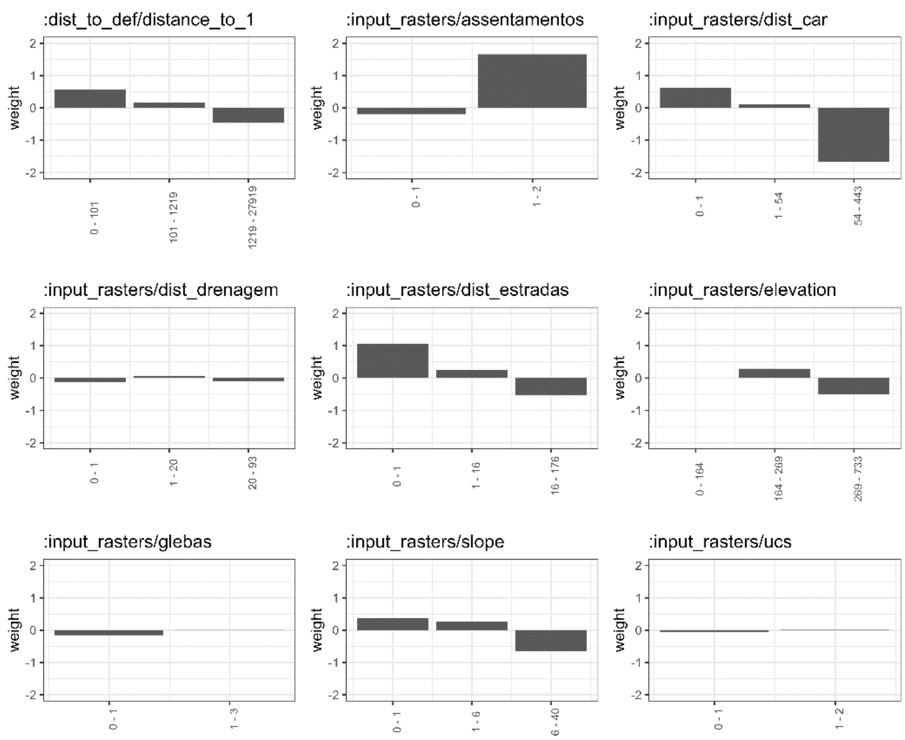

Figure 13: Spatial drivers analyzed and their respective values of influence on deforestation (Evidence weights).

3.1.4.2.4. Identification of underlying causes of deforestation

a) **Short Description:** Among the factors underlying deforestation are: i) population growth that increases the demand for food and makes producers expand the cultivation and pasture areas by advancing over the existing forest areas in the municipalities; ii) Low level of education of the population of the municipalities, which limits the use of new production techniques, in addition to those culturally passed on over time; iii) Low family income of residents in the municipality and in the project area, in which income is concentrated between 1 or 2 minimum wages, with a high number of people receiving Brazil Aid, a social benefit from the federal government; iv) The increase in beef consumption in the national and international market, is another factor that explains, in part, deforestation and has increased due to the growing volume of live cattle exported, frozen meat and soybeans consumed by importing countries, with effects on the increase in prices in the domestic market.

b) **Impact on the decision of the group of agents to deforest:** There is a combination of factors that lead to periodic increases in deforestation, such as: population growth that increases the demand for food and makes producers expand the areas of cultivation and pasture, advancing over the existing forest areas in the vicinity of the community. Assuming that the environmental impacts of food production are created by producers of different sizes and who grow various crops and produce animals, especially cattle. The main environmental risks of livestock farming to achieve environmental sustainability arise from changes in land use, resulting from deforestation and the conversion of natural ecosystems into cultivated areas, and from the degradation of cultivated areas, caused by inadequate management practices. The low level of education among farmers, in which 85% of the community members interviewed have completed or incomplete elementary school, limits the use of new production techniques beyond those culturally passed on over time. The level of education of land managers is a factor that influences the implementation of information technologies for digitization and monitoring of production and the health condition of animals. It is necessary that conservation practices need to be expanded by producers so that they can leave a traditional management and be inserted in a management based on technological and information standards and biotechnology transfers. Low family income is another determinant factor of deforestation by small producers, since 67% of families receive between one and two minimum wages per month, the low level of income means that producers have to increase cattle production to create value from the sale of the surplus produced. The increase in the consumption of beef in the national and international markets is another factor that partly explains deforestation and has increased due to the growing volume of live cattle exported, frozen meats and soybeans consumed by importing countries, reflecting the increase in prices in the domestic market. In summary, there is a lack of public policies that can induce the development of communities in order to raise the level of income, schooling, agricultural production such as technical assistance and rural extension, and commercialization based on the inclusion of small producers in value chains of agricultural products.

c) **Likely future development:** The results of the research point to the increase in the population residing in the area of interest of the project, largely explained by the expansion of cattle production areas. Small producers will continue to produce cattle in a traditional and extensive manner as has been done for many years and deforest areas as a way to expand the size of the pasture, without applying sustainable animal management techniques and area turnover, which compromises productivity and degrades deforested areas. Tenants and land grabbers will continue to appropriate land from the APA and use the land irregularly. The practice that has been perpetuated since the 1970s continues in the face of the absence of inspection by the government and the accelerated growth of deforestation, induced by the opening of secondary roads, without effective mitigation action by the institutions responsible for maintaining the forest area.

d) **Measures that will be implemented:** Given the conditions of income, education, lack of adequate infrastructure in the community, the fact that producers are not part of food value chains, and considering the traditional production system of small producers, some actions should be implemented as a way to increase family income, maintenance in rural areas from practices related to low-carbon agriculture and livestock, including: i) transfer of scientific knowledge and technologies to raise the quality of the herd via genetic improvement, semen and sale of embryos to farmers in the region, and thus improve the income of establishments. ii) structure the production chain of the main crops in the municipalities, especially cocoa, and include small producers in individual or cooperative practices that develop low-carbon rural production, since there is evidence of community representation (Leader), which will contribute to the process of forming relationships of trust, reciprocity and cooperation. The effect of this type of actions is to retain families in the countryside, induce them to adopt sustainable practices and thus reduce deforestation and recover degraded areas. iii) use the agroforestry system as a way to structure land use. iv) develop projects to reduce environmental liabilities and enable land and environmental security of properties. v) stimulate social organization in communities. vi) enable technical assistance and rural extension to small producers.

3.1.4.2.5. Analysis of the chain of events leading to deforestation

The chain of events can be thus established according to the historical information of the items below:

i) The initial step was the development process of the Amazon promoted by the federal government in its different development plans for the region since the 1960s, which had as their central axis the logistical infrastructure and integration of the region with the other regions of Brazil and that would enable the exploitation of mineral and forest resources creating foreign exchange for the country.

ii) Prior to the 1970s, the predominant activities in São Félix do Xingu were extractive and small-scale agriculture, with small fields (corn, cassava, beans) of up to 1 ha, with no livestock activity. Unlike other micro-regions that were not located on the banks of rivers, these areas already had a large number of inhabitants, including riverside dwellers, Indians, rubber tappers, fishermen. Although there were potentially good soils in the region, traditional populations were more dedicated to extractive activities.

iii) From 1970, loggers, mining companies, small farmers, road construction, cattle ranching front start their activities in the forest areas in the municipality of São Félix do Xingu, in a new economic cycle marked by mineral exploration and logging. Attracted by the large amount of mineral deposits (cassiterite, iron, gold, etc.) and wood with high market value (mahogany), mining and logging companies begin to structure themselves, creating a new scenario. This phase paved the way for the development of livestock in the region.

iv) The São Félix do Xingu region had 10% of fertile purple land, more than 700,000 ha of land suitable for annual and perennial crops, and about 200,000 ha suitable for pastures. The race of the big land speculators and intense mining companies. In 1973, 2,000 land requisitions had been registered with state land agencies. In 1975, hundreds of licenses for mineral exploration were filed by national and multinational companies. In the late 1970s, the state agency responsible for much of the area of the São Félix do Xingu microregion began to auction land of up to 3,000 ha, allowing the purchase by companies and farmers only.

v) The predominant pattern in the region is that of large farms. The process of land appropriation in the municipality began in 1985, forming a model of occupation with a predominance of large properties, exclusively focused on livestock. Although farmers who own more than 90% of properties do not have titled land, the properties are recognized and consolidated by livestock.

vi) In the 1990s, the livestock front was configured and mineral exploration intensified, with an important influx of small rural producers. The groups of migrants who arrived came from different places. In 13 years, the municipality of São Félix do Xingu reached the first place in herds in Pará and the fourth place in Brazil. The prospect of growth in activity is measured by the amount of land available for livestock. The expansion is towards Mato Grosso, and there is a prospect of the herd tripling in the municipality over the next few years.

3.1.4.2.6. Conclusion

The conclusion of this study is supported by primary data on the factors linked to deforestation in communities and summarized by the Project (Peabiru, 2023). [64]The reflections considering determinant vectors of deforestation indicate the maintenance of the growing dynamics of deforestation, either due to the fragility of the command and control policies that have been disarticulated in the last four years; the high impetus of the squatters and land grabbers to expand the areas of deforested forest since they found no resistance; the economic and social shortage of small producers continues to be used by large producers to expand pasture areas either by leasing or renting, or by force and violence.

Cattle production has a strong positive correlation with deforestation, indicating that increasing deforestation values are associated with increasing cattle herd values. This dynamic should be reproduced in the coming years and its intensity should increase if no deforestation mitigation policy is implemented. This logic should be maintained because it considers:

- That the traditional production system, with extensive dynamics, will be reproduced in the coming years, as it has a low level of technology;
- The greatest accumulation of income will continue to be from large producers, as they have gains in scale in the production and purchase of inputs, and in the pressure to lease the areas of neighbors with small areas;
- The low level of education makes small producers vulnerable, either to define the allocation of pasture areas, to the detriment of permanent crops such as cocoa, or because they have little information for decision making in the process of buying and selling inputs and products;
- The low effectiveness of governments to solve the problems of land and environmental regularization and rural assistance, potentiate the effects on deforestation, and the low efficiency or absence of education, health and sanitation actions reinforces the context.

3.1.4.3. Step 4 of VM0015 – Projection of Future Deforestation

3.1.4.3.1. Projection of the amount of future deforestation

The Reference Region was not stratified, as the characteristics of the agents, drivers and causes of deforestation are the same throughout its extension.

3.1.4.3.2. Baseline Approach Selection

The VM0015 methodology suggests three methods to project the amount of future deforestation: (a) the historical average of deforestation; (b) deforestation as a function of time and (c) modeling of the deforestation rate.

It was conclusive that annual deforestation rates in the region clearly increased from 2019, as demonstrated in the topic “Results of the analysis of historical land use and land cover changes” and “Step 3 of VM0015 – Analysis of agents, drivers and underlying causes of deforestation and its likely future development”. After the 2019 peak, there was a slight decrease in the rate in the last three years of the series, however, still remaining above the average for the period from 2013 to 2019. In addition, we adjusted a simple regression model to the annual deforestation time series, which although highly significant ($p<0.01$) had a moderate R^2 (73%), indicating a certain trend of increase in annual deforestation rates. In view of these results and to ensure a balance between conservatism of the projections, but in order not to overestimate the rates, we considered the method a) "function of time" of sub-step 4.1.1 of the VM0015 methodology to draw the baseline of deforestation.

[64] PEABIRU. 2023. Ateles REDD+ Project: Social and environmental diagnosis. Final report, 266 p.

3.1.4.3.3. Quantitative projection of future deforestation

As mentioned in the previous item, we used the "b" method (function of time) to estimate future deforestation rates that were later spatially distributed in the Reference Region. For this, we use an exponential smoothing model to project the remaining forest area each year. Simple exponential smoothing models are flexible because they can be used to adjust time series with or without trend and with or without seasonality[65]. These models fit the data giving more importance to the most recent points in the series and less importance to the oldest points. The use of this class of models to model future deforestation is a middle ground because it considers the recent increase in deforestation rates observed, without necessarily assuming an increasing trend.

We adjust the model using the "forecast" package[66] in R. The *holt* function of the package allows you to adjust models automatically, selecting optimal parameters for trend and seasonality, if present in the series. The adjustment resulted in the following parameters: $a = 0.9999$, $\beta = 0$ and $\lambda = 0.0001$ equivalent to a steady decrease in forest cover over time. Figure 13 shows the values adjusted together with the deforestation time series data, while Table 22 and Figure 14 show the deforestation values observed in the historical period and future forecasts in the reference region:

Table 22: Projection of future deforestation in the Reference Region.

Year	Deforestation (ha)	
	Observed	Predicted
2013	8,513	18579
2014	7,907	18594
2015	8,566	18596
2016	16,616	18593
2017	11,174	18597
2018	11,753	18593
2019	24,475	18593
2020	32,459	18590
2021	28,667	18586
2022	25,880	18587
2023		18579
2024		18594
2025		18596
2026		18593
2027		18597
2028		18593
2029		18593
2030		18590
2031		18586
2032		18587

[65] Hyndman, Rob J., and George Athanasopoulos. 2018. Forecasting: Principles and Practice. OTexts.
[66] Hyndman, Rob J., and Yeasmin Khandakar. 2008. "Automatic Time Series Forecasting: The Forecast Package for R". Journal of Statistical Software 27 (July): 1–22. https://doi.org/10.18637/jss.v027.i03.

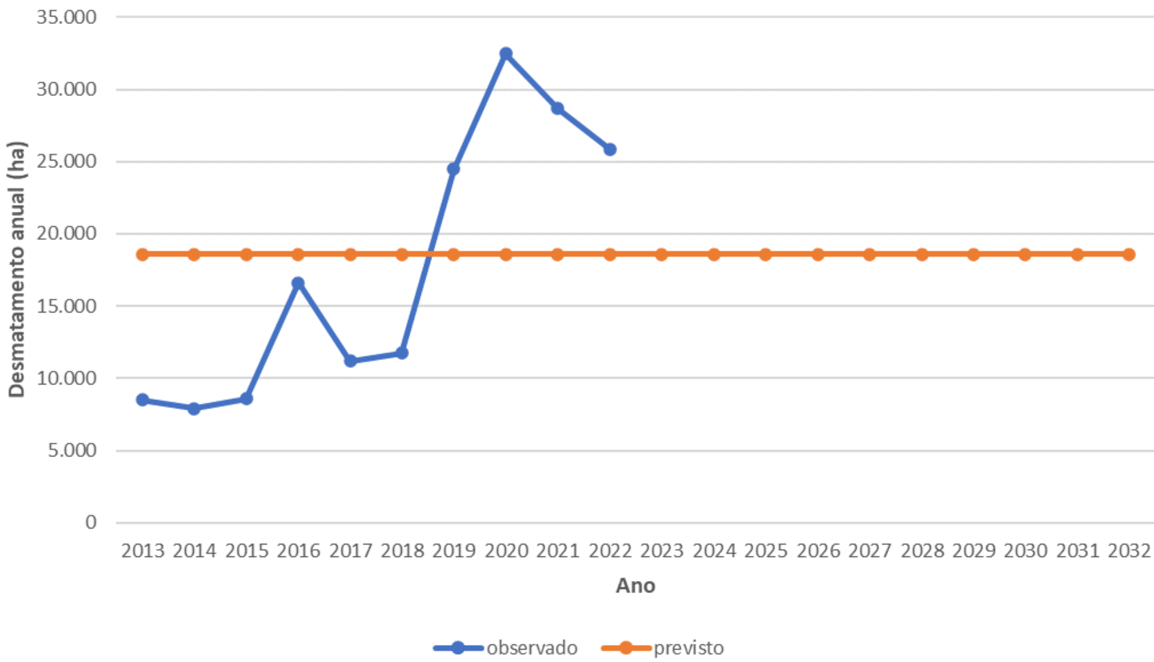

Figure 14: Evolution of deforestation in the historical period and future forecasts in the reference region.

Then, the time function equation used to obtain the future predictions was:

$$ABSLRR_{i,t} = y_t - y_{t-1}$$
$$y_{t+h|t} = l_t + hb_t$$
$$l_t = \alpha y_t + (1-\alpha)(l_{t-1} + b_{t-1})$$
$$b_t = \beta^*(l_t - l_{t-1}) + (1-\beta^*)b_{t-1}$$

Where:

deforestation $\mathrm{ABSLRR}_{i,t}$: area in year t (t = 2023, 2024, ..., 2032) in stratum i of the reference region;

y_t: forest area in year t;

l_t: estimate of the time series level in year t;

h: difference between the forecast year and the year t;

b_t: estimated trend in year t;

α e β^*: exponential smoothing model parameters;

3.1.4.3.4. Projection of annual baseline deforestation areas in the Project Area and Leak Belt

Annual deforestation was distributed spatially throughout the Reference Region (next section). Baseline deforestation in the Project Area and Leak Belt corresponds to the baseline deforestation allocated in these regions. The projected deforestation values for the period 2023 to 2032 in the Reference Region (Table 23), Project Area (Table 24) and Leakage Belt (

Table 25are presented. The total deforestation projected for the Project Area in the accreditation period was 12,631 ha, with an annual average of 1,263 ha.

Table 23: Projected deforestation for the Reference Region (Table 9a of the VM0015 methodology).

Project year *t*	Stratum i in the reference region 1 ABSLRRi,t ha	Total	
		Annual $ABSLRR_t$ ha	Cumulative $ABSLRR_t$ ha
2023	18,422	18,422	18,422
2024	18,425	18,425	36,847
2025	18,422	18,422	55,269
2026	18,421	18,421	73,690
2027	18,412	18,412	92,102
2028	18,417	18,417	110,519
2029	18,418	18,418	128,937
2030	18,422	18,422	147,359
2031	18,419	18,419	165,778
2032	18,407	18,407	184,185

Table 24: Projected deforestation for the Project Area (Table 9b of the VM0015 methodology).

Project year *t*	Stratum i in the reference region 1 ABSLPAi,t ha	Total	
		Annual $ABSLPA_t$ ha	Cumulative $ABSLPA_t$ ha
2023	1,318	1,318	1,318
2024	856	856	2,174
2025	1,127	1,127	3,301
2026	1,199	1,199	4,500
2027	1,322	1,322	5,822
2028	813	813	6,635
2029	1,457	1,457	8,092
2030	1,553	1,553	9,645
2031	1,323	1,323	10,968
2032	1,663	1,663	12,631

Table 25: Deforestation designed for the Leakage Belt (Table 9c of the VM0015 methodology).

Project year *t*	Stratum i in the reference region 1 ABSLLBi,t ha	Total	
		Annual $ABSLLB_t$ ha	Cumulative $ABSLLB_t$ ha
2023	4,200	4,200	4,200
2024	4,011	4,011	8,211
2025	4,205	4,205	12,416
2026	5,178	5,178	17,594
2027	4,443	4,443	22,037
2028	4,514	4,514	26,551
2029	4,283	4,283	30,834
2030	4,364	4,364	35,198

2031	4,571	4,571	39,769
2032	4,509	4,509	44,278

3.1.4.3.5. Projection of the Location of Future Deforestation

Dynamic EGO software was used to project the location of future deforestation. This software is indicated by the VM0015 methodology as appropriate for baseline modeling of REDD+ projects to prevent unplanned deforestation (AUD). The use of Dinamica EGO is justified by the following reasons: a) it is a model available in scientific publications[67]; b) it has a transparent process for input and output of data and parameters processed with an easy-to-understand graphical interface; c) it incorporates the use of appropriate data to explain the location of deforestation; d) it has an adequate instrument to assess uncertainties[68].

The main steps taken in this step were:

1. Organization of georeferenced maps of land use and land cover and georeferenced maps with the explanatory factors of deforestation.
2. Calibration of the model by determining the weights of evidence (WoE presented in Figure 13) and analysis of the correlation between the variables;
3. Evaluation of the accuracy of the model (Figure of Merit - FOM);
4. Development of deforestation base scenarios.

- **Preparation of factor maps**

To carry out this step, the empirical approach was used to create factorial maps (spatial variables that explain the location of deforestation). Studies on deforestation in the Amazon show that maps of distances from spatial attributes (roads, localities, etc.) and ecological aspects of the landscape (relief, soils and vegetation, etc.) have a high correlation with the location of new deforestation[69].

To elaborate the risk map and calibrate the future deforestation projection model, the Dinamica EGO software requires that the input spatial variables be independent before using them. Six independent spatial variables were used to produce the deforestation risk map (Table 23). In the Dynamic EGO, spatial data were processed with a pixel size of 30 x 30 meters (same resolution as PRODES rasters), GeoTiff format (Geodetic coordinate system) and dimensions of 1566 rows by 1860 columns.

[67] SOARES-FILHO, B. et al. Modeling conservation in the Amazon Basin. Nature 440, pp.520-523, 2006
[68] Hagen, Alex. 2003. Fuzzy set approach to assessing similarity of categorical maps. International Journal of Geographical Information Science 17 (3): 235–49. https://doi.org/10.1080/13658810210157822.
[69] BARRETO, P., BRANDÃO Jr., A., MARTINS, H., SILVA, D., SOUZA Jr., C., SALES, M., & FEITOSA, T. 2011. Risk of Deforestation Associated with the Belo Monte Hydroelectric Power Plant (p. 98). Belém: Imazon

Table 26: List of maps, variables and factorial maps (Table 10 VM0015).

Factor Maps		Source	Variable represented		Meaning of categories or pixel values		Maps used to create		Algorithm or equation used
ID	**File name (.tif)**		**Unit**	**Description**	**Range**	**Meaning**	**ID**	**File name**[70]	
1	distance_to_1	INPE[71]	Meters	Continuous data		Old deforestation distance.	1	PRODES 2021_plus_2022pri_raster.tif	Euclidean distance (Quantum GIS 3)
2	settlements	INCRA[72]	Unitless	Categorical date		1, if inside a settlements. 0 otherwise	2	Assentamento Brasil.shp	Rasterization (QuantumGIS 3)
3	dist_car	CAR[73]	Meters	Continuous		Distance from private properties smaller than 1000 ha (if registered in the Brazilian Rural Environmental Registry).	3	car_sicar_08022021.shp	Euclidean distance (QGIS 3)
4	dist_drenagem	ANA[74]	Meters	Continuous		Distance from rivers and other drainage	4	vw_drenagem.gpkg	Euclidean distance (QGIS 3)

[70]Consult raster files (.tiff) in the "...\baseline\1_variables" folder.
[71]http://terrabrasilis.dpi.inpe.br/download/dataset/legal-amz-prodes/raster/PDigital2000_2021_AMZ_raster_v20211118.zip
[72]https://certificacao.incra.gov.br/csv_shp/export_shp.py
[73]https://www.car.gov.br/publico/municipios/downloads
[74] https://metadados.snirh.gov.br/geonetwork/srv/por/catalog.search#/metadata/0f57c8a0-6a0f-4283-8ce3-114ba904b9fe

Factor Maps		Source	Variable represented		Meaning of categories or pixel values		Maps used to create		Algorithm or equation used
ID	**File name (.tif)**		**Unit**	**Description**	**Range**	**Meaning**	**ID**	**File name**[70]	
5	dist_roads	DNIT[75], ImazonGeo[76]	Meters	Continuous		Distance to official and unofficial roads	5. 6	roads_final_16032020.shp; State Highways_vw_cide_rod_2021.gpkg	Euclidean distance (QGIS 3)
6. 7	elevation, slope	NASA[77]	Meters, Degrees	Continuous		Terrain height and slope	7	Terrain.tif	none
8	glebas	SFB[78]	Unitless	Binary		1 if inside a public, non-destinated area (Gleba), 0 otherwise	8	CNFP_2020_PA.shp	Rasterization (QuantumGIS 3)
9	ucs	MMA[79], FUNAI[80]	Unitless	Binary		1 if inside a Conservation Unit or Indigenous Lands (Protected Areas), 0 otherwise	9-10	ucstodas.shp, ti_sirgas_2021.shp	Rasterization (QuantumGIS 3)

[75]https://servicos.dnit.gov.br/vgeo/
[76] https://www.imazongeo.org.br/
[77]h https://www.earthdata.nasa.gov/sensors/srtm
[78] https://snif.florestal.gov.br/pt-br/cadastro-nacional-de-florestas-publicas
[79] https://cnuc.mma.gov.br/map
[80] https://www.gov.br/funai/pt-br/atuacao/terras-indigenas/geoprocessamento-e-mapas

- **Elaboration of deforestation risk maps**

Deforestation risk maps show the regions with the highest (risk close to or equal to 1) or lowest conditions for deforestation to occur (risk close to or equal to 0). In this baseline study, the risk map was produced using the evidence weights method (Bonham-Carter, 1994) available in Dinamica EGO 7. This method calculates the probability of transition from forest to deforested area at each pixel in the reference region, based on the sum of all evidence weights that overlap at a given pixel, and dependent on combinations of all static and dynamic maps[81].

The result of applying the weight of evidence method in Dynamics EGO is a deforestation risk map that identifies areas with a higher (1.0) and lower (0.0) probability of deforestation (Figure 15). The spatial variables presented in Table 26, together with the deforestation risk map, are the starting point for production of future deforestation baseline scenarios.

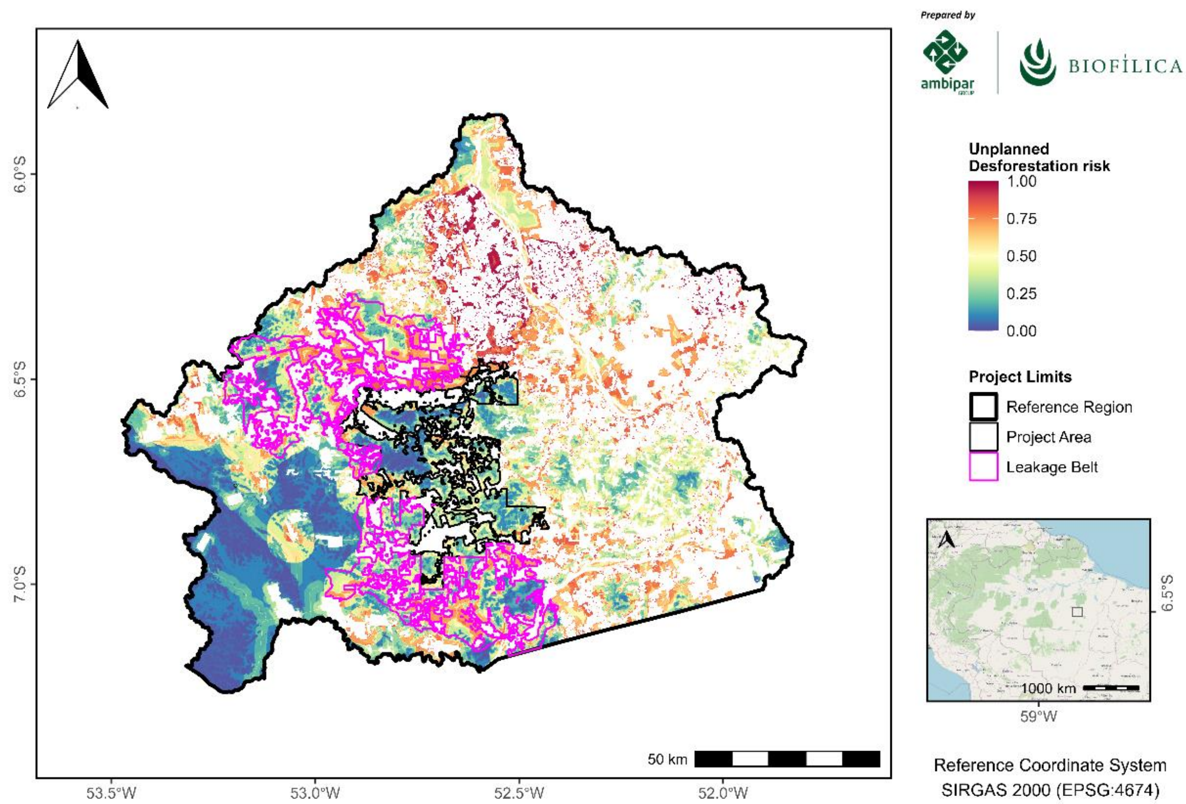

Figure 15: Map of deforestation risk in the Reference Region.

- **Selection of the most accurate deforestation risk map (Calibration and Validation of the model)**

To evaluate the quality of the model produced, the option "a" available in the VM0015 version 1.1 methodology was used: *calibration and confirmation through two historical subperiods*. Data on deforestation between 2013 and 2017 and the variables listed were used to calibrate the model, while the deforestation map mapped by Prodes in 2022 was used for the confirmation process. In this process, a 2022 deforestation map was simulated from the data observed between 2013 and 2017.

[81] SOARES-FILHO, B. et al. Modeling conservation in the Amazon Basin. Nature 440, pp.520-523, 2006.

The FOM (Figure of Merit) technique was used to evaluate the accuracy of the simulated map in 2020. The FOM result is the ratio of the intersection of the observed changes (changes between the reference map at time 1 and time 2) and the simulated changes (changes between the reference map at time 1 and the reference map at time 2) for the union of the observed change and the predicted variation, as defined in equation 9 of the VM0015 methodology.

Methodology VM0015 indicates that the minimum threshold for the best fit measured by the FOM should be defined by the net change observed in the reference region for the model calibration period. The observed net change shall be calculated as the modeled total area of change in the reference region during the calibration period (percentage of the total area of the reference region), and the FOM value shall be at least equivalent to this value. If the FOM value is below this threshold, the project proponent must demonstrate that at least three models have been tested (three deforestation risk maps), and the one with the best FOM should be used.

The net change observed in the Reference Region was 1.7%, and the FOM value obtained by applying Equation 9 of VM0015 was 6%. Thus, the FOM of the risk map produced is greater than the required limit (Step 4.2.4 of VM0015). Thus, the deforestation risk map developed in the calibration stage (Figure 15: Map of deforestation risk in the Reference Region.) offers good performance to spatially project changes in land use by 2032 in the Reference Region of the Ateles REDD+ Project.

- **Spatial projection of future deforestation**

The procedure for selecting the pixels with the highest risk of deforestation and preparing the deforestation base maps was carried out automatically by Dinamica EGO for the 10-year period, starting in 2022. The results are presented with deforestation in the Reference Region planned for the first 10 years of the Project. In the developed scenario, the Farm Lagoa do Triunfo will have a total deforestation of 12,631 hectares in the first 10 years of the Project.

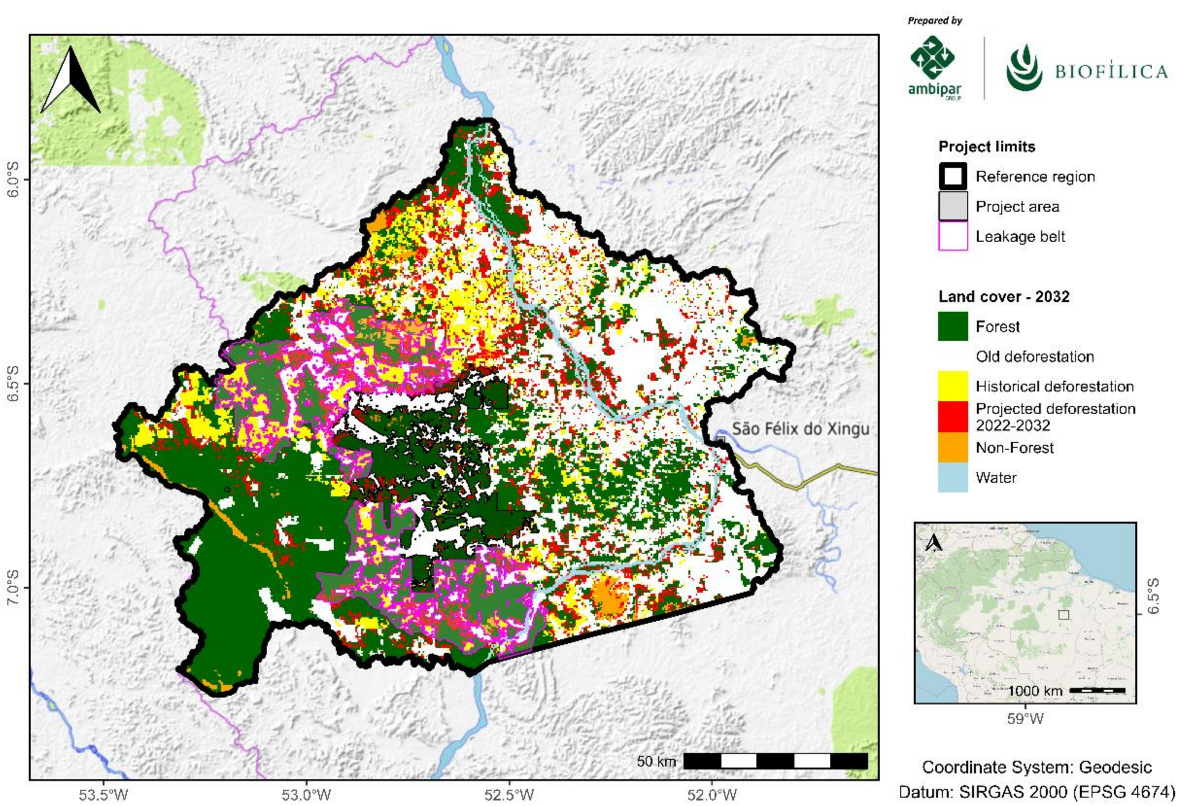

Figure 16: Projected deforestation in 10 years in the Reference Region.

3.1.5. Additionality

The additionality of the Project was analyzed according to the tool approved by VCS “VT0001 – Tool for the Demonstration and Assessment of Additionality in VCS Agriculture, Forestry and Other Land Use (AFOLU) Project Activities”, version 3.0, of February 1, 2012.

The conditions of applicability of the tool are met, because:

- AFOLU activities are the same as or similar to the proposed activities of the Project, within their respective limits, whether or not registered as an AFOLU VCS Project, and do not lead to violation of any applicable law even if this law is not applied; and
- • VM0015 baseline methodology provides a step-by-step approach to justify the determination of the most plausible baseline scenario (see “Part 2 – Methodology Steps for ex-ante estimate of GHG emissions reductions” of VM0015).

The tool establishes four steps for demonstrating additionality, which will be followed below:

STEP 1: Identification of alternative land use scenarios to the AFOLU project activity

STEP 2: Investment analysis to determine that the proposed project activity is not the most economically or financially attractive of the identified land use scenarios

STEP 3: Barriers analysis

STEP 4: Common practice analysis

Step 1: Identification of land use scenarios alternative to those proposed by the VCS AFOLU Project activity

<u>Sub-step 1a. – Identify alternative land use scenarios credible to the proposed VCS AFOLU Project activities</u>

The scenarios described in this item were based on the data collected by the socioeconomic study carried out in November 2022, which relied on the use of secondary (literature review based on public institutions) and primary data, through participant observation techniques; application of a questionnaire with a script of quantitative and qualitative questions; interviews with a script of semi-structured questions; and the construction of a timeline for the purpose of contextualizing these interactions within the APA Triunfo do Xingu with farmers with areas overlapping those of the AgroSB company's Farms in São Félix do Xingu/PA and; individual interviews with the workers of the Lagoa do Triunfo farm.

Among the alternative realistic and credible land use scenarios that would occur within the Project boundaries in the absence of the AFOLU Project activity registered in the VCS, the following were considered:

1) Continuation of land use prior to the Project (baseline scenario):

The municipality of São Félix do Xingu has undergone different occupation processes over the last decades. The expansion of agricultural frontiers in the Amazon occurs around the mid-1960s when the Amazon region is included in the country's planning with the national integration plans (IANNI, 1979[82]), in which the large projects arise, the financial incentives to large private companies that started to develop extensive agriculture for beef production (SCHINEIDER, 1993[83]) and the mining-metallurgical projects of

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[82] Ianni, O. The Struggle for Land: História Social da Terra e da Luta pela Terra na Área da Amazônia, 2nd ed.; Vozes: Petrópolis, RJ, USA, 1979.
[83] SCHNEIDER, Robert et al. Land Abandonment, property rights, and agricultural sustainability in the Amazon. World Bank, Latin America Technical Department, Environment Division, 1993.

the large Carajás project. This integration process was made possible by the expansion of the road network with the construction of the Belém-Brasília and Transamazônica highways.

In the 1980s, through government incentive policies, livestock began to be developed on a large scale with the concentration of land by landowners, in fact there was the emergence of land conflicts between the owners of large tracts of land and small producers, traditional and indigenous communities that had their rights ignored by the state (Peabiru, 2023). However, the penetration of agriculture in the Xingu territory also contributed substantially to the municipality becoming a symbol of deforestation in the region. According to data from INPE (2022[84]), the municipality of São Félix occupies the first place in the ranking of the most deforested in Brazil since 2001.

An additional factor that stimulates this dynamic in the region is the fragility and low governance by state and federal governments, showing no sign of change, resulting in impunity for illegal practices and enhancing the occurrence of deforestation in native forest fragments in the region.

Corroborating the above, although the project area is part of the Triunfo do Xingu Environmental Protection Area created in order to protect all the socio-biodiversity that makes up the ecological corridor of the Xingu river basin from anthropogenic pressure arising mainly from illegal logging and livestock expansion (COSTA et al., 2017), 35% of the primary forest (or old growth) within the APA was lost between 2006 and 2021, which is equivalent to more than 533,000 hectares viewed on the Global Forest Watch platform (Mongabay, 2022[85]), placing it among the three most deforested areas on the list of protected reserves in Brazil. A scenario that is perpetuated until today, where, in 2023, the APA Triunfo do Xingu occupies the 1st place of the most deforested conservation unit, having lost 60 km² of forest between January and June, which is equivalent to a thousand football fields per month (Imazon, 2023[86]).

Therefore, with the continuation of this dynamic, for the next 10 years (2023-2032), a loss of 185,908 hectares is projected in this scenario, of which 14,106 hectares are expected to be deforested in the Project area. In the scenario described, such dynamic tends to remain until large part of the forest cover is altered, generating an inestimable impact on local biodiversity, and further deepening social and economic problems. Thus, this scenario can be classified as the common practice scenario in the region, or business as usual scenario.

2) Scenario with REDD+ Project, conservation activities and sale of carbon credits:

In order to develop several synchronized actions that induce a change in the local context, avoiding rural exodus and ensuring the permanence of families in the countryside, in addition to strengthening value chains, stimulating sustainable practices, reducing invasion attempts in the forest areas of Fazenda Lagoa do Triunfo, as well as biodiversity conservation actions combined with the perspective of sustainable development, actions are implemented in the development of the Ateles REDD+ Project, such as:

I) structuring of patrimonial surveillance at Fazenda Lagoa do Triunfo, through remote and continuous monitoring, combined with terrestrial surveillance, in order to avoid invasions, deforestation and any type of degradation in forest areas;

[84] INPE - National Institute for Space Research. **Project for Monitoring Deforestation in the Legal Amazon by Satellite (PRODES)**. Available at: http://www.dpi.inpe.br/prodesdigital/prodesmunicipal.php.

[85] KIMBROUGH, Liz. Pasture replaces large tract of intact primary forest in Brazilian protected area. Mongabay News and Inspiration from nature's frontline, [S.l.], 13 May 2022. Available at: <https://news.mongabay.com/2022/05/pasture-replaces-large-tract-of-intact-primary-forest-in-brazilian-protected-area/\>. Access on: July 24th, 2023.

[86] https://imazon.org.br/imprensa/desmatamento-da-amazonia-tem-queda-de-60-no-primeiro-semestre/

II) develop and strengthen value chains of the main crops in the region, especially livestock, encouraging social and productive organization, means of valuing products and market access, as a way to encourage the autonomy and socioeconomic development of the parties involved;

III) promote to family farmers and small rural producers agricultural practices that direct sustainable production, allowing to retain families in the field, reduce deforestation and guarantee income and food security;

IV) structure a biodiversity conservation program, promoting environmental education actions and awareness campaigns on topics such as the presence of domestic animals in the project area, and identifying possible impacts of deforestation on local water resources, in addition to continuous monitoring of the biodiversity of Fazenda Lagoa do Triunfo;

V) strengthening local governance, establishing and expanding new partnerships and strengthening social capital in communities, in addition to ensuring collective engagement, and transparency in the execution of the activities proposed by the Project, as well as monitoring the risks and improvements about them

Given this scenario, considering that activities that promote changes in local conditions and guarantee forest conservation compromise and burden AgroSB Agropecuária, an additional revenue resulting from the commercialization of verified credits for specific investments in containment and monitoring initiatives in the Fazenda Lagoa do Triunfo region becomes vital, through the promotion of patrimonial surveillance, strengthening of local governance, strengthening of sustainable agricultural practices, development and strengthening of value chains and biodiversity conservation program, actions mentioned above that will promote sustainable socioeconomic development and reduce emissions from deforestation and forest degradation.

Therefore, the aggregation of additional revenue from the commercialization of verified credits and registrations would provide net positive impacts to the preservation of forest cover and maintenance of the carbon stock, as well as co-benefits to biodiversity and communities in the Project region.

<u>Sub-step 1b. – Consistency of credible land use scenarios with applicable laws and regulations</u>

Of the proposed scenarios, scenario (ii) is in compliance with all applicable legal and regulatory requirements and only the practices contained in scenario (i) are not in accordance with mandatory legislation and regulations.

Illegal or unauthorized deforestation is systematic and widespread in the Legal Amazon. According to the Annual Report on Deforestation in Brazil (2022)[87], released by the MapBiomas Project, in the Amazon biome, about 5% of the areas deforested in 2022 had Vegetation Suppression Authorization (ASV) by the responsible government agencies. In addition, the document also exposes the difficulty of the competent environmental agencies in supervising and punishing illegal suppressions in states of the Legal Amazon. For example, between January 2018 and May 2023, IBAMA embargoed and/or assessed only 13% of the areas with irregular forest clearing in the state of Pará. These numbers reinforce the government's limitation to ensure compliance with laws and regulations that were created to prevent deforestation.

Strengthening the above scenario, a study carried out by the Institute of Man and Environment of the Amazon (IMAZON, 2013)[88] showed that for the years 2011 to 2012, 78% of logging was unauthorized and even though of these 78%, the majority (67%) was located in private, vacant or disputed areas. This data corroborates previous studies by the same organization that identified the “private, unused and unclaimed” land categories as the main stages of illegal/ unauthorized deforestation. Reinforcing this information, more

[87] Annual Report on Deforestation in Brazil 2022 - São Paulo, Brazil - MapBiomas, 2023 - 82 pages. Available at: http://alerta.mapbiomas.org
[88] IMAZON. (2013). Imazon: illegal deforestation grows 151% in Pará. Amazônia , P. 1- 3.

recently, the MapBiomas report (2021) revealed that between 2019 and 2020 there was a 15% increase in deforestation in areas under Sustainable Forest Management Plans in force, places where any type of intervention involving shallow cuts of tree individuals and land use conversions is prohibited, until the management period is completed.

In this context, considering the municipality of São Félix do Xingu, which includes Fazenda Lagoa do Triunfo, as well as the current environmental regulations (Law 12.651/2012 - Native Vegetation Protection Law[89] - Normative Instruction No. 02, of July 6, 2015[90]), which deals with the suppression of native vegetation in rural properties or cutting of trees in urban areas, it is understood that any activity that requires the suppression of native vegetation is subject to the issuance of ASV by the governmental agency, whether state or federal. Since 2018, authorizations granted by states must be issued or registered in the National System for the Control of Origin of Forest Products (SINAFLOR) which is managed by IBAMA. However, Pará is not fully integrated into the system yet, and verification of the existence of authorization is carried out on the state basis.

Finally, the municipality of São Félix do Xingu, which belongs to Fazenda Lagoa do Triunfo, occupies the first place of most deforested municipality in Brazil since 2001 and was the fifth that deforested the most in 2022. Between 2019 and 2022, 22,787 hectares of forests were suppressed in the city. In addition, the APA Triunfo do Xingu is the most deforested conservation unit in Brazil. These numbers, obtained through the Amazon monitoring system carried out by INPE[91], demonstrate the historical trend of growth in forest clearing, not only for that municipality, but also for the entire region of the Project.

In addition, the municipality of São Félix has the largest herd of cattle in Brazil, according to IBGE data (2021), with about 2.4 million units and studies indicate that the expansion of livestock is the main driver of deforestation in the Brazilian Amazon, accounting for about 80% of deforestation in the region. And livestock farming has a significant impact on intensifying deforestation, especially in remote areas and on properties with a high percentage of remaining forest, such as the project area (Skidmore et al., 2021)[92].

Sub-step 1c. – Selection of the baseline scenario

Described in Section 3.1, specified in item 3.1.4 Baseline Scenario.

Step 2: Investment Analysis

Under development.

Sub-step 2a. Determining suitable method of analysis

Under development.

Step 3: Barrier Analysis

VCS "VT0001 – Tool for the Demonstration and Assessment of Additionality in VCS Agriculture, Forestry and Other Land Use (AFOLU) Project Activities - requires Investment Analysis (Step 2) or Barrier Analysis (Step 3). In this case, we opted for the Investment Analysis, already described in Step 2.

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89 BRAZIL. Law No. 12.651, of May 25, 2012. Available at: L12651 (planalto.gov.br)

90 GOVERNMENT OF THE STATE OF PARÁ. Normative Instruction No. 2, of July 6, 2015. . Belém-PA, July 8, 2015. Available at: https://www.semas.pa.gov.br/legislacao/files/pdf/176.pdf.

91 INPE - National Institute for Space Research. **Project for Monitoring Deforestation in the Legal Amazon by Satellite (PRODES)**. Available at: http://www.dpi.inpe.br/prodesdigital/prodesmunicipal.php.

92 SKIDMORE, M. E., MOFFETTE, F., RAUSCH, L., CHRISTIE, M., MUNGER, J., & GIBBS, H. K. (2021). Cattle ranchers and deforestation in the Brazilian Amazon: Production, location, and policies. Global Environmental Change, 68, 102280.

Step 4: Common Practice Analysis

The fourth stage of the additionality analysis considers the evaluation of areas similar to the model proposed by the REDD+ project to identify common practices. For this verification, the geographical delimitation of the Reference Region of the Ateles REDD+ Project was considered.

As the region where the REDD+ project to be implemented has differentiated characteristics compared to other regions of the state of Pará, it was decided to restrict the analysis to the Reference Region, instead of expanding to other regions of the state. For this similarity analysis, basic assumptions associated with the category of land use, area size, economic activities developed, regulatory structure, environmental characteristics and context of action of deforestation agents and vectors were applied.

As explained in the previous analyses of this document (section 3.1.4), it is evident that the annual deforestation rates in the region showed a significant increase from 2019. After reaching a peak that year, there was a slight reduction in the rate in the last three years of the series. However, it is important to note that this decline still remains above the average recorded in the period between 2013 and 2019. This scenario expresses the current situation of lack of government supervision and control to curb unplanned deforestation throughout the Amazon region of Brazil. This trend was initiated due to the political and economic instabilities of 2016 and intensified during the Bolsonaro administration. It is critical to note that recent deforestation, dating from 2012, is concentrated mainly in the north and northwest areas. The multiple incidences of recent deforestation adjacent to the project area underscore the increasing pressure exerted on internal forests.

As already described in other sections of this document, this pressure comes largely from small family farmers, who carry out deforestation fractions of forest areas on their properties, up to two hectares annually, to plant subsistence crops, products such as corn and cassava used as inputs in cattle breeding. In addition to medium and large rural producers seeking new areas to expand pasture areas for extensive cattle breeding, it has a high level of income and greater legal bargaining power (Peabiru, 2022[93]). According to COSTA et al., 2017[94], deforestation in the region presents itself with a specific dynamic, starting with the conversion of the forest cover to pasture implantation that, over time, suffers a process of degradation due to overcrowding of cattle, and the producer is forced to acquire and/or lease new land.

In addition, from the point of view of the land issue and in an attempt to identify possible areas similar to the REDD+ project, a search was made for private properties registered in the Land Management System (SIGEF[95]), available on the INCRA website. An analysis was then carried out, highlighting which properties had areas close to the size of the project area (Table 27), which is one of the premises for being considered a similar activity, according to the tool "VT0001 - Tool for the Demonstration and Assesment of Addicionality in VCS Agriculture, Forestry and Other Land Use (AFOLU) Project Activities", version 3.0. In addition, the land situation of the property was also assessed, considering the areas regularized according to the land data, thus limiting itself to the certified parcels, which the accredited person or the land registry officer confirms or updates the registration and owner data.

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[93] PEABIRU. 2022. REDD+ ATEKES PROJECT: SOCIOECONOMIC AND ENVIRONMENTAL DIAGNOSIS. Final report. 266p.

[94] Costa, A. F. et al. Challenges for land regularization in the Triunfo do Xingu Environmental Protection Area, Pará, Brazil. Ciência Agrar Magazine, v. 60, n. 1, p. 96-102, Jan./Mar. 2017. Available at: <http://dx.doi.org/10.4322/rca.60105>. Access on: August 8, 2023.

[95] INCRA - Instituto Nacional de Colonização e Reforma Agrária. Sistema de Gestão Fundiária (SIGEF). Disponível em: https://sigef.incra.gov.br/.

Name of the area	Area (ha)	Date of approval
FAZENDA BARRA DO TRIUNFO	42.844,78	13/12/2013
FAZENDA UMUARAMA DO XINGU	37.546,49	13/03/2019
FAZENDA BITUVA GRANDE E FAZENDA MATÃO	6.790,70	2017/08/31
FAZENDA CONFIANÇA	5.849,09	2015/09/16
FAZENDA RECREIO II E III	4.945,81	2023/01/03
LOTE 110	4.784,09	2023/07/09
FAZENDA RIO GRANDE	4.745,62	2023/03/16
FAZENDA SANTA BARBARA II	4.722,70	2023/03/16
LOTE 06 E 07 (PARTE) SETOR A	4.418,96	2023/03/16
LOTE 38	4.400,84	2023/03/16
FAZENDA TERRA BOA	4.363,86	2023/03/16
FAZENDA DIVISA I	4.358,50	2023/03/16
FAZENDA SANTA INHÁ CHICA	4.358,32	2014/12/17
FAZENDA TIBORNA	4.356,86	2014/11/27
FAZENDA SERINGAL PORTO SEGURO	4.356,42	2022/08/12
FAZENDA OURO VERDE	4.351,89	2021/08/17
FAZENDA CINCO ESTRELAS	4.342,30	2022/11/10
GLEBAS 1 E 1-A	4.325,71	2023/01/03
FAZENDA BUCAINA LOTE 78	4.302,75	2022/01/11
FAZENDA SANTA TEREZA	4.270,12	2023/01/12
FAZENDA MONTREAL	4.212,10	2014/11/27
FAZENDA MUNDO NOVO	4.182,35	2023/01/04
FAZENDA SERINGAL JAPONESA	4.143,69	2020/05/15
FAZENDA EL SHADAI	4.096,56	2023/01/13
FAZENDA TIVORNA I	4.075,74	2023/01/03
FAZENDA SERRA DOURADA	3.824,48	2022/08/27
FAZENDA LAGO RICO	3.700,78	2023/01/12
FAZENDA MORRO VERDE	3.433,41	2023/09/15
FAZENDA SÃO PEDRO - LOTE	3.241,73	2022/01/11
FAZENDA MARINGÁ	3.043,11	2022/08/04
LOTE 18 SETOR A	2.974,96	2014/08/19
LOTE 11 SETOR A	2.930,66	2023/03/16
FAZENDA PARIRI	2.901,94	2023/01/03
FAZENDA SANTA CRUZ II	2.879,31	2019/12/27
FAZENDA ELOHIM	2.778,19	2022/05/18
LOTE 109	2.764,67	2022/08/04
LOTE 22 SETOR D	2.739,63	2022/08/29
LOTE 14 SETOR A	2.730,99	2022/08/29
FAZENDA NOSSA SENHORA APARECIDA	2.635,85	2022/08/29
FAZENDA BAÚ LOTE 18 SETOR E	2.604,84	2022/08/29
LOTE 43 SETOR D LOTEAMENTO SÃO FÉLIX DO XINGU	2.578,29	2022/08/29
LOTE 10 SETOR A	2.564,60	2016/08/24
GLEBA SÃO FELIX DO XINGU LOTE 01 SETOR A	2.522,50	2022/08/04
FAZENDA BURITI III	2.517,71	2022/08/04
FAZENDA NOVO MÉXICO 2	2.512,81	2021/08/17

Name of the area	Area (ha)	Date of approval
FAZENDA SANTA LUCIA	2.501,38	2022/08/29
FAZENDA PROGRESSO	2.467,90	2022/08/04
FAZENDA BURITI V	2.448,74	2022/01/31
LOTE 42 SETOR D LOTEAMENTO SÃO FÉLIX DO XINGU	2.437,13	2017/03/21
FAZENDA MONTE DAS OLIVEIRAS	2.424,14	2016/05/18
FAZENDA BURITI I	2.401,01	2015/04/09
FAZENDA EL SHADDAY I	2.397,45	2017/03/21
FAZENDA LUA NOVA	2.391,86	2022/07/28
FAZENDA EL SHADDAY	2.390,45	2022/01/31
LOTE 44 SETOR D LOTEAMENTO SÃO FÉLIX DO XINGU	2.354,33	2022/01/31
FAZENDA BAÚ	2.272,54	2015/10/22
FAZENDA CALIFORNIA PA 02	2.150,72	2013/11/20
FAZENDA AÇAÍ DO PARÁ	2.098,80	2013/11/20
FAZENDA DR BRASÍLIO RAMOS CAIADO	2.083,49	2013/11/20
FAZENDA ANGLO AMÉRICA	2.081,97	2013/11/20
FAZENDA BURITI IV	2.080,88	2013/11/20
NOSSA SENHORA DE FATIMA L	2.054,31	2017/09/19
FAZENDA SANTA BARBARA I	2.053,77	2023/09/15
GLEBA SÃO FELIX DO XINGU LOTE 09 SETOR A	2.036,69	2013/11/20
FAZENDA BURITI II	2.028,64	2022/12/16
FAZENDA BOA ESPERANCA II	2.020,33	2013/11/20
FAZENDA COLORADO	2.015,26	2013/11/20
FAZENDA SANTA BARBARA I	1.983,88	2017/11/15
FAZENDA BELA MOÇA I	1.952,35	2023/09/06
LOTES 04 E 13	1.906,53	2022/11/10
FAZENDA NOSSA SENHORA APARECIDA	1.878,56	2013/11/20
FAZENDA GAMELA	1.864,82	2017/03/21
FAZENDA PACHECÃO LOTE 38 SETOR D	1.853,10	2023/03/22
FAZENDA MUNDO NOVO	1.848,81	2023/06/29
FAZENDA ÁREA GRANDE	1.833,34	2023/06/29
FAZENDA BELA MOÇA II	1.808,43	2014/11/27
FAZENDA SÃO JOSÉ	1.786,75	2023/06/29
FAZENDA BURITI VI	1.782,48	2013/11/20

According to the results, in the context of the REDD+ Ateles Project Reference Region, the private properties observed in the vicinity of the Project differ significantly from the Project Area as they are considerably smaller in scale. In addition, data from the 2017[96] Agricultural Census shows that only 7% of agricultural establishments in the municipality of São Félix do Xingu have an area like or larger than the area of Fazenda Lagoa do Triunfo. For the municipality, 86% of establishments are considered smallholdings (properties with an area between the Minimum Land Fraction and 4 fiscal modules - between 4 ha and 300ha). Thus, in the context of the Reference Region, the areas belonging to AgroSB Agropecuária represent a unique situation.

Although it is difficult to find similar characteristics in relation to the size of the property, it is possible to notice an increase in the development of REDD+ Projects in the State of Pará. As a comparison, we have

[96] https://censoagro2017.ibge.gov.br/

in the south of the State a project in the municipality of Altamira, which despite having a smaller scale (about 10% of the area of the Ateles REDD+ Project), has similar environmental characteristics and premises for the sustainable exploitation of forest resources (Triunfo do Xingu Grouped REDD+ Project – ID 3738), under VCS certification in validation and CCB under development, which aims to preserve the forest within the project area, direct and indirect investments in communities related to the project areas and the stabilization or increase of the populations of all threatened species identified in the project areas; making any and all types of common practice similarity assessment impossible.

In this sense, all areas, including the project area described above, do not represent the "business as usual" scenario in the Reference Region of the project and, as a conclusion of this analysis, it was found that there is no common practice for the Ateles REDD+ Project in the geographic region analyzed.

This understanding demonstrates the relevance of the REDD+ Project for the evaluated region, considering the particular environmental and socioeconomic characteristics of Fazenda Lagoa do Triunfo and its immediate surroundings. The activities proposed by the project, in addition to directly helping to preserve the standing forest, will also mitigate land pressure on native forest fragments, providing co-benefits to biodiversity and communities in the project region.

3.1.6. Methodology Deviations

No methodology deviation was applied in this Project.

3.2. Quantification of GHG Emission Reductions and Removals

3.2.1. Baseline Emissions

3.2.1.1. VM0015 Step 5 – Baseline Land Use and Land Cover Change Component Definition

3.2.1.1.1. Calculation of baseline activity data by forest class (Step 5.1 VM0015)

The result of the baseline projections indicates deforestation of 12,631hectares for the project area (Table 27) and 44,278 hectares for the leakage belt (Table 28) between 2022 and 2032.

Table 27: Annual deforested areas by ICL forest class within the Project Area in the case of the baseline (baseline activity data by forest class) (Table 11b of Methodology VM0015).

Area deforested per forest class *icl* within the project area		Total baseline deforestation in the project area	
ID$_{icl}$>	icl1	ABSLPA$_t$	ABSLPA
Name>	Forest	annual	cumulative
Project year$_t$	ha	ha	ha
2023	1,318	1,318	1,318
2024	856	856	2,174
2025	1,127	1,127	3,301
2026	1,199	1,199	4,500
2027	1,322	1,322	5,822
2028	813	813	6,635
2029	1,457	1,457	8,092
2030	1,553	1,553	9,645
2031	1,323	1,323	10,968
2032	1,663	1,663	12,631

Table 28: Annual deforested areas by forest class icl within the Leak Belt area in the case of the baseline (baseline activity data by forest class) (Table 11c of Methodology VM0015).

Deforestaded area per forest class *icl* within the Leakage Belt		Total deforetation at baseline in Leakage Belt	
ID_{icl}>	icl1	$ABSLLK_t$	ABSLLK
Name>	Forest	Annual	Accumulated
Year of Projeto	ha	ha	ha
2023	4,200	4,200	4,200
2024	4,011	4,011	8,211
2025	4,205	4,205	12,416
2026	5,178	5,178	17,594
2027	4,443	4,443	22,037
2028	4,514	4,514	26,551
2029	4,283	4,283	30,834
2030	4,364	4,364	35,198
2031	4,571	4,571	39,769
2032	4,509	4,509	44,278

3.2.1.1.2. Calculation of post-deforestation baseline activity data by forest class (Step 5.2 VM0015)

Method 1 available in methodology VM0015 was used to define the class that will replace the forest cover in the baseline of the Project (anthropic vegetation in balance). A Table 29 shows the zone 1 area, which covers the Project Area, the Leakage Belt and the Leakage Management Areas, and the corresponding areas of each post-deforestation land use/land use change class.

Table 29: Zone of the Reference Region covering the potential post-deforestation LU/LC class (Table 12 of Methodology VM0015).

Zone		Name		Total of all other LU/LC classes present in the zone		Total area of each Zone	
		Zone 1					
		ID_{fcl}	1				
		Area	% of Zone	Area	% of Zone	Area	% of Zone
IDz	Name	ha	%	ha	%	ha	%
1	Zone 1		100				100
Total area of each class *fcl*			100				100

The area projected to be deforested is reported in (project area) and (leakage belt).

Table 30: Annual deforested areas in each zone within the Project Area in the case of the baseline (Table 13b of Methodology VM0015).

Area established after deforestation per zone within the project area		Total baseline deforestation in the project area	
IDz>	1		
Name>	Zone 1	$ABSLPA_t$	ABSLPA
Project $year_t$	ha	ha	ha
2023	1,318	1,318	1,318

Area established after deforestation per zone within the project area		Total baseline deforestation in the project area	
IDz>	1		
Name>	Zone 1	$ABSLPA_t$	ABSLPA
Project year$_t$	ha	ha	ha
2024	856	856	2,174
2025	1,127	1,127	3,301
2026	1,199	1,199	4,500
2027	1,322	1,322	5,822
2028	813	813	6,635
2029	1,457	1,457	8,092
2030	1,553	1,553	9,645
2031	1,323	1,323	10,968
2032	1,663	1,663	12,631

Table 31: Annual deforested areas in each zone within the Leakage Belt in the case of the baseline (Table 13c of Methodology VM0015).

Area established after deforestation per zone within the Leakage Belt		Total deforestation at baseline in Leakage Belt	
IDz>	1		
Name>	Zone 1	$ABSLLK_t$	ABSLLK
Year of the Project	ha	ha	ha
2023	4,200	4,200	4,200
2024	4,011	4,011	8,211
2025	4,205	4,205	12,416
2026	5,178	5,178	17,594
2027	4,443	4,443	22,037
2028	4,514	4,514	26,551
2029	4,283	4,283	30,834
2030	4,364	4,364	35,198
2031	4,571	4,571	39,769
2032	4,509	4,509	44,278

3.2.1.1.3. Calculation of baseline activity data by land use change and land cover category (Step 5.3 VM0015)

Not applicable because method 2 was not performed.

3.2.1.2. VM0015 Step 6: Estimation of changes in carbon stocks in the baseline scenario and emissions of non-CO2 gases.

3.2.1.2.1. Estimation of changes in carbon stocks from baseline scenario (Step 6.1 VM0015)

The estimate of the carbon stock for the forest class was obtained through the forest inventory carried out by the technical team of Biodendro – Consultoria Ambiental, in 2022, in partnership with Biofílica Ambipar Environment. The main results found in this study will be described below, and more information can be obtained in the Final Carbon Inventory Report of the company Biodendro.

Below, we present the synthesis of carbon stocks in the Project Area.

Estimation of the average carbon stocks of each land use and land cover class (Step 6.1.1 VM0015)

Sampling

The estimated biomass stocks of the Lagoa do Triunfo farm were carried out through a field survey and forest inventory carried out between November and December 2022. The forest inventory was carried out through the cluster sampling process, with each cluster consisting of four plots with an individual area of 2,500 m² (Figure 17). The plots were subdivided into four 625m² subunits to measure trees with different sizes. Regarding the size of the trees, 4 DBH classes (10-20cm; 20-30cm; 30-50 cm and >50 cm) were stipulated for the inventory. In all of them, 33 clusters were installed, making up 132 plots of 2,500 m² and a sampling area of 330,000m² or 33 hectares. These plots were used to measure live trees and standing dead trees.

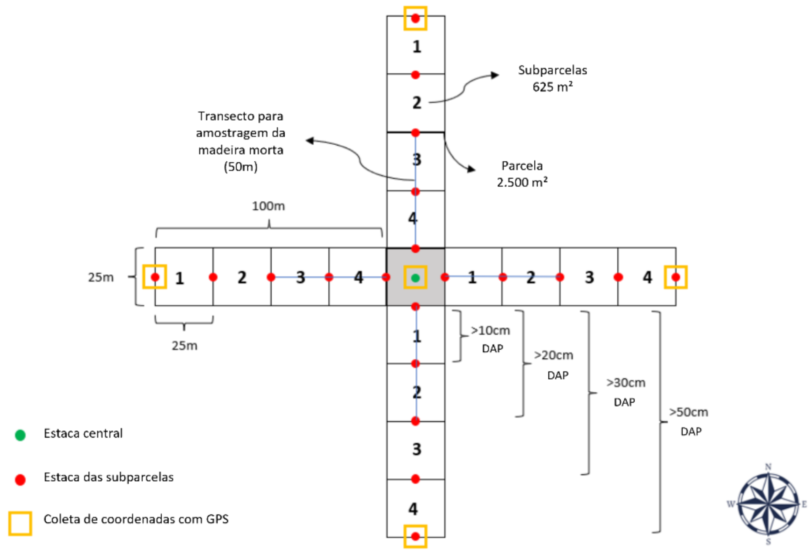

Figure 17: Schematic drawing of a conglomerate used in the forest carbon inventory. Source: Report "Forest Carbon Stock Ateles REDD+ Project".

The cluster of plots were randomly distributed throughout the sampled area ("Project Area"). Considering the prior knowledge of the existence of areas considered as explored and in areas not explored in the "Project Area", the distribution of the plots was carried out in order to have a good representation of the particular relief characteristics of the area. Figure 18 shows the location map of the plots.

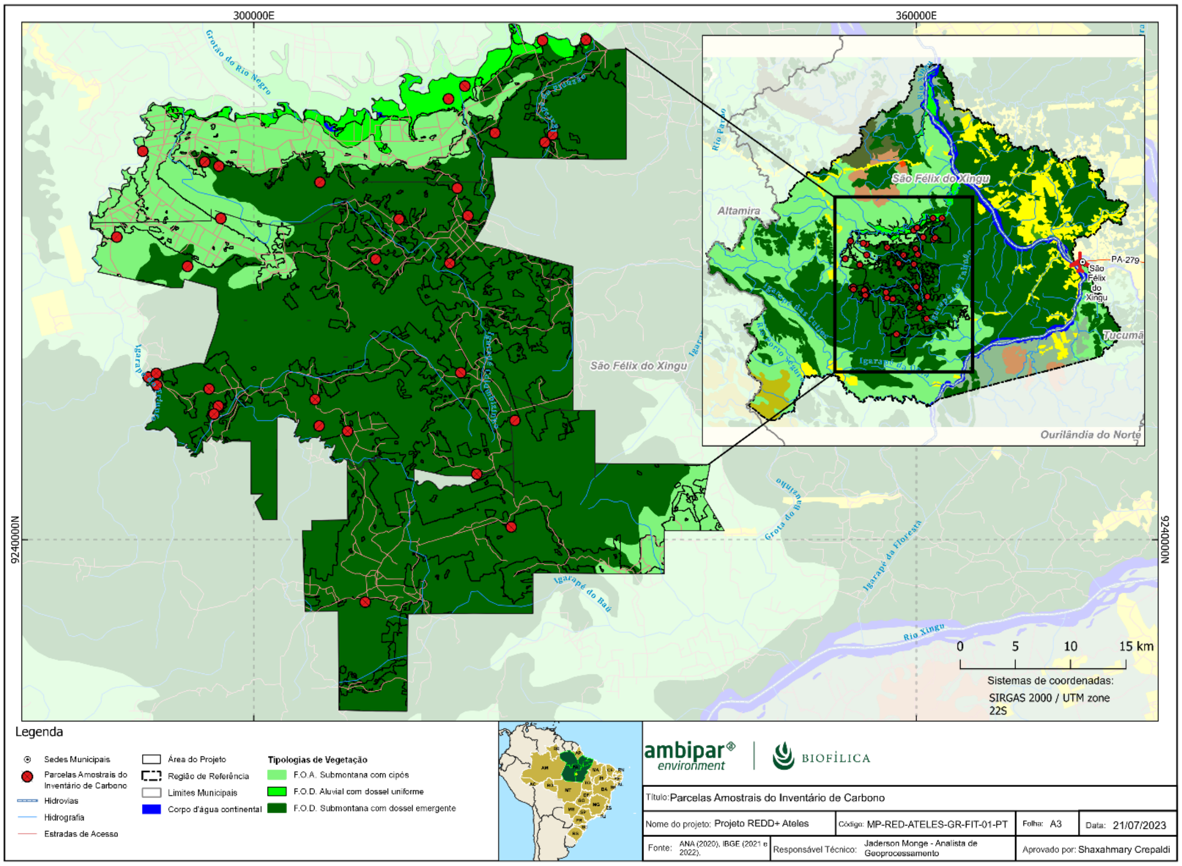

Figure 18: Location of the sample plots of the carbon inventory.

The plots were subdivided into 4 sub-units (sub-plots) of 625 m² (25 x 25 m), named sub-plots 1, 2, 3 and 4. This subdivision was adopted to better represent the larger trees that, due to their lower density of individuals per area, require a larger plot size than trees with smaller DBH. Thus, the DBH classes used as inclusion criteria in the sub-plots were the following: a) Sub-plot 1: all trees with DBH ≥ 10cm; b) Sub-plot 2: all trees with DBH ≥ 20cm; c) Sub-plot 3: all trees with DBH ≥ 30 cm; d) Sub-plot 4: all trees with DBH ≥ 50cm.

In this way, the different diametric classes had different useful plot area, as reported below:

- Class 1 - trees with DBH ≥ 10 cm and < 20 cm – 625 m^2 (measurement only in subplot 1)
- Class 2 - trees with DBH ≥ 20 cm and < 30 cm – 1,250 m^2 (measurement in subplots 1 and 2)
- Class 3 - trees with DBH ≥ 30 cm and < 50 cm – 1,875 m^2 (measurement in subplots 1, 2 and 3)
- Class 4 - trees with DBH ≥ 50 cm– 2,500 m^2 (measurement in all subplots)

Estimates of forest inventory parameters

The estimates of the carbon stock obtained from the cluster sampling process and the statistics of the parameters related to carbon quantification were calculated according to forest inventory procedures presented in Pélico Netto & Brena (1997[97]). The equations below were used to estimate the mean and

[97] PÉLLICO NETTO, S; BRENA, D.A. Inventário Florestal. v.1, Curitiba, PR, 1997. 316p.

standard error for each component of the inventoried biomass (live trees, standing dead trees and dead trees on the ground).

The estimate of the average population per subunit is given by:

$$\bar{x} = \frac{\sum_{i=1}^{n}\sum_{j=1}^{M} X_{i,j}}{n \times M}$$

Where:

M = number of cluster sub-units;

n = number of sampled clusters;

Xi,j = variable of interest.

The estimate of the average of sub-units per cluster is given by:

$$\bar{x}_i = \sum_{j=1}^{M} \frac{X_{i,j}}{M}$$

The estimate of population variance per sub-unit is given by:

$$s_x^2 = \frac{1}{nM-1}\sum_{i=1}^{n}\sum_{j=1}^{M}\left(X_{i,j} - \bar{x}\right)^2$$

In cluster sampling, the total variance S_x^2 is divided into two components of variation: within (S_d^2) and between clusters (S_e^2):

$$S_x^2 = S_e^2 + S_d^2$$

The estimates of S_e^2 e S_d^2 are obtained by analysis of variance, which has the following mathematical hopes:

$$E(MS_{between}) = MS_e^2 + S_d^2$$

$$E(MS_{within}) = S_d^2$$

Where $MS_{within} = \frac{1}{nM-1}\sum_{i=1}^{n}\sum_{j=1}^{M}\left(X_{i,j} - \bar{x}_i\right)^2 = s_d^2$, and $MS_{between} = \frac{\sum_{i=1}^{n} M(\bar{x}_i - \bar{x})^2}{n-1}$

Then S_e^2 it is estimated given by:

$$s_e^2 = \frac{MS_{between} - MS_{within}}{M}$$

And finally, the total variance is estimated by:

$$S_x^2 = S_e^2 + S_d^2 = \frac{(MS_{between} + (M-1) \times MS_{within})}{M}$$

The variance of the overall mean is estimated by:

$$s_{\bar{x}} = \sqrt{\frac{s_x^2}{n \times M}[1 + r(M-1)]}$$

Where r is the intra-cluster correlation coefficient:

$$r = \frac{s_e^2}{s_e^2 + s_d^2}$$

Finally, the sampling error is given by:

$$e = t \times s_x$$

Where t is the value of the *student* t-distribution with n-1 degrees of freedom.

Estimates of average carbon stocks per hectare

The average total carbon stock per hectare (= carbon density) in a land-use and land-cover (LU/LC) class was estimated by the following equation:

$$Ctot_{cl} = Cab_{cl} + Cbb_{cl} + Cdw_{cl}$$

Where:

$Ctot_{c,l}$ = Average carbon stock per hectare in all carbon pools accounted for in class LU/LC cl, in tCO_2eq ha^{-1};

$Cab_{c,l}$ = Average carbon stock per hectare in the aboveground biomass carbon pool of class LU/LC cl, in tCO_2eq ha^{-1};

$Cbb_{c,l}$ = Average carbon stock per hectare in the belowground biomass carbon pool of class LU/LC cl, in tCO_2eq ha^{-1};

$Cdw_{c,l}$ = Average carbon stock per hectare in the LU/LC cl class dead wood carbon pool, in tCO_2qe ha^{-1}.

Measurements

Estimate of carbon stocks in living biomass carbon pools

The estimation of dry biomass of the trees included in the sampling was made by allometry using the prediction equation developed by Nogueira et al. (2008)[98]. This equation used was adjusted adopting the DBH as input variable in a simple model that calculates the dry biomass above ground (in Kg).

The equation proposed by Nogueira et al. (2008) for forests in Southern Amazonia and used for the determination of aboveground dry biomass of the whole tree, is as follows:

$$ln(Peso\ seco) = -1{,}716 + 2{,}413\ ln(DAP)$$

Where:

Dry weight = Weight of the tree's dry biomass above ground, in Kg. DBH = Tree trunk diameter measured at breast height, in cm.

The individual tree biomass prediction equation presented by Nogueira et al. (2008) was adjusted considering a sample set of 262 trees, using samples with DBH ranging from 5 to 124 cm. The adjusted coefficient of determination of the equation (adjusted R2) was 0.964 and the standard error of the absolute estimate was 0.306. The application of the equation for predicting dry biomass of the sampled trees showed good accuracy in the prediction of prediction, with bias of -1% for the prediction of the biomass of sampled trees and -0.05% for the normalized dry biomass per hectare.

To make the total of the plot, considering the different sampling areas applied to the different classes of DBH, the biomass data of the trees in the subplots were transformed to the hectare area unit, according to Table 32.

[98] NOGUEIRA, E.M., P.M. FEARNSIDE, B.W. NELSON, R.I. BARBOSA & E.W.H. KEIZER. Estimates of forest biomass in the Brazilian Amazon: New allometric equations and adjustments to biomass from wood-volume inventories. Forest Ecology and Management (published online 17-SEP-2008) doi:10.1016/j.foreco.2008.07.022

Table 32: Plot size by DAP class.

DBH Class Number	DBH range	Plot Area (PA) (m^2)
1	10cm ≤ DBH < 20cm	625
2	20cm ≤ DBH < 30cm	1,250
3	30cm ≤ DBH < 50cm	1,875
4	DBH >50cm	2,500

Standardization of the size of sampling areas by DBH class was carried out by transforming data into the hectare area unit. After this transformation, the total biomass value of the plot per hectare was obtained by adding the biomass of the different DBH classes, using the following equation:

$$BAb_{pl} = \sum_{cd=1}^{CD} \left(BAb_{cd} \times \frac{10000}{AP} \right)$$

Where:

Above-ground BAb_{pl} biomass of the trees in the plot, in kg.ha-1;

Above-ground BAb_{cd} biomass of diameter class cd, 1,2,3, 4, in kg.ha^{-1};

Cd: DBH classes 1, 2, 3 and 4, dimensionless; Plot expansion factor, dimensionless; Plot area, in m^2

PA: Plot Area, in m^2

The aboveground biomass carbon stock is calculated by:

$$TCab_{tr} = TBab_{tr} \times CF$$

Where $TCab_{tr}$ is the carbon stock in the above-ground biomass of the *tr tree;* $TBab_{tr}$ is the aerial biomass of the tr tree and CF is the carbon fraction, adopted as 0.47 (IPCC, 2006)[99];

The calculation of the carbon stock in the above-ground biomass was done per plot and per area, adding the above-ground carbon stock of all trees within each plot and multiplying by an expansion factor of the portion proportional to the area of the plot measured for each DAP class (table xx).

Carbon stock in the biomass of living trees below ground

Estimate of the carbon stock of the belowground living biomass was obtained using the root/shoot ratio and the aboveground carbon stock:

$$Cbb_{cl} = TCab_{tr} \times R \times CF$$

Where Cbb_{cl} is the carbon stock in the biomass below the ground of the *tr tree;* $TCab_{tr}$ is the aerial biomass of the tr tree; R is the root-to-air ratio, equal to 0.24 and CF is the carbon fraction, adopted as 0.47 (IPCC, 2006);

The calculation of the carbon stock in the below-ground living biomass was done per plot and per area, adding the below-ground carbon stock of all trees within each plot and multiplying by an expansion factor of the plot proportional to the area of the plot measured for each DAP class (table xx).

Carbon stock in the biomass of standing dead trees

[99] IPCC. (2006). Agriculture, Forestry and Other Land Use. In: Eggleston H.S., Buendia L., Miwa K., Ngara T. and Tanabe K. (eds). *2006 IPCC Guidelines for National Greenhouse Gas Inventories*, *Vol. 4*. IGES, Japan.

The same plots and procedures used for sampling living trees were used for standing dead trees. For this component, additional information was used to characterize the sampled individuals, dividing the samples into four categories of shape, described below and exemplified in the illustrations in figure xy.

Category 1 - Tree with branches, with architecture similar to a living tree, except for the absence of leaves

Category 2 - Tree without thin branches, but with persistent small and large branches;

Category 3 - Tree with large branches only;

Category 4 - Trees with trunk only, without the crown or branches.

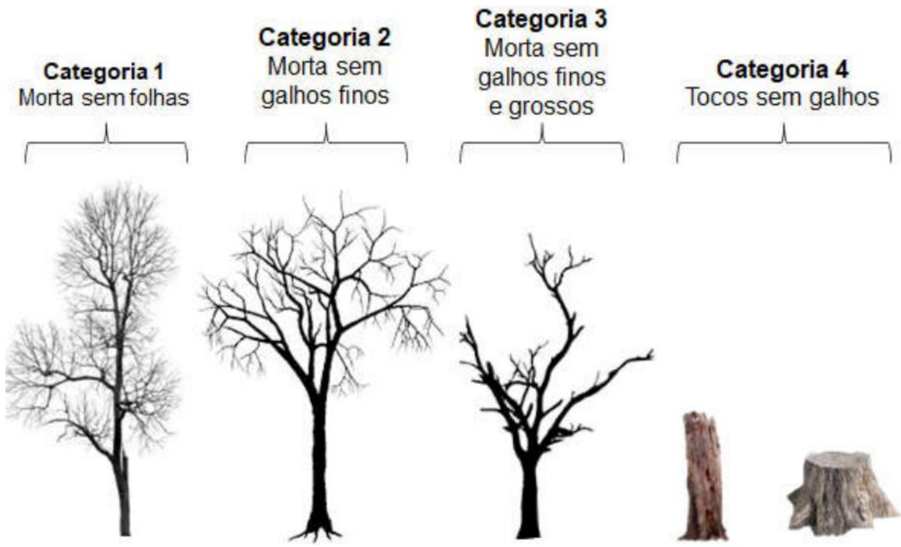

Figure 19: Illustration to exemplify the categories of standing dead trees.

Although there was a differentiation of the four categories of dead trees standing in the field, the calculation of the biomass stock was performed only for two categories, Category 1 and the grouping of Categories 2, 3 and 4, according to the module VMD0002 - Estimation of carbon stocks in the dead-wood pool.

For Category 1 wood from standing dead trees, biomass was estimated using an allometric equation and wood density was assumed to be comparable to that of the living tree.

In categories 2, 3, and 4, biomass estimation was limited to the main trunk (trunk) of the tree, with biomass being calculated from the volume converted to biomass using the appropriate deadwood density class.

The volume has been estimated as the volume of a cone, where the top diameter is assumed to be zero.

For classes 2, 3 the biomass of standing dead trees is estimated when the diameter of the top is not measured by the equation:

$$B_{SDWl,sp,i} = \frac{1}{3} * \pi * \left(\frac{BDia_{SDWl,sp,i}}{200}\right) * H_{SDWl,sp,i} * D_{DWdc}$$

Where:

$B_{SDWl,sp,i}$ = Biomass of the standing dead tree *l* of the sample plot/sp point in stratum i; d.m.;

$BDia_{SDWl,sp,i}$ = Basal diameter of the standing dead tree *l* of the sample plot/sp point in stratum i; cm;

$H_{SDWl,sp,i}$ = Height of the standing dead tree l of the sample plot/sp point in stratum i; m;

$D_{DW,dc}$ = Average density of dead wood in the density class (dc) – intact (1), intermediate (2) and rotten (3); In this project the value of 0.483 t m^{-3} was used (Keller et al., 2004) sp = 1, 2, 3, … sample plots/points Pi in stratum i;

i = 1, 2, 3, … M strata in the project scenario.

Carbon stock in the biomass of dead trees on the ground

The dead tree lying (under the ground) was sampled using the line intersection method. A 50-meter line was established in each sample plot and the deadwood diameters over the soil (≥ 10 cm in diameter) crossing the lines were measured. Trees were categorized into one of three density states (intact, intermediate and rotten) using the 'machete test' as recommended by the IPCC Good Practice Guide to LULUCF (2003), Section 4.3.3.5.3.

The carbon biomass of dead trees on the forest soil per unit area was calculated as follows

$$Bldw_i = \sum_{dc=1}^{3} \left(\frac{\pi^2 * \left(\sum_{n=1}^{N} Dia_{dc,n,i}^2 \right)}{8 * L} \right) \times D_{DWdc}$$

Where:

B_{LDWi} Dead wood biomass over soil per unit area in stratum i; d.m.ha^{-1};

$Day_{n,i,t}$ Deadwood n-piece diameter along transect in stratum i; cm

n 1, 2, 3, … N sequence number of wood pieces in density class dc crossing the transect

L Transect length; 50 m

deadwood density class dc – intact (1), intermediate (2) and rotten (3); dimensionless;

D_{DWdc} Average density of dead wood in density class (dc) – intact (1), intermediate (2) and rotten (3); d.m. m^{-3};

dc deadwood density class dc – intact (1), intermediate (2) and rotten (3); dimensionless;

i 1, 2, 3, … M strata in the project scenario.

The average carbon stock in dead wood for each stratum was then calculated as the sum of dead wood components standing and lying, converted to carbon dioxide equivalents

$$C_{DWi} = \left((B_{SDWi} + B_{LDWi}) * CF_{DW} \right) * \frac{44}{12}$$

Where:

C_{DWi} Average carbon stock of dead wood in stratum i; t CO2-e ha^{-1};

B_{SDWi} Standing deadwood biomass in stratum i; d.m. ha^{-1};

B_{LDWi} Dead wood biomass at rest in stratum i; d.m. ha^{-1};

CF_{DW} Carbon fraction of dry matter in dead wood; tCt^{-1} d.m.;

i 1. M strata

44/12 Molecular weight ratio of CO_2 to carbon, t CO_2-e t C^{-1}.

Total carbon stocks

Considering the different components of living trees above and below ground, dead trees on the ground and standing, the forests of the Ateles Project contain 50.3 million tons of CO2eq, with 44.7 million tons in the aerial part of the living trees, representing the largest compartment in stock (88%). Also, the underground part of living trees represents 17% of the total stock, with 8.5 million tons of CO2eq, 4.2 million tons of CO2eq in dead wood on the ground and 1.3 million tons of CO2eq in dead wood in standing trees totaling 2.6% of the total stock.

The relative sampling error of the average estimates referring to the total carbon stock per hectare in the project area was 9.9%, below the 10% established in the VM0015 methodology.

To calculate the values of Carbon Equivalent (COe, sometimes also called CO_2e) presented in Table 33, the reduced emissions were calculated by multiplying the estimated inventory stock by 3.6667, since CO_2 mass = 44 and C mass = 12; 44/12 = 3.6667. Values were calculated in Carbon equivalent to comply with Table 15 of methodology VM0015.

Table 33: Carbon stock per hectare for the initial class of ICL existing in the project area and leakage belt (table 15a of VM0015).

Initial forest class *icl* Name: Forest ID_{icl} 1 Average carbon stock per hectare + 90% CI							
Cab_{icl}		Cbb_{icl}		Cdw_{icl}		$Ctot_{icl}$	
C stock $tCO_2e\ ha^{-1}$	± 90% CI $tCO_2e\ ha^{-1}$	C stock $tCO_2e\ ha^{-1}$	± 90% CI $tCO_2e\ ha^{-1}$	C stock $tCO_2e\ ha^{-1}$	± 90% CI $tCO_2e\ ha^{-1}$	C stock $tCO_2e\ ha^{-1}$	± 90% CI $tCO_2e\ ha^{-1}$
325,8	33,1	69,7	7,6	50,4	13,6	445,9	17,9

Where:

Cabicl : Average carbon equivalent stock per hectare for the above-ground biomass reservoir for the initial forest class;

Cbbicl : Average carbon equivalent stock per hectare for the biomass reservoir below ground for the initial forest class;

Cdwicl : Average carbon equivalent stock per hectare for the dead biomass reservoir for the initial forest class;

Ctoticl : Average carbon stock per hectare for the total biomass reservoir for the initial forest class

Designed post-deforestation classes for the Project area and leak belt in the baseline scenario and existing non-forest classes in the spill management areas

The VM0015 methodology allows estimates from local studies and, therefore, the value of 61.2 tCO2e $^{ha-1}$ was taken as a reference for the carbon stock of anthropogenic vegetation in the equilibrium class, a class that was designed to exist in the Project Area and Leakage Belt in the Project scenario. This carbon stock estimate was obtained by (WANDERLLI; FEARNSIDE, 2015), through a long-term study of the landscape and the average vegetation composition in deforested areas of the Brazilian Amazon, which consists of a matrix composed of pastures, small-scale agriculture and secondary vegetation, usually found in a post-deforestation scenario in the Amazon.[100]

Wanderlli & Fearnside (2015) is a peer-reviewed scientific literature, and represents one of the most current studies for the Brazilian Amazon on carbon stock in deforested areas, satisfying the requirements of Section 4.5.6 of the VCS Standard:

1. Data were not collected directly from primary sources;
2. Data were collected from secondary sources, by researchers from INPA (renowned research institute on the subject in Brazil), published in an international and reputable scientific journal (Forest Ecology and Management);
3. The data are from a period that accurately reflects the current practice available for the determination of carbon stock;
4. No sampling was applied to these data;
5. The data is available to the public through the website: http://www.ppginpa.eco.br/documents/teses_dissertacoes/wandelli-fearnside-2015-for-ecol-man_Land-use-history-and-capoeira-growth.pdf
6. They are available for independent evaluation of VCSA and VVB;
7. The data is suitable for the geographical scope of VM0015,
8. No expert analysis was required; and
9. Data is not only kept in a central storage repository.

Calculation of carbon stock variation factors (Step 6.1.2 VM0015)

In the baseline scenario, the Project considers that the change in the carbon stock of the forest cover is replaced by a type of vegetation that can be pasture areas, small-scale agricultural plantations or plantations (temporary or permanent). AFOLU requirements require that the decomposition of carbon stock into soil carbon, subsoil biomass, deadwood, and harvested wood products, in the case of the baseline, be considered. To calculate this reduction in carbon stock, version VM0015 1.1 applies a standard linear function to explain the reduction in carbon stock in the initial forest classes (ICL) and the increase in carbon stock in the post-deforestation use classes.

Table 34 and Table 35 summarize how the carbon stock change factor was calculated.

[100] WANDERLLI, E.V.; FEARNSIDE, P.M. Secondary vegetation in the central Amazon: effects of land use history on aerial biomass. Ecology and Forest Management, v. 347, n. 11, p. 140 – 148, 2015.

Table 34: Carbon stock variation factors for initial forest classes ICL (Method 1) (Table 20a of Methodology VM0015).

Year after deforestation		$\Delta Cab_{icl,t}$	$\Delta Cbb_{icl,t}$	$\Delta Cdw_{icl,t}$	$\Delta Ctot_{cl,t}$
0	t*	325,8	7,0	5,0	337,8
1	t*+1	0	7,0	5,0	12,0
2	t*+2	0	7,0	5,0	12,0
3	t*+3	0	7,0	5,0	12,0
4	t*+4	0	7,0	5,0	12,0
5	t*+5	0	7,0	5,0	12,0
6	t*+6	0	7,0	5,0	12,0
7	t*+7	0	7,0	5,0	12,0
8	t*+8	0	7,0	5,0	12,0
9	t*+9	0	7,0	5,0	12,0
10	t*+10				
11	t*+11				
12	t*+12				
13	t*+13				
14	t*+14				
15	t*+15				
16	t*+16				
17	t*+17				
18	t*+18				
19	t*+19				
20-T	t*+20...				

Table 35: Factors of change of carbon stock for the final classes zones fcl or z (Method 1) (Table 20b of Methodology VM0015).

Year after deforestation		$\Delta Ctot_{fcl,t}$
0	t*	0,0
1	t*+1	6,1
2	t*+2	6,1
3	t*+3	6,1
4	t*+4	6,1
5	t*+5	6,1
6	t*+6	6,1

Year after deforestation		$\Delta Ctot_{fcl,t}$
7	t*+7	6,1
8	t*+8	6,1
9	t*+9	6,1
10	t*+10	0,0
11	t*+11	0
12	t*+12	0
13	t*+13	0
14	t*+14	0
15	t*+15	0
16	t*+16	0
17	t*+17	0
18	t*+18	0
19	t*+19	0
20-T	t*+20...	

Calculation of changes in baseline carbon stocks (Step 6.1.3 VM0015)

Method 1 (activity data available for the classes) was used to calculate the variation of the carbon stock from the total baseline in the Project Area (Table 36) and in the leakage belt in table 37 according to Equation 10 on page 72 of VM0015 version 1.1:

$$\begin{aligned}\Delta CBSLPA_t = \sum_{p=1}^{P}\Bigg(&\sum_{icl=1}^{Icl} ABSLPA_{icl,t} * \Delta Cp_{icl,t=t*} - \sum_{z=1}^{Z} ABSLPA_{z,t} * \Delta Cp_{z,t=t*}\\ &+ \sum_{icl=1}^{Icl} ABSLPA_{icl,t-1} * \Delta Cp_{icl,t=t*+1} - \sum_{z=1}^{Z} ABSLPA_{z,t-1} * \Delta Cp_{z,t=t*+1}\\ &+ \sum_{icl=1}^{Icl} ABSLPA_{icl,t-2} * \Delta Cp_{icl,t=t*+2} - \sum_{z=1}^{Z} ABSLPA_{z,t-2} * \Delta Cp_{z,t=t*+2} \quad + \cdots\\ &+ \sum_{icl=1}^{Icl} ABSLPA_{icl,t-19} * \Delta Cp_{icl,t=t*+19} - \sum_{z=1}^{Z} ABSLPA_{z,t-19} * \Delta Cp_{z,t=t*+19}\Bigg)\end{aligned}$$

Where:

ΔCBSLPAt: Total variation of the baseline carbon stock in the Project Area in year t (tCO2-e);

ABSLPAicl, t: Area of the initial forest class icl deforested at time t within the Project Area in the case of the baseline (ha);

ABSLPAicl, t-1: Area of the initial forest class icl deforested at time t-19 within the Project Area in the case of the baseline (ha);

ABSLPAicl, t = t-19: Initial class area of icl forest cleared at time t-19 within the Project Area in case of baseline (ha);

ΔCpicl, t = t *: The average variation factor of the carbon stock for the carbon stock fixes the initial forest class icl applicable at time t (according to Table 20.a) (tCO2-e.ha-1);

ΔCpicl, t = t * + 19: The average variation factor of the carbon stock for the carbon pool fixes the initial forest class icl applicable at time t = t * + 19 (20th year after deforestation, (according to Table 20.a VM0015) (tCO2-e. ha-1);

ABSLPAz, t: Area of the "deforested" z-zone at time t within the Project Area in the case of the baseline (ha);

ABSLPAz, t-1: Area of the "deforested" z-zone at time t- 1 in the Project Area in the case of the baseline (ha);

ABSLPAz, t-19: Area of the "deforested" z-zone at time t-19 in the Project Area in the case of the baseline (ha);

ΔCpz, t = t * : Average variation factor of the carbon stock for the z carbon stock applicable at time t = t * (according to table 20.b VM0015) (tCO2-e.ha-1);

ΔCpz, t = t + 1: Average variation factor in the carbon stock for the applicable carbon stock at time t = t * + 1 ((= second year after deforestation, according to Table 20.b VM0015) (tCO2-e.ha-1);

ΔCpz, t = t * + 19: Average variation factor in the carbon stock for the applicable carbon stock at time t = t * + 19 ((= 20 years after deforestation, according to Table 20.b VM0015) (tCO2-e.ha -1).

Table 36: Baseline of carbon stock variation in the Project Area (Table 21b of Methodology VM0015).

Carbon stock changes per initial forest class *icl*		Total carbon stock change of initial forest class in the project area	
ID_{icl}>	1	$\Delta CBSLPA_{icl,t}$	$\Delta CBSLPA_{icl}$
Name>	Forest	annual	cumulative
Project Year *t*	tCO_2-e	tCO_2-e	tCO_2-e
2023	445.233,6	445.233,6	445.233,6
2024	304.994,5	304.994,5	750.228,1
2025	406.821,6	406.821,6	1.157.049,7
2026	444.679,2	444.679,2	1.601.728,9
2027	500.629,8	500.629,8	2.102.358,8
2028	344.561,8	344.561,8	2.446.920,5
2029	571.875,5	571.875,5	3.018.796,0
2030	621.803,9	621.803,9	3.640.599,9
2031	562.759,1	562.759,1	4.203.359,0
2032	693.503,7	693.503,7	4.896.862,7

Carbon stock changes per post-deforestation zone z		Total carbon stock change of post-deforestation zones in the project area	
ID_{iz}>	1	$\Delta CBSLPA_{z,t}$	$\Delta CBSLPA_{z}$
Name>	Zone 1	annual	cumulative
Project Year *t*	tCO_2-e	tCO_2-e	tCO_2-e
2023	0,0	0,0	0,0
2024	8.062,1	8.062,1	8.062,1
2025	13.298,2	13.298,2	21.360,3
2026	20.191,9	20.191,9	41.552,2
2027	27.526,1	27.526,1	69.078,4
2028	35.612,7	35.612,7	104.691,1
2029	40.585,8	40.585,8	145.276,8
2030	49.498,1	49.498,1	194.774,9
2031	58.997,7	58.997,7	253.772,6
2032	67.090,4	67.090,4	320.862,9

Total net carbon stock change of the project area	
$\Delta CBSLPA_{t}$	$\Delta CBSLPA$
annual	cumulative
tCO_2-e	tCO_2-e
445.233,6	445.233,6
296.932,4	742.166,0
393.523,4	1.135.689,5
424.487,3	1.560.176,7
473.103,7	2.033.280,4
308.949,1	2.342.229,4
531.289,8	2.873.519,2
572.305,7	3.445.825,0
503.761,4	3.949.586,4
626.413,4	4.575.999,7

Table 37: Baseline of carbon stock variation in the leakage belt area (Table 21c of Methodology VM0015).

Changes in Carbon stock per forest initial classes *icl*		Changes in total cabon storage at initial forest class within Leakage Belt		Changes in carbon storage per zone post deforestation		Changes in the total carbon storage of zones post deforestation within Leakage Belt		Changes in carbon net storage within Leakage Belt	
ID_{icl}>	1	$\Delta CBSLLK_{icl,t}$	$\Delta CBSLLK_{icl}$	ID_{iz}>	1	$\Delta CBSLLK_{z,t}$	$\Delta CBSLLK_{z}$	$\Delta CBSLLK_{t}$	$\Delta CBSLLK$
Name>	Forest	Annual	Accumulated	Name>	Zone 1	Annual	Acnumulated	Annual	Acnumulated
Project Year *t*	tCO_2-e	tCO_2-e	tCO_2-e	Project Year *t*	tCO_2-e	tCO_2-e	tCO_2-e	tCO_2-e	tCO_2-e
2023	1.418.802,0	1.418.802,0	1.418.802,0	2023	0,0	0,0	0,0	1.418.802,0	1.418.802,0
2024	1.405.397,9	1.405.397,9	2.824.199,9	2024	25.691,1	25.691,1	25.691,1	1.379.706,9	2.798.508,9
2025	1.519.105,2	1.519.105,2	4.343.305,1	2025	50.226,0	50.226,0	75.917,1	1.468.879,1	4.267.388,0
2026	1.898.296,3	1.898.296,3	6.241.601,4	2026	75.947,7	75.947,7	151.864,7	1.822.348,7	6.089.736,7
2027	1.712.193,8	1.712.193,8	7.953.795,2	2027	107.621,1	107.621,1	259.485,8	1.604.572,7	7.694.309,4
2028	1.789.538,7	1.789.538,7	9.743.333,9	2028	134.798,5	134.798,5	394.284,3	1.654.740,2	9.349.049,6
2029	1.765.717,7	1.765.717,7	11.509.051,6	2029	162.410,3	162.410,3	556.694,6	1.603.307,5	10.952.357,0
2030	1.844.519,2	1.844.519,2	13.353.570,8	2030	188.609,0	188.609,0	745.303,6	1.655.910,1	12.608.267,2
2031	1.966.857,5	1.966.857,5	15.320.428,3	2031	215.303,3	215.303,3	960.606,9	1.751.554,2	14.359.821,4
2032	2.000.811,0	2.000.811,0	17.321.239,3	2032	243.263,7	243.263,7	1.203.870,6	1.757.547,3	16.117.368,7

Baseline non-CO2 emissions from forest fires (step 6.2 VM0015)

Non-CO2 emissions were not considered and accounted for for the Ateles REDD+ Project, due to the low risk in the Project Area.

3.2.2. Project Emissions

3.2.2.1.1. Step 7 of VM0015 - Ex-ante estimate of actual variations of carbon stock and non-CO2 emissions in the Project area

Non-CO2 emissions were not considered and accounted for the Ateles REDD+ Project.

Ex-ante estimate of actual changes in carbon stocks (Step 7.1 VM0015)

Ex-ante estimate of actual changes in carbon reserves due to planned activities (Step 7.1.1 VM0015)

The Ateles REDD+ Project Area does not have any planned activities, that is, no activities are carried out in the project area such as timber and non-timber forest management, nor is it intended to have any planned activities in the future. In this sense, the planned wood extraction, charcoal production and firewood collection were not considered and accounted for the Ateles REDD+ Project since these activities are not performed in the Project Area.

Optional accounting for increased carbon stocks

The ex-ante estimate of the increase in carbon stock by regeneration after management activities was not considered by conservative measure.

Ex-ante estimate of carbon stock changes due to unplanned unavoidable deforestation within the Project Area (Step 7.1.2 VM0015).

Unavoidable and significant unplanned deforestation is not expected in the Project scenario, due to the implementation of effective monitoring of forest cover, the strengthening of the degree of governance in the area due to the management activity, the activities foreseen by the Project and the greater alignment with the communities, with this, the project is expected to reach high levels of effectiveness during its 30 years of duration.

However, some unplanned deforestation may occur in the Project Area, depending on the effectiveness of the proposed activities, which cannot be measured ex-ante. The ex post measurements prepared for the monitoring report will be important in determining actual emission reductions.

To allow ex-ante projections, we started from a conservative assumption about the effectiveness of the activities proposed to define the Effectiveness Index (EI). The estimated IE value is used to multiply the base projections by the factor (1 - IE) and the result was considered to be the estimated ex-ante emissions from unplanned deforestation in the case of the Project. To calculate the actual ex-ante variation in carbon stock due to unplanned inevitable deforestation, equation 16 of VM0015 Methodology version 1.1, presented below.

$$\Delta CUDdPAt = \Delta CBSLt * (1 - EI)$$

Where:

∆CUDdPAt: Total ex-ante variation of the actual carbon stock due to unplanned and unavoidable deforestation in year t in the Project Area (tCO2-e);

∆CBSLt: Total variation in the baseline carbon stock in the year, in the Project Area (tCO2- e);

IE: Estimated Ex-ante Effectiveness Index;

t: 1, 2, 3 ... T, year of the credit period of the proposed project (dimensionless)

Based on the history of deforestation in the area before the start of the project, the Effectiveness Index (EI) of the project activities was conservatively assumed to be 90% in the first five years of implementation, and that this value will gradually increase with its efficiency over the years.

Ex-ante estimate of effective net changes in carbon stocks in the project area (Step 7.1.3 VM0015)

The changes in the carbon stock related to the planned activities and the effectiveness of the Project are presented in Table 38.

Table 38: Ex-ante estimates of the net carbon reduction in the Project Area over the Project scenario (Table 27 of VM0015).

Project Year t	Total carbon stock decrease due to planned activities		Total carbon stock increase due to planned activities		Total carbon stock decrease due to unavoided unplanned deforestation		Total carbon stock change in the project case	
	annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative
	$\Delta CPAdPA_t$	$\Delta CPAdPA$	$\Delta CPAiPA_t$	$\Delta CPAiPA$	$\Delta CUDdPA_t$	$\Delta CUDdPA$	$\Delta CPSPA_t$	$\Delta CPSPA$
	tCO_2e	tCO_2e	tCO_2e	tCO_2e	tCO_2e	tCO_2e	tCO_2e	tCO_2e
2023	0,0	0,0	0,0	0,0	44.523,4	44.523,4	44.523,4	44.523,4
2024	0,0	0,0	0,0	0,0	29.693,2	74.216,6	29.693,2	74.216,6
2025	0,0	0,0	0,0	0,0	39.352,3	113.568,9	39.352,3	113.568,9
2026	0,0	0,0	0,0	0,0	42.448,7	156.017,7	42.448,7	156.017,7
2027	0,0	0,0	0,0	0,0	47.310,4	203.328,0	47.310,4	203.328,0
2028	0,0	0,0	0,0	0,0	24.715,9	228.044,0	24.715,9	228.044,0
2029	0,0	0,0	0,0	0,0	42.503,2	270.547,1	42.503,2	270.547,1
2030	0,0	0,0	0,0	0,0	40.061,4	310.608,5	40.061,4	310.608,5
2031	0,0	0,0	0,0	0,0	35.263,3	345.871,8	35.263,3	345.871,8
2032	0,0	0,0	0,0	0,0	37.584,8	383.456,6	37.584,8	383.456,6

Ex-ante estimate of actual non-CO2 emissions from forest fires (Step 7.2 VM0015)

Non-CO2 emissions from fire were not accounted for in the baseline scenario.

Total ex-ante estimates for the project area (Step 7.3 VM0015)

Table 39 presents the expected net changes and non-CO2 emissions in the Project Area. If these emissions occur during the development of the Project activities, they will be monitored and reported to verify if there will be an increase in the projected emissions in the Project scenario.

Table 39: Total ex-ante estimate of net changes in carbon stock and non-CO2 emissions in the project area (Table 29 of VM0015).

Project Year t	Total ex ante carbon stock decrease due to planned activities		Total ex ante carbon stock increase due to planned activities		Total ex ante carbon stock decrease due to unavoided unplanned deforestation		Total ex ante net carbon stock change		Total ex ante estimated actual non-CO_2 emissions from forest fires in the project area	
	annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative
	$\Delta CPAdPA_t$	$\Delta CPAdPA$	$\Delta CPAiPA_t$	$\Delta CPAiPA$	$\Delta CUDdPA_t$	$\Delta CUDdPA$	$\Delta CPSPA_t$	$\Delta CPSPA$	$EBBPSPA_t$	$EBBPSPA$
	tCO_2e	tCO_2e	tCO_2e	tCO_2e	tCO_2e	tCO_2e	tCO_2e	tCO_2e	tCO_2e	tCO_2e
2023	0,0	0,0	0,0	0,0	44.523,4	44.523,4	44.523,4	44.523,4	0,0	0,0
2024	0,0	0,0	0,0	0,0	29.693,2	74.216,6	29.693,2	74.216,6	0,0	0,0
2025	0,0	0,0	0,0	0,0	39.352,3	113.568,9	39.352,3	113.568,9	0,0	0,0
2026	0,0	0,0	0,0	0,0	42.448,7	156.017,7	42.448,7	156.017,7	0,0	0,0
2027	0,0	0,0	0,0	0,0	47.310,4	203.328,0	47.310,4	203.328,0	0,0	0,0
2028	0,0	0,0	0,0	0,0	24.715,9	228.044,0	24.715,9	228.044,0	0,0	0,0
2029	0,0	0,0	0,0	0,0	42.503,2	270.547,1	42.503,2	270.547,1	0,0	0,0
2030	0,0	0,0	0,0	0,0	40.061,4	310.608,5	40.061,4	310.608,5	0,0	0,0
2031	0,0	0,0	0,0	0,0	35.263,3	345.871,8	35.263,3	345.871,8	0,0	0,0
2032	0,0	0,0	0,0	0,0	37.584,8	383.456,6	37.584,8	383.456,6	0,0	0,0

3.2.3. Leakage

3.2.3.1. VM0015 Step 8 - Ex-ante leak estimation

Ex-ante estimate of the decrease in carbon stock and the increase in GHG emissions due to leak prevention measures (Step 8.1 VM0015)

The leak prevention measures will take place within the limits of the leak management areas. As described in section 2.1.11, three activities proposed by the Project will contribute as management measures for the spill: "Monitoring of deforestation and hot spots", "Heritage surveillance program" and "Sustainable production practices". Thus, no activities are planned that encourage the opening of new areas for agricultural or pasture production, or any other activities that reduce carbon stocks and increase GHG emissions compared to the baseline scenario.

The monitoring of the activities developed that act as leak management will be performed as indicated and will be reported in all Project verification events.

Changes in carbon stock due to activities implemented in leak management areas (Step 8.1.1 VM0015)

Table 30c of VM0015 is not applicable, as no reduction in carbon stocks is expected due to the execution of activities. In case of significant changes in carbon stock, these activities will be monitored, accounted for and reported.

Ex-ante estimation of methane (CH4) and nitrous oxide (N20) emissions from grazing animals (step 8.1.2 VM0015)

As mentioned above, the development of activities that generate a significant increase in CH4 and N2O emissions from grazing animals are not foreseen within the Project activities. Therefore, tables 31 and 32 of VM0015 are not applicable.

Estimated ex-ante total of carbon stock changes and increased greenhouse gas emissions due to leak prevention measures (Step 8.1.3 VM0015)

Table 33 of VM0015 does not apply (justifications presented above).

Ex-ante estimate of decreased carbon stocks and increased GHG emissions due to activity displacement leakage (Step 8.2 VM0015)

Activities that will cause deforestation within the Project Area in the case of the baseline may be shifted outside the project boundaries due to the implementation of the AUD project activity. A greater decrease in carbon stocks within the leak range during the project scenario than those predicted ex-ante would indicate displacement of deforestation activities due to the project.

The ex-ante activity displacement leak was calculated based on the predicted combined effectiveness of the proposed leak prevention measures and project activities. As explained above, the Project will seek to prevent deforestation through the activities of "Monitoring deforestation and hot spots", "Heritage surveillance program" and "Sustainable production practices".

The "Monitoring of deforestation and hot spots" activity aims to remotely monitor the entire area of the farm and its surroundings, through the processing and analysis of satellite images, to assess human pressure, and prepare bulletins informing the points of deforestation and hot spots to strengthen surveillance actions. Through the activity called "Heritage Surveillance Program", it is intended to structure the inspection activities inside the farm, by training surveillance teams to contain illegal deforestation and invasions, as well as the maintenance of forest cover. The activity related to "Sustainable production practices", on the other hand, aims to enable allied food security, diversification of income sources, strengthening of the value chain and forest conservation through the strengthening of sustainable agricultural practices.

Although the Project aims to reach 100% of agents at baseline, it was conservatively considered a “Leak Displacement Factor”. To calculate the ex-ante variation of the actual carbon stock due to unplanned unavoidable deforestation, an equation similar to equation 16 of the VM0015 Methodology version 1.1 presented in Step 7.1.2 was used; however, make an adaptation by multiplying the changes of the estimated base carbon stock for the Project Area by a "Displacement Leakage Factor" (DLF) that represents the percentage of deforestation that is expected to be displaced outside the project boundaries, starting with an index of 10% and giving it over the life of the project. The equation is presented below:

$$\Delta CADLKt = \Delta CBSLPAt * DLF$$

Where:

ΔCADLKt: Total decrease in carbon stock due to displacement of deforestation in year t (tCO2e);

ΔCBSLPAt: Total change in carbon stock from baseline in Project Area in year t (tCO2e);

DLF: Leakage displacement factor (%).

Thus, a displacement factor of 10% was adopted in the first five years. So, the reduction of the leakage displacement factor is gradual, already considering the influence of the project in this context. Thus, the displacement factor due to leakage tends to approach zero during the 40 years of project implementation. Ex-ante estimates of leakage due to change of activity for the first fixed reference period are found in Table 40 and Table 41.

Table 40: Ex-ante leakage estimated due to activity displacement (Table 34 of Methodology VM0015 version 1.1).

Project Year t	Total ex ante estimated decrease in carbon stocks due to displaced deforestation		Total ex ante estimated increase in GHG emissions due to displaced forest fires	
	annual $\Delta CADLK_t$ tCO_2e	cumulative $\Delta CADLK$ tCO_2e	annual $EADLK_t$ tCO_2e	cumulative $EADLK$ tCO_2e
2023	44.523,4	44.523,4	0,0	0,0
2024	29.693,2	74.216,6	0,0	0,0
2025	39.352,3	113.568,9	0,0	0,0
2026	42.448,7	156.017,7	0,0	0,0
2027	47.310,4	203.328,0	0,0	0,0
2028	24.715,9	228.044,0	0,0	0,0
2029	42.503,2	270.547,1	0,0	0,0
2030	40.061,4	310.608,5	0,0	0,0
2031	35.263,3	345.871,8	0,0	0,0
2032	37.584,8	383.456,6	0,0	0,0

Table 41: Total ex-ante leakage estimate (Table 35 of the VM0015 Methodology version 1).

Project Year *t*	Total ex ante GHG emissions from increased grazing activities		Total ex ante increase in GHG emissions due to displaced forest fires		Total ex ante decrease in carbon stocks due to displaced deforestation		Carbon stock decrease due to leakage prevention measures		Total net carbon stock change due to leakage		Total net increase in emissions due to leakage	
	annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative
	$EgLK_t$	EgLK	$EADLK_t$	EADLK	$\Delta CADLK_t$	ΔCADLK	$\Delta CLPMLK_t$	ΔCLPMLK	ΔCLK_t	ΔCLK	ELK_t	ELK
	tCO_2e	tCO_2e	tCO_2e	tCO_2e	tCO_2e	tCO_2e	tCO_2e	tCO_2e	tCO_2e	tCO_2e	tCO_2e	tCO_2e
2022	0,0	0,0	0,0	0,0	44.523,4	44.523,4	0,0	0,0	44.523,4	44.523,4	0,0	0,0
2023	0,0	0,0	0,0	0,0	29.693,2	74.216,6	0,0	0,0	29.693,2	74.216,6	0,0	0,0
2024	0,0	0,0	0,0	0,0	39.352,3	113.568,9	0,0	0,0	39.352,3	113.568,9	0,0	0,0
2025	0,0	0,0	0,0	0,0	42.448,7	156.017,7	0,0	0,0	42.448,7	156.017,7	0,0	0,0
2026	0,0	0,0	0,0	0,0	47.310,4	203.328,0	0,0	0,0	47.310,4	203.328,0	0,0	0,0
2027	0,0	0,0	0,0	0,0	24.715,9	228.044,0	0,0	0,0	24.715,9	228.044,0	0,0	0,0
2028	0,0	0,0	0,0	0,0	42.503,2	270.547,1	0,0	0,0	42.503,2	270.547,1	0,0	0,0
2029	0,0	0,0	0,0	0,0	40.061,4	310.608,5	0,0	0,0	40.061,4	310.608,5	0,0	0,0
2030	0,0	0,0	0,0	0,0	35.263,3	345.871,8	0,0	0,0	35.263,3	345.871,8	0,0	0,0
2031	0,0	0,0	0,0	0,0	37.584,8	383.456,6	0,0	0,0	37.584,8	383.456,6	0,0	0,0

3.2.4. Net GHG Emission Reductions and Removals

3.2.4.1. VM0015 Step 9 – Reduction of net anthropogenic emissions ex-ante

Significance assessment (Step 9.1 VM0015)

Using the latest document "EB-CDM approved "Tool for testing significance of GHG emissions in A/R CDM Project activities “it was possible to verify that above-ground biomass will contribute 74% of the expected emissions in the base scenario, below-ground biomass with 13% and dead wood with 13%. Therefore, they all represent significant sources of emission (above 5%).

Calculation of ex-ante estimates of total net GHG emission reductions (Step 9.2 VM0015)

The equation below was used as suggested by the VM0015 version 1.1 methodology to estimate the net ex-ante reduction of Project emissions. The result is presented in (Table 36 of the Methodology version VM0015 1.1).

$$\Delta REDDt = (\Delta CBSLPAt + EBBBSLPAt) - (\Delta CPSPAt + EBBPSPAt) - (\Delta CLKt + ELKt)$$

Where:

∆REDDt: Reduction of ex-post anthropogenic GHG emissions attributed to the project's AUD activity in year t (tCO2e);

∆CBSLPAt: Sum of changes in the basal carbon stock in the Project Area in year t (tCO2e);

EBBBSLPAt: Sum of basal emissions caused by biomass burning in the Project Area in year t (tCO2e);

∆CPSPAt: Sum of the variations of the ex-post carbon stock in the Project Area in year t (tCO2e);

EBBPSPAt: Sum of ex post emissions caused by the burning of biomass in the Project Area in year t (tCO2e);

∆CLKt: Sum of ex post variations of carbon stock due to leaks in year t (tCO2e);

ELKt: Sum of ex post emissions from leakage in year t (tCO2e);

t: 1, 2, 3 ... T, one year of the proposed credit period (no size).

Calculation of ex-ante Verified Carbon Units (VCUs) (Step 9.3 VM0015)

Equation 20 of Methodology VM0015 was used to estimate the number of VCU’s. The Risk Factor parameter was estimated using the Non Permanence Risk Tool VCS AFOLU, resulting in 10%. The result is presented in the below (Table 36 of the VM0015 Methodology version 1.1).

$$\Delta VCUt = \Delta REDDt - VBCt$$

$$VBCt = (\Delta CBSLPAt - \Delta CPSPAt) * RFt$$

Where:

VCUt: Number of Verified Carbon Units that can be traded in year t (tCO2e);

∆REDDt: Reduction of ex-post anthropogenic GHG emissions attributed to the project's AUD activity in year t (tCO2e);

VBCt: Buffer credit number deposited in buffer VCS in year t (t CO2-e);

∆CBSLPAt: Sum of changes in the basal carbon stock in the Project Area in year t (tCO2e);

∆CPSPAt: Sum of the ex post variations of the carbon stock in the Project Area in year t (tCO2e);

RFt: Risk factor used to calculate the VCS credits buffer (%);

t: 1, 2, 3 ... T, one year of the proposed credit period (no size).

Table 42: Ex-ante estimation of net anthropogenic GHG emission reductions (ΔREDDt) and Verified Carbon Units (VCUt) (Table 36 of Methodology VM0015).

Project Year t	Baseline carbon stock changes		Baseline GHG emissions		Ex ante project carbon stock changes		Ex ante project GHG emissions		Ex ante leakage carbon stock changes		Ex ante leakage GHG emissions		Ex ante net anthropogenic GHG emission reductions		Ex ante VCUs tradable		Ex ante buffer credits	
	annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative	annual	cumulative
	$\Delta CBSLPA_t$	$\Delta CBSLPA$	$\Delta EBBSLPA_t$	$\Delta EBBSLPA$	$\Delta CPSPA_t$	$\Delta CPSPA$	$EBBPSPA_t$	$EBBPSPA$	ΔCLK_t	ΔCLK	ELK_t	ELK	$\Delta REDD_t$	$\Delta REDD$	VCU_t	VCU	VCB_t	VCB
	tCO_2-e	tCO_2-e	tCO_2-e	tCO_2-e	tCO_2e	tCO_2e	tCO_2e	tCO_2e	tCO_2e	tCO_2e	tCO_2e	tCO_2e	tCO_2e	tCO_2e	tCO_2e	tCO_2e	tCO_2e	tCO_2e
2023	445234	445234	0	0	44523	44523	0	0	44523	44523	0	0	356187	356187	308102	308102	48085	48085
2024	296932	742166	0	0	29693	74217	0	0	29693	74217	0	0	237546	593733	205477	513579	32069	80154
2025	393523	1135689	0	0	39352	113569	0	0	39352	113569	0	0	314819	908552	272318	785897	42501	122654
2026	424487	1560177	0	0	42449	156018	0	0	42449	156018	0	0	339590	1248141	293745	1079642	45845	168499
2027	473104	2033280	0	0	47310	203328	0	0	47310	203328	0	0	378483	1626624	327388	1407030	51095	219594
2028	308949	2342229	0	0	24716	228044	0	0	24716	228044	0	0	259517	1886142	225409	1632439	34108	253702
2029	531290	2873519	0	0	42503	270547	0	0	42503	270547	0	0	446283	2332425	387629	2020068	58654	312357
2030	572306	3445825	0	0	40061	310609	0	0	40061	310609	0	0	492183	2824608	428314	2448382	63869	376226
2031	503761	3949586	0	0	35263	345872	0	0	35263	345872	0	0	433235	3257843	377015	2825397	56220	432446
2032	626413	4576000	0	0	37585	383457	0	0	37585	383457	0	0	551244	3809086	480584	3305981	70659	503105

3.3. Monitoring

3.3.1. Data and Parameters Available at Validation

Data / Parameter	$Ctot_{cl}$
Data unit	t CO_2-and ha_{-1}
Description	Average stock of CO_{2-e} per hectare in all forest class carbon pools used in the baseline scenario
Source of data	Calculated by allometric equations, conversion factors from the literature and data measured in the field
Value applied	445.9
Justification of choice of data or description of measurement methods and procedures applied	Estimates of aboveground, belowground and dead wood carbon stock were obtained using forest inventory data and allometric equations[101] developed in areas similar to the Project Area and adopting standard values from the literature or recommended by VCS VM0015
Purpose of data	• Determination of the baseline scenario • Baseline emissions calculation • Project emission calculation • Leakage calculation
Comments	See document: Forest Carbon Stock Ateles REDD+ Project

Data / Parameter	Cab_{cl}
Data unit	t CO_2-and ha_{-1}
Description	Average stock per hectare of CO_{2-e} biomass in the aboveground pool
Source of data	Calculated by allometric equations, conversion factors from the literature and data measured in the field
Value applied	325.8
Justification of choice of data or description of measurement methods and procedures applied	Estimates of aboveground carbon stock were obtained using forest inventory data and allometric[22] equations developed in areas similar to the Project Area and adopting standard values from the literature or recommended by VCS VM0015

[101] Nogueira, E.M.; Fearnside, P.M.; Nelson, B.W. Barbosa, R.I.; Keizer, E.W.H. Estimates of forest biomass in the Brazilian Amazon: New allometric equations and adjustments to biomass from wood-volume inventories. Forest Ecology and Management, v. 256, n. 11, p. 1853-1867, 2008.

Purpose of data	• Determination of the baseline scenario • Baseline emissions calculation • Project emission calculation • Leakage calculation
Comments	See document: Forest Carbon Stock Ateles REDD+ Project

Data / Parameter	Cbb_{cl}
Data unit	t CO_2-and ha_{-1}
Description	Average stock per hectare of $CO_{2\text{-}e}$ biomass in the underground pool
Source of data	Calculated by allometric equations, conversion factors from the literature and data measured in the field
Value applied	69.7
Justification of choice of data or description of measurement methods and procedures applied	Estimates of belowground carbon stock were obtained using forest inventory data and allometric[22] equations developed in areas similar to the Project Area and adopting standard values from the literature or recommended by VCS VM0015
Purpose of data	• Determination of the baseline scenario • Baseline emissions calculation • Project emission calculation • Leakage calculation
Comments	See document: Forest Carbon Stock Ateles REDD+ Project

Data / Parameter	Cdw_{cl}
Data unit	t CO_{2}-and ha_{-1}
Description	Average stock per hectare of $CO_{2\text{-}e}$ biomass in standing and ground dead wood
Source of data	Calculated by allometric equations [22], density and volume equations, literature conversion factors, and forest inventory data
Value applied	50.4
Justification of choice of data or description of measurement methods and procedures applied	Carbon stock estimates in standing and ground deadwood were obtained using forest inventory data, allometric equations [22], and density and volume equations recommended by VCS VM0015

Purpose of data	• Determination of the baseline scenario • Baseline emissions calculation • Project emission calculation • Leakage calculation
Comments	See document: Forest Carbon Stock Ateles REDD+ Project

Data / Parameter	DAP
Data unit	cm
Description	Diameter at breast height (130 cm) for each tree
Source of data	Measured in field by Biodendro Consultoria Florestal
Value applied	See worksheet with field data
Justification of choice of data or description of measurement methods and procedures applied	Requirement demanded by VCS methodology VM0015. Forest inventory data collected less than 10 years ago in multiple plots located in wide spatial distribution. Parameter measured only in trees with a diameter at breast height equal to or greater than 5 cm
Purpose of data	Variable used in the allometric equations for calculating above and below ground biomass and in the volume equation for calculating standing deadwood biomass
Comments	

Data / Parameter	H
Data unit	meters
Description	Tree height
Source of data	Measured in field by Biodendro Consultoria Florestal
Value applied	See worksheet with field data
Justification of choice of data or description of measurement methods and procedures applied	Requirement demanded by VCS methodology VM0015. Forest inventory data collected less than 10 years ago in multiple plots located in wide spatial distribution
Purpose of data	Variable used in the volume equation to calculate standing dead wood biomass

Comments	

Data / Parameter	$Volume_{dc}$
Data unit	m^3
Description	Volume of dead wood on the ground
Source of data	Measured in laboratory b Biodendro Consultoria Florestal
Value applied	See worksheet with field data
Justification of choice of data or description of measurement methods and procedures applied	Requirement demanded by VCS methodology VM0015. Calculation of the amount of dead wood biomass on the ground in rotten, intermediate and hard samples with a diameter greater than 10 cm.
Purpose of data	Calculation of the average stock per hectare of carbon biomass in dead wood on the ground
Comments	

Data / Parameter	D_{dc}
Data unit	t m_{-3}
Description	Deadwood Density
Source of data	According to a study by Keller et al., 2004.
Value applied	0.483
Justification of choice of data or description of measurement methods and procedures applied	Requirement demanded by VCS methodology VM0015. Calculation of the amount of dead wood biomass on the ground in rotten, intermediate and hard samples with a diameter greater than 10 cm.
Purpose of data	Calculation of the average stock per hectare of carbon biomass in dead wood on the ground
Comments	

Data / Parameter	R_j
Data unit	dimensionless
Description	Root-shoot ratio

Source of data	Literature: GOFC-GOLD, 2008[102] and VCS VM0015, 2012
Value applied	0.24
Justification of choice of data or description of measurement methods and procedures applied	Suggested default value for tropical rainforest regions with aboveground biomass greater than 125 t ha-1 (GOFC-GOLD, 2008; VCS Methodology VM0015, p. 140, Table -2)
Purpose of data	Calculation of the average stock per hectare of carbon biomass in below ground pools
Comments	

Data / Parameter	CF
Data unit	t C
Description	Carbon fraction in biomass
Source of data	Literature: IPCC, 2006[103]
Value applied	0.47
Justification of choice of data or description of measurement methods and procedures applied	IPCC default value
Purpose of data	Determination of the carbon biomass fraction in the above and below ground pools and in dead wood
Comments	

Data / Parameter	44/12
Data unit	t CO_2-and
Description	Conversion factor from mass of carbon to mass of $CO_{2\text{-e}}$
Source of data	Literature: IPCC, 2006[25]
Value applied	44/12

[102] http://www.gofcgold.wur.nl/redd/sourcebook/GOFC-GOLD_Sourcebook.pdf
[103] IPCC. (2006). Agriculture, Forestry and Other Land Use. In: Eggleston H.S., Buendia L., Miwa K., Ngara T. and Tanabe K. (eds). 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Vol. 4. IGES, Japan.

Justification of choice of data or description of measurement methods and procedures applied	IPCC default value
Purpose of data	Determination of the average stock of $CO_{2\text{-}e}$ biomass in above and below ground pools and in dead wood
Comments	

3.3.2. Data and Parameters Monitored

The selected data and parameters that are contemplated and described are directed to respond and measure the effectiveness of the activities developed for the General and Climate scope of the Project, defined in Section 2.1.11. For the social scope, the chosen data and parameters that will be collected in the checks have been entered in Section 4.4.1 and those related to the biodiversity scope have been included in Section 5.4.1.

Data to be collected for the Project Climate scope

Data / Parameter	ABSLPAicl,t
Data unit	Hectare (ha)
Description	Forest cover areas converted to non-forest cover areas within the Project area in year $_t$ of the Ateles REDD+ Project
Source of data	Calculated through remote sensing and scientifically available data
Description of measurement methods and procedures to be applied	Monitoring of the forest component through satellite images and scientifically proven data
Frequency of monitoring/recording	Annual
Value applied	Average annual deforestation in the Project Area during the calculated baseline: 2.212 ha
Monitoring equipment	Remote sensing images of digital processing program and geographic information systems
QA/QC procedures to be applied	In mapping changes in forest cover and defining land use classes, data obtained at medium spatial resolution (between 10m and 100m) will be used. Subsequently, for validation and refinement of the described mapping, data obtained by high resolution sensors (up to 5m pixels) will be used. The minimum accuracy of the land use and land cover classification map is 80%
Purpose of data	Calculation of emissions in the Project Area

Calculation method	If unplanned deforestation areas are detected, the Forest Coverage Benchmark Map will be updated by map algebra
Comments	-

Data / Parameter	∆CUDdPAt
Data unit	tCO2e
Description	Total change of actual carbon stock due to unplanned deforestation unavoidable in year $_t$ in the Ateles REDD+ Project Area
Source of data	Calculated through the detected areas of forest loss by unplanned deforestation in the Project Area and the average carbon stock
Description of measurement methods and procedures to be applied	∆BSLPAt indicator follow-up to later calculation of carbon stock change originated from unplanned and unavoided deforestation
Frequency of monitoring/recording	Annual
Value applied	To be accounted for after the start of the Project
Monitoring equipment	Emissions spreadsheet
QA/QC procedures to be applied	Good practices applied to ABSLPAt calculation
Purpose of data	Calculation of emissions in the Project Area
Calculation method	The parameter is estimated from the multiplication of unplanned deforestation areas by the estimated average carbon stock value for the initial forest class. The sum of carbon stock residual emissions under the soil and in dead timber is also considered, as such pools have 1/10 annual decay, causing emissions along the years. Finally, it is subtracted from this result, carbon stock value estimated to Reference Region in a post-deforestation scenario, achieving carbon stock net value that was reduced by unplanned and unavoided deforestation
Comments	-

Data / Parameter	AUFPAicl,t
Data unit	Hectare (ha)
Description	Areas affected by forest fires in class icl where carbon stock recovery occurs in year $_t$ of the Ateles REDD+ Project
Source of data	Adequate sources of detection of forest fires and scars caused for identification and classification of affected areas

Description of measurement methods and procedures to be applied	Identification and classification of affected areas from appropriate sources of forest fire detection and scarring
Frequency of monitoring/recording	Whenever the occurrence of forest fires is identified
Value applied	To be accounted for after Project commencement and when any forest fire occurs
Monitoring equipment	Remote sensing images of digital processing program and geographic information system
QA/QC procedures to be applied	In validating and refining the mapping of areas affected by fires, data or images obtained from high-resolution sensors (up to 5m pixels) will be used. Minimum mapping accuracy is 80%
Purpose of data	Calculation of emissions in the Project Area
Calculation method	If affected areas are detected, the Forest Coverage Reference Mark Map will be updated by map algebra
Comments	-

Data / Parameter	∆CUFdPAt
Data unit	tCO2e
Description	Total reduction in carbon stock due to unplanned forest fires in year $_t$ in the Ateles REDD+ Project Area
Source of data	Calculated through the areas affected by forest fires in the Project Area and the average carbon stock
Description of measurement methods and procedures to be applied	Monitoring of the AUFPAicl indicator,t for later calculation of the carbon stock change from the areas affected by forest fires
Frequency of monitoring/recording	Whenever the occurrence of forest fires is identified
Value applied	To be accounted for after Project commencement and when any forest fire occurs
Monitoring equipment	Emissions spreadsheet
QA/QC procedures to be applied	Good practices applied in the calculation of AUFPAicl,t
Purpose of data	Calculation of emissions in the Project Area

Calculation method	The variation in carbon stock is estimated by multiplying the area affected by the forest fire and the average carbon stock per unit area
Comments	-

Data / Parameter	ACPAicl,t
Data unit	Hectare (ha)
Description	Analysis Area within the Ateles REDD+ Project Area affected by catastrophic events in class icl in year $_t$
Source of data	High-resolution satellite imagery
Description of measurement methods and procedures to be applied	Realization of photointerpretation of high-resolution satellite images identifying areas of forest cover affected by catastrophic events
Frequency of monitoring/recording	Whenever the occurrence of a catastrophic event is identified
Value applied	To be accounted for after Project commencement and when a catastrophic event occurs
Monitoring equipment	Remote sensing images and geographic information system
QA/QC procedures to be applied	In validating and refining the mapping of areas affected by catastrophic events, data or images obtained from high-resolution sensors (up to 5m pixels) will be used. Minimum mapping accuracy is 80%
Purpose of data	Calculation of emissions in the Project Area
Calculation method	If affected areas are detected, the Forest Coverage Reference Mark Map will be updated by map algebra
Comments	-

Data / Parameter	∆CUCdPAt
Data unit	tCO_2e
Description	Total reduction in carbon stock due to catastrophic events in year $_t$ in the Ateles REDD+ Project Area
Source of data	Calculated through the areas affected by catastrophic events in the Project Area and the average carbon stock
Description of measurement methods	Monitoring of ACPAicl indicator,t for subsequent calculation of carbon stock change from areas affected by catastrophic events

and procedures to be applied	
Frequency of monitoring/recording	Whenever a catastrophic event occurs
Value applied	To be accounted for after Project commencement and when a catastrophic event occurs
Monitoring equipment	Emissions spreadsheet
QA/QC procedures to be applied	Good practices applied in the calculation of ACPAicl,t
Purpose of data	Calculation of emissions in the Project Area
Calculation method	The carbon stock variation is estimated by multiplying the area affected by catastrophic events and the average carbon stock per area unit.
Comments	-

Data / Parameter	∆BSLLKicl,t
Data unit	Hectare (ha)
Description	Forest cover areas converted to non-forest cover areas within the leakage belt in year $_t$ of the Ateles REDD+ Project
Source of data	Calculated through remote sensing and scientifically available data
Description of measurement methods and procedures to be applied	Monitoring of the forest component through satellite images and scientifically proven data
Frequency of monitoring/recording	Annual
Value applied	To be accounted for after the start of the Project
Monitoring equipment	Remote sensing images of digital processing program and geographic information system
QA/QC procedures to be applied	In mapping changes in forest cover and defining land use classes, data with medium spatial resolution (between 10m and 100m) will be used. Subsequently, for the validation and refinement of the described mapping, data obtained by high-resolution sensors (up to 5m pixels) will be used. The minimum accuracy of the land use and land cover classification map is 80%
Purpose of data	Calculation of emissions in the Project Area
Calculation method	If unplanned deforestation areas are detected, the Forest Coverage Benchmark Map will be updated by map algebra

Comments	-

Data / Parameter	∆CADLKt
Data unit	tCO2e
Description	Total reduction in carbon stocks due to deforestation displaced in year $_t$ in the Leakage Belt of the Ateles REDD+ Project
Source of data	Calculated through the detected areas of forest loss in the Leakage Belt, the average carbon stock and the estimated loss in carbon stock at the baseline for the Leakage Belt
Description of measurement methods and procedures to be applied	Monitoring of the indicator ∆ BSLLKt for subsequent calculation of the change in carbon stock from unplanned and unavoidable deforestation in the Leakage Belt
Frequency of monitoring/recording	Annual
Value applied	To be accounted for after the start of the Project
Monitoring equipment	Emissions spreadsheet
QA/QC procedures to be applied	Good practices applied in the calculation of ∆BSLLKt
Purpose of data	Calculation of emissions in the Project Area
Calculation method	The parameter is estimated from the multiplication of areas of forest loss by the average carbon stock value estimated for the initial forest class. The sum of carbon stock residual emissions under the soil and in dead timber is also considered, as such pools have 1/10 annual decay, causing emissions along the years. Then, the value of the estimated carbon stock for the Reference Region in a post-deforestation scenario is subtracted from this result, obtaining the net value of the carbon stock that was reduced by displaced deforestation. Finally, the estimated loss of carbon stock in the Leakage Belt projected by the baseline is subtracted from this value.
Comments	-

Data / Parameter	∆CLPMLKt
Data unit	tCO2e
Description	Decrease in carbon stock due to leak prevention measures in year $_t$

Source of data	Monitoring report of project activities that have been implemented and other records related to leak prevention activities
Description of measurement methods and procedures to be applied	Monitoring of grazing activities following the guidelines of section 8.1.1 of the VM0015 v1.1 methodology
Frequency of monitoring/recording	Annual
Value applied	To be accounted for after the start of the Project
Monitoring equipment	Emissions spreadsheet
QA/QC procedures to be applied	To be defined when it is necessary to be used
Purpose of data	Calculation of emissions in the Leakage Belt
Calculation method	Emissions will be calculated using the guidelines in section 8.1.1 of methodology VM0015 v1.1
Comments	No leak prevention activities are foreseen that generate a decrease in the carbon stock

Data / Parameter	∆CADLKt
Data unit	tCO2e
Description	Total reduction in carbon stocks due to deforestation displaced in year $_t$ in the Leakage Belt of the Ateles REDD+ Project
Source of data	Calculated through the detected areas of forest loss in the Leakage Belt, the average carbon stock and the estimated loss in carbon stock at the baseline for the Leakage Belt
Description of measurement methods and procedures to be applied	Monitoring of the indicator ∆ BSLLKt for subsequent calculation of the change in carbon stock from unplanned and unavoidable deforestation in the Leakage Belt
Frequency of monitoring/recording	Annual
Value applied	To be accounted for after the start of the Project
Monitoring equipment	Emissions spreadsheet
QA/QC procedures to be applied	Good practices applied in the calculation of ∆BSLLKt
Purpose of data	Calculation of emissions in the Leakage Belt

Calculation method	Carbon stock variation is estimated by multiplying the detected area of forest loss in the Leakage Belt and the average carbon stock per unit area, minus the estimated carbon stock variation at baseline for the Leakage Belt
Comments	-

Data / Parameter	EgLKt
Data unit	tCO2e
Description	Emissions from animals in pastures in spill management areas in year $_t$
Source of data	Existing records on the practice of grazing
Description of measurement methods and procedures to be applied	Monitoring of grazing activities following the guidelines of section 8.1.2 of the VM0015 v1.1 methodology
Frequency of monitoring/recording	Annual
Value applied	To be accounted for after the start of the Project
Monitoring equipment	Emissions spreadsheet
QA/QC procedures to be applied	To be defined when performed
Purpose of data	Calculation of emissions in the Leakage Belt
Calculation method	Emissions will be calculated using the guidelines in section 8.1.2 of methodology VM0015 v1.1
Comments	No activities involving grazing are foreseen.

Data / Parameter	∆REDDt
Data unit	tCO2e
Description	Net reduction in anthropogenic greenhouse gas emissions attributable to the AUD Project activity in the year $_t$
Source of data	Calculated by subtracting the carbon stock rates in the baseline scenario from changes in carbon stock throughout the Project
Description of measurement methods and procedures to be applied	The calculation of net anthropogenic GHG emissions reductions attributable to Project activities will be calculated using equation 19 and using table 36 of the VM0015 v1.1 methodology

Frequency of monitoring/recording	Annual
Value applied	To be accounted for after the start of the Project
Monitoring equipment	Emissions spreadsheet
QA/QC procedures to be applied	Good practices applied to Project emission calculations
Purpose of data	This parameter will be used to measure the efficiency of the Project when calculating the net reductions of anthropogenic emissions by the Project over the years by comparing the baseline scenario
Calculation method	Emissions will be calculated using the guidelines in section 9.2 of methodology VM0015 v1.1
Comments	-

Data / Parameter	VCU,t
Data unit	tCO2e
Description	Number of Verified Carbon Units (VCUs) to be made available for commercialization in year $_t$
Source of data	Calculated by subtracting the net GHG emission reductions from the buffer
Description of measurement methods and procedures to be applied	VCU's calculation will be calculated using equations 20 21 and 22 and using table 36 of the VM0015 v1.1 methodology
Frequency of monitoring/recording	Annual
Value applied	To be accounted for after the start of the Project
Monitoring equipment	Emissions spreadsheet
QA/QC procedures to be applied	Good practices applied to Project emission calculations
Purpose of data	This parameter will be used to measure the amount of marketable carbon credits (VCU's) for the Project
Calculation method	Emissions will be calculated using the guidelines in section 9.3 of methodology VM0015 v1.1
Comments	-

Parameters monitored in the Project General scope

Data / Parameter	Number of Reports
Data unit	Number
Description	This parameter will be responsible for accounting for the amount of all material produced, in the form of a report, designed with considerations, clarifications and contents containing information regarding the implementation, monitoring and evaluation of the activities developed as well as the contents prepared to strengthen the governance of the Project
Source of data	Accounting through reports, meeting minutes, monitoring guides and memoranda developed by the Project, with emphasis on matters related to Project management (governance, implementation, monitoring and evaluation of activities)
Description of measurement methods and procedures to be applied	All documents that can be considered as “reports” produced for the Project will be stored in digital files throughout the Project's accreditation period. Thus, these reports from the activity of "Implementation, monitoring and evaluation of the activities developed by the Project" and "Strengthening of governance" will be monitored and accounted for.
Frequency of monitoring/recording	Annual
Value applied	To be accounted for after registration of the Project
Monitoring equipment	Not applicable
QA/QC procedures to be applied	The information systematized in these documents will be validated between the bidders, allowing greater reliability and quality of the data. In addition, the Project will undergo continuous evaluation of the information generated, allowing the identification of improvements in the collection and registration processes, and the incorporation of these in the strategic planning of the Project when they are identified
Purpose of data	Not applicable
Calculation method	Not applicable
Comments	-

Data / Parameter	Number of procedures and/or protocols
Data unit	Number
Description	This parameter will be responsible for accounting for the amount of all material produced, in the form of procedures and/or protocols, which will be established to promote improvements in governance,

	implementation, monitoring and evaluation of the activities developed by the Project
Source of data	Documents with procedures and protocols developed and established to guide and improve the activities of the general scope of the Project "Implementation, monitoring and evaluation of the activities developed by the Project" and "Strengthening of governance"
Description of measurement methods and procedures to be applied	All documents that can be considered protocols and/or procedures prepared for the Project will be stored in digital files throughout the Project's accreditation period. Thus, any and all procedures and protocols to be developed for the activity of "Implementation, monitoring and evaluation of activities developed" and "Strengthening of Governance" by the Project will be monitored and accounted for.
Frequency of monitoring/recording	Annual
Value applied	To be accounted for after registration of the Project
Monitoring equipment	Not applicable
QA/QC procedures to be applied	The systematized information will be validated between the proponents, allowing greater reliability and quality of the data. In addition, the Project will undergo continuous evaluation of the information generated, allowing the identification of improvements in the collection and registration processes, and the incorporation of these in the strategic planning of the Project when they are identified
Purpose of data	Improve and monitor the adaptive management of the Project on issues related to governance, implementation, monitoring and development of activities over the 30 years of the Project
Calculation method	Not applicable
Comments	-

Parameters monitored in the Climate scope

Data / Parameter	Number of reports
Data unit	Number
Description	This parameter will be responsible for accounting for the amount of all material produced, in the form of a report, designed to monitor deforestation as well as promote improvements in the equity surveillance of the Project
Source of data	Calculated through reports, meeting minutes, monitoring guides and bulletins developed and focused on matters related to the

	Project's climate scope (deforestation monitoring and patrimonial surveillance program)
Description of measurement methods and procedures to be applied	All documents that can be read as "reports" produced for the Project will be stored in digital files throughout the Project's accreditation period. Thus, these reports from the activity of "Monitoring deforestation" and the "Asset Surveillance Program" will be monitored and accounted for
Frequency of monitoring/recording	Annual
Value applied	To be accounted for after the start of the Project
Monitoring equipment	Not applicable
QA/QC procedures to be applied	Information systematized in reports will be validated among the proponents, allowing for greater reliability and quality of data. In addition, the Project will undergo continuous evaluation of the information generated, allowing the identification of improvements in the collection and registration processes, and the incorporation of these in the strategic planning of the Project when they are identified
Purpose of data	Not applicable
Calculation method	Not applicable
Comments	-

Data / Parameter	Number of procedures and/or protocols
Data unit	Number
Description	This parameter will be responsible for accounting for the quantity of all material produced, in the form of procedures and protocols, which will be established to develop and improve deforestation monitoring and asset surveillance of the Project
Source of data	Documents with procedures and protocols developed to guide and improve climate activities for the Project: "Monitoring of deforestation" and the "Asset Surveillance Program"
Description of measurement methods and procedures to be applied	All documents that can be read as procedures and protocols produced for the Project will be stored in digital files throughout the Project's accreditation period. Thus, these reports from the activities of "Monitoring deforestation" and the "Asset Surveillance Program" will be monitored and accounted for
Frequency of monitoring/recording	Annual
Value applied	To be accounted for after the start of the Project

Monitoring equipment	Not applicable
QA/QC procedures to be applied	Information systematized in the procedures and protocols will be validated among the proponents, allowing for greater reliability and quality of data. In addition, the Project will undergo continuous evaluation of the information generated, allowing the identification of improvements in the collection and registration processes, and the incorporation of these in the strategic planning of the Project when they are identified
Purpose of data	Not applicable
Calculation method	Not applicable
Comments	-

Data / Parameter	Number of trainings and/or interventions
Data unit	Number
Description	This parameter aims to measure the number of all courses and/or interventions carried out that will be defined and implemented throughout the Project, specifically for the "Asset Surveillance Program" activity. It is important to note that although the frequency is established to be accounted for annually, it is expected to decrease over time, as this indicator is associated with short and medium term actions for the Project
Source of data	Reports (e.g. follow-up report of project activities that were implemented), attendance lists of participants, contracts, photos, among other documents
Description of measurement methods and procedures to be applied	All reports and documents produced will be stored in digital files for the entire duration of the Project. Thus, the performance of training/interventions linked to the activity of "Asset Surveillance Program within the farm" will have records of its developments either by reports, attendance lists, contracts, photos, among other documents, which will be monitored and accounted for
Frequency of monitoring/recording	Annual
Value applied	To be accounted for after registration of the Project
Monitoring equipment	Not applicable
QA/QC procedures to be applied	Systematized training and/or intervention information will be validated among proponents, allowing greater reliability and quality of data. In addition, the Project will undergo continuous evaluation of the information generated, allowing the identification of improvements in the collection and registration processes, and the

	incorporation of these in the strategic planning of the Project when they are identified
Purpose of data	Demonstrate the courses and interventions the Project is undertaking to improve the property surveillance of the farm
Calculation method	Not applicable
Comments	-

Data / Parameter	Frequency of asset surveillance operations
Data unit	Number
Description	This parameter will account for the recording frequency of the number of surveillance operations carried out on the farm during the monitoring period
Source of data	There is currently no official record to account for the frequency of control and surveillance actions within the Project Area. Ideally, the counting of this data will be done by the reports (e.g. monitoring report of the project activities that have been implemented), monitoring sheets, occurrence records, etc.
Description of measurement methods and procedures to be applied	All documents produced will be stored in digital files for the entire duration of the Project. Thus, the performance of surveillance and patrol operations related to the activity of "Improvement of asset surveillance within the farm" will have records that will be monitored and accounted for
Frequency of monitoring/recording	Every 6 months
Value applied	To be accounted for after registration of the Project
Monitoring equipment	Not applicable
QA/QC procedures to be applied	To be established after Project registration
Purpose of data	Improve asset surveillance in containing unplanned deforestation of the Project Area and illegal activities
Calculation method	Not applicable
Comments	The monitoring sheets and reports of the patrimonial surveillance on the farm will be implemented from the validation of the Project

3.3.3. Monitoring Plan

The Climate Impacts Monitoring Plan will encompass key issues for demonstrating the reduction of emissions from deforestation and degradation due to avoided unplanned deforestation, in accordance with the applied methodology VM0015. Thus, the main objective is to monitor changes in the carbon stock throughout the Project's life cycle, resulting from changes in land use within the Project Area and in the Leakage Belt.

The monitoring plan consists of two main parts:

i) Monitoring of changes in carbon stocks and GHG emissions considering periodic checks that will take place within a fixed baseline period (PART 1);

ii) Monitoring of key parameters for baseline reassessment at the end of a fixed baseline period (PART 2).

PART 1. MONITORING CHANGES IN CARBON STOCKS AND GHG EMISSIONS FOR PERIODIC VERIFICATIONS

1.1 Monitoring actual changes in carbon stock and GHG emissions within the Project Area

Monitoring actual changes in carbon stock and GHG emissions within the Project Area involves four main scopes, which are:

i) project implementation;

ii) land use and land cover change;

iii) carbon stocks and non-CO_2 emissions, and

iv) impacts from natural disturbances and other catastrophic events.

The procedures applied to this monitoring plan contemplate what is developed and applied within the perspective of the project, therefore, within the scope, non-CO_2 emissions (iii) were not considered, as emissions were not considered in the baseline derived from biomass burning.

Details on the monitoring of the four scopes are presented below.

a) Technical description of monitoring tasks

Changes in carbon stock due to conversion of forested areas to non-forested areas by unplanned deforestation will be monitored. Likewise, changes in carbon stock due to uncontrolled forest fires and other catastrophic events will be monitored and considered under the Project scenario where they are significant.

As explained in Section 2.1.11, the proponents will develop the activity “Strengthening asset surveillance”, which consists of remote and continuous monitoring of the Project Area in order to allow agility in identifying and checking deforestation events, in addition to improvements in procedures for communication, approach offenders and verification of occurrences.

b) Data to be collected

Data/Parameter	Description	Unit	Source	Frequency
ABSLPAicl,t	Forest cover areas of initial icl forest class converted to non-forest cover areas within the Project Area in year t	ha (hectare)	Calculated using remote sensing and data available from reliable sources	Annual

Data/Parameter	Description	Unit	Source	Frequency
ΔCUDdPAt	Total change in actual carbon stock due to unavoidable unplanned deforestation in year t in the Project Area	tCO_2-e	Calculated from the detected areas of forest loss due to unplanned deforestation and estimated average carbon stock for the initial forest class	Annual
AUFPAicl,t	Areas affected by forest fires in the initial icl forest class in which carbon stock recovery occurs in year t	ha (hectare)	Calculated using remote sensing and data available from reliable sources	Annual
ΔCUFdPAt	Total reduction in carbon stock due to unplanned (and planned - when applicable) forest fires in year t in the Project Area	tCO_2-e	Calculated from the areas affected by forest fires and the estimated average carbon stock for the initial forest class	Annual
ACPAicl,t	Area affected by catastrophic events in the icl class in year t in Project Area	ha (hectare)	Calculated using remote sensing and data available from reliable sources	Annual
ΔCUCdPAt	Total reduction in carbon stock due to catastrophic events in year t in the Project Area	tCO_2-e	Calculated through the areas affected by forest fires in the Project Area and the estimated average carbon stock for the initial forest class	Annual

c) Summarized description of data collection procedures

Monitoring the implementation of Project activities

Monitoring the implementation of REDD+ activities will be carried out through schedules, activity performance reports, registration of indicators, financial reports, attendance lists, minutes of meetings, plans, established procedures and protocols, updated forest cover maps, among other relevant documents. As described in Section 2.1.11, the proposed activity “Implementation, monitoring and evaluation of the activities developed by the Project” will promote an efficient management of the Project over the years, ensuring effectiveness in the execution of activities so that positive impacts are achieved.

Monitoring of land use change and land cover within the Project Area

This monitoring will be carried out by mapping the forest cover of the Project Area, using qualified and scientifically recognized data sources, such as PRODES and DETER, databases developed through the National Institute for Space Research, data made available by MapBiomas (collaborative network formed by NGO’s, universities and technology startups), among other qualified and recognized sources that can be used in the future. The choice of methodology for identifying and quantifying changes in land use must meet the requirements of quality of data and minimum accuracy, according to the indications of Methodology VM0015.

In addition, different classification techniques and visual interpretation can be used during the course of the Project, both to identify deforestation and forest degradation and to validate and refine secondary data, such as photointerpretation using high-resolution satellite images, alternative sensors and data collected in field.

After collecting deforestation data, these will be compared with the baseline scenario, and the emission reduction values for the monitored period will be based on the comparison between expected deforestation and actual deforestation.

<u>Monitoring of carbon stock changes</u>

It is expected that the ex-ante estimate of the carbon stock for the forest class does not change during the baseline period. However, Methodology VM0015 requests monitoring of the carbon stock in the Project Area subject to relevant decrease in the Project scenario in accordance with the ex-ante assessment due to deforestation of areas subject to unplanned and significant decrease in the carbon stock in the Project scenario. Thus, the monitoring of changes (reduction) in the carbon stock of these areas will be carried out as follows:

These areas of unplanned deforestation are multiplied by the average carbon stock value in the initial forest class (Ctot) established as an indicator in the validation of the Project. The sum of carbon stock residual emissions under the soil and in dead timber is also considered, as such pools have 1/10 annual decay, causing emissions along the years. Finally, the value of the carbon stock estimated for the Reference Region in a post-deforestation scenario is subtracted from this result, obtaining the net value of the carbon stock that was reduced by unplanned and unavoidable deforestation. If there is a significant reduction in the carbon stock due to deforestation in the Project Area, this reduction will be presented in the verification processes using Table 29 of Approved Methodology VM0015 version 1.

<u>Monitoring of non-CO2 emissions</u>

Monitoring of non-CO2 emissions will be based on sources of secondary data from reliable bases, such as hot spots or fire scars. Complementarily, the technique of photointerpretation of high-resolution images will be used for validation of secondary data, classification and quantification of affected areas. In order to verify the damage and recovery of vegetation over time, NDVI analyses will be carried out, and, when necessary, field checks of points of interest to assess vegetation in situ. If forest areas are affected, the possible reduction in the carbon stock caused by forest fires will be evaluated based on the multiplication of the mapped area of forest loss by the average forest carbon stock. If change is significant, it will be reported in the verification processes using Tables 25e, 25f and 25g of methodology VM0015 version 1.1.

<u>Monitoring of natural disturbances and other catastrophic events</u>

Carbon stock reduction and increase in GHG emissions, as well as significant carbon stock reductions caused by natural disturbances or catastrophic events will be tracked, monitored and reported similarly to non-CO2 emissions in the Project Area. Therefore, if there is a significant decrease in the carbon stock due to natural disturbances or catastrophic events, this reduction will be reported in the verification processes using Tables 25e, 25f and 25g of Approved Methodology VM0015 version 1.1.

d) Quality control and quality assurance procedures

In order to monitor the activities of Ateles REDD+ Project, the activity of “Implementation, monitoring and evaluation of the activities developed” is foreseen, which will allow continuous monitoring of the Project, accompanied by evaluation processes, allowing the incorporation of learning and improvements and, consequently, quality assurance for the Project.

As described in the previous items, changes in carbon stock due to conversion of forested areas to non-forested areas by unplanned deforestation will be monitored. Likewise, changes in carbon stock due to uncontrolled forest fires and other catastrophic events will be monitored and discounted against the project scenario where they were significant. Monitoring quality control and assurance of these parameters will be carried out through the accuracy process indicated by methodology VM0015 version 1.1, which will be the same regardless of the type of data used in monitoring.

An analysis of general accuracy and the kappa index obtained from a confusion matrix such as Congalton's (1999) will be performed[104]. Through a geographic information system, at least 100 points will be generated and randomly distributed in the area of interest. Validation will be performed using high spatial resolution satellite images and/or data collected in field. The minimum mapping accuracy, according to VM0015, for each class or category in the land use and land cover map, must be 80%.

In addition to the accuracy process carried out, when necessary, field verifications will also be carried out in areas where unplanned deforestation events, uncontrolled forest fires and catastrophic events are identified.

e) Data archiving

The Biofílica Ambipar Environment will store all data and reports of the Ateles REDD+ Project in digital files throughout the duration of the Project. All documents relating to Project monitoring will be made available to auditors at each verification event.

f) Organization and responsibilities of the parties involved in all of the above

The procedures described will be the responsibility of the Project proponents: Biofílica Ambipar Environment Investments and AgroSB Agropecuária.

1.2 Leakage monitoring

Leakage monitoring by the Project involves two main scopes, which are:

i) changes in carbon stocks and GHG emissions associated with leakage prevention activities; and,

ii) changes in carbon stocks and GHG emissions associated with leakage from displacement of activities.

The procedures applied to this monitoring plan contemplate what is developed and applied within the perspective of the project, thus, within the scope ii) the monitoring of changes in GHG emissions derived from the burning of biomass was not contemplated, as it was not considered on the baseline.

Details on the monitoring of the two scopes are presented below.

a) Technical description of monitoring tasks

It is not expected that there will be changes in the carbon stock and GHG emissions associated with spill prevention activities, since no activity is planned, such as intensive agriculture, pasture area management or forage production, capable of changing the carbon stock and increasing GHG emissions when compared to the baseline scenario.

[104] CONGALTON, R. G.; KASS GREEN. Assessing The Accuracy Of Remotely Sensed Data: Principles And Practices. New York – CRC Press, 1999.

However, although no stock reduction in leakage prevention activities is foreseen, should they prove necessary during Project implementation, the ex-ante changes in carbon stock and GHG emissions associated with these activities will be estimated according to step 8 of Methodology VM0015. If results are significant, they will be monitored and data will be made available to verifiers at each verification event using Tables 30b, 30c, 31, 32 and 33 of Methodology VM0015 version 1.1.

Changes in carbon stock and GHG emissions associated with leakage from displacement of activities will be monitored using the same technique applied in monitoring changes in carbon stock due to conversion of forested areas to non-forested areas by unplanned deforestation in the Project Area.

b) Data to be collected

Data/Parameter	Description	Unit	Source	Frequency
$\Delta BSLLK_{icl,t}$	Forest cover areas of the initial icl forest class converted to non-forest cover areas within the Leakage Belt in year t	ha (hectare)	Calculated using remote sensing and data available from reliable sources	Annual
$\Delta CADLK_{t}$	Total reduction in carbon stocks due to displaced deforestation in year t in the Leakage Belt	tCO_2-e	Calculated from the detected areas of forest loss in the Leakage Belt, the average carbon stock and the estimated loss in carbon stock projected by the baseline	Annual
$APSLK_{fcl,t}$	Portion of area within Leakage Management Areas with decreasing carbon stock in year t	ha (hectare)	Monitoring report of project activities and other records related to leak prevention activities	Only when applicable
$\Delta CLPMLK_{t}$	Decrease in carbon stock due to leak prevention measures in year t	tCO_2-e	Calculated from area quantitative value in Leakage Handling Areas, with decreasing carbon stock, of initial carbon average stock and carbon stock loss estimated in Leakage Belt designed by baseline	Only whenever applicable
$EgLK_{t}$	Emissions from animals in pastures in spill management areas in year t	tCO_2-e	Existing records on the practice of grazing	Only when applicable

Data/Parameter	Description	Unit	Source	Frequency
$EADLK_t$	Emissions from forest fires displaced to the Leak Belt in year t of the Ateles REDD+ Project	tCO_2-e	Calculated using the areas affected by forest fires in the Leakage Belt and estimated average carbon stock for the initial land use class	Only when applicable

c) Summarized description of data collection procedures

Changes in carbon stocks and GHG emissions associated with leakage prevention activities

As explained in item a), it is not expected that there will be changes in the carbon stock and GHG emissions associated with leak prevention activities, since no activity capable of changing the carbon stock and increasing GHG emissions is foreseen when compared to the baseline scenario. However, should such activities prove necessary, ex-ante changes in carbon stock and GHG emissions associated with these activities will be monitored and data will be made available to auditors at each verification event using Tables 30b, 30c, 31, 32 and 33 of Methodology VM0015 version 1.1.

Monitoring, considering data collection procedures, will consider the following activities:

- List of leakage prevention activities.
- Production of a map showing the areas of intervention and the type of intervention.
- Recognition of areas where leakage prevention activities have an impact on the carbon stock.
- The existing non-forest classes in these areas in the baseline case will be identified.
- Carbon stocks in the identified classes will be measured or a literature conservative estimate will be used.
- Changes in carbon stock in the leakage management areas under the project scenario will be reported using Table 30b of VM0015.
- Calculation of net changes in carbon stock caused by leakage prevention measures during the fixed period of the project baseline and crediting period.
- The calculation results will be reported by Table 30c of the approved Methodology VM0015.

Changes in carbon stock and GHG emissions associated with leakage from displacement of activities

These will be monitored through the same methods applied to monitor the conversion of forest areas to non-forest areas due to unplanned deforestation in the Project Area, that is, qualified and scientifically recognized sources will be used, such as PRODES, DETER and MapBiomas, in which quality of data and accuracy requirements will be used for evaluation. If in the Leakage Belt there is a deforestation event greater than expected for the baseline scenario and is attributed to deforestation agents in the Project Area, losses in carbon stock will be accounted for and reported using either Table 22c or Table 21c of the VM0015 Methodology version 1.1.

d) Quality control and quality assurance procedures

Quality control and assurance in relation to the monitoring of changes in the carbon stock and GHG emissions associated with leakage prevention activities will be determined according to the activity, if implemented. As for changes in the carbon stock and in the GHG emissions associated with leakage due to displacement of activities, they will be carried out through accuracy analysis, as indicated by methodology VM0015 version 1.1.

The classification accuracy analysis will be carried out through the analysis of general accuracy and the kappa index obtained from a confusion matrix such as that of Congalton and Green (2008), in which at least 100 points distributed randomly in relation to the analyzed area will be generated through a geographic information system. Validation will be performed using high spatial resolution satellite images and/or data collected in field. The minimum mapping accuracy, according to VM0015, for each class or category in the land use and land cover map, must be 80%.

e) Data archiving

The Biofílica Ambipar Environment will store all data and reports of the Ateles REDD+ Project in digital files throughout the duration of the Project.

All documents relating to Project monitoring will be made available to auditors at each verification event.

f) Organization and responsibilities of the parties involved in all of the above

The procedures described will be the responsibility of the Project proponents: Biofílica Ambipar Environment and AgroSB Agropecuária.

1.3 Monitoring ex-post reductions in net anthropogenic GHG emissions

Details on monitoring are presented below.

a) Technical description of monitoring tasks

In the verification procedures, the results will be represented using Table 36 of the VM0015 Methodology version 1.1, together with spatial data (deforestation maps, when available).

A map showing the cumulative areas credited within the Project Area will be updated and presented to VVB at each verification event.

b) Data to be collected

Data/Parameter	Description	Unit	Source	Frequency
ΔREDD,t	Reductions in net GHG emissions attributable to Project AUD activities year t	tCO_2-e	Calculated by subtracting ex post carbon stock changes from the baseline scenario	Annual
VCU,t	Number of Verified Carbon Units (VCU's) to be made available for commercialization in year t	tCO_2-e	Calculated by subtracting ex post Project net GHG emission reductions from the buffer	Annual

PART 2. MONITORING BASELINE PROJECTIONS IN THE FUTURE

2.1 Updating information on agents, drivers and underlying causes of deforestation

The Project baseline will be updated and used in revising the baseline projections after a fixed period of 6 years, in addition to statistical and spatial data, studies and information on agents, motivations and underlying causes of deforestation necessary to carry out the Steps 2 and 3 of Approved Version of Methodology VM0015.

2.2 Updating the baseline land use change and land cover component

The Project will monitor updates regarding the national and sub-national baselines, and thus, will apply if improvements compatible with the rigor applied to the Project are verified. Otherwise, step 4 of Methodology VM0015 will be redone considering the period of the last 6 years and using updated variables on the agents, drivers and underlying causes of deforestation in the Reference Region. The area of annual deforestation and the location of deforestation in the baseline are the two main components to be reviewed.

Assumptions and hypotheses considered in modeling the dynamic component of future deforestation (population data) as well as data used in the spatial projection (update of roads, location and distance of new deforestation) will be reviewed and updated.

2.3 Baseline carbon component update

According to the results generated during changes in the carbon stock monitoring processes throughout the Project, the spatial estimate of the carbon component can be revised in Methodology VM0015 version 1.1, Part 3, item 1.1.3. Thus, if there are more accurate estimates, from the use of techniques such as LIDAR or SAR interferometric data, they will be applied are baseline revisit period

3.3.4. Dissemination of Monitoring Plan and Results (CL4.2)

The monitoring plan, as well as the results obtained by monitoring the Ateles REDD+ Project, will be made available to the public through a page on the official website of Biofílica Ambipar Environment Investments. The summary documents related to the monitoring plan and results, as well as relevant information, will be made available to communities and stakeholders through feedback, meetings, lectures and by physical means on the premises of Fazenda Lagoa do Triunfo, belonging to AgroSB.

3.4. Optional Criterion: Climate Change Adaptation Benefits

Not applicable. Ateles REDD+ Project does not intend to be validated for the gold level of this section.

3.4.1. Regional Climate Change Scenarios (GL1.1)

Not applicable

3.4.2. Climate Change Impacts (GL1.2)

Not applicable

3.4.3. Measures Needed and Designed for Adaptation (GL1.3)

Not applicable

4. COMMUNITY

4.1. Without-Project Community Scenario

4.1.1. Descriptions of Communities at Project Start (CM1.1)

Formerly known as Casa de Tábua, Novo Horizonte, it is a village located in the peripheral area of the headquarters of the municipality of São Félix do Xingu (Figure 18), in the environmental protection area (APA Triunfo do Xingu). Vila Novo Horizonte is home to settlers who arrived in the region from the early 2000s. Most came from states in the northeast, midwest and southeast of the country, such as Goiás, Tocantins, São Paulo, Maranhão and Minas Gerais.

Settlers in Novo Horizonte face the insufficiency or inexistence of public services that would be the responsibility of municipal, state or federal management in the region. Such as lack of road maintenance, lack of a health system to treat more complex diseases, lack of investment in education, water supply, sanitation, security and electricity. However, living this experience for more than two decades, the settlers have organized themselves and use their own resources to provide their essential needs in these aspects.

These community members declare themselves as family farmers, small producers, squatters, farmers or medium producers. Regarding income generation, the main activities are focused on the production and marketing of cattle, milk and cheese, the raising of small animals and land maintenance and preparation services on neighboring farms.

Some villagers have small orchards near their homes, as well as ornamental and medicinal plants. Some families show interest in specializing in cocoa and livestock farming. In addition, they produce from small fields of beans, corn, cassava and other annual crops, on a smaller scale, for their own consumption, they practice the extraction of Brazil nuts and açaí (Sousa et al, 2016).

Most of the community members are organized around an Association, known as AGRICATU. The Association of Farmers of the Calumbi and Tucunaré Region (AGRICATU), allowed the settlers access to Pronaf, enabled social security insurance, access to credit, promotes the facilitation of the sale of production, technical guidance in productive activity, assistance in the regularization of the land, in addition to strengthening the bonds of the associates.

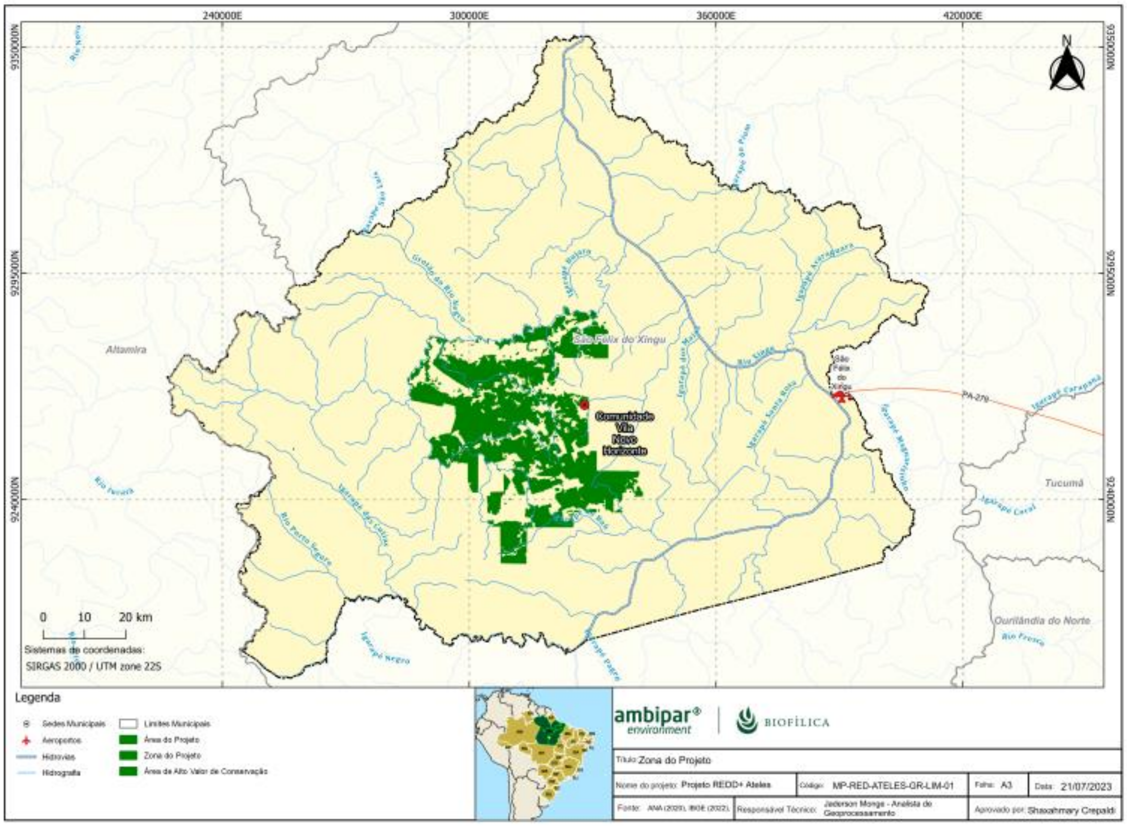

Figure 20: Location map of Vila Novo Horizonte.

Historical social occupations of the territory

The occupation of São Félix do Xingu, as well as other micro-regions, was driven by the exploration and use of natural resources, and this dynamic established conflicts between different groups at the time. The political situation contributed to the intensification of territorial disputes between 1970 and 1980. In Brazil under the tutelage of a military government, a discourse of occupation for exploitation was strengthened. In this context, highways were built, encouraging the migration of large farmers and ranchers, interested in the exploitation of mahogany and in the activity of raising cattle. In line with the exploration project of the region, Highway PA-279 was built, which enabled the accelerated advance of territorial dynamics and redefined the modes of land occupation in the Municipality of São Félix do Xingu and surroundings (SCHMINK et al, 2017[105]; CASTRO et al, 2002[106]). Unlike other border areas of the Amazon, the territorial limits of São Félix do Xingu border the states of Mato Grosso and Tocantins and have considerable

[105] SCHMINK, Marianne et al. From contested to 'green' frontiers in the Amazon? A long-term analysis of São Félix do Xingu, Brazil. The Journal of Peasant Studies, v. 46, n. 2, p. 377-399, 2019.

[106] Castro, Edna Ramos; MONTEIRO, Raimunda; CASTRO, Carlos Potiara. Actors and social relationships in new frontiers in the Amazon. Novo Progresso, Castelo de Sonhos and São Félix do Xingu. 2002. Doctoral Thesis. World Bank Groupe-Banque Mondiale.

proximity to the state of Goiás. With this location and the ease of transport routes with the center of the country, in São Félix do Xingu there was a favoring of demographic growth over time (KAHWAGE, 2002[107]).

The advance of livestock in São Félix do Xingu had its rhythm influenced, directly and indirectly, by government incentive policies that transformed the area into one of the largest agricultural centers in the country. Despite this, due to the late opening of the road, the development of livestock in the municipality is seen as recent and a little different from the "nobler" livestock areas of the southern center of the country. Livestock in São Félix do Xingu was able to trigger other segments of the urban sector economy by moving the trade in seeds, fertilizers, fences, etc. However, the penetration of agriculture in the Xingu territory left in its wake high rates of deforestation (Castro and Campos, 2015[108]).

Currently, it is understood that the main cause of deforestation in São Félix do Xingu is livestock, the municipality is considered one of the largest beef producers in the country. Livestock farming is the most important economic activity in the municipality, and the expansion of this activity has led to the destruction of large forest areas. Although São Félix do Xingu has reached the municipal position holding the largest cattle herd in Pará and Brazil, with 2.46 million head of cattle, according to the IBGE Municipal Livestock Survey for 2021,13 other vectors that drive its economic growth are in agricultural production. In this productive environment, there are cocoa and banana crops, in addition to temporary crops of corn, beans and cassava. The mining industry is centered on the extraction of cassiterite, nickel and cobalt (State of Pará, 2017[109]).

Land insecurity is also a relevant contributor to deforestation. Since many areas of the municipality are occupied by squatters and land grabbers, who do not have property titles and, therefore, do not have incentives to preserve the forest (KAHWAGE, 2002). In addition, the lack of land regularization makes it difficult to control and supervise economic activities in the region. The municipality is also marked by land disputes between large rural landowners, small farmers and traditional communities. These conflicts often result in deforestation, since large rural landowners use forest clearing as a way to pressure other groups (Schmink and Wood, 1992[110]).

In summary, deforestation in São Félix do Xingu is a complex and multifaceted problem that requires coordinated and integrated action by different social and governmental actors. Deforestation in São Félix do Xingu has significant impacts on the environment and local society. The destruction of the Amazon rainforest contributes to increased greenhouse gas emissions, biodiversity loss and climate change. In addition, deforestation directly affects traditional communities and small farmers, who depend on the forest for their livelihoods.

4.1.2. Interactions between Communities and Community Groups (CM1.1)

Most of the community members living in the project area have been affiliated with the Association of Farmers of Calumbi and Tucunaré (AGRICATU) for an average of 12 years. AGRICATU and its board of

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[107] KAHWAGE, Claudia. Political organization of peasants in São Félix do Xingu – PA: Strategies, Identities and Associations. Master's Dissertation in Family Farming and Sustainable Development. Federal University of Pará, Belém, 2002, p. 23.
[108] CASTRO, Edna Ramos de; CAMPOS, Índio. Socioeconomic Formation of the Amazon, 2015, p. 448-449. 2015, p. 448-449.
[109] STATE OF PARÁ. PLAMUS: Sustainable Urban Mobility Plan. Municipality of São Félix do Xingu, 2021, p. 14-15. Available at: www.sfxingu.pa.gov.br. Accessed on Wednesday, December 14, 2022.
[110] SCHMINK, Marianne; WOOD, Charles H. Contested frontiers in Amazonia. Columbia University Press, New York, 1992, 398 p.

directors state that they have a relationship and interaction with other groups and social actors, as well as their own advice and consultancy to support them in their demands.

Among the settlers mapped until the beginning of the Project, 67% stated that there was no relationship with other community groups and/or institutions. Among the 29% who said they had relations with other institutions, the Association for the Development of Family Agriculture in the Upper Xingu ADAFAX was mentioned, from which meetings were held to discuss activities regarding the rights of owners and other matters of particular interest of an individual nature.

Also in the project area are the employees of Fazenda Lagoa do Triunfo, who historically had little or no relationship with the settlers and AGRICATU. However, within the scope of the project's development, it is hoped that these relations will be strengthened and that there will be greater interaction between these groups. Both are affected by the absence of the state in the area where they live and may benefit from the actions proposed in section 2.1.11, since they have their source of livelihood and family constitution in the project development area.

The Ateles REDD+ Project aims to strengthen existing interactions and provide opportunities for new relationships between community groups and other social actors. AGRICATU is a great opportunity for the development of social activities for the Project, since this interest group is well articulated and with a high level of trust among community members.

4.1.3. High Conservation Values (CM1.2)

The concept of High Conservation Values (HCV) was developed by the Forest Stewardship Council (FSC, 1996)81 for the certification of timber products from responsible forest management, according to standardized Principles and Criteria that reconcile environmental and ecological safeguards with social benefits and economic viability (FSC, 2015[111]).

Every forest has some environmental and social value. The values that forests contain may include, but are not limited to, the presence of rare species, recreation areas, or resources collected by local population. When these values are considered exceptional or of critical importance, the forest area can be defined as a High Conservation Value Forest (HCVF). The key to the concept of High Conservation Value Forests is the identification of High Conservation Value attributes: it is these values that are important and that need to be protected. High Conservation Value Forests are simply forest areas where these values are found or, more precisely, the forest area that must be managed appropriately in order for the identified values to be maintained or increased.

In summary, a High Conservation Value Forest is the forest area required to maintain or increase a High Conservation Value attribute. By identifying key values and ensuring that they are maintained or increased, it is possible to make rational management decisions that are consistent with protecting the important environmental and social values of a forest area.

Of the six criteria listed by the FSC, one of them is directly related to the forest and surrounding communities:

HCV 4: Basic ecosystem services in critical situations, including protection of springs and erosion control in vulnerable soils and slopes.

[111] Forest Stewardship Council (FSC). International Generic Indicators. FSC International Center, Policy and Standards Unit, Charles-de-Gaulle-Str. 5. Bonn, Germany

Indicator used: Remote and/or poor rural areas where people rely directly on natural resources to meet their main needs, including water supply.

For this definition, the concepts defined in the General Guide for the Identification of High Conservation Values (HCVA) (Brown *et al*., 2013[112]) were used, which is a guide in the interpretation of HCVA definitions and their identification in practice, to achieve standardization in the use of this approach. The document can also help developers of National HCV Interpretations by providing a reference to adapt definitions, data sources and examples in national contexts.

High Conservation Value	Areas of native vegetation in the Ateles REDD+ Project Area
Qualifying Attribute	Areas with native vegetation cover that provide ecosystem support services and essential regulations to the surrounding communities, mainly helping to maintain the quantity and quality of water, since the community depends on rivers and drinking water channels to provide clean water suitable for consumption. In addition, these forest areas help maintain soil quality and nutrient cycling, contribute to climate regulation, improve air quality and thermal comfort.
Focal Area	All areas of native vegetation in the Project Area positively assist in maintaining the water cycle for the communities' subsistence areas, and other essential ecosystem services. Therefore, they need continuous monitoring to ensure the maintenance of the structure of the environment and its environmental services. The surveillance and monitoring actions that will be carried out on the farm will help to contain the deforestation of these areas and the illegal extraction of their resources.

4.1.4. Without-Project Scenario: Community (CM1.3)

Current socioeconomic indicators characterize the region with low welfare conditions for communities, low education, lack of representative organization and few productive economic alternatives. From this, some scenarios can lead to the advance of deforestation of the region's remaining forests. The scenarios in the absence of the REED+ Ateles Project were described considering political-legal, economic, social and environmental factors.

- Political-legal: The low level of application of environmental legislation resulting from the lack of governance results in an increase in environmental liabilities; In view of the maintenance of the absence of mechanisms that enable territorial belonging and land security, communities residing in the project area will continue to lack a sense of responsibility for the environmental liability of deforestation, without access to public financing and reproducing in a cyclical manner the logic of the ousted-burning production system, expanding the fractions of deforestation focused on extensive livestock activity; The predatory production of natural resources by producers tends to

[112] Brown, E., N. Dudley, A. Lindhe, D.R. Muhtaman, C. Stewart, and T. Synnott (eds.). 2013 (October). General guide for identification of High Conservation Values. HCV Resource Network.

continue to go against environmental legislation and the very Management Plan of the Triunfo do Xingu APA that is being prepared.

- Economic: Small and medium-sized rural producers will continue to reproduce the logic of the felling-burn system with difficulties of inclusion in the value chains of the livestock economy; The low educational level of small producers will make it difficult to transfer technologies; The poor conditions of secondary roads will make it difficult to access goods and services, in addition to hindering the flow of agricultural and livestock production; Community producers will remain without access to technical assistance and rural extension; Small, medium and large producers will continue in the process of expanding their areas with the planting of pastures and the production of cattle with continuous pressure on the forest; Livestock producers will continue to suffer market restrictions, without being able to unlink their production to the occurrence of illegal deforestation.
- Social: Maintenance and aggravation of tensions between community members and AgroSB resulting from land conflicts and illegal deforestation that will continue to occur in the areas of Lagoa do Triunfo farm; Permanence of a frayed relationship, without spaces for dialogue and closer ties between community members and AgroSB, further straining territorial disputes; Low level of income, education and employment opportunities among small producers; Community members will continue to lack access to policies that guarantee land security, technical assistance and rural extension; Maintenance of the situation of social vulnerability of small farmers; Computer technology and communication infrastructure will remain precarious; Community members will continue to be made invisible by the responsible public sector, and with precarious access to basic health, infrastructure and education services.
- Environmental: livestock production, especially among small and medium producers, will continue to have a low technological profile based on the extensive system that promotes deforestation; Given the land conditions in the region, deforestation will continue to be used as a tool to claim land ownership rights in the region; The non-use of more efficient and sustainable production systems, and low access to credit and technical assistance and rural extension make it impossible for small and medium producers to participate more in global food chains; Production practices that result in soil and forest degradation, GHG emissions, non-compliance with environmental legislation and the promotion of conflict in the field will remain.

In the scenario with the presence of the Ateles REDD+ Project, there is a positive impact on the socioeconomic context of the communities, achieving improvements in the relationship between stakeholders, allowing the construction of safe spaces for dialogue and management of the needs of these communities, in addition to the development of profitable agricultural production and reducing environmental impacts, greater access to public policies, innovation and technologies. Resulting in social well-being and a favorable and sustainable business environment economically, environmentally and socially.

4.2. Net Positive Community Impacts

4.2.1. Expected Community Impacts (CM2.1)

Community Group	Community members residing in the Project area
Impact(s)	- Strengthening human capital through access to training, training and technical assistance to encourage sustainable and adaptive agricultural and livestock production practices and models.

	- Strengthening local governance. - Development and integration into value chains. - Environmental awareness and permanence of families on their lands. - Land security and social welfare. - Improvements in income generation and diversification.
Type of Benefit/Cost/Risk	Expected benefits: - Direct and indirect positive impact on the community. Costs: - Human and physical resources related to training actions, training and technical assistance. Risks: - Lack of community engagement and the continuity of unsustainable agricultural and livestock practices.
Change in Well-being	- Improvement of income. - Food security. - Technical improvement of economic activities focused on agriculture and livestock. - Improvement in local governance. - Territorial belonging.

4.2.2. Negative Community Impact Mitigation (CM2.2)

As mentioned in the section above (Section 4.1.34.3.1) the Ateles REDD+ Project is not intended to provide negative impacts to the well-being of local communities. However, some potential risks must be monitored carefully to avoid future losses to the development of the Project, among them we can mention:

- The possible lack of settler engagement, due to the history of territorial and environmental conflicts, with low interaction, communication and few opportunities for mediation of interests.
- Public agents and class representatives are not sensitized to participate in the project;.
- Lack of buy-in from other stakeholders; and low acceptance for developing more sustainable production models.
- Since the project proposes to reduce deforestation, it may result in limited access to forest areas, greater control of activities related to the expansion of useful production areas and, consequently, the reduction, in the short term, of the supply of some resources.

In order to mitigate these risks, some measures will be adopted, such as the creation of communication channels aimed at community demands, the development of safe and open spaces for dialogue, the consolidation of involvement between all stakeholders in the decision-making processes, through public consultation and other collective face-to-face activities during the construction and development phases of the Project, in addition to involvement in the training and training proposed in section 2.1.11, providing an improvement in the well-being of the population and reducing the environmental impacts of activities traditionally developed in the project area.

As for the possible low adherence to more sustainable production practices, a mitigating measure is to prioritize the involvement of community members in the development and strengthening of value chains for agricultural and livestock products, which aim to guarantee income and food security, while minimizing possible negative impacts and reducing pressure for deforestation.

Regarding the strengthening of patrimonial surveillance, with actions that prevent deforestation and forest degradation, such as remote and continuous monitoring and the presence of forest guards, are intended to assist in the maintenance of forest cover in the Project Area and in the generation of ecosystem services essential for the survival of communities in the area.

Finally, it is worth noting that the project is committed to identifying and mitigating any signs of risk or negative impact regarding the working conditions associated with the project, as well as the safety of Lagoa do Triunfo farm employees, community members, women and girls from the local community and impacts related to the environment.

4.2.3. Net Positive Community Well-Being (CM2.3, GL1.4)

The Ateles REDD+ Project proposes a series of activities that encourage socioeconomic and sustainable development for the communities involved, providing training, training and technical assistance to promote more sustainable practices and production models. Efforts will be invested in the development and strengthening of value chains and engagement through a mechanism of transparency, communication and encouragement of collective participation in the actions developed by the Project.

In the without-project scenario, as described in Section 4.1.4, the context of low income, lack of access to public policies and other services, causes families belonging to the communities to seek alternatives to increase their income, from economic and subsistence activities practiced in an unsustainable and unplanned way.

Based on this scenario, the Project foresees creating opportunities for community members, generating a series of positive net impacts, such as those listed below:

- Develop actions that encourage the adoption of diversified and adaptive agricultural and livestock production models, which integrate protection of forest resources with sustainable production, ensuring the food security of families and maintenance of ecosystem services;
- Increase community engagement based on their participation in the Project's activities, in addition to promoting actions to strengthen local governance and the empowerment of the existing local association;
- Strengthening skills, knowledge and human capabilities related to sustainable economic activities, management and productive organization, in order to develop and strengthen value chains;
- Increase levels of knowledge about sustainable practices, such as agrosilvopastoral activities, promoting the protection and conservation of forest cover and biodiversity, stimulating alternative means of subsistence and income generation for impacted families;
- Permanence of families in the communities, ensuring the settlers' land security;
- Implementation of strategic partnerships for the on-site execution of the proposed actions.

The main problems that will be faced in this context are:

- Low access to public policies related to goods, basic services and infrastructure;
- Unsustainable economic activities, with low technification, productivity and little assistance;
- Difficulty in mobility and access;

The Project aims to positively influence the social relations and living conditions of the communities residing in the Project area, in order to reduce the social vulnerability and conflicts generated by the situation of land insecurity of these families, providing improvement in the quality of life and income stability, in addition to allowing conditions for access to goods and services that promote economic and social well-being.

4.2.4. High Conservation Values Protected (CM2.4)

So far, during the preliminary evaluation conducted with the DSEA (socioeconomic and environmental diagnosis) studies, no impacts have been identified on the attributes of high value for conservation related to social issues (HCVAs 5 and 6 – Section 4.1.3) caused by the implementation of the Ateles REDD+ Project. However, if these are identified in the future, measures must be taken to ensure that there are no net negative impacts on the attributes.

In order to preserve the well-being of communities and avoid any negative impacts related to HCVA, the Ateles REDD+ Project will adopt measures and actions, already described in detail in Section 2.1.11, in order to ensure the protection and conservation of forest preservatives. These activities have been carefully planned to ensure the maintenance of crucial ecosystem services for the communities benefiting from the project.

4.3. Other Stakeholder Impacts

4.3.1. Impacts on Other Stakeholders (CM3.1)

The Ateles REDD+ Project does not foresee significant negative impacts on other stakeholders. However, although unlikely to happen, the following risks were mapped to the Project:

- Lack of engagement of communities and other actors in Project activities and other articulations;
- Failure to communicate the Project's actions and establishment of possible conflicts arising from the implementation and conduct of activities.
- -Resurgence of relational conflicts between the project proponent and the community;
- However, it is worth noting that the positive impacts of the project can bring a number of benefits and well-being to other actors, such as:
- All local communities, as well as other actors residing in the project region, whether or not participating in project activities, will benefit from all the positive impacts related to the conservation and protection of forest cover and biodiversity;
- All communities and other actors will benefit from sustainable development, as well as from the opportunities generated by the Project's activities, improving the quality of life and well-being;
- All stakeholders in the region will benefit not only from project activities, but also from the expected positive impacts such as strengthening local governance and increasing social and human capital;

4.3.2. Mitigation of Negative Impacts on Other Stakeholders (CM3.2)

As previously mentioned, there is no forecast of relevant negative impacts for other actors involved in this Project. However, as a form of prevention, transparency mechanisms and communication and consultation channels will be implemented, aiming to identify and involve possible stakeholders that will not be directly affected by the Project.

4.3.3. Net Impacts on Other Stakeholders (CM3.3)

As described and detailed in Section 4.3.1, significant negative impacts on the well-being of other groups of local actors are not expected, since the activities outlined provide for articulation with government agencies and other local institutions, in order to encourage the relationship between the parties, in addition to promoting improvements in local living conditions, greater access to public policies and strengthening governance. Thus, the activities developed and proposed for this Project aim to promote only positive impacts, which provide inclusion and well-being to communities and other parties involved, while minimizing possible risks to other stakeholders and the communities themselves directly involved.

4.4. Community Impact Monitoring

4.4.1. Community Monitoring Plan (CM4.1, CM4.2, GL1.4, GL2.2, GL2.3, GL2.5)

The monitoring of the Project's impacts on communities and other actors is an important management tool, enabling the evaluation and measurement of the effectiveness of the proposed activities, in order to achieve positive changes in the well-being of the population involved in the Ateles REDD+ Project. In this sense, it is suggested to develop a monitoring system for the Project, based on the goals foreseen for the construction of the indicators to be collected, on the verification tools and on the procedures for analysis and evaluation of results, to, when necessary, indicate the essential measures to improve the intended advances.

The monitoring of benefits to communities presents four main components, structured based on what was proposed in section 2.1.11, which addresses the theory of change and the activities of the Project:

<u>Strengthening local governance:</u> The actions implemented will be monitored in order to establish and expand new partnerships and strengthen social capital in the communities, in addition to ensuring collective engagement, and transparency in the execution of the activities proposed by the Project, as well as monitoring the risks and improvements about them;

<u>Developing and strengthening value chains:</u> The activities developed within the scope of strengthening the value chain will be monitored, based on sustainable practices, as well as the promotion of strategies to facilitate the process of commercialization and valorization of local production, and the establishment of strategic partnerships to achieve these objectives.

<u>Promotion of sustainable production practices:</u> Actions aimed at the development of the community's human capital will be monitored, involving technical training activities for more sustainable production models, rural extension and the establishment of strategic partnerships to achieve these objectives.

<u>Land security and social welfare:</u> Actions aimed at achieving land security and social well-being of the communities benefited by the project will be monitored, through the strengthening of relations with the government, investment in infrastructure improvements, also helping to design the useful and productive areas of these families, thus allowing a restructuring of the social fabric of this community and strengthening the sense of territorial belonging.

Finally, the plan will seek to protect High Conservation Value Areas (HCVA) through socio-environmental activities (described in Section 2.1.11) that seek to contain deforestation and degradation of these areas, but also by organizing and enhancing alternative traditional economic and livelihood activities, such as the promotion of sustainable practices. As explained in Section 4.1.3, the high conservation value attribute HCVA4 was identified for the communities: Basic ecosystem services in critical situations, including protection of springs and erosion control in vulnerable soils and slopes. The evaluation of the effectiveness of the measures adopted to maintain and improve HCVA4 is linked to the remote and on-site monitoring of forest areas within the Project Area that offer ecosystem services to communities and the promotion of alternative activities to deforestation and forest degradation contemplated by the indicator "Area with adoption of sustainable practices and models of agricultural and livestock production".

The Community Impact Monitoring Plan covers the process indicators and part of the result indicators, referring to the activities developed for the project. Whenever necessary, this initial monitoring plan will be revisited and complemented, always counting on the evaluation and validation of the same by the interested parties.

a) Data to be collected

Data / Parameter	Number of partnerships contacted
Data unit	Number
Description	This parameter aims to measure the number of negotiations, approaches and partnerships signed that the Project carried out and/or tried to carry out throughout its life cycle, in order to contribute to the development and improvement of actions and activities linked to the Project's social activities
Source of data	Reports (e.g. follow-up report of project activities that have been implemented), contracts, memos, emails, meeting minutes and/or other supporting documents as evidence that a partnership has been established and built
Description of measurement methods and procedures to be applied	All documents produced will be stored in digital files throughout the crediting period of the Project. In this way, the creation of partnerships linked to the Project's social activities will be monitored and accounted for
Frequency of monitoring/recording	At every verification event
Value applied	To be accounted for after the start of the Project
Monitoring equipment	Not applicable
QA/QC procedures to be applied	Systematized information from partnerships established for social activities will be validated by proponents, allowing greater reliability and quality of data. In addition, the Project will undergo continuous evaluation of information generated, through the identification of improvements in the collection and registration processes and, when applicable, the incorporation of adjustments in the strategic planning of the Project
Purpose of data	Demonstrate that the Project is dedicating itself and expanding its activities in its social scope through partnerships carried out
Calculation method	Not applicable
Comments	

Data / Parameter	Number of qualifications
Data unit	Number
Description	This parameter aims to measure the number of courses, qualifications and training carried out in the social activities proposed by the Project
Source of data	Reports (e.g. monitoring report of project activities that have been implemented), attendance lists of participants, contracts, photos, promotional materials, among other documents
Description of measurement methods and procedures to be applied	All documents produced will be stored in digital files throughout the crediting period of the Project. In this way, the qualifications, courses and training carried out by the activities of the Social Scope of the Project will be monitored and accounted for
Frequency of monitoring/recording	Annual
Value applied	To be accounted for after registration of the Project
Monitoring equipment	Not applicable
QA/QC procedures to be applied	The systematized information on qualifications, courses and training proposed by the social activities will be validated among the proponents, allowing for greater reliability and quality of data. In addition, the Project will undergo continuous evaluation of information generated, through the identification of improvements in the collection and registration processes and, when applicable, the incorporation of adjustments in the strategic planning of the Project
Purpose of data	Not applicable
Calculation method	Not applicable
Comments	

Data / Parameter	Number of people benefited
Data unit	Number
Description	This data will account for all people or groups of people who have benefited from the Project, through the planned actions and activities related to the Social Scope, over the years of implementation and monitoring.

Source of data	Reports (e.g. activity monitoring report), interviews, feedback on results and/or consultations, attendance lists, presentations, and other documents that corroborate as evidence that a person, whether community member or not, has benefited from the Project
Description of measurement methods and procedures to be applied	All documents produced will be stored in digital files throughout the crediting period of the Project. In this way, information about the people benefited by the activities of the Social Scope of the Project will be monitored and accounted for.
Frequency of monitoring/recording	Annual
Value applied	To be accounted for after registration of the Project
Monitoring equipment	Not applicable
QA/QC procedures to be applied	The systematized information of the people benefited by the social activities will be validated by proponents, allowing for greater reliability and quality of data. In addition, the Project will undergo continuous evaluation of the information generated, through the identification of improvements in the collection and registration processes and, when relevant, the incorporation of adjustments in the Project's strategic planning.
Purpose of data	Not applicable
Calculation method	Not applicable
Comments	

Data / Parameter	Community Engagement and Adherence to the Project
Data unit	Number
Description	This parameter aims to measure community adherence and involvement in the project's proposals and activities, as a direct impact of the actions provided for in the Project's Social Scope.
Source of data	Reports (e.g. activity monitoring report), interviews, feedback on results and/or consultations, attendance lists, presentations, and other documents that corroborate as evidence that a person, whether community member or not, has benefited from the Project
Description of measurement methods and procedures to be applied	All documents produced will be stored in digital files throughout the crediting period of the Project. Thus, information on the elevation of the social organization of

	these communities, based on the development of the actions of the Social Scope of the Project, will be monitored and accounted for.
Frequency of monitoring/recording	At every verification event
Value applied	To be accounted for after the start of the Project
Monitoring equipment	Not applicable
QA/QC procedures to be applied	The systematized information on the elevation of the social organization of these community members will be validated among the proponents, allowing greater reliability and quality of the data. In addition, the Project will undergo continuous evaluation of the information generated, through the identification of improvements in the collection and registration processes and, when relevant, the incorporation of adjustments in the Project's strategic planning.
Purpose of data	Not applicable
Calculation method	Not applicable
Comments	

Data / Parameter	Design of productive useful areas, with the adoption of sustainable practices and models.
Data unit	m^2
Description	This parameter aims to measure the amount of area in which practices occur and the development of sustainable models of agricultural and livestock production, which were implemented or expanded through the activities provided for in the Social Scope of the Project
Source of data	Reports (e.g. activity monitoring report), contracts, maps, photographs, geographic data, meeting minutes and/or other documents that corroborate as evidence that sustainable agricultural practices and production models are being implemented
Description of measurement methods and procedures to be applied	All documents produced will be stored in digital files throughout the crediting period of the Project. In this way, the areas with the adoption of sustainable practices and production models linked to the Project's social activities will be monitored and accounted for
Frequency of monitoring/recording	At every verification event
Value applied	To be accounted for after the start of the Project

Monitoring equipment	Not applicable
QA/QC procedures to be applied	The systematized information on the areas with the adoption of sustainable production practices and models will be validated among the proponents, allowing for greater reliability and quality of the data. In addition, the Project will undergo continuous evaluation of information generated, through the identification of improvements in the collection and registration processes and, when applicable, the incorporation of adjustments in the strategic planning of the Project
Purpose of data	Not applicable
Calculation method	Not applicable
Comments	This parameter also evaluates the effectiveness of the Project in promoting alternative activities to deforestation with the intention of protecting and maintaining the ecosystem services offered in the Project Area, contributing to the improvement and maintenance of HCVA 4.

Data / Parameter	Infrastructure, goods and services implemented
Data unit	Number
Description	This parameter aims to measure the amount of improvements and investments made in infrastructure, acquisition of goods and provision of services to the community, which were made possible through the activities provided for in the Project's social scope
Source of data	Reports, contracts, projects, invoices, photographs, meeting minutes and/or other documents that corroborate as evidence that there have been improvements and investments in infrastructure, goods and services for the communities involved in the project.
Description of measurement methods and procedures to be applied	All documents produced will be stored in digital files throughout the crediting period of the Project. In this way, information on improvements and investments in infrastructure, goods and services for communities will be monitored and accounted for.
Frequency of monitoring/recording	At every verification event
Value applied	To be accounted for after the start of the Project
Monitoring equipment	Not applicable

QA/QC procedures to be applied	Systematized information on improvements and investments in infrastructure, goods and services for communities will be validated among proponents, allowing greater reliability and quality of data. In addition, the Project will undergo continuous evaluation of the information generated, through the identification of improvements in the collection and registration processes and, when relevant, the incorporation of adjustments in the strategic planning of the Project
Purpose of data	Not applicable
Calculation method	Not applicable
Comments	

b) Summary of the data collection procedure

Data will be collected during and after activities with stakeholders, as well as whenever a dialogue is established with a possible local partner responsible for developing an on-site activity. This information will be systematized and presented through the Project's social activity reports.

c) Quality control and assurance procedures

The data collected and portrayed in the reports will be presented and validated during the meetings between the bidders and possible partners, as well as in the meetings with the interested parties, in order to present the results achieved by the Project, receive suggestions and criticisms to improve the actions.

d) Data archiving

All data and reports produced by the Ateles REDD+ Project will be stored by Biofílica Ambipar Environment Investments through digital files during the life cycle of the Project. Original (physical) reports, meeting minutes and field records produced during the execution of social activities will be digitized and subsequently stored by AgroSB, as well as by a possible local partner who will act in locu . Biofílica Ambipar Environment Investments will receive and keep on file a copy of these documents in digital format throughout the Project.

e) Organization and responsibilities of the parties involved in the above

All monitoring activities are the responsibility of Biofílica Ambipar Environment Investments, AgroSB and a potential local contracted partner.

4.4.2. Monitoring Plan Dissemination (CM4.3)

The monitoring plan, as well as the results obtained by monitoring the Ateles REDD+ Project, will be made available to the public through a page on the official website of Biofílica Ambipar Environment Investments. The summary documents related to the monitoring plan and results, as well as relevant information, will be made available to communities and stakeholders through feedback, meetings, lectures and by physical means on the premises of Fazenda Lagoa do Triunfo, belonging to AgroSB.

4.5. Optional Criterion: Exceptional Community Benefits

Not applicable. The Ateles REDD+ Project does not intend to be validated for gold level of this section.

4.5.1. Exceptional Community Criteria (GL2.1)

Not applicable.

4.5.2. Short-term and Long-term Community Benefits (GL2.2)

Not applicable.

4.5.3. Community Participation Risks (GL2.3)

Not applicable.

4.5.4. Marginalized and/or Vulnerable Community Groups (GL2.4)

Not applicable.

4.5.5. Net Impacts on Women (GL2.5)

Not applicable.

4.5.6. Benefit Sharing Mechanisms (GL2.6)

Not applicable.

4.5.7. Benefits, Costs, and Risks Communication (GL2.7)

Not applicable.

4.5.8. Governance and Implementation Structures (GL2.8)

Not applicable.

4.5.9. Smallholders/Community Members Capacity Development (GL2.9)

Not applicable.

5. BIODIVERSITY

5.1. Without-ProjecT BIODIVERSITY SCENARIO

5.1.1. Existing CoNDITIONS (B1.1)

The Amazon rainforest covers a vast area of plains and plateaus from the Andes to the Atlantic coast, covering approximately 6 million square kilometers (HAFFER, 2008)[113]. It corresponds to about 40% of the world's tropical rainforests and is home to one of the highest levels of biodiversity on the planet (LAURENCE, 2001)[114]. The forest plays important roles in the water and carbon cycle and acts in climate regulation on a regional and global scale (HOUGHTON et al, 2000)[115].

However, the historical and current use of Amazon Forest resources causes physical and environmental impacts, making the protection of the Brazilian Amazon one of the greatest challenges for global

[113] HAFFER, J. 2008. Hypotheses to explain the origin of species in Amazonia. Braz. J. Biol 68:917–947.
[114] LAURANCE, W. F., M. a COCHRANE, S. BERGEN, P. M. FEARNSIDE, P. DELAMÔNICA, C. BARBER, S. D'ANGELO, and T. FERNANDES. 2001. Environment. The future of the Brazilian Amazon. Science (New York, N.Y.) 291:438–439.
[115] Houghton, R., Skole, D., Nobre, C. et al. Annual fluxes of carbon from deforestation and regrowth in the Brazilian Amazon. Nature 403, 301–304 (2000). https://doi.org/10.1038/35002062

conservation objectives (Skidmore et al., 2021)[116]. During the last decades, the Brazilian Amazon has suffered from different pressures of anthropogenic origin, mainly deforestation linked to the expansion of agricultural frontiers (OLIVEIRA et al. 2014). The creation of pastures for livestock is the main factor and responsible for up to 80% of deforestation (Skidmore et al., 2021). This statement is especially pronounced in the project area, since the municipality of São Félix do Xingu is the first place in the ranking of the largest cattle herd in Brazil, with 2.4 million head of cattle (IBGE, 2022[117]), and, since 2001, occupies the worrying first place as the most deforested municipality in Brazil, a title that reaffirms the intensity of threats to the region. In this context, sustainable forest management is presented as an alternative to minimize damage to forest ecosystems.

The natural plant formations present in the Fazenda Lagoa do Triunfo complex, where the Project is located, are the Submontane Dense Ombrophilous Forest with emergent canopy, the Submontane Open Ombrophilous Forest with vines and the Alluvial Dense Ombrophilous Forest with uniform canopy. In addition, the Forest Reserves areas of AgroSB S/A are located in the eastern portion of the Amazon, in one of the eight Amazon Endemism Centers, the Tapajós Endemism Center (Figure 21). According to Braz et al. (2017)[118], areas of endemism are important because they are considered the smallest geographical units for historical biogeography analysis and house sets of unique and irreplaceable species. This Center is bounded by the right bank of the Tapajós River and the left bank of the Xingu River, where about 20% of its original primary vegetation had already been deforested by 2017 (Braz et al., 2017). In this context, the Forest Reserve areas of the properties of Fazenda Lagoa do Triunfo - AgroSB S/A are important for the maintenance of several populations of species endemic to this interfluve.

[116] SKIDMORE, M. E., MOFFETTE, F., RAUSCH, L., CHRISTIE, M., MUNGER, J., & GIBBS, H. K. (2021). Cattle ranchers and deforestation in the Brazilian Amazon: Production, location, and policies. Global Environmental Change, 68, 102280.

[117] https://cidades.ibge.gov.br/brasil/pa/sao-felix-do-xingu/pesquisa/18/16459?tipo=ranking&indicador=16533

[118] BRAZ, L. C., PEREIRA, J. L. G., FERREIRA, L. V., & THALÊS, M. C. (2017). the Situation of Amazon Endemic Areas in Relation to Deforestation and Protected Areas. Geography Bulletin, 34(3), 45. https://doi.org/10.4025/bolgeogr.v34i3.30294

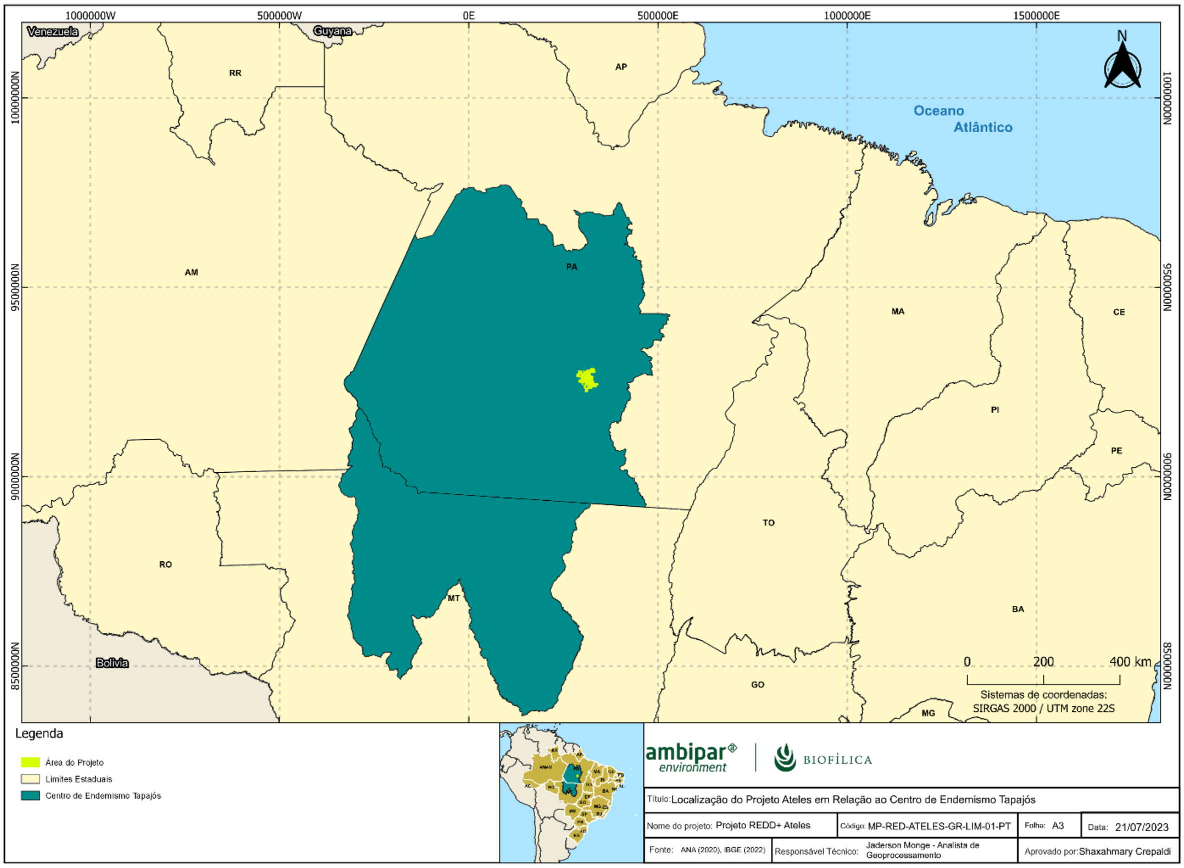

Figure 21: Location of the Tapajós Endemism Center in relation to the location of the Ateles REDD+ Project.

In addition, the Ateles REDD+ Project area is part of the Triunfo do Xingu Environmental Protection Area, which, according to Imazon data, is the most deforested Conservation Unit in Brazil. The project is also adjacent to the Terra do Meio Ecological Station, another Conservation Unit that is on the list of the most deforested, occupying the eighth place (Figure 22). Given the above, we can state the importance of preserving the forest remnants of the Lagoa do Triunfo farm for local biodiversity.

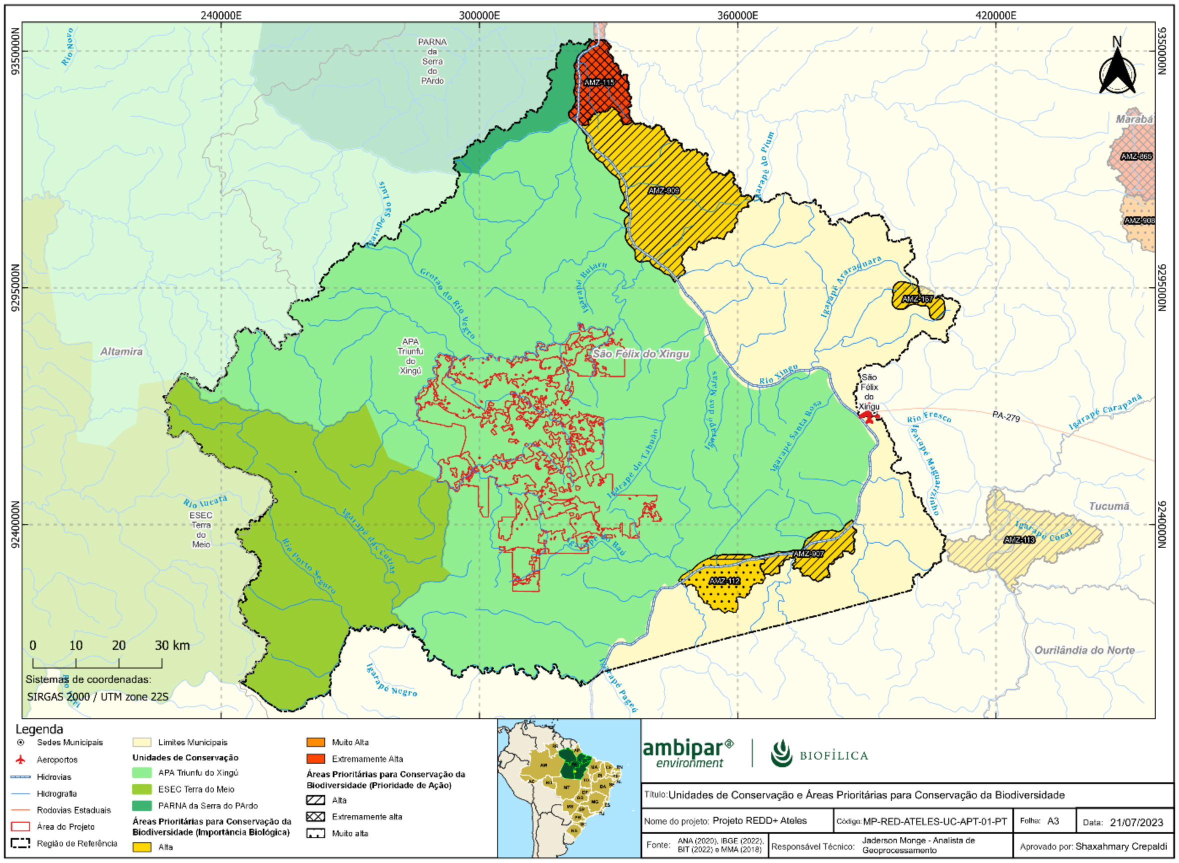

Figure 22: Degree of conservation priority in the Project Area and nearby Conservation Units. (Adapted from MMA, 2022).

Flora

For plant characterization, an analysis was carried out based on the results of the forest inventory carried out for carbon estimation by Biodendro Consultoria Florestal, obtained from the sampling technique with identification of popular names (vernaculars), in the field visit carried out in November 2022 in the project area. The studies were carried out with data obtained in 16 plots of 625 m² (25 x 25) corresponding to subplot 1 (all trees with a Breast Height Diameter greater than or equal to 10cm) of the sampling plots of the conglomerates intended for the quantification of biomass and carbon stocks.

The parameters used in the phytosociological evaluation were density, dominance and frequency of the species, in absolute and relative terms. The Coverage Value index was calculated by summing the relative values of density and dominance, while the importance value index was calculated by summing the relative values (%) of density, frequency and dominance, according to Mueller Dombois and Ellemberg (1974[119]).

In this survey, 34 botanical families and 110 taxa were identified. The classification of the individuals was made through the popular name of the species by the field team, formed by people who live in communities in the middle of the forest and have experience as woodsmen. And, later, compared with the Brazilian Tree

[119] MUELLER DOMBOIS, D.; ELLENBERG, H. Aims and methods of vegetation and ecology. New York: J. Wiley & Sons, 1974. 547 p.

Catalog[120], the main reference of popular names of Amazonian tree species, and debugged using the Reflora project database as a reference[121]. From these analyzes, the following table was prepared.

Table 43: Flora species recorded in the Project Area.

Family	Species	Popular Name
Anacardiaceae R. Br.	Astronium lecointei Ducke	maracatiara
Anacardiaceae R. Br.	Spondias mombin L.	taperebá
Annonaceae Juss.	Guatteria citriodora Ducke	laranjinho
Annonaceae Juss.	Guatteria procera R.E.Fr.	envira-preta
Annonaceae Juss.	Unonopsis guatterioides (A.DC.) R.E.Fr.	envira
Annonaceae Juss.	Xylopia benthamii R.E.Fr.	envira-amarela
Apocynaceae Juss.	Aspidosperma excelsum Benth.	sapopema
Apocynaceae Juss.	Aspidosperma Mart. & Zucc.	carapanã
Apocynaceae Juss.	Himatanthus articulatus (Vahl) Woodson	janaúba, sucuba
Apocynaceae Juss.	Lacmellea arborescens (Müll.Arg.) Markgr.	Pau-de-colher
Araliaceae Juss.	Didymopanax morototoni (Aubl.) Decne. & Planch.	morototó
Arecaceae Schultz Sch.	Astrocaryum aculeatum G.Mey.	tucumã
Arecaceae Schultz Sch.	Attalea Kunth	palmeira inajá
Arecaceae Schultz Sch.	Attalea speciosa Mart. ex Spreng.	babaçu
Arecaceae Schultz Sch.	Euterpe precatoria Mart.	açaí
Arecaceae Schultz Sch.	Socratea exorrhiza (Mart.) H.Wendl.	paxiúba
Arecaceae Schultz Sch.	Syagrus inajai (Spruce) Becc.	jareua
Bignoniaceae Juss.	Handroanthus Mattos	ipê
Bignoniaceae Juss.	Handroanthus serratifolius (Vahl) S.Grose	ipê-amarelo
Bignoniaceae Juss.	Jacaranda copaia (Aubl.) D.Don	caranapaúba, para-pará
Burseraceae Kunth	Protium altissimum (Aubl.) Marchand	breu-vermelho
Burseraceae Kunth	Protium Burm.f.	breu; breu-amarelo; breu-branco
Caryocaraceae Voigt	Caryocar glabrum (Aubl.) Pers.	piquiarana
Caryocaraceae Voigt	Caryocar villosum (Aubl.) Pers.	piquiá
Chrysobalanaceae R.Br.	Couepia paraensis (Mart. & Zucc.) Benth.	uxirana
Chrysobalanaceae R.Br.	Licania Aubl.	macuco
Clusiaceae Lindl.	Garcinia madruno (Kunth) Hammel	bacuri
Combretaceae R.Br.	Terminalia corrugata (Ducke) Gere & Boatwr	cuiarana
Combretaceae R.Br.	Terminalia tetraphylla (Aubl.) Gere & Boatwr.	tanimbuca

[120] Camargos, J.A.A.; Czarneski, CM. ; Meguerditchian, I.; Iliveira, D. 1996. Catalog of trees from Brazil. IBAMA/ LPF, Brasília, Federal District. 888p.
[121] https://floradobrasil.jbrj.gov.br/reflora/listaBrasil/ConsultaPublicaUC/ConsultaPublicaUC.do#CondicaoTaxonCP

Family	Species	Popular Name
Coulaceae Tiegh.	Minquartia guianensis Aubl.	quariquara
Dichapetalaceae Baill.	Tapura singularis Ducke	pau-de-bicho
Elaeocarpaceae Juss.	Sloanea guianensis (Aubl.) Benth.	urucurana
Euphorbiaceae Juss.	Hevea brasiliensis (Willd. ex A.Juss.) Müll.Arg.	seringueira
Fabaceae Lindl.	Abarema jupunba (Willd.) Britton & Killip	fava-amargoso
Fabaceae Lindl.	Amburana acreana (Ducke) A.C.Sm.	cerejeira
Fabaceae Lindl.	Apuleia leiocarpa (Vogel) J.F.Macbr.	garapeira
Fabaceae Lindl.	Cedrelinga cateniformis (Ducke) Ducke	cedro mara
Fabaceae Lindl.	Copaifera duckei Dwyer	copaíba
Fabaceae Lindl.	Dalbergia spruceana Benth.	jacarandá
Fabaceae Lindl.	Dialium guianense (Aubl.) Sandwith	tamarindo
Fabaceae Lindl.	Diplotropis purpurea (Rich.) Amshoff	sucupira
Fabaceae Lindl.	Dipteryx odorata (Aubl.) Forsyth f.	cumaru
Fabaceae Lindl.	Hymenaea courbaril L.	jatobá
Fabaceae Lindl.	Hymenolobium Benth.	angelim-amargoso
Fabaceae Lindl.	Hymenolobium petraeum Ducke	angelim-pedra
Fabaceae Lindl.	Inga Mill.	ingá; ingá-pena; ingá-xixi
Fabaceae Lindl.	Inga paraensis Ducke	ingarana
Fabaceae Lindl.	Parkia nitida Miq.	fava
Fabaceae Lindl.	Peltogyne paniculata Benth.	roxinho
Fabaceae Lindl.	Pentaclethra macroloba (Willd.) Kuntze	paracaxi
Fabaceae Lindl.	Platymiscium trinitatis Benth.	macacauba
Fabaceae Lindl.	Pterocarpus officinalis Jacq.	sangue-de-dragão
Fabaceae Lindl.	Schizolobium parahyba (Vell.) Blake	bandarra
Fabaceae Lindl.	Tachigali glauca Tul.	taxi
Goupiaceae Miers	Goupia glabra Aubl.	cupiuba
Humiriaceae A.Juss.	Endopleura uchi (Huber) Cuatrec.	uxi
Hypericaceae Juss.	Vismia guianensis (Aubl.) Choisy	lacre
Lauraceae Juss.	Aniba canelilla (Kunth) Mez	preciosa
Lauraceae Juss.	Licaria armeniaca (Nees) Kosterm.	louro-pimenta
Lauraceae Juss.	Licaria brasiliensis (Nees) Kosterm.	louro
Lauraceae Juss.	Nectandra cuspidata Nees	louro-tamanco
Lauraceae Juss.	Ocotea Aubl.	louro-caraíba; louro-passarinho
Lauraceae Juss.	Ocotea longifolia Kunth	louro-abacate
Lecythidaceae A.Rich.	Bertholletia excelsa Bonpl.	castanheira
Lecythidaceae A.Rich.	Couratari multiflora (Sm.) Eyma	tauari
Lecythidaceae A.Rich.	Eschweilera coriacea (DC.) S.A.Mori	matamata
Lecythidaceae A.Rich.	Lecythis idatimon Aubl.	ripeira
Lecythidaceae A.Rich.	Lecythis pisonis Cambess.	sapucaia
Malvaceae Juss.	Apeiba albiflora Ducke	pente-de-macaco
Malvaceae Juss.	Ceiba pentandra (L.) Gaertn.	sumaúma

Family	Species	Popular Name
Malvaceae Juss.	Guazuma ulmifolia Lam.	periquiteira
Malvaceae Juss.	Matisia cordata Humb. & Bonpl.	sapateira
Malvaceae Juss.	Theobroma cacao L.	cacau
Malvaceae Juss.	Theobroma speciosum Willd. ex Spreng.	cacauí
Malvaceae Juss.	Theobroma sylvestre Mart.	cacau-do-mato
Meliaceae A.Juss.	Carapa guianensis Aubl.	andiroba
Meliaceae A.Juss.	Cedrela sp.	cedro
Meliaceae A.Juss.	Guarea guidonia (L.) Sleumer	jataúba
Moraceae Gaudich.	Brosimum acutifolium Huber	moruré
Moraceae Gaudich.	Brosimum lactescens (S.Moore) C.C.Berg	muiratinga
Moraceae Gaudich.	Brosimum parinarioides Ducke	amapá
Moraceae Gaudich.	Brosimum Sw.	amapaçu
Moraceae Gaudich.	Brosimum utile (Kunth) Pittier	garrote
Moraceae Gaudich.	Clarisia racemosa Ruiz & Pav.	Quariuba
Moraceae Gaudich.	Ficus donnell-smithii Standl.	apuí
Moraceae Gaudich.	Ficus L.	figueira
Moraceae Gaudich.	Perebea guianensis Aubl.	moratinga
Myrtaceae Juss.	Psidium L.	goiaba-araçá; goiabinha
Peraceae Klotzsch	Pera glabrata (Schott) Baill.	tamanqueira
Rubiaceae Juss.	Calycophyllum spruceanum (Benth.) K.Schum.	pau-mulato
Rubiaceae Juss.	Capirona macrophylla (Poepp.) Delprete	mulatinheira
Rutaceae A.Juss.	Euxylophora paraensis Huber	pau-amarelo
Rutaceae A.Juss	Galipea trifoliata Aubl.	canela-de-velho
Rutaceae A.Juss	Pilocarpus Vahl	cutieira
Sapindaceae Juss.	Talisia esculenta (Cambess.) Radlk.	pitomba
Sapotaceae Juss.	Chrysophyllum amazonicum T.D.Penn.	abiurana
Sapotaceae Juss.	Chrysophyllum lucentifolium Cronquist	guajara-abiu
Sapotaceae Juss.	Manilkara bidentata (A.DC.) A.Chev.	maçaranduba
Sapotaceae Juss.	Manilkara paraensis (Huber) Standl.	maparajuba
Sapotaceae Juss.	Gardnerian micropholis (A.DC.) Pierre	catuaba
Sapotaceae Juss.	Pouteria Aubl.	abiu-ferro; guajara-faia
Sapotaceae Juss.	Pouteria guianensis Aubl.	guajara
Sapotaceae Juss.	Pouteria oblanceolata Pires	abiu
Sapotaceae Juss.	Pouteria oppositifolia (Ducke) Baehni	guajara-bolacha
Simaroubaceae DC.	Simarouba had loved Aubl.	marupá
Urticaceae Juss.	Cecropia sciadophylla Mart.	embaúba-branca, embaúba-vermelha
Urticaceae Juss.	Pourouma guianensis Aubl.	embaubarana
Verbenaceae J.St.-Hil.	Citharexylum macrochlamys Pittier	jacatauba

Family	Species	Popular Name
Vochysiaceae A.St.-Hil.	Erisma laurifolium Spruce ex Warm.	jaboti moita
Vochysiaceae A.St.-Hil.	Erisma uncinatum Warm.	cedrinho

Among the identified species, twelve were classified as endemic and thirteen species were categorized under some level of threat. Effectively threatened are nine, those in the categories (EN, VU and NT). Endangered species are, for the most part, timber species that experience population decline due to habitat reduction and forest exploitation.

Table 44: Endangered and endemic species registered in the flora inventory of Fazenda Lagoa do Triunfo, São Félix do Xingu – PA. Threat grades: LC: Least Concern; NT: Near Threatened; VU: Vulnerable; EN: Endangered; and Endemic: END.

Family	Species	Popular Name	IUCN	END
Arecaceae Schultz Sch.	Attalea speciosa Mart. ex Spreng.	babaçu	-	END
Bignoniaceae Juss.	Handroanthus serratifolius (Vahl) S.Grose	ipê-amarelo	EN	-
Bignoniaceae Juss.	Jacaranda copaia (Aubl.) D.Don	caranapaúba, para-pará	NT	-
Combretaceae R.Br.	Terminalia corrugata (Ducke) Gere & Boatwr	cuiarana	-	END
Combretaceae R.Br.	Terminalia tetraphylla (Aubl.) Gere & Boatwr.	tanimbuca	-	END
Dichapetalaceae Baill.	Tapura singularis Ducke	pau-de-bicho	LC	END
Fabaceae Lindl.	Amburana acreana (Ducke) A.C.Sm.	cerejeira	VU	-
Fabaceae Lindl.	Apuleia leiocarpa (Vogel) J.F.Macbr.	garapeira	VU	-
Fabaceae Lindl.	Hymenaea courbaril L.	jatobá	LC	-
Fabaceae Lindl.	Hymenolobium petraeum Ducke	angelim-pedra	LC	-
Fabaceae Lindl.	Peltogyne paniculata Benth.	roxinho	LC	-
Fabaceae Lindl.	Copaifera duckei Dwyer	copaíba	-	END
Humiriaceae A.Juss.	Endopleura uchi (Huber) Cuatrec.	uxi	-	END
Lauraceae Juss.	Aniba canelilla (Kunth) Mez	preciosa	-	END
Lecythidaceae A.Rich.	Bertholletia excelsa Bonpl.	castanheira	VU	-
Lecythidaceae A.Rich.	Lecythis idatimon Aubl.	ripeira	-	END
Lecythidaceae A.Rich.	Lecythis pisonis Cambess.	sapucaia	-	END
Moraceae Gaudich.	Brosimum lactescens (S.Moore) C.C.Berg	muiratinga	LC	-
Rutaceae A.Juss.	Euxylophora paraensis Huber	pau-amarelo	EN	END
Sapotaceae Juss.	Manilkara paraensis (Huber) Standl.	maparajuba	NT	END

Family	Species	Popular Name	IUCN	END
Sapotaceae Juss.	Pouteria oppositifolia (Ducke) Baehni	guajara	-	END

The *Handroanthus serratifolius*, ipê amarelo, is threatened by forest exploitation that has caused significant population decline. It is one of the most vulnerable forest species in the Amazon, due to its low natural density and slow growth (IUCN, 2023[122]). *Jacaranda copaia*, on the other hand, has a localized population decline due to deforestation activities throughout its range, but the species is able to regenerate in deforested areas, so these are not considered major threats to this species.

The cerejeira, *Amburana acreana*, has been widely exploited for its wood, used for the manufacture of luxury furniture, with continuous decline in area, extent and/or quality of habitat (IUCN, 2023[123]). The main threat to the garapeira, *Apuleia leiocarpa*, is logging, since it is widely used in construction.

While the castanheira-do-pará, *Bertholletia excelsa*, experiences a major population decline due mainly to deforestation. As well as *Pouteria oppositifolia, Manilkara paraensis* and *Euxylophora paraensis*, which are threatened due to the continuous decline in area, extent and/or quality of their habitat and the reduction in the number of mature individuals by forest exploitation.

The embaúba branca (Cecropia sciadophylla) was the species with the highest value of absolute density, highest frequency and dominance, so it stands out as the tree with the highest index of coverage value and importance. As well as Inga sp, Protium sp.e louro abacate (Ocotea longifolia).

Finally, the botanical families that best represent the forests of the Lagoa do Triunfo farm are Fabaceae Lindl., Urticaceae Juss., Lauraceae Juss., Sapotaceae Juss., Malvaceae Juss., Burseraceae Kunth, Annonaceae Juss., Moraceae Gaudich. and Chrysobalanaceae R.Br.

<u>Fauna</u>

For the development of conservation strategies, it is essential to understand the habitat requirements of the resident fauna and determine which of them are most vulnerable to the disturbances of human activities. In this perspective, a field faunal survey was carried out in the area of Lagoa do Triunfo farm comprising three groups: Avifauna, Mastofauna and Herpetofuna. These groups were chosen due to the representativeness that these individuals have in the composition of the local fauna, as well as the availability of methodological procedures already established in the literature for the diagnosis. In addition, a survey was also carried out with secondary data for the Ichthyofauna and Entomofauna of the project area.

With regard to the collection of primary data, the survey was carried out in the spring, between November 20 and 26, 2022, only in the Project Area. Secondary data were obtained from searches in technical-scientific information bases (Scielo and Google Scholar) by fauna studies carried out in areas close to AgroSB, or even in AgroSB itself, such as the conservation units, Terra do Meio Ecological Station, Serra do Pardo National Forest and Rio Iriri Extractive Reserve, as well as in the municipalities of São Felix do Xingu, Tucumã, Parauapebas, PA. In all, 570 bird species were recorded in the base data, distributed in 77 families, 42 species of medium and large mammals, 15 species representing herpetofauna, 12 representing amphibians and 03 the reptile group, 71 fish species, in addition to 876 species representing entomofauna, distributed in 90 families.

[122] https://www.iucnredlist.org/species/61985509/145677076#threats
[123] https://www.iucnredlist.org/species/34426/9867560

Avifauna

To complement the characterization made with secondary data of the avifauna of the Study Area, the quantitative technique of MacKinnon lists (Ribon 2010) was used[124], indicated to maximize the number of species recorded in a location. To this end, a systematic sampling was carried out, selecting three types of vegetation: dense ombrophilous forest, riparian forest and palm plantation interface with forest edge. These areas were chosen because they represent the variation of existing environments within the reserves and because they house large remnants of native vegetation.

During the bird sampling, 40 MacKinnon lists were made, in which 152 bird species were recorded, distributed in 48 families. All species recorded in the field were also found in the bibliographic data. Two species recorded during sampling are classified as endemic to the Southeast Amazon or the Tapajós Endemism Center, namely: the Jacupiranga (Penelope pileata) and the Araçari-de-pescoço-vermelho (Pteroglossus bitorquatus).

Figure 23: Endemic species registered in the Project Area: A. Araçari-de-pescoço-vermelho (Pteroglossus bitorquatus), B. jacupiranga (Penelope pileate).

Six bird species recorded during sampling are mentioned in some list of threatened fauna (Table 45). All of them are mentioned in the global list of threatened fauna (IUCN, 2022). The Arara-azul (Anodorhynchus hyacinthinus) is cited as a species "Vulnerable" to extinction in the global list and in the state of Pará. The jacupiranga (Penelope pileata) is cited as a "Vulnerable" species in the national and global threat lists. The red-necked jaguar (Pteroglossus bitorquatus) and the Mutum-de-penacho are cited in the global list. In addition, azulona (Tinamus tao) is cited in the national lists and the Hellmayr's Tribe (Pyrrhura amazonum) is cited in the national list as a "Vulnerable" species and as "near-threatened" in the global list (Table 45).

[124] Ribon, R. 2010. Sampling of birds by the Mackinnon lists method. In: Ornitol. and Conserv. Apl. Science Pesqui. and Levant. Techniques (eds. . Von Matter, S., Straube, F., Accordi, I., Piacentini, V. & Cândido-Junior, J.). Technical Books, Rio de Janeiro, pp. 33–44

Table 45: List of species mentioned in the lists of threatened fauna recorded during field sampling in the reserve areas of AgroSB – Lagoa do Triunfo (six species). End: endemic species of the Amazon; Tapajós: Endemic species of the Tapajós Endemism Center. IUCN: Species listed on the Global List of Endangered Fauna (IUCN, 2022) – degrees of threat: endangered species (EN), vulnerable species (VU), near-threatened species (NT).

Taxon	Popular Name	END	PA	BR	IUCN
Anodorhynchus hyacinthinus	arara-azul		VU		VU
Crax fasciolata	mutum-de-penacho				VU
Penelope pileata	jacupiranga	Tapajós		VU	VU
Pteroglossus bitorquatus	araçari-de-pescoço-vermelho	Amazon			EN
Pyrrhura amazonum	tiriba-de-hellmayr			VU	NT
Tinamus tao	azulona			VU	VU

The collector curve did not reach a horizontal plateau, indicating that more bird species should be found with increasing sampling effort (Figure 24). In fact, the number of species recorded in the field (152) represented 70% of the number of species estimated by the first-order Jackknife estimator (210 species). The richness recorded during the samplings represented 26% of the total richness found in the bibliographic data (570 species).

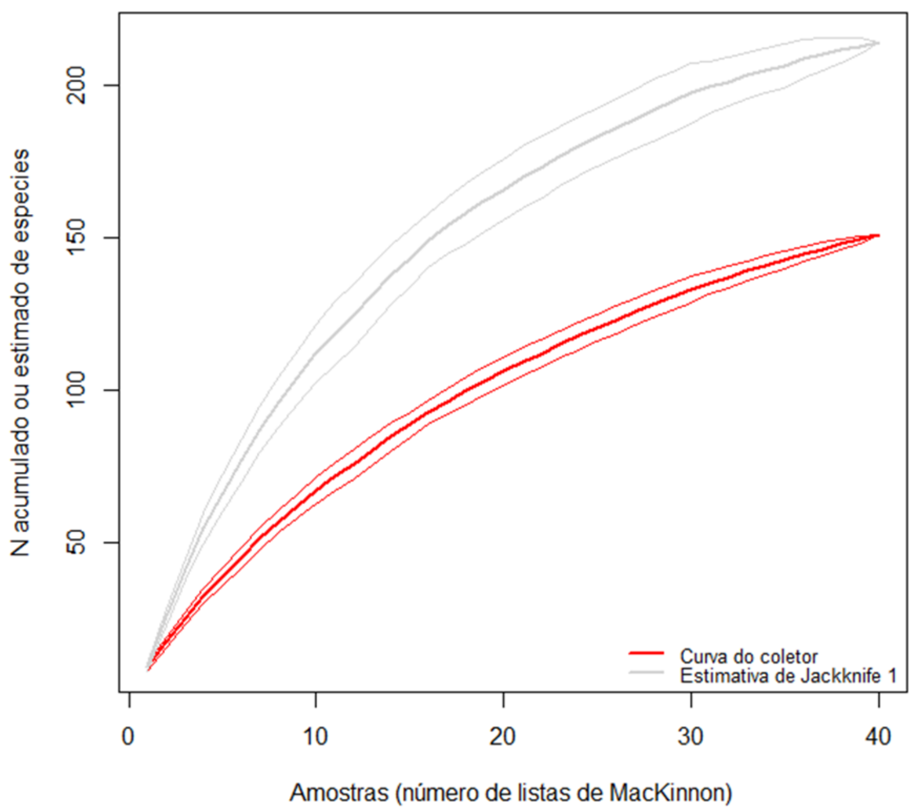

Figure 24: Collector curve (red line) and first order Jackknife richness estimation (gray line) of poultry sampling through the MacKinnon lists technique performed in the reserve areas of AgroSB S/A (November 2022). Vertical bars represent standard deviations.

Of the 152 bird species recorded in the reserve areas of AgroSB, 112 (74% of the total) are classified as dependent on forest environments, such as the papa-formiga-de-bando (Microrhopias quixensis), the maitaca-de-cabeça-azul (Pionus menstruus) and the choquinha-de-olho-branco (Epinecrophylla leucophthalma). While 12% (18 spp.) of the species are classified as not dependent on forest environments.

In addition, 22 species occupy aquatic environments, the Anhuma (Anhima cornuta), the arapapá (Cochlearius cochlearius) and the picaparra (Heliornis fulica).

Regarding sensitivity to environmental changes, 27 (18% of the total) species are classified as highly sensitive to changes in their habitat, among them the vissiá (Rhytipterna simplex), the ipecuá (Thamnomanes caesius) and the pomba-botafogo (Patagioenas subvinacea). On the other hand, 40% (61 spp.) of the species have average sensitivity to changes, including the inhambu-pixuna (Crypturellus cinereus), the periquito-de-asa-dourada (Brotogeris chrysoptera) and the benedito-de-cabeça-vermelha (Melanerpes cruentatus). 63 species have low sensitivity to changes in their habitat.

Mammalian Fauna

To characterize the medium and large mammals of the Study Area, bibliographic data (literature review) and field sampling were used, through active search techniques, traversing the demarcated trails and the installation of camera traps in the areas selected for each treatment.

The active search method through paths was used aiming at direct visual contact of individuals or visualization of traces such as footprints, feces, burrows and sniffing. The work was carried out from 6 am to 11 am and from 3:30 pm to 8 pm. Trails were established proportional to the types of environments present in the study areas, taking advantage, as far as possible, of roads, stream banks and pre-existing trails. These trails were all georeferenced with a manual GPS device. The routes were traveled at a slow pace with a constant speed of approximately 2 km/h, at least two repetitions were performed for each route.

According to the data obtained in the literature, 42 species were recorded, belonging to 23 families and 9 orders. On the other hand, through the data obtained in this first monitoring campaign, 16 species of medium and large wild mammals were recorded, distributed in 14 families and 6 orders (Table 46).

Table 46: Preliminary list of mammal species recorded during the 1st monitoring campaign for medium and large mammals. End: endemic species of the Amazon biome. Tolerance to human presence (Tol.): S: synanthropic: very tolerant or exclusive of anthropic areas; P: perianthropic: tolerant to low densities of human presence; A: alloanthropic – little tolerant to human presence. Dependence on forest formations (DEP): ND: non-dependent, SD: semi-dependent; D: dependent. Type of record (Reg): RE: direct observation; R: footprints or traces; PA: camera trap; VOC: vocalization. Degree of threat according to: BP: Threatened species in the state of Pará (IDERFLOR-BIO, 2016 and according to COEMA Resolution no. 54, of 10/24/2007); BR: Threatened species in Brazil (MMA, 2018); IUCN: Globally threatened species (IUCN, 2023). Threat grades: DD= data deficient; LC= least concern; NT= near threatened; VU=vulnerable; EN= endangered; CR= critically endangered. Exo = exotic species.

Family / Species	Popular Name	Reg	PA	BR	IUCN	End	Dep	Tol
Dasypodidae								
Dasypus novemcinctus	tatu-galinha	R	-	-	-		SD	P
Dasypus kaapleri	tatu-quinze-quilos	OD	-	-	-		SD	P
Pithecidae								
Callicebus moloch	guigó-vermelho	VOC	-	-	-	X	D	A
Cebidae								
Sapajus apella	macaco-prego	VOC	-	-	-		D	P
Atelidae								
Ateles marginatus	macaco-aranha-de-cara-branca	OD; VOC	VU	EN	EN	X	D	A
Felidae								
Leopardus pardalis	jaguatirica	AF	-	-	-		D	A
Panthera onca	onça-pintada	OD; R	VU	VU	NT		D	A

Family / Species	Popular Name	Reg	PA	BR	IUCN	End	Dep	Tol
Canidae								
Cerdocyon thous	cachorro-do-mato	OD	-	-	-		SD	P
Tapiridae								
Tapirus terrestris	anta	OD; R	-	VU	VU		D	A
Tayassuidae								
Tayassu pecari	queixada	OD; AF; R	-	VU	VU		D	A
Suiidae								
Sus sp.	javaporco	OD	Exo	Exo	Exo	Exo	ND	Exo
Cervidae								
Mazama americana	veado-mateiro	AF	-	-	DD		D	A
Sciuridae								
Sciurus aestuans	caxinguele	OD	-	-	-		SD	P
Caviidae								
Hydrochoerus hydrochaeris	capivara	OD; R	-	-	-		A	P
Cuniculidae								
Cuniculus paca	paca	AF	-	-	-		D	A
Dasyproctidae								
Dasyprocta leporin	cutia	OD	-	-	-		D	P

Most of the species registered have low tolerance to environmental changes and are dependent on forest formations, that said, it is worth reinforcing the need to maintain areas with native vegetation in order to provide a favorable place for the maintenance of biodiversity. Another important point is that the presence of endemic species sensitive to environmental changes is a strong indication that the fauna community is well established, since we observed the presence of top carnivorous species of the food chain, such as the onça-pintada (Panthera onca) and jaguatirica (Leopardus pardalis). In addition, we recorded several potential seed dispersal species, such as the cutia (Dasyprocta leporina), paca (Cuniculus paca), anta (Tapirus terrestris), in addition to the primates macaco-aranha-de-cara-branca (Ateles marginatus), guigó-vermelho (Callicebus moloch) and macaco-prego (Sapajus appella). Mammals play an important role in the recovery processes of altered areas, due to their ability to disperse and predate seeds and seedlings of several plant species (Bascompte & Jordano, 2007)[125]. Medium and large mammals can disperse many seeds over long distances (Stoner et al., 2007)[126]. These plant-animal interactions are fundamental for the biodiversity of the Amazon Forest, assisting in the recovery of fauna diversity and associated ecosystem services, which are fundamental for natural regeneration processes in degraded forests (Bowen et al., 2007)[127]. According to some studies, the community of medium and large mammals reflects the availability of food provided by the environment, so that frugivorous and granivorous species are more frequent in

[125] BASCOMPTE, J. Jordano, P. Plant-animal mutualistic networks: the architecture of biodiversity. Annual Review of Ecology, Evolution, and Systematics. 2007; 38:567–93.
[126] STONER, K.E., Riba-Hernandez, P., Vulinec, K., Lambert, J.E. The role of mammals in creating and modifying seed shadows in tropical forests and some possible consequences of their elimination. Biotropics. 2007; 39(3):316–27.
[127] BOWEN, M.E. McAlpine, C.A.; house, A.P., Smith, G.C. Regrowth forests on abandoned agricultural land: a review of their habitat values for recovering forest fauna. Biological Conservation. 2007; 140(3):273–96.

places with little or no anthropic alteration (Parry, Barlow & Peres, 2007[128]). Given this, there is a need to maintain extensive forest areas to ensure the survival of several species of wildlife.

Another important factor is the presence of endangered species recorded during this campaign, such as the onça-pintada (Panthera onca) and the macaco-aranha-de-cara-branca (Ateles marginatus) present in the three lists consulted, both classified as "Vulnerable (VU)" in the state list, while in the national list, the primate was classified as "Endangered (EN)", while the onça-pintada was classified as "Vulnerable (VU)" and in the IUCN global list, the macaco-aranha-de-cara-branca was classified as "Endangered (EN)" and the onça-pintada as "Near Threatened (NT)". Also referring to the data collected in the field, two species classified as "Vulnerable (VU)" were registered in the national and global lists, the anta (Tapirus terrestris) and the queixada (Tayassu pecari) and finally, the veado-mateiro (Mazama americana) was classified as "Data deficient (DD)" in the global list of endangered fauna. Regarding the data obtained through the available literature, some species were found in the lists of endangered fauna consulted, the species present in the three lists are the tatu-canastra (Priodontes maximus), tamanduá-bandeira (Myrmecophaga tridactyla) classified as "Vulnerable (VU)" in all. Other species present in the three lists included are: margay cat, otter and vinegar dog, in addition to the species mentioned referring to field collections. Already, the bugio-de-mãos-ruivas (Alouatta belzebul), the anta (Tapirus terrestris) and the queixada (Tayassu pecari) are present in the national and global lists of the IUCN, classified as and "Vulnerable (VU)". The onça-parda (Puma concolor) was classified as "Vulnerable (VU)" in the state and national lists. Finally, the veado-mateiro (Mazama americana) was classified as "Data deficient (DD)" and the tapiti (Sylvilagus brasiliensis) classified as "Near Threatened (NT)" by the IUCN global list.

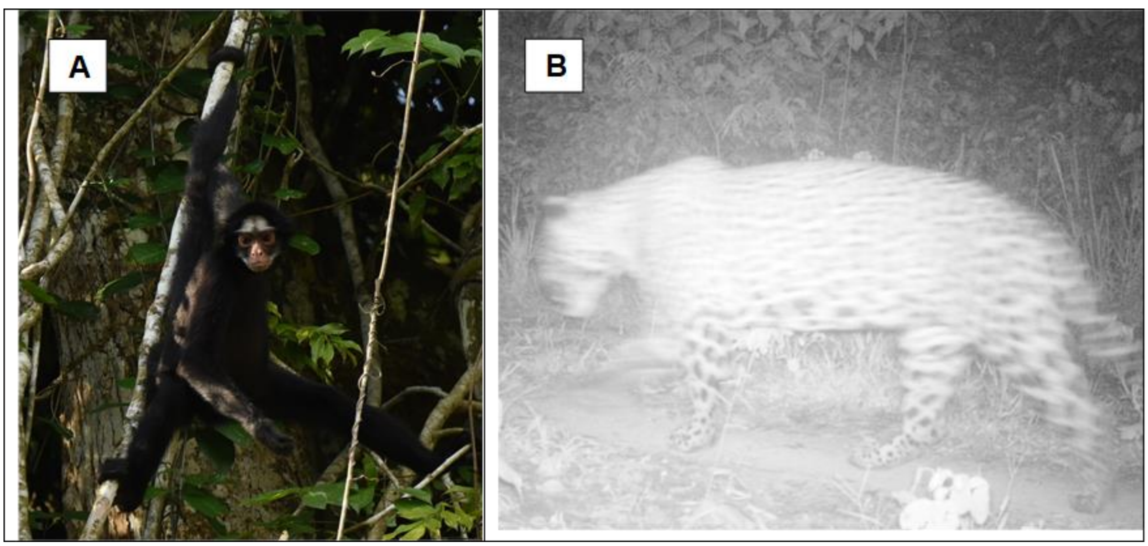

Figure 25: Figure A2 -. Endangered species recorded in the Project Area: Macaco-aranha-de-cara-branca(Ateles marginatus), B.onça-pintada (Panthera onca).

However, a factor that can influence wildlife species is the presence of exotic species in the forest areas, because in addition to the presence of wild boars, we also observed the presence of domestic dogs and cats in places close to small villages. The presence of exotic species in native forests can influence wild species in several aspects, for example through competition for food, shelters, among others. That said, invasion by alien species can be so severe, it is considered the second leading cause of biodiversity loss worldwide.

[128] Parry, L., Barlow, J., & Peres, C. (2007). Large-vertebrate assemblages of primary and secondary forests in the Brazilian Amazon. Journal of Tropical Ecology, 23(6), 653-662. doi:10.1017/S0266467407004506

Herpetofauna

The methodology used in field work was active search, that is, visual and auditory contacts. The active search carried out during the day (6:00 am to 11:00 am) is directed to records of reptiles and possibly amphibians, since the period of greatest amphibian activity is the afternoon (5:00 pm to 7:00 pm) and night (7:00 pm to 10:00 pm). In this method, visual inspections are carried out in the places with the greatest potential to house species representative of these groups, following pre-existing trails and water bodies.

The search for reptiles occurred in forests and their surroundings, cultivated areas, in litter, under fallen trunks, in the branches of shrubs and trees, in tree hollows, in bromeliads, under stones and in the inspection of burrows (Vanzolini & Papavero, 1967)[129]. During the sampling, possible traces of reptiles, such as feces and traces, were also searched.

In addition to the auditory detections of anuran amphibians, performed through visual and vocal recognition in the field, as well as later through the analysis of the material obtained (vocal and photographic records). The taxonomy and nomenclature used were based on Costa & Bérlis (2018; reptiles)[130] and Segalla et. al (2021; amphibians)[131].

During this campaign, a total of 15 species representing herpetofauna were recorded, 12 representing amphibians and 03 the reptile group. In relation to the CITES list (Convention on International Trade in Endangered Species of Wild Flora and Fauna), the species Paleosuchus trigonatus is listed in Annex II in the database consulted (https://checklist.cites.org/). According to Article 8 of Decree No. 3,607, of September 21, 2000, the species included in Annex II of CITES are those that, although currently not necessarily in danger of extinction, may reach this situation, unless the trade in specimens of such species is subject to strict regulation, and their commercialization may be authorized by the Administrative Authority, by granting a License or issuing a Certificate.

Most recorded herpetofauna species inhabit both open and forested areas and can use temporary puddles throughout the year to reproduce in the case of amphibians (Haddad & Prado, 2005[132]; Haddad et al., 2013[133]). Thus, they have a wide distribution and prolonged reproductive behavior and do not require exclusively forest environments for reproduction or feeding.

The strictly forest species recorded during this campaign are Physalaemus ephippifer, Leptodactylus knudseni, Boana boans and Dendropsophus melanargyreus, highlighting the need to maintain preserved areas close to the study area. In forest areas, there is less anthropogenic interference, greater availability of resources for the survival of herpetofauna representatives, such as food supply, reproductive sites and shelters.

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[129] VANZOLINI, P.E.; PAPAVERO,N. (ORG.).1967. Manual of collection and preparation of terrestrial and freshwater animals. Department of Zoology of the University of São Paulo and Department of Agriculture of the State of São Paulo. São Paulo, Brazil. 222pp.
[130] COSTA, H.C.; BÉRNILS, R.S. 2018. Brazilian Reptiles. Brazilian Herpetology. Volume 4. Number 3. April 2018.
[131] SEGALLA, M.V.; BERNECK, B.; CANEDO, C.; CARAMACHI, U.; CRUZ, C.A.G.; GARCIA, P.C.A.; GRANT,T.; HADDAD,C.F.B.; LOURENÇO, A.C.C.; MANGIA,S.; MOTT,T.; NASCIMENTO,L.B.; TOLEDO, L.F.; WERNECK,F.P.; LANGONE, J.A. 2021. . Brazilian Amphibians: List of species. Brazilian Herpetology 5(2): 34-46
[132] HADDAD, C.F.B.; PRADO, C.P.A. 2005. Reproductive modes in frogs and their unexpected diversity in the Atlantic forest of Brazil. BioScience 55:207-217.
[133] HADDAD, C.F.B. ; TOLEDO, L.F. ; PRADO, CPA ; LOEBMANN, D. ; GASPARINI, J.L. ; SAZIMA, I . 2013. Atlantic Forest Amphibian Guide: Diversity and Biology. 1. ed. São Paulo: Anolis Books, v. 1. 543p.

Ichthyofauna

Fish are one of the largest faunal groups represented in the Amazon basin, however, despite this mega diversity and the attention it attracts, knowledge about the diversity and geographical distribution of Amazonian fish is not yet synthesized in a general framework that allows broad generalizations, as well as there are few studies for some specific regions. When it comes to freshwater fish species, Brazil is the country with the highest biodiversity, with at least 2,587 valid species, and this high diversity is mainly due to the presence of several hydrographic systems with considerable ichthyofaunistic distinction between them, as well as the interspecific, phylogenetic and zoogeographic complexity (AGOSTINHO, et al 2005; BUCKUP et al., 2007; ROSA & LIMA, 2018).

Some studies suggest that the Amazon basin has a number of species between 1,300 and 3,500 species, but further studies are still needed for the region (for example, Géry, 1969; Lundberg et al., 2000, 2010; Junk et al., 2007; Albert et al., 2011; van der Sleen and Albert, 2017). Some data published in scientific articles and journals were consulted to prepare the list of species of probable occurrence for the region.

Through the searches for data available for the ichthyofauna of the region of the REDD+ AgroSB project, 71 species were found, distributed in 21 families and 5 orders. Among the species studied, only one is mentioned in the lists of endangered fauna consulted, the amarelinho (Baryancistrus xanthellus), cited in the Brazilian list as "CR (Critically Endangered)" in the IUCN global list as "NT (Near Threatened). The yellowing fish is a species of shellfish of the Loricaridae family, a species that has wide interest for the aquarium trade, one of the great threats to the region's ichthyofauna. Regarding endemism, 43 species are endemic to the Amazon basin, which represents 60.5% of the species of possible occurrence for the region.

Entomofauna

The Amazon is one of the regions with the greatest diversity of insects in the world, housing a variety of habitats and ecosystems that favor the presence of numerous groups of insects (Addis & Junk, 2002). The diversity of lepidopteran species in the Amazon region is immense, with estimates of more than 50,000 species present in the area (Brown Jr. et al., 2018). This high diversity may be related to the wide variety of habitats and their complex geological history, which resulted in speciation and migration processes (Zanella et al., 2016).

The state of Pará, as part of the Amazon, has a great wealth of insect species, many of which are still unknown to science. Recent studies have shown that the region hosts a great diversity of species of moths, beetles, ants and bees, among other groups of insects (Silva et al., 2016; Santos et al., 2018; Souza et al., 2020).

The characterization of the entomofauna was carried out by consulting information available in surveys already carried out in the surrounding areas, in search of species with probable occurrence in the area. The Brazilian Biodiversity Information System (SiBBr - https://www.sibbr.gov.br/) was used to compose the list of species of probable occurrence in the area based on specimens collected from the State of Pará and cataloged in scientific collections.

According to the data consulted, 876 species representing the entomofauna were obtained, distributed in 90 families and 12 orders of possible occurrence in the study area.

5.1.2. High Conservation Values (B1.2)

High Conservation Values (HCV)[134] are a set of criteria to identify and evaluate areas and elements that are of significant importance for biodiversity conservation and human well-being. These criteria were developed by the HCV Resource Network, a global network of organizations working to promote the conservation and responsible use of natural resources.

HCVs are based on six main categories of conservation values, such as significant concentration of biodiversity, seasonal concentration of species, threatened and rare ecosystems, presence of endangered species, provision of essential ecosystem services, social, historical and cultural values, among others.

Through the diagnoses carried out in the project area, Fazenda Lagoa do Triunfo was classified as having high values 1 and 2, which deal with the concentration of endemic, rare and endangered species (HCV 1) and extensive ecosystems and mosaics at the landscape level, capable of maintaining viable populations of the vast majority of naturally occurring species (HCV 2).

High Conservation Value	HCV 1: according to diagnostics carried out in the project area, it was possible to identify 14 endemic and 10 threatened species of fauna. For the flora, there are 9 endangered and 12 endemic species. HCV 2: due to the presence of a significant population of castanheira trees in the Submontane Dense Ombrophilous Forest at Lagoa do Triunfo farm, in the position of an emerging tree, of very long life, of undisputed socioeconomic importance, of great ecological importance and threatened with extinction, forests containing this important attribute should be considered as areas with high conservation value.
Qualifying Attribute	The forest fragments and remnants of the Project Area and its area of influence are extremely important for the conservation of regional fauna, presenting a High Value for Conservation. They are home to species or subspecies of many faunal groups that have a restricted distribution area to the Tapajós EC, as well as several endangered species of fauna and flora.
Focal Area	It is noteworthy, based on the evidence pointed out, that the entire Project Area must be preserved, as it promotes the maintenance of fauna and flora communities on a regional scale, housing several endangered and endemic species. Populations of endangered and endemic species will be monitored with techniques that provide time series of abundance and/or occupation of said species.

[134] https://hcvnetwork.org/

5.1.3. Without-project Scenario: Biodiversity (B1.3)

In the Brazilian Amazon, the main responsible for deforestation is the removal of the forest for conversion into pasture and agricultural areas (DOMINGUES & BERMANN 2012). São Felix do Xingu, in the state of Pará, is located on the agricultural frontier and at the edges of the arc of deforestation in the Amazon. The municipality is annually among the highest deforestation rates and is among the ten that have lost the most vegetation since 2008 (INPE 2023; SCHINEIDER et al 2015). It was once the municipality with the largest cattle herd in Brazil (IBGE 2002) and high deforestation rates are related to the significant expansion of pasture areas (SCHINEIDER et al 2015).

The area of Fazenda Lagoa do Triunfo is part of the State Conservation Unit APA Triunfo do Xingu, which extends over 1.68 million hectares, with more than 2/3 in the municipality of São Félix do Xingu, according to State Decree No. 2612, of December 4, 2006 of the government of Pará[135]. APA Triunfo do Xingu is the conservation unit most pressured by deforestation in the Amazon (IMAZON, 2022)[136].

São Félix do Xingu presents serious problems related to illegal deforestation and the land issue, with many conflicts over land in the region. The vast majority of land comes from the purchase of lots and land grabbing. Due to conflicts and impunity in the region, São Félix do Xingu leads the list of municipalities with the highest numbers of murders linked to land conflicts in the national territory (Kawakubo et al. 2013).

According to data from MapBiomas[137] in 2020 in the list of Brazilian municipalities with the highest amount of deforestation, São Félix do Xingu (PA) was in the 2nd position, devastating 45,587 hectares of forests. In view of the widespread trend of conversion of land use from forests to pastures in the municipality, forest vegetation is under pressure and threats that cast doubt on the long-term permanence of forests in the region's landscape, whether on small, medium or large rural properties.

The trend in the region is illegal deforestation and conversion of land use from forests to pastures. This represents loss of biodiversity at all levels (ecosystems, species and genes). In a scenario WITHOUT the project, there may be conversion of land use from forest to pastures, degradation of ecosystem services related to the quantity, regularity of flow and quality of the water that drains from the hydrographic network of the Lagoa do Triunfo farm and increased greenhouse gas emissions will occur.

In addition, due to the coexistence of jaguars with cattle ranching operations in the region where the project is located, the conflict between this predator and cattle ranchers is frequent. As a consequence, despite being a species protected by law, stealth slaughter in retaliation for the damage that this cat causes is common practice, causing a significant reduction in its populations outside protected areas. In the municipality of São Félix do Xingu, this is the most serious threat to the conservation of the onça-pintada (Oliveira et al., 2013[138]). Data provided by Fazenda Lagoa do Triunfo I indicate the loss of cattle (115 heads) due to onça-pintada predation. This situation will also be addressed within this REDD+ project as referenced in section 2.1.11.

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[135] http://www.semas.pa.gov.br/2006/12/04/9672/
[136] https://imazon.org.br/imprensa/apa-triunfo-do-xingu-no-para-foi-a-area-protegida-da-amazonia-mais-pressionada-pelo-desmatamento-no-primeiro-trimestre/
[137] https://mapbiomas.org/regiao-norte-lidera-desmatamento-no-brasil
[138] Oliveira, T.; Ramalho, E.E.; De Paula, R.C. Amazon. In: Desdiez, A. et al. 2013. National action plan for the conservation of the jaguar. De Paula, R.C.; Desdiez, A.; Cavalcanti, S. (eds.). Brasília: Chico Mendes Institute for Biodiversity Conservation, ICMBio.

5.2. Net Positive Biodiversity Impacts

5.2.1. Expected Biodiversity Changes (B2.1)

Expectations of changes to biodiversity as a result of the Project were estimated using the theory of change method, also known as causal model, better defined in Section 2.1.1 in which all activities were described. From this definition, we can better glimpse the cause-and-effect relationship between the Project activities defined so far, the actions involved, their expected results and impacts in the short, medium and long term.

That said, all activities were defined with a view to promoting the long-term preservation of forest cover in the Project Area and, consequently, promoting changes in the future expectation of the region. For biodiversity, it is evident that the greatest changes, based on the scenario without a Project, are associated with the preservation of natural habitats, providing the maintenance and improvement of floristic and faunal species, in which they would be threatened by unplanned deforestation, and which, without the intervention of a REDD+ mechanism, tend to increase for the Amazon region.

Thus, ensuring forest protection through activities and through the REDD+ mechanism will enable the predicted changes to biodiversity to be positive. Still, as an essential tool for this, the biodiversity monitoring plan will make it possible to verify and ascertain whether any changes are impacting biological diversity in a negative way, portraying the photograph of the local biota throughout the life cycle of the Project, providing greater lucidity about the population dynamics of species, including endemic and vulnerable ones, as well as the conflicts generated by the coexistence of man and nature.

As a result, the expected changes to the Project's biodiversity are:

- Maintenance and addition of species identified in the scenario without Project.
- Ensure the conservation of habitats and species in the Brazilian Amazon.
- Reduce illegal activities through Project activities.
- Increase awareness of environmental issues in local communities, reducing conflicts between communities and biodiversity.
- Ensure the conservation of endangered and endemic plant and animal species.
- Promote an ecologically balanced environment.
- Maintain landscape connectivity to biological diversity.

General Element	Ateles REDD+ Project Activities – General Scope
Estimated Change	Strengthened management and monitoring of Project implementation
Justification of Change	Partnerships to support biodiversity conservation, identification of bottlenecks, threats and negative impacts of the Project, iteration and mutual collaboration between all stakeholders in favor of biodiversity and joining efforts between various actors to maintain biodiversity in the region.

Climate Element	Ateles REDD+ Project Activities – Climate Scope

Estimated Change	Reduction of deforestation and forest degradation
Justification of Change	Maintenance of forest cover, satellite image deforestation monitoring, development of heritage surveillance, maintenance of carbon stocks, reduction of loss of landscape connectivity and promotion and maintenance of ecologically balanced habitats and environments.

Community Element	Ateles REDD+ Project Activities – Community Scope
Estimated Change	Resource alternatives and environmental awareness
Justification of Change	Increased environmental awareness about hunting, fishing and illegal activities for local communities, courses and training increasing the educational base, improvement and development of more sustainable practices applied to production bases, exploitation of non-timber forest products in a conscious way, improvement of the management of natural resources by local communities and reduction of human activities that do not comply with biodiversity conservation.

Biodiversity Element	Ateles REDD+ Project Activities – Biodiversity Scope
Estimated Change	Biodiversity maintenance
Justification of Change	Monitoring of population dynamics, greater long-term knowledge of regional megadiversity, identification of abnormal patterns and more assertive interventions if necessary.

5.2.2. Mitigation Measures (B2.3)

The data collected for studies related to biodiversity were satisfactory in order to evaluate the current context of biodiversity conservation in the Project Zone, surroundings, and focusing on the Project Area. However, longer-term studies are needed to elucidate the variations that occur in the biotic community throughout forest changes, whether arising from the decrease in forest area in the Project Zone and external to the Project Area, climate change or management activities, in order to better understand its dynamics (HENRIQUES et al., 2008).

Also, according to the Brazilian Diagnosis of Biodiversity & Ecosystem Services, the main vectors of threat to biodiversity are: a) climate change, which alters the configuration and functioning of ecosystems, and land use changes — in other words: deforestation, or any activity that involves the conversion of native vegetation areas to other uses, such as agriculture, livestock or mining; and b) the introduction and spread of invasive alien species that lead to the loss of native species and changes in interspecific relationships, ecological processes and the provision of ecosystem services.

In order to seek improvement in the population conditions of the species and mitigation of the impacts caused by internal and external factors exposed above, all activities of the Project, especially "biodiversity monitoring", were designed and structured to act as mitigating measures of the main threats to biodiversity, in addition to mitigating against negative adverse factors in the conservation and maintenance of HCVAs.

We can cite as an example the presence of the Boar, an exotic and invasive species, which will be monitored during the course of the project to understand the most effective action to control the species, avoiding damage to local biodiversity. Understanding the dynamics of the Boar is very important to avoid the negative impacts resulting from a hasty decision, such as the search for authorization to hunt boars – a release that may end up encouraging the hunting of other animals within the project area.

With this in mind, although no negative impacts on biodiversity are expected, a constant evaluation of the results obtained through the monitoring of activities ensures that the project will be able to take faster action if negative impacts are identified.

5.2.3. Net Positive Biodiversity Impacts (B2.2, GL1.4)

The REDD+ project is the most favorable for wildlife conservation among terrestrial carbon initiatives. In addition to the main objective of reducing the impacts of climate change, effective actions to protect against deforestation and forest degradation (REDD+) bring benefits to ecosystem services, such as biodiversity conservation and benefits on a local and regional scale (MILES & DICKSON, 2010; PITMAN, 2011).

As a positive effect of the implementation of the project, we can mention the preservation of the forest areas of the forest areas of the Lagoa do Triunfo farm. Species with records for the study area are mostly dependent on conserved forest formations, in addition to having extensive living areas. That said, it is providential to maintain these large forest areas, in addition to ecological corridors to ensure the connection between the forested areas, thus assisting in the flow of individuals and regional gene flow.

Another positive impact is the monitoring of the fauna that are associated with the implementation of the REDD+ project, allowing the monitoring of population trends of species that occur within the area and its surroundings. Thus, it is possible to choose species that respond quickly to changes in their habitat and are useful to measure the efficiency of the project in maintaining forested areas and "High Conservation Value" attributes for fauna.

5.2.4. High Conservation Values Protected (B2.4)

The Project Area has two attributes of High Conservation Value related to biodiversity, which have already been described in Section 5.1.2 and is related to i) areas that contain significant concentrations of biodiversity values on a global, regional or national scale and ii) large areas on a global, regional or national level, in which there are viable populations of most, if not all naturally occurring species, in natural patterns of distribution and abundance.

The proposed measures to guarantee integrity of HCV's and thus maintain and improve these attributes, ensuring that they are not negatively affected by the Project, were considered and incorporated by the defined activities of the Project. Therefore, no negative impact was foreseen for these areas and, furthermore, the proposed activities as well as their implementation throughout the life of the Project will generate positive impacts on these attributes.

Thus, it is not expected that the attributes of HCV's will be negatively affected by the Project. On the contrary, by reducing deforestation and forest degradation in the Project Area, what is expected is the preservation of intact and appropriate habitats for the entire biotic community, including the recovery of ecological niches for endemic, rare or threatened species.

5.2.5. Species Used (B2.5)

In the project area there is the presence of settlers who have small orchards, ornamental and medicinal plants in their surroundings, as well as families that specialize in cocoa and livestock farming, in addition to other forest and white crop species, such as corn, cassava and bananas. In addition, these communities practice small-scale extraction of Brazil nuts and açaí, mainly for their own consumption.

The Ateles REDD+ Project seeks to encourage these economic activities in a sustainable way, adopting practices that minimize environmental impacts and promote the regeneration of the local forest. Proper management of orchards and ornamental and medicinal plants can contribute to biodiversity conservation, as well as provide food and medicine for local communities. Specialization in cocoa and livestock, combined with the practice of agroforestry, can increase the productivity and profitability of properties, while contributing to forest conservation.

5.2.6. Invasive Species (B2.5)

In the Flora diagnosis carried out in the Project Area, showing the without-project scenario for biodiversity, described in Section 5.1.1. of this document, it is possible to infer the non-occurrence of invasive exotic forest species in the area, since there was no record of exotic forest species with a consolidated popular name such as eucalyptus, pine, Australian acacia, leucene, among others, associated with the fact that the evaluated remnant has a well-conserved forest structure, which acts as a filter, preventing the entry of opportunistic exotic plant species.

It is worth mentioning that, within Fazenda Lagoa do Triunfo, there are some areas dedicated to beef cattle in which non-native grasses are used. Such species, without proper management, can act as invasive species. Due to the premises of the Project, such areas were withdrawn, but it is evident that they are very close to the massive forests that make up the Project Area. To date, there is no indication or evidence that such species affect and or act as invasive species beyond the areas intended for livestock within the limits of Fazenda Lagoa do Triunfo.

However, invasive alien species were identified for the fauna in the field campaign carried out in the project area, and it is possible to verify the frequent presence of javali (Sus scrofa), a species from Europe, Asia and North Africa, introduced in Brazil in the 1960s for meat production and consumption. The wild boar is classified as one of the hundred worst invasive alien species in the world by the International Union for Conservation of Nature. Its aggressiveness and ease of adaptation are characteristics that, associated with uncontrolled reproduction and the absence of natural predators, result in a series of environmental and socioeconomic impacts, especially for small farmers and other species that compete for the same resources such as native pigs, catetus (Pecari tajacu) and queixada(Tayassu pecari). In addition, wild boar poses a health and environmental challenge to society, as it is a large reservoir of diseases that can compromise both human and wildlife health, in addition to imposing massive pressures on the environment in the form of predation of native species of plants and animals and through contamination and destruction of springs and watercourses in the region.

In addition, the presence of domestic dogs and cats was also observed in places close to small villages. These animals are also considered exotic species in native forests and can influence wild species in several aspects, for example through competition for food, shelters, among others.

That said, it is worth mentioning that these exotic animals are included in the monitoring of the Project, so that, based on more information, it is possible to make the most assertive decision in order to guarantee the safety of the biodiversity existing in the project area.

In addition, it is not foreseen, as well as the previous section and based on the proposed activities, to incorporate exotic species in the Project Area, since they are globally known as the second main threat to biodiversity and may compromise the balance of the ecosystem in which they are inserted. Therefore, in the same way as Section 2.1.11, the Project activities will not foster the inclusion of exotic species and the

Project Area will not be affected by the introduction of this type of species, not resulting in any threat or increase as a result of the Project.

5.2.7. Impacts of Non-native Species (B2.6)

As specified in Section 4, the Ateles REDD+ Project Area has communities, formed by family farmers, small producers, squatters, farmers or medium producers, who use non-native species, mainly for subsistence. However, the main sources of income for producers are mostly based on the production and marketing of cattle, milk and cheese; the raising of small animals; and land maintenance and preparation services on neighboring farms.

The few non-native species are, however, used by local communities, that is, on a small scale and do not have an adverse impact on the environment. These species have been cultivated in the form of small orchards, near their homes, as well as ornamental plants and medicine, and, until then, even without the insertion of a REDD+ Project, they have not resulted in negative implications for the area, given the great resilience of the forest. In addition, the use of non-native species will not be encouraged by the Ateles REDD+ Project. It is worth mentioning that this applies only to the Project Zone, and exotic and invasive species are not foreseen or incorporated in the Project's activities.

However, Fazenda Lagoa do Triunfo has some areas for beef cattle, in which non-native grass species are used, but which are managed correctly, not extending beyond their limits. Nevertheless, throughout the implementation of the Project this should be monitored and, if there is the identification and adverse effects of these species on the forest component, they will be reported in the monitoring report.

Therefore, it can be said that no impact is foreseen in relation to non-native species.

5.2.8. GMO Exclusion (B2.7)

Through the Ateles REDD+ Project it is ensured that no genetically modified organisms (GMOs) will be used. The reduction or removal of greenhouse gas emissions will be achieved by reducing deforestation and forest degradation.

5.2.9. Inputs Justification (B2.8)

In the Ateles REDD+ Project Zone, where the activities are intended, there is no intention to use any chemical pesticide, biological control agent or other types of chemical inputs in the activities to be implemented.

In case there is the application of any chemical compound, or the use of biological control agents or any other type of input by the responsible parties, they will be reported in the monitoring report.

5.2.10. Waste PRODUCTS (B2.9)

Waste management in São Félix do Xingu, municipalities where the project area is located, presents significant challenges, as pointed out by data from the National Sanitation Information System (SNIS) of the Ministry of Cities in 2020. The municipality lacks a Municipal Plan for Integrated Solid Waste Management (PMGIRS), which is contrary to Federal Law No. 12.305/2010, which establishes the National Solid Waste Policy. The absence of such a plan may hinder the implementation of effective waste management practices, including selective collection. In addition, the distance from the project area to the municipality headquarters makes it even more difficult to dispose of various types of waste.

Nevertheless, AgroSB Agropecuária, through its own means, manages waste in the units through the basic principles of minimizing waste generation, identifying and describing the actions related to its proper management, taking into account the aspects related to all stages, including generation, segregation, packaging, identification, collection, internal transport, temporary storage, internal treatment, external storage, external collection and transport, external treatment and final disposal of waste.

In the units/farms, the waste generated in greater quantity and with polluting potential, are stored in duly adequate places to avoid contamination and destined as provided for by the legislation.

For pesticide packaging, storage is done in the pesticide warehouse with a separate place for the temporary storage of empty packaging. The return of these packages is made directly to the receiving stations located in the region registered with the National Institute of Empty Packaging Processing - INPEV.

The waste generated from the oil change of the machines is temporarily stored in suitable containers and destined according to the demand and quantity for commercialization with recycling companies, with service providers for the collection of hazardous waste or for recycling cooperatives.

Below is a summary table of the main waste generated in the units and their final destination:

	Waste 1	Waste 2
Waste Generated	Used oil, contaminated packaging, filters and tow	Defensive Packaging
Generation Point	Workshops	Agriculture/Livestock
Packaging	Aerial tanks and specific dumpsters	Stacked on pallets
Storage	Distributed in the workshops	Deposits of pesticides
Collection Frequency	To be performed according to the quantity packaged	To be performed according to the quantity packaged
Final Disposal	Recycling / Marketing Cooperative/ Outsourced company	Receiving stations or Centers

In addition, all employees involved in waste management, from segregation to final disposal, are properly trained and undergo regular medical examinations, according to NR 07, focusing on the prevention of diseases such as tetanus, tuberculosis and hepatitis. Practices that ensure the safety of professionals are adopted, following local guidelines and cleaning standards. And strict hygiene and safety measures are implemented, providing an effective working environment and minimizing occupational risks, with the provision of Personal Protective Equipment (PPE) to employees, such as gloves, masks and aprons, to prevent accidents.

For the activities proposed by the Project, no additional generation of waste is foreseen to those normally generated by the operations already performed by AgroSB. However, if there is the production of any material that needs to be properly disposed of, the PGRS will be followed and all environmental laws will be considered.

5.3. Offsite Biodiversity Impacts

5.3.1. Negative Offsite Biodiversity Impacts (B3.1) and Mitigation Measures (B3.2)

In general, a REDD+ Project, based on the premises of the VM0015 methodology, has as its central objective the reduction of emissions from unplanned deforestation and forest degradation. That said, it is undeniable that the conservation provided by this mechanism is considered the largest mitigating measure, benefiting the Project Zone as a whole.

Specifically, to the Ateles REDD+ Project, the promotion of greater control in the occurrence of anthropogenic disorders that would negatively impact biodiversity, such as habitat loss due to deforestation for the practice of predominant economic activities in the region such as agriculture, livestock and timber extraction, would act in an even more representative way in mitigating negative adverse effects. However, based on the strengthening of measures aimed at conserving the forest and its resources, it may be that these disturbances and activities will naturally move to areas outside the Project Zone, where they are more vulnerable to such events.

Regarding the mitigation measures arising from the Project, we can mention the permanence and strengthening in the Project Area of alternative economic activities that generate income and employment, such as the promotion of sustainable practices, and education and awareness of the importance of biodiversity, that is, the social activities of the project were elaborated in order to mitigate possible negative impacts, with emphasis on environmental education, forest appreciation and the sustainable use of forest resources for local communities.

It is worth mentioning that all Project activities were discussed and refined in discussions (Section 2.3.7) with community members for their validation and understanding of conservation efforts.

In addition, although negative impacts on off-site biodiversity are unlikely, the Project will seek mitigation through the articulation of proponents, in which they must practice adaptive management and collectively deal with any additional negative impact on off-site biodiversity that is identified later, articulating and promoting partnerships that increase such efforts.

5.3.2. Net Offsite Biodiversity Benefits (B3.3)

The main positive impact expected outside the Project Zone is the favoring of biodiversity by maintaining large forest remnants that can function as ecological corridors, which will serve as refuge and protection for threatened species and ecosystems and ecological processes can occur without any human intervention or only from sustainable use. Thus, despite the possible displacement of activities and disturbances outside the Project Zone, which can cause occasional negative impacts, landscape connectivity and favoring gene flow between forest areas, habitat maintenance and in situ monitoring of threatened species are benefits that justify the presence of the Ateles REDD+ Project in the region.

5.4. Biodiversity Impact Monitoring

5.4.1. Biodiversity Monitoring Plan (B4.1, B4.2, GL1.4, GL3.4)

Biological diversity, or biodiversity, is characterized by the variety of life on Earth, encompassing all species of plants, animals and microorganisms, all genetic variability among species, and also all the diversity of terrestrial and aquatic ecosystems – continental and marine – and the ecological complexes of which they are part (Secretariat of the Convention on Biological Diversity 2010)[139].

Although Brazil, especially the Amazon biome, has a rich biodiversity, the exact knowledge and identification of all species are still quite scarce. That said, monitoring the entire biodiversity of a given region or location is impossible in logistical, financial and knowledge terms available and adequate. According to the "Proposal for a Brazilian Biodiversity Monitoring System"[140], a reasonable way to remedy this problem, making biodiversity monitoring more practical and feasible, is to monitor specific groups of animals and plants in which they respond in a calculable way to environmental changes. These groups of animals and plants that have such potential are called biological indicators and, for them to be considered good indicators of fact, three characteristics are taken into account: high rationality, high performance and high possibility of implantation.

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[139] Secretariat of the Convention on Biological Diversity (2010) Global Biodiversity Outlook 3. Montreal.
[140] COSTA-PEREIRA, R.; ROQUE, F. O.; CONSTANTINO, P. A. L.; SABINO, J.; UEHARA-PRADO, M. **In situ monitoring of Biodiversity: Proposal for a Brazilian Biodiversity Monitoring System.** 1. ed. Brasília: ICMBio. v. 1, 2013. 61p.

The methodological script considers four reasonable groups of biological indicators to be monitored: tree plants, selected groups of birds, medium and large mammals and frugivorous butterflies. It is important to remember that to define such groups, their sensitivity to climate change issues was taken into account.

a) Technical description of monitoring tasks

Taking the aforementioned roadmap into account, the biodiversity monitoring plan of the Ateles REDD+ Project will focus on biological indicators that should be monitored and evaluated in the face of forest modifications and anthropogenic actions, whether arising from the decrease in forest area in the Project Zone or from climate change, in order to better understand its dynamics and track the impact of these elements on biological diversity.

In the scenario without the Project, described in Section 5.1, the diagnosis considered different sampling areas. Therefore, this plan will seek, whenever possible, to carry out expeditions to monitor biodiversity considering the same conditions and methodologies used to diagnose the without-project scenario, in which it is possible to compare, observe and measure the effects of the Project's activities on biodiversity over time.

That said, for the monitoring of fauna, the Project will carry out expeditions (to be established), considering the periods of low and high rainfall, so that the presence of migratory species and reproductive periods are considered. As for the monitoring of the flora, considering that the dynamics and floristic changes occur more slowly, the expeditions will be made in periods close to each verification of the Project in order to evaluate any changes that may arise in the structure and composition of the floristic species.

In addition, it is expected that species of relevance, whether endemic, threatened or rare, will be prioritized in the expeditions and collected during the monitoring campaigns, especially the key species, considered of great biological importance and high degree of threat, according to the IUCN criteria. In addition, we emphasize that exotic species registered in the biodiversity diagnosis, such as wild boar and domestic animals, will be duly considered in the Project monitoring process. Although so far we have not identified negative impacts arising from their presence, we recognize that these species pose a potential risk to local diversity. Therefore, it is essential to closely monitor its interaction with native biodiversity.

Finally, the plan will seek to protect High Conservation Value (HCV) attributes by stimulating and enhancing knowledge about local biodiversity through continuous in situ monitoring. The evaluation of the effectiveness of the measures adopted to maintain and improve HCVs is covered by the tasks described, especially in the monitoring of indicators related to species richness of both fauna and flora. Thus, the monitoring plan allows monitoring the permanence of relevant species, whether threatened, rare or endemic in these areas defined as priorities for conservation.

b) Data to be collected

Data / Parameter	Number of documents focused on biodiversity scope
Data unit	Number
Description	This parameter will be responsible for accounting for the amount of all material produced, from presentations or didactic tool, designed to implement and monitor the Project's biodiversity scope
Source of data	Calculated through reports, meeting minutes, monitoring guides and memos, focused on issues related to the biodiversity scope of the Project (biodiversity monitoring)

Description of measurement methods and procedures to be applied	All documents produced for the Project will be stored in digital files throughout the Project's accreditation period. Thus, these reports from the "Biodiversity monitoring" activity will be monitored and accounted for
Frequency of monitoring/recording	Annual
Value applied	To be accounted for after registration of the Project
Monitoring equipment	Not applicable
QA/QC procedures to be applied	The information systematized in these documents will be validated between the bidders, allowing greater reliability and quality of the data. In addition, the Project will undergo continuous evaluation of the information generated, allowing the identification of improvements in the collection and registration processes, and the incorporation of these in the strategic planning of the Project when they are identified
Purpose of data	Not applicable
Calculation method	Not applicable
Comments	-

Data / Parameter	Morpho-species richness of the monitored plant community
Data unit	Number
Description	Number of morphospecies found in the monitoring being the fundamental unit for the evaluation of the homogeneity of the monitored environment. Due to the great difficulty of identifying an individual at a specific level, the parameter is focused on morpho-species, in which they are defined by morphological similarity.
Source of data	Field sheets, forest inventories, monitoring reports, additional records
Description of measurement methods and procedures to be applied	To be established (ideally similar to what was done in the initial diagnoses)
Frequency of monitoring/recording	One year before a verification event
Value applied	To be accounted for after registration of the Project
Monitoring equipment	Not applicable
QA/QC procedures to be applied	To be established
Purpose of data	Datasheet
Calculation method	-

Comments	Field sheets, forest inventories, monitoring reports, additional records

Data / Parameter	Percentage of threatened species monitored
Data unit	Percentage
Description	Monitoring of species of relevance to the Project's fauna in relation to their status on the IUCN List of Endangered Species, with emphasis on species cited as critically endangered (CR) or endangered (E). In addition to species considered exotic, such as wild boar and domestic animals
Source of data	Field worksheets, photographs, monitoring reports, additional records
Description of measurement methods and procedures to be applied	Systematization and comparison of data and information collected in fauna expeditions with the IUCN Official List, available at: https://www.iucnredlist.org/
Frequency of monitoring/recording	
Value applied	To be established.
Monitoring equipment	Not applicable
QA/QC procedures to be applied	Not applicable
Purpose of data	Comparison of the different sources of information available: empirical survey, secondary data, official lists and traditional knowledge
Calculation method	Datasheet
Comments	-

Data / Parameter	Percentage of actions taken or materials developed for awareness and sanitation measures with domestic animals
Data unit	Percentage
Description	This parameter measures the number of initiatives implemented or materials produced to raise awareness among residents of the region about the importance of adopting adequate sanitary measures and promoting the health and well-being of domestic animals. These actions may include awareness campaigns, educational workshops, distribution of informational materials, creation of good practice guides, among others.

Source of data	Calculated through reports, minutes of meetings, guides to good practices, attendance list of workshops, memoranda and informational materials focused on matters related to the scope of "Sanitary precautions with domestic animals"
Description of measurement methods and procedures to be applied	All documents produced for the Project will be stored in digital files throughout the Project's accreditation period. Thus, these reports from the activity of "Sanitary precautions with domestic animals" will be monitored and accounted for
Frequency of monitoring/recording	Annual
Value applied	To be accounted for after registration of the Project
Monitoring equipment	Not applicable
QA/QC procedures to be applied	The information systematized in these documents will be validated between the bidders, allowing greater reliability and quality of the data. In addition, the Project will undergo continuous evaluation of the information generated, allowing the identification of improvements in the collection and registration processes, and the incorporation of these in the strategic planning of the Project when they are identified
Purpose of data	Not applicable
Calculation method	Not applicable
Comments	-

Data / Parameter	Percentage of materials developed to reduce cattle predation per ounce
Data unit	Percentage
Description	This parameter measures the amount of materials produced with proposals to decrease cattle predation per ounce.
Source of data	Calculated through reports, best practice guides, memos and informational materials focused on subjects related to the scope of “Alternatives to cattle predation”
Description of measurement methods and procedures to be applied	All documents produced for the Project will be stored in digital files throughout the Project's accreditation period. In this way, these reports from the activity of "Alternatives to livestock predation" will be monitored and accounted for
Frequency of monitoring/recording	Annual
Value applied	To be accounted for after registration of the Project

Monitoring equipment	Not applicable
QA/QC procedures to be applied	The information systematized in these documents will be validated between the bidders, allowing greater reliability and quality of the data. In addition, the Project will undergo continuous evaluation of the information generated, allowing the identification of improvements in the collection and registration processes, and the incorporation of these in the strategic planning of the Project when they are identified
Purpose of data	Not applicable
Calculation method	Not applicable
Comments	-

5.4.2. Biodiversity Monitoring Plan Dissemination (B4.3)

The monitoring plan, as well as the results obtained by monitoring the Ateles REDD+ Project, will be made available to the public through a page on the official website of Biofílica Ambipar Environment Investments. The summary documents related to the monitoring plan and results, as well as relevant information, will be made available to communities and stakeholders through feedback, meetings, lectures and by physical means on the premises of Fazenda Lagoa do Triunfo, belonging to AgroSB.

5.5. Optional Criterion: Exceptional Biodiversity Benefits

5.5.1. High Biodiversity Conservation Priority Status (GL3.1)

As they are located within the Tapajós Endemism Center (CE), the forest fragments and remnants of the Project Area and its area of influence are extremely important for the conservation of regional fauna, housing species or subspecies of many fauna groups that have a restricted distribution area to this CE. Such areas therefore have a High Value for Conservation, since, due to the enormous rates of deforestation in the region, many of these species are mentioned in the state, national and global lists of endangered fauna.

That said, the presence of threatened species of flora and fauna found in the without-project scenario was verified according to the IUCN Red List of Threatened Species:

Flora

- **Endangered (EN)** – Handroanthus serratifolius (Ipê-amarelo) and Euxylophora paraensis (Pau-amarelo)

Fauna

- **Endangered** - Ateles marginatus (Macaco-aranha-da-cara-branca) and Pteroglossus bitorquatus (araçari-de-pescoço-vermelho)

5.5.2. Trigger Species Population Trends (GL3.2, GL3.3)

The key species and their respective population trends for the Ateles REDD+ Project can be found below.

- Flora:

Trigger Species	Handroanthus serratifolius - Ipê-amarelo.

Population Trend at Start of Project	The *Handroanthus serratifolius*, Ipê-amarelo, is threatened by forest exploitation that has caused significant population decline. It is one of the most vulnerable forest species in the Amazon, due to its low natural density and slow growth (IUCN, 2023[141]).
Without-project Scenario	Without a carbon project (REDD+) to protect the forests where Handroanthus serratifolius (Yellow Ipe) is found, CNCFlora (2022) highlights that deforestation and forest exploitation would continue to be the main threats to the species. The region of São Félix do Xingu, in the state of Pará, has a history of deforestation and is one of the most critical areas for the conservation of the Amazon.
With-project Scenario	The Ateles REDD+ Project with its purpose of mitigation, reduction of deforestation and forest degradation will promote the increase or at least the population maintenance of the species. The conservation of the forest massif should also contribute to the connectivity of the landscape, favoring the gene flow between the forest areas in the vicinity.

Trigger Species	Euxylophora paraensis – Pau-amarelo
Population Trend at Start of Project	Popularly known as Pau-amarelo, this species occurs naturally at low densities in preserved forests, and as a result of logging and deforestation for settlements and agricultural activities, population decline is suspected in the last three generations (about 20 years each generation) equal to or greater than 50% (IUCN, 2023[142]).
Without-project Scenario	The wood of this species, of high commercial value and with several applications, is intensively exploited in Pará (CNCFlora, 2022)99. Forest fragmentation caused by increased deforestation without the implementation of the Project tends to cause a drastic reduction in species richness, whose density and distribution is lower in small fragments.
With-project Scenario	The Ateles REDD+ Project with its purpose of mitigation, reduction of deforestation and forest degradation will promote the increase or at least the population maintenance of the species. The conservation of the forest massif should also contribute to the connectivity of the landscape, favoring the gene flow between the forest areas in the vicinity.

- Avifauna:

Trigger Species	Pteroglossus bitorquatus - araçari-de-pescoço-vermelho

[141] https://www.iucnredlist.org/species/61985509/145677076#threats
[142] https://www.iucnredlist.org/species/61958701/176125940

Population Trend at Start of Project	The species is considered Endangered (EN) by the IUCN (2016) and is restricted to the Belém Endemism Center, meeting the criteria defined by the CCB standard. There was a significant reduction in their populations due to habitat loss. This species was recorded during the field samplings carried out for the Environmental Diagnosis of the Project Area.
Without-project Scenario	Forest fragmentation caused by increased deforestation without the implementation of the project tends to cause a drastic reduction in species richness, whose density and distribution is lower in small fragments, mainly affecting more specialized taxa (LAURANCE and VASCONCELOS, 2009)100, many of which are threatened, endemic or have restricted distribution.
With-project Scenario	The Ateles REDD+ Project with its purpose of mitigation, reduction of deforestation and forest degradation will promote the increase or at least the population maintenance of the species. The conservation of the forest massif should also contribute to the connectivity of the landscape, favoring the gene flow between the forest areas in the vicinity.

- Mastofauna:

Trigger Species	*Ateles marginatus* – Macaco-aranha-da-cara-branca
Population Trend at Start of Project	The species is considered Endangered (EN) by the IUCN (2019) due to the decrease in its habitat and population declines due to hunting.[143]
Without-project Scenario	According to ICMBio/MMA, the São Felix do Xingu region has shown a significant deforestation rate in recent years, which indicates that the natural habitat of the white-faced spider monkey has been degraded. Habitat loss and forest fragmentation negatively affect the survival of the species, reducing the availability of food, shelter and breeding areas. The human presence in the area can also affect the species, as hunting and capture of wild animals are common practices in the region.
With-project Scenario	The Ateles REDD+ Project with its purpose of mitigation, reduction of deforestation and forest degradation will promote the increase or at least the population maintenance of the species. The conservation of the forest massif should also contribute to the connectivity of the landscape, favoring the gene flow between the forest areas in the vicinity.

Trigger Species	Tayassu pecari – Queixada

[143] https://www.iucnredlist.org/species/2282/191689524

Population Trend at Start of Project	This species has been considered Vulnerable (VU) to extinction by the IUCN since 2012, because of the ongoing population reduction estimated at around 30% in the last three generations (18 years) due to habitat loss, poaching, competition with livestock and epidemics. During the campaign to diagnose biodiversity, flocks were found, which justifies their choice as a trigger species by the CCB.
Without-project Scenario	Describe the most likely changes in the non-design land use scenario
With-project Scenario	The Ateles REDD+ Project with its purpose of mitigation, reduction of deforestation and forest degradation will promote the increase or at least the population maintenance of the species. The conservation of the forest massif should also contribute to the connectivity of the landscape, favoring the gene flow between the forest areas in the vicinity.